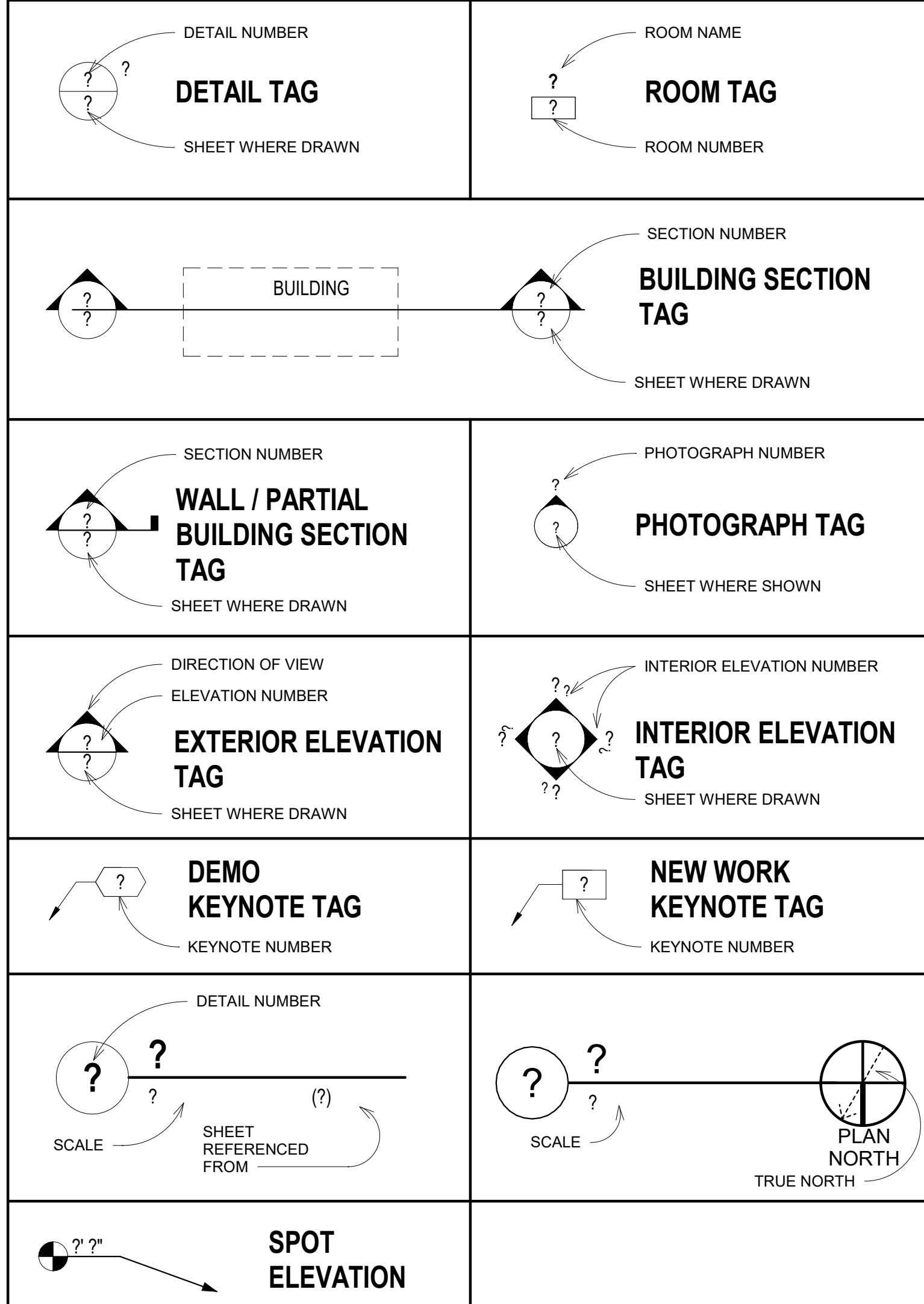


MECHANICAL LEGEND

- | | |
|-------|---|
| | THERMOSTAT |
| | HUMIDITY SENSOR |
| | BOILER SHUTDOWN SWITCH |
| | ATFP EMERGENCY SHUTDOWN SWITCH |
| 12x10 | RECTANGULAR DUCT DIMENSIONS (INCHES) |
| 12/10 | FLAT OVAL DUCT DIMENSIONS (INCHES) |
| 12"Ø | ROUND DUCT DIMENSION |
| | FLEXIBLE ROUND DUCT (5'-0" MAX LENGTH) |
| | DOWNWARD DIRECTION OF SLOPED PIPING |
| | SUPPLY DUCT, SUPPLY DUCT DOWN/UP |
| | RETURN DUCT, RETURN DUCT DOWN/UP |
| | GATE VALVE |
| | GLOBE VALVE |
| | BUTTERFLY VALVE |
| | CHECK VALVE |
| | PRESSURE REDUCING VALVE |
| | MANUAL CALIBRATED BALANCING VALVE |
| | DRAIN VALVE WITH HOSE CONNECTION |
| | 2-WAY CONTROL VALVE |
| | 3-WAY CONTROL VALVE |
| | RELIEF VALVE |
| | BACKFLOW PREVENTER |
| | MANUAL AIR VENT |
| | AUTOMATIC AIR VENT |
| | VACUUM BREAKER |
| | WYE STRAINER |
| | VALVE IN VERTICAL PIPING |
| | STRAINER WITH BLOWDOWN VALVE |
| | CLEAN OUT |
| | ECCENTRIC REDUCER |
| | PIPE REDUCER |
| | UNION |
| | THERMOMETER IN PIPE WELL |
| | PRESSURE GAUGE WITH SHUTOFF COCK |
| | PIPE CAP |
| | FLANGE |
| | FLEXIBLE PIPE CONECTOR |
| | RISE (R)/DROP (D) IN DIRECTION OF FLOW |
| | FIRE DAMPER |
| | FIRE/SMOKE DAMPER |
| | SMOKE DAMPER |
| | ACCESS DOOR (AD), ACCESS PANEL (AP) |
| | PARALLEL BLADE CONTROL DAMPER, MOTOR OPERATED |
| | OPPOSED BLADE CONTROL DAMPER, MOTOR OPERATED |
| | MANUAL BALANCE DAMPER |
| | BACKDRAFT DAMPER |
| | AIRFLOW MEASURING STATION |

GENERAL PROJECT SYMBOLS



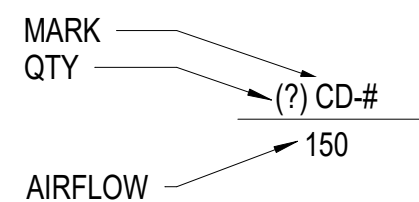
HVAC DESIGN CRITERIA

LOCATION: MCAS CHERRY POINT, NC	LATITUDE: 34.90 N	CLIMATE ZONE 3A		
	LONGITUDE: 76.88 W			
STATION ID# ICAO_KNKT	ELEVATION: 29 FT.			
STANDARD OUTDOOR DESIGN CONDITIONS:				
WINTER DESIGN DRY BULB (99.6%):	27°F	1% HR/MCWB DRY BLUB 84.0°F		
SUMMER DESIGN DRY BULB (0.4%):	93°F	1% HR/MCWB WET BLUB 78.7°F		
SUMMER DESIGN WET BULB (0.4%):	78°F			
CONDENSING UNIT (0.4% DB + 5°F):	98°F			
ROOM TYPE:	STANDARD INDOOR DESIGN CONDITIONS:			
	SUMMER DESIGN DRY BULB:	SUMMER DESIGN HUMIDITY:	WINTER DESIGN DRY BULB:	WINTER DESIGN HUMIDITY:
TYPICAL SPACES: OFFICES, AIR TERMINAL, ETC.	78°F	50% RH	68°F	NC*
ATC CRITICAL SPACES	75°F	60% RH	71°F	35% RH
RANGE EQPT	72°F	50% RH	70°F	30% RH
RANGE OPS	72°F	50% RH	70°F	NC*
FIRE / MECH / ELEC ROOMS	90°F	38% RH	50°F	UC*
UPS ROOMS	80°F	50% RH	50°F	30%
STAIR	80°F	UC*	50°F	UC*
ELEV MACHINE RM	90°F	60% RH	70°F	35% RH
*UC - UNCONTROLLED				
*NC - HUMIDITY NOT DIRECTLY CONTROLLED				

MECHANICAL PIPING LEGEND

CHWR	CHILLED WATER RETURN PIPING
CHWS	CHILLED WATER SUPPLY PIPING
D	CONDENSATE DRAIN PIPING
DCW	DOMESTIC COLD WATER PIPING
G	NATURAL GAS PIPING
HHWR	HEATING HOT WATER RETURN PIPING
HHWS	HEATING HOT WATER SUPPLY PIPING
HRHWR	HEAT RECOVERY HOT WATER RETURN PIPING
HRHWS	HEAT RECOVERY HOT WATER SUPPLY PIPING
STM	STEAM PIPING
SC	STEAM CONDENSATE PIPING

AIR DISTRIBUTION DEVICE IDENTIFICATION



MECHANICAL ABBREVIATIONS

°	DEGREES
Ø	DIAMETER
AD	ACCESS DOOR
AFF	ABOVE FINISHED FLOOR
AHU	AIR HANDLING UNIT
AP	ACCESS PANEL
APD	AIR PRESSURE DROP
AS	AIR SEPARATOR
ATFP	ANTITERRORISM FORCE PROTECTION
BD	BACKDRAFT DAMPER
BHP	BRAKE HORSEPOWER
BOD	BOTTOM OF DUCT
BTUH	BRITISH THERMAL UNITS PER HOUR
BSS	BOILER SHUTDOWN SWITCH
C	COMMON
CA	COMBUSTION AIR
CD	CEILING DIFFUSER
Cv	LOSS COEFFICIENT
CFM	CUBIC FEET PER MINUTE
CLG	CEILING
CND	CONDENSER
COP	COEFFICIENT OF PERFORMANCE
CRAH	COMPUTER ROOM AIR HANDLING UNIT
CU	CONDENSING UNIT
CUH	CABINET UNIT HEATER
CHW	CHILLED WATER
DALT	DUCT AIR LEAKAGE TESTING
DB	DRY BULB
DDC	DIRECT DIGITAL CONTROL
DEG	DEGREE
DOAH	DEDICATED OUTSIDE AIR HANDLING UNIT
DX	DIRECT EXPANSION
EAT	ENTERING AIR TEMPERATURE
EC	ELECTRONICALLY COMMUTATED
ECM	ELECTRONICALLY COMMUTATED MOTOR
EER	ENERGY EFFICIENCY RATIO
EF	EXHAUST FAN
EFF	EFFECTIVENESS
EG	EXHAUST GRILLE
ESP	EXTERNAL STATIC PRESSURE
EWI	ENTERING WATER TEMPERATURE
F	FARENHEIT
FCU	FAN COIL UNIT
FD	FIRE DAMPER
FT	FEET
FT WG	FEET WATER GAUGE
FV	FLUE VENT
GPM	GALLONS PER MINUTE
H	HUMIDIFIER
HD	HEAD
HHW	HEATING HOT WATER
HHWC	HEATING HOT WATER COIL
HGT	HEIGHT
HP	HEAT PUMP, HORSEPOWER
HR	HOUR
HW	HOT WATER
HX	HEAT EXCHANGER
IAW	IN ACCORDANCE WITH
IWG	INCHES WATER GAUGE
kW	KILOWATT
LAT	LEAVING AIR TEMPERATURE
LBS	POUNDS
LV	LOUVER
LWT	LEAVING WATER TEMPERATURE
MAX	MAXIMUM
MIN	MINIMUM
NC	NORMALLY CLOSED, NOISE CRITERIA
NO	NUMBER, NORMALLY OPEN
OA	OUTSIDE AIR
PH	PHASE
PSI	POUNDS PER SQUARE INCH
QTY	QUANTITY
RA	RETURN AIR
RG	RETURN GRILLE
RH	RELATIVE HUMIDITY
RL	REFRIGERANT LIQUID
RS	REFRIGERANT SUCTION
SA	SUPPLY AIR
SC	STEAM CONDENSATE
SEER	SEASONAL ENERGY EFFICIENCY RATIO
SF	SHOT FEEDER
SG	SUPPLY GRILLE
SIM	SIMILAR
SP	STATIC PRESSURE
STM	STEAM
SQ	SQUARE
TDH	TOTAL HEAD
TYP	TYPICAL
UH	UNIT HEATER
UOI	UNLESS OTHERWISE INDICATED
V	VOLTS
VA	VENTILATION AIR (TREATED OUTSIDE AIR)
VAV	VARIABLE AIR VOLUME
VFD	VARIABLE FREQUENCY DRIVE
W	WIDTH
WB	WET BULB
WG	WATER GAUGE
WPD	WATER PRESSURE DROP

GENERAL MECHANICAL NOTES

1. THESE GENERAL NOTES APPLY TO ALL MECHANICAL DRAWINGS IN THIS SET. SHEET SPECIFIC NOTES ARE INDICATED ON EACH SHEET UNDER GENERAL SHEET NOTES.
2. PROVIDE ALL MATERIALS AND EQUIPMENT AND PERFORM ALL LABOR REQUIRED TO INSTALL COMPLETE AND OPERABLE MECHANICAL SYSTEMS AS INDICATED ON THE DRAWINGS, AS SPECIFIED AND AS REQUIRED BY CODE.
3. INSTALL ALL MECHANICAL EQUIPMENT AND APPURTENANCES IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS, THIS CONTRACT'S DOCUMENTS, AND APPLICABLE CODES AND REGULATIONS.
4. PROVIDE VIBRATION ISOLATORS FOR ALL MECHANICAL EQUIPMENT TO PREVENT TRANSMISSION OF VIBRATION TO BUILDING STRUCTURE.
5. PROVIDE ACCESS PANELS IN WALLS AND CEILINGS, WHERE REQUIRED, TO SERVE FIRE DAMPERS, VALVES, SMOKE DETECTORS, AND OTHER CONCEALED MECHANICAL EQUIPMENT THAT REQUIRES ROUTINE MAINTENANCE. ACCESS PANELS MUST BE LARGE ENOUGH TO ALLOW PROPER CLEARANCE FOR ACCESS DOORS.
6. ALL OPENINGS IN FIRE WALLS DUE TO DUCTWORK, PIPING, CONDUIT, ETC., SHALL BE FIRE STOPPED.
7. LOCATE ALL NEW HORIZONTAL DUCTWORK ABOVE THE SUSPENDED CEILINGS UNLESS OTHERWISE INDICATED.
8. PROVIDE PIPE SLEEVES FOR ALL PIPING PENETRATIONS THRU WALLS AND FLOORS. EACH PIPE SHALL BE SEPARATELY SLEEVED. PROVIDE CONTINUOUS SEALANT JOINT AROUND PIPE SLEEVE, BOTH SIDES OF WALL, WATERTIGHT.
9. PROVIDE P-TRAPS IN CONDENSATE DRAIN PIPING AT EACH AHU UNLESS THE AHU IS EQUIPPED WITH AN INTERNAL TRAP.
10. MOUNT THERMOSTATS, HUMIDISTATS, BSS, AND ESS 48 INCHES ABOVE THE FINISHED FLOOR UNLESS OTHERWISE INDICATED. REFER TO ELECTRICAL DRAWINGS FOR LOCATION OF LIGHT SWITCHES AND COORDINATE.
11. CONDENSATE DRAIN PIPING SHALL HAVE CLEANOUTS INSTALLED AT EACH CHANGE IN DIRECTION GREATER THAN 45 DEGREES. WHERE MORE THAN ONE CHANGE OF DIRECTION OCCURS IN A RUN OF PIPING, ONLY ONE CLEANOUT SHALL BE REQUIRED FOR EACH 45 FEET OF DEVELOPED LENGTH. COMBINATION WYE AND 1/8 BENDS SHALL BE USED IN LIEU OF TEES.
12. AFTER THE PIPING IS INSULATED, PROVIDE FACTORY-FABRICATED PIPE LABELS AS SPECIFIED, WITH DIRECTION OF FLOW ARROWS, ON THE FOLLOWING PIPING, AT 10 FOOT ON CENTER ON THE PIPING MAINS AND 5 FOOT ON CENTER ON THE BRANCH RUNOUTS FROM THE MAINS.
13. PIPE RUNOUTS SHALL BE SIZED AS INDICATED ON EQUIPMENT SCHEDULES.
14. DUCT RUNOUT SIZE TO DIFFUSER, REGISTER, OR GRILLE SHALL BE SIZE OF DIFFUSER, REGISTER, OR GRILLE NECK UNLESS OTHERWISE INDICATED.
15. UTILIZE RADIUS TYPE ELBOWS WITH A CENTERLINE RADIUS OF 1.5 TIMES THE WIDTH OR DIAMETER OF THE DUCT WHERE SPACE PERMITS. OTHERWISE, FLAT OVAL AND RECTANGULAR ELBOWS SHALL HAVE FACTORY FABRICATED TURNING VANES.
16. CONDENSATE DRAIN PIPING SHALL BE SLOPED 1/8 INCH PER FOOT DOWNWARD IN THE DIRECTION OF FLOW. HHW AND CHW PIPING DO NOT REQUIRE SLOPING.
17. PIPING SHOWN ON DRAWINGS IS SCHEMATIC IN NATURE AND DOES NOT SHOW ALL FITTINGS, ELBOWS, AND COMPONENTS FOR DETAILED CONNECTION. CONTRACTOR IS RESPONSIBLE FOR PROVIDING ANY ADDITIONAL FITTINGS, ELBOWS, AND COMPONENTS TO PROVIDE A COMPLETE OPERABLE SYSTEM. PIPING CONNECTIONS SHALL ALLOW FOR ROUTINE MAINTENANCE INCLUDING REMOVAL AND REPLACEMENT OF FILTERS AND COILS.
18. WHERE PIPING AND DUCTWORK CROSS OVER TELECOM CABLETRAY, THE BOTTOM OF THE PIPING AND DUCTWORK SHALL BE A MINIMUM OF 8" ABOVE CABLE TRAY. COORDINATE WITH TELECOMMUNICATIONS CONTRACTOR.
19. INSTALL PLENUM CONNECTIONS TO LOUVERS AT A MINIMUM 1/4" / 12" SLOPE SO PLENUMS DRAIN ANY WATER BACK TO AND OUT OF LOUVERS.
20. BLANK OFF UNUSED PORTIONS OF LOUVERS WITH INSULATED METAL BLANK OFF SECTIONS.

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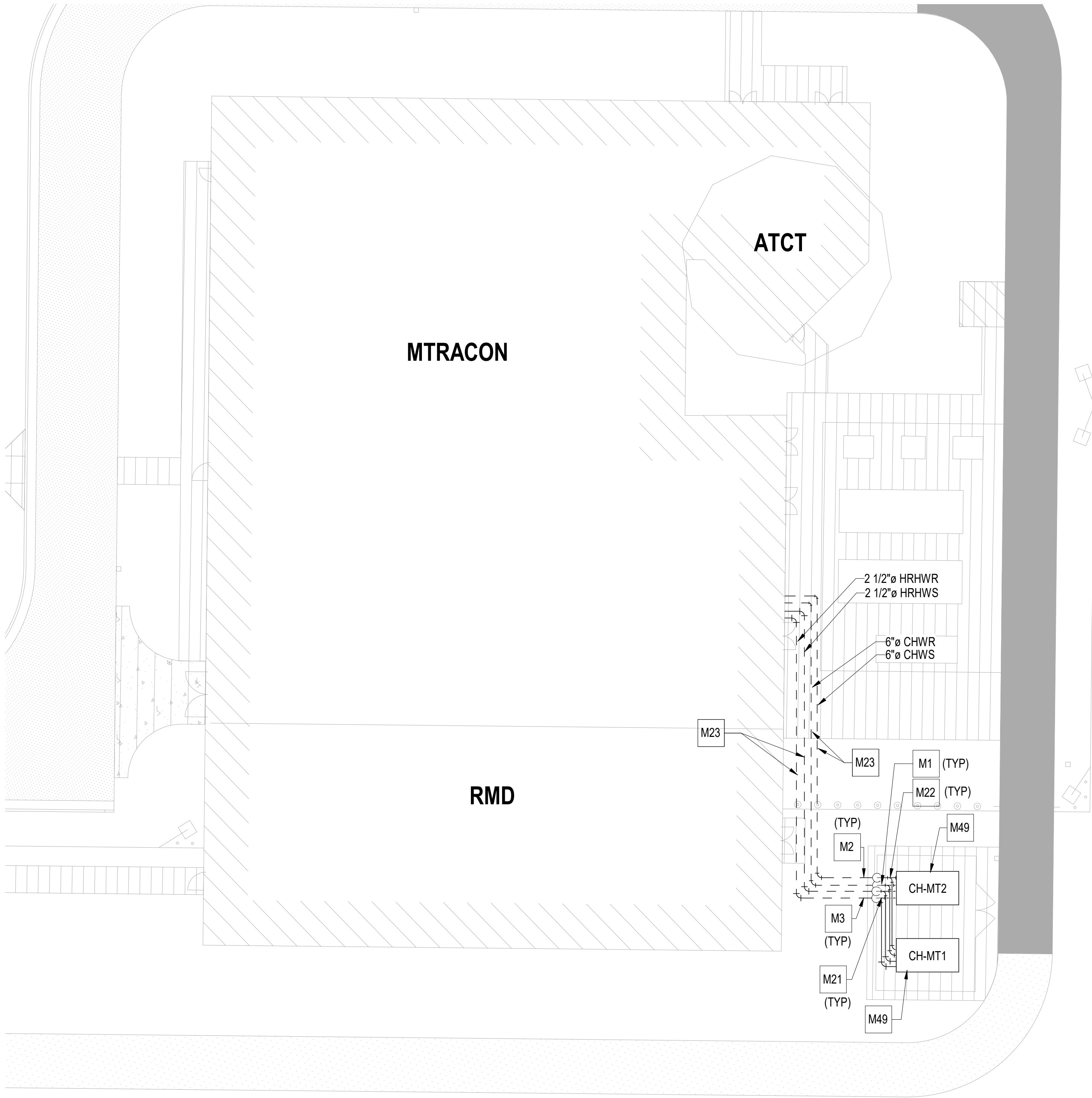
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B

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A1 MECHANICAL SITE PLAN - MTRACON / ATCT

1/16" = 1'-0"



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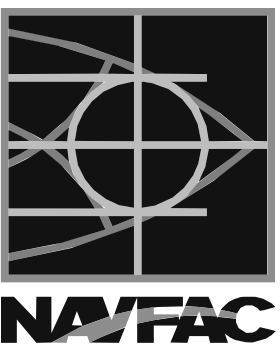
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GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

WORK NOTES

- M1 EXTERIOR ABOVE GROUND PIPING SHALL HAVE ELECTRIC HEAT TRACE.
M2 CHILLED WATER PIPING SHALL TURN DOWN AND BE ROUTED 4'-0" BELOW GRADE.
M3 HEAT RECOVERY HOT WATER PIPING SHALL TURN DOWN AND BE ROUTED 4'-0" BELOW GRADE.
M21 6"Ø CHWS/R AND 2 1/2" HRHWS/R CHILLER PIPING CONNECTIONS.
M22 UTILIZE LATERAL TEE FOR CHILLER BRANCH CONNECTIONS TO AVOID BULLHEAD TEE.
M23 BELOWGRADE PIPING SHALL BE ROUTED IN ACCORDANCE WITH SECTION 33 61 14 EXTERIOR BURIED PREINSULATED WATER PIPING .
M49 PROVIDE FINAL COORDINATION OF CHILLER PAD AND ENCLOSURE WITH CHILLERS PROVIDED. ENCLOSURE/CHILLER SEPARATION SHALL BE COORDINATED WITH CHILLERS PROVIDED TO AVOID DE-RATE OF CHILLER CAPACITY.



BURNS
MEDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED: *[Signature]*

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES CWS DRW CWS CHK RJ

PMIDM CJC/KDB

BRANCH MANAGER J.L.H.

CHIEF ENGINEER EJA

FIRE PROTECTION HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

MECHANICAL SITE PLAN - MTRACON / ATCT

MS102

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12801696

SHEET 436 OF 686

DRAWING REVISION: 12 APRIL 2018

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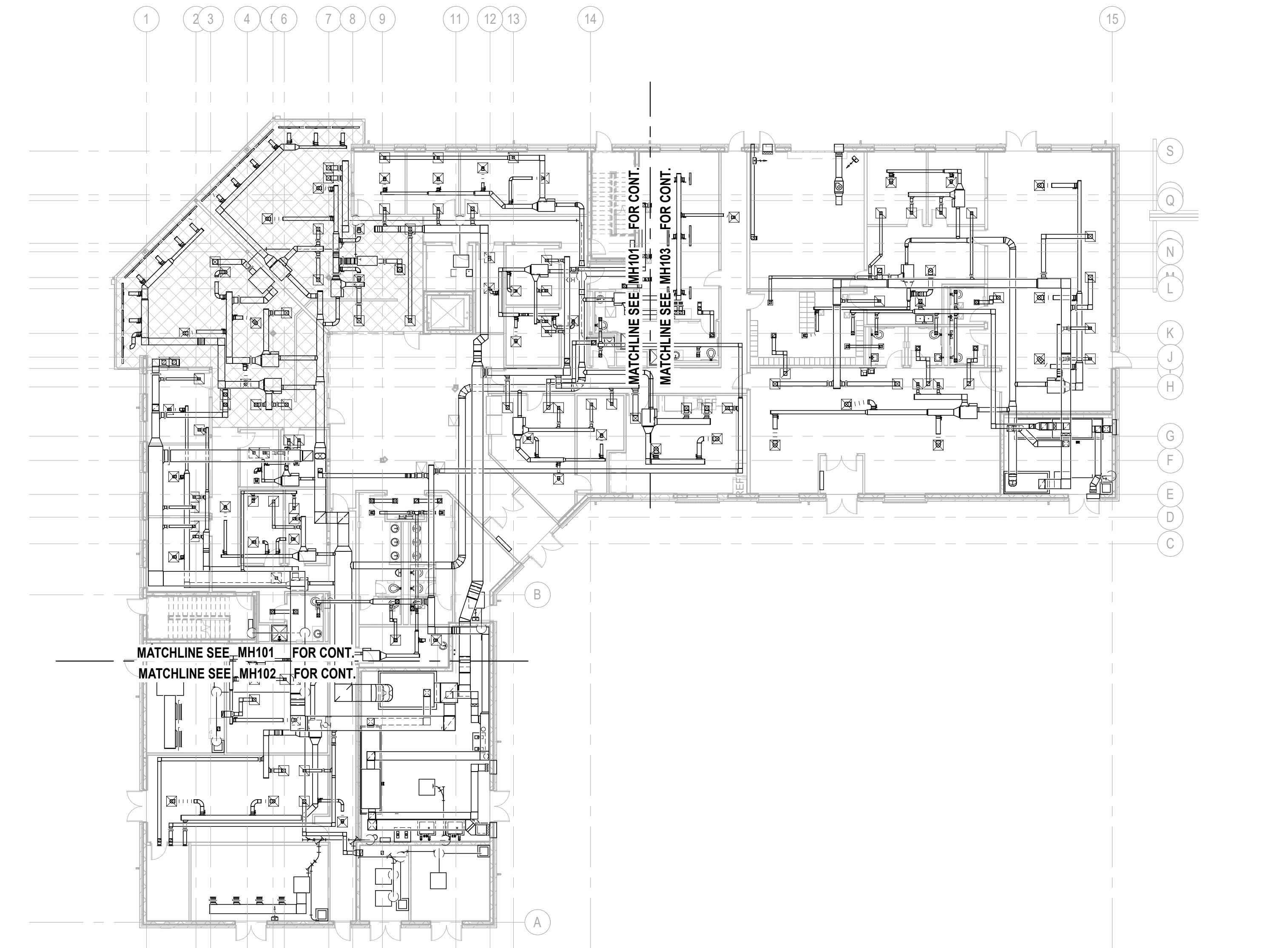
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UNCLASSIFIED



A1

AIR OPS - FIRST FLOOR MECHANICAL DUCTWORK PLAN - OVERALL
1/16" = 1'-0"

1/16" = 1'-0"

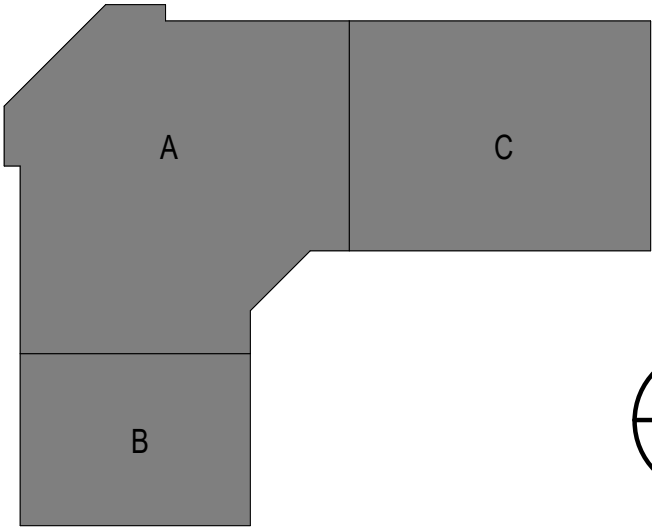
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PLAN NORTH

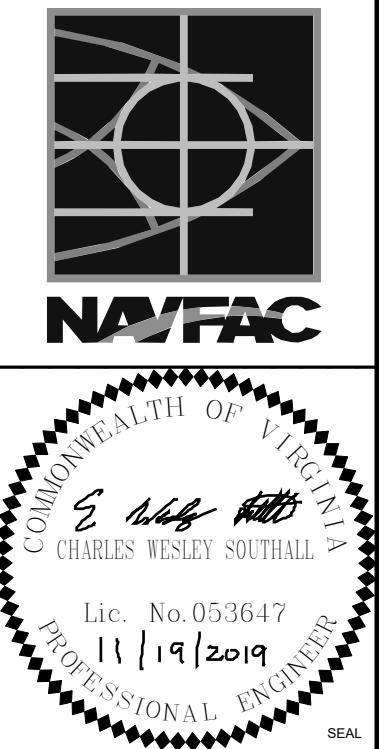
GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

KEY PLAN



SYN	DESCRIPTION	DATE	APPR
0	ISSUED FOR CONSTRUCTION	11/19/19	



BURNS MEDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED	
FOR COMMANDER NAVFAC	
ACTIVITY	
SATISFACTORY TO	DATE
DES FYA	DRW FYA
CHK	RJ
PMIDM	CJCKDB
BRANCH MANAGER	TLH
CHIEF ENGINEER	EJA
FIRE PROTECTION	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
MARINE CORPS IPT
MCAS CHERRY POINT
CHERRY POINT, NC
FY20 P-228 ATC TOWER AND AIR OPERATIONS
AIR OPS - FIRST FLOOR MECHANICAL DUCTWORK PLAN - OVERALL

SCALE: AS NOTED
PROJECT NO.: 1596300
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12801697
SHEET 437 OF 686
MH100

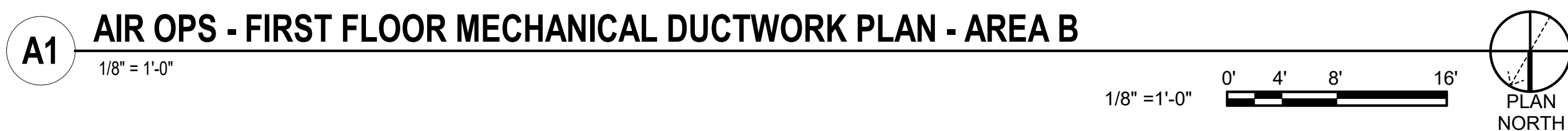
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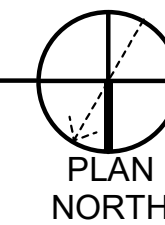
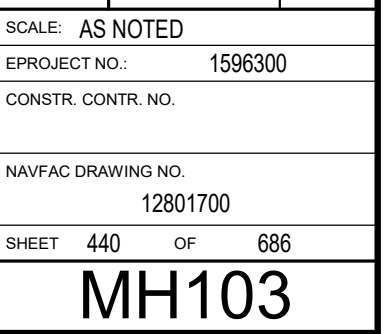
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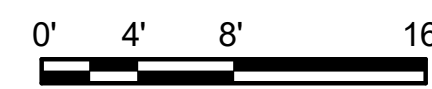
1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
2. AIRFLOW BALANCING TO BE PERFORMED TWICE. ONCE AT MINIMUM PRIMARY FLOW TO BALANCE VAV SUPPLY AND RETURN AIRFLOWS, AND ONCE AT MAXIMUM PRIMARY AIRFLOW TO BALANCE AHU RETURN AIRFLOWS.

M25	BALANCING DAMPER IN VERTICAL.
M29	INSTALL MOTORIZED PARALLEL BLADE DAMPER IN VERTICAL FOR 24/7 RETURN AIR ISOLATION.
M33	8"x6" EA UP TO EF-A2. MOUNT EG AT 16'0" AFF. INSTALL BALANCING DAMPER AND MOTORIZED DAMPER IN VERTICAL.
M41	24/7 RETURN AIR ISOLATION DAMPER.
M47	8"x6" EA UP TO EF-A3. MOUNT EG AT 16'0" AFF. INSTALL BALANCING DAMPER AND MOTORIZED DAMPER IN VERTICAL.
M48	MOUNT TG-2 TO DUCT 18" AFF.



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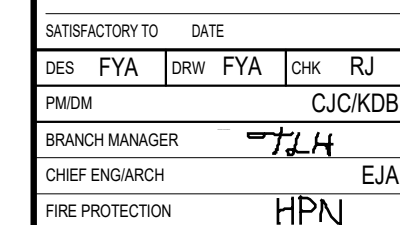
PLAN
NORTH


$$1/8" = 1'-0"$$


PLAN
NORTH

PLAN
NORTH

- | | |
|-----|--|
| M7 | 26"16" SA UP FROM FLOOR BELOW. PROVIDE FIRE DAMPER AT FLOOR PENETRATION. SEE MH101 FOR CONTINUATION. |
| M8 | 26"x18" RA TRANSITION TO 24"x24" RA IN VERTICAL AND DOWN TO FLOOR BELOW. PROVIDE FIRE DAMPER AT FLOOR PENETRATION. SEE MH101 FOR CONTINUATION. |
| M9 | 10"x8" EA DOWN TO FLOOR BELOW. SEE MH101 FOR CONTINUATION. |
| M50 | DAMPER SHALL BE NORMALLY CLOSED AND SHALL ONLY BE OPENED WHEN VAV-A205 IS IN UNOCCUPIED MODE AND UNIT GOES INTO HEATING MODE. |



MCAS CHERRY POINT
CHERRY POINT, NC
FY20 P-228 ATC TOWER AND AIR OPERATIONS

HAIR OPS - SECOND FLOOR MECHANICAL DUCTWORK PLAN - AREA A

SCALE: AS NOTED	
EPROJECT NO.:	1596300
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	
12801701	
SHEET 441	OF 686

MH104

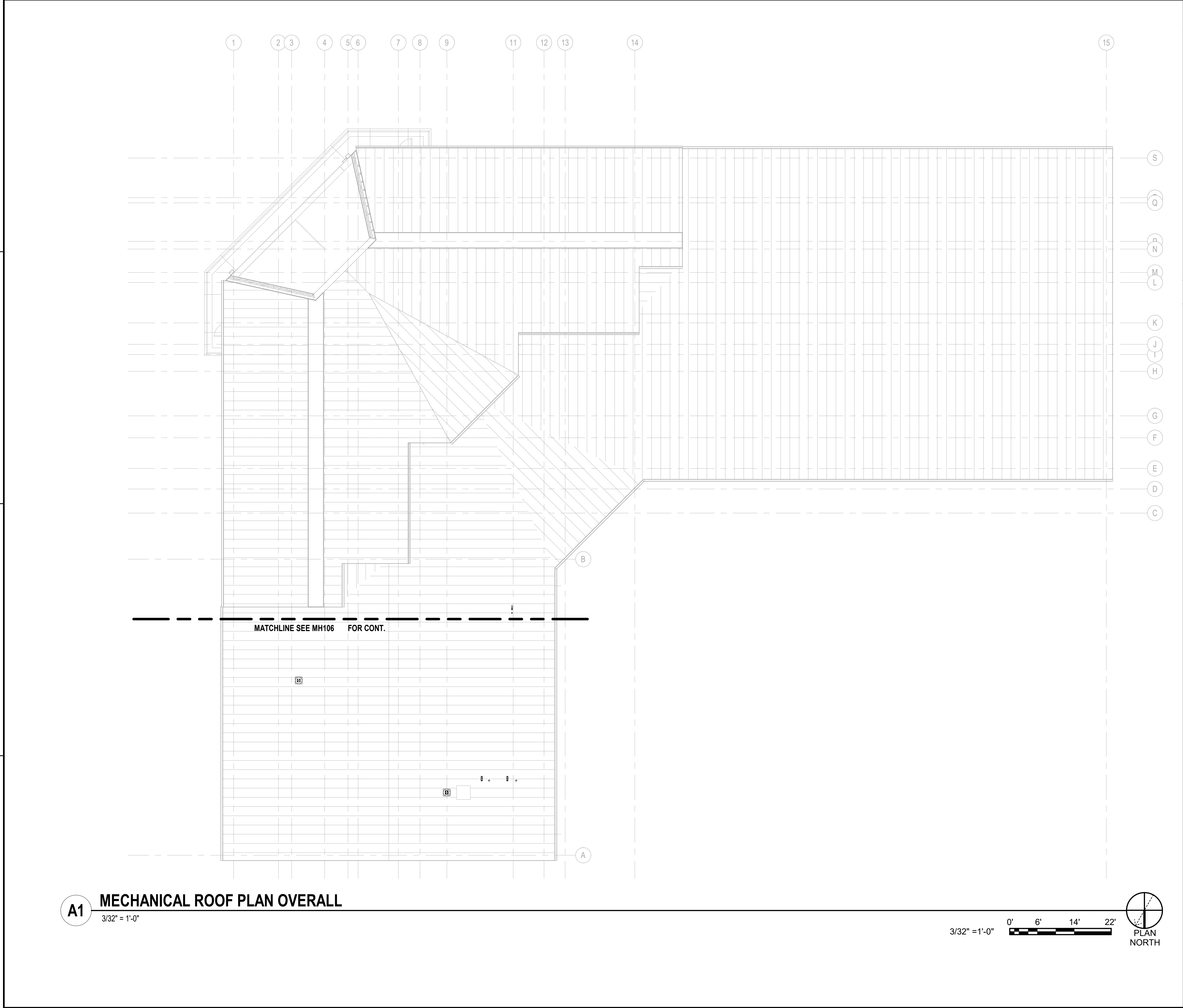
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AIR OPS - SECOND FLOOR MECHANICAL DUCTWORK PLAN - AREA A

DATE	11/19/19	APP'D	
CHECKED BY	RJ	DRAWN BY	
SCALE	AS SHOWN	SHEET NO.	33320
TITLE	AIR OPS - SECOND FLOOR MECHANICAL DUCTWORK PLAN - AREA A		

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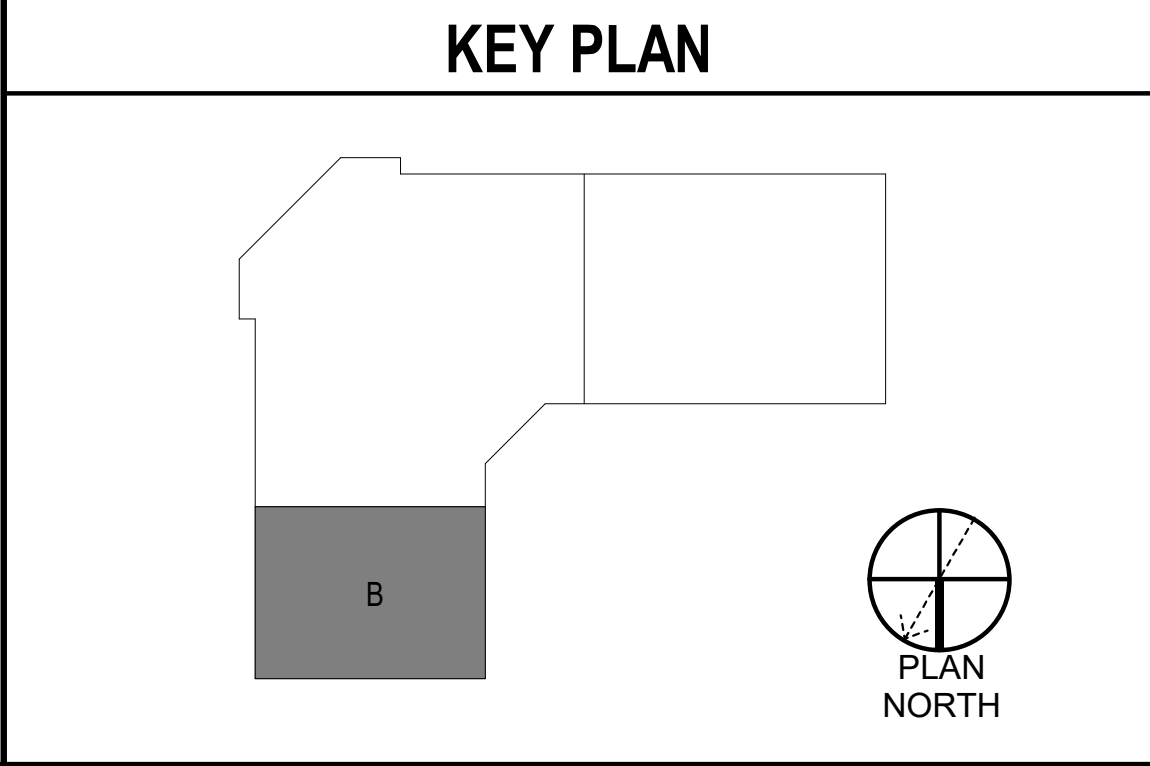
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GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

WORK NOTES



APPR	
11/19/19	DATE
0	ISSUED FOR CONSTRUCTION
SYM	DESCRIPTION

BURNS & MCDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED:
FOR COMMANDER NAVFAC
ACTIVITY

SATISFACTORY TO		DATE	
DES	FYA	DRW	FYA
CHK	RJ	CHK	RJ
PM/DM		CJC/KDB	
BRANCH MANAGER		TJH	
CHIEF ENGINEER		EJA	
FIRE PROTECTION		HPN	

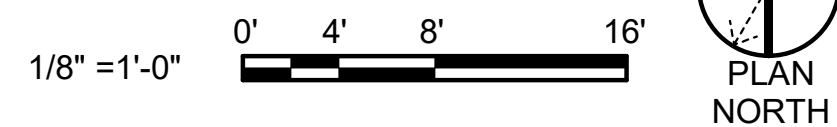
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
MARINE CORPS IPT
CHERRY POINT, NC
CHERRY POINT, NC
FY20 P-228 ATC TOWER AND AIR OPERATIONS
MECHANICAL ROOF PLAN OVERALL

SCALE: AS NOTED	
PROJECT NO.: 1596300	
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO. 12801702	
SHEET 442	OF 686

MH105

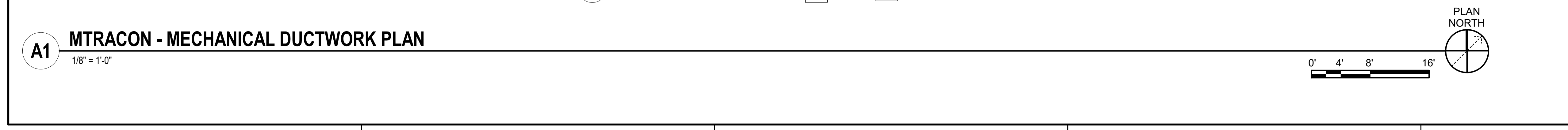
DRAWING REVISION: 12 APRIL 2018

UNCLASSIFIED



PLAN
NORTH

PLAN
NORTH



DRAWING REVISION: 12 APRIL 2018

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UNCLASSIFIED

A1

MTRACON - MECHANICAL DUCT ROOF PLAN

1/8" = 1'-0"

3ø CA
3ø FV

M27

C3
M-311

EF-M1

4ø CA
4ø FV
4ø FV

M27 (TYP)

0' 4' 8' 16'

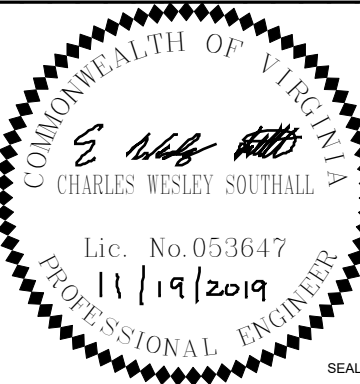
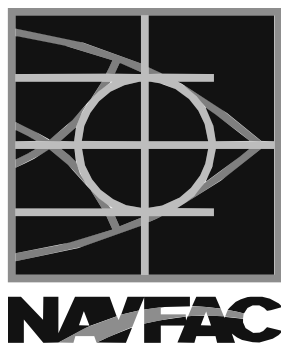


GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

WORK NOTES

M27 FOLLOW MANUFACTURER'S INSTALLATION INSTRUCTIONS TO PROVIDE REQUIRED SEPERATION BETWEEN CA AND FV.



**BURNS
MCDONNELL**
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED:

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DES: CWS BRW CWS CHK: RJ

PMIDM: CJC/KDB

BRANCH MANAGER:

CHIEF ENGINEER: EJA

FIRE PROTECTION: HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

MTRACON - MECHANICAL DUCTWORK ROOF PLAN

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

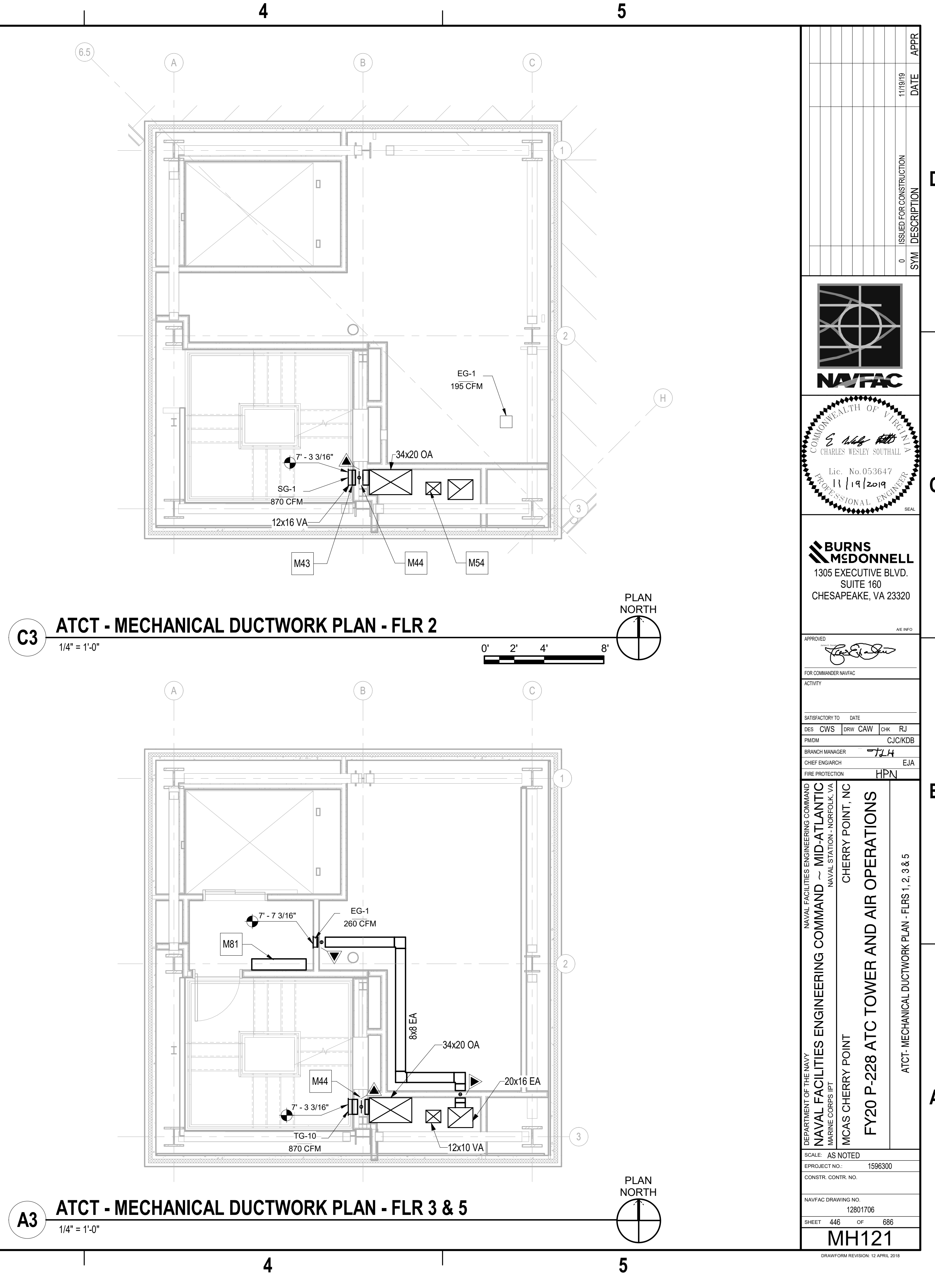
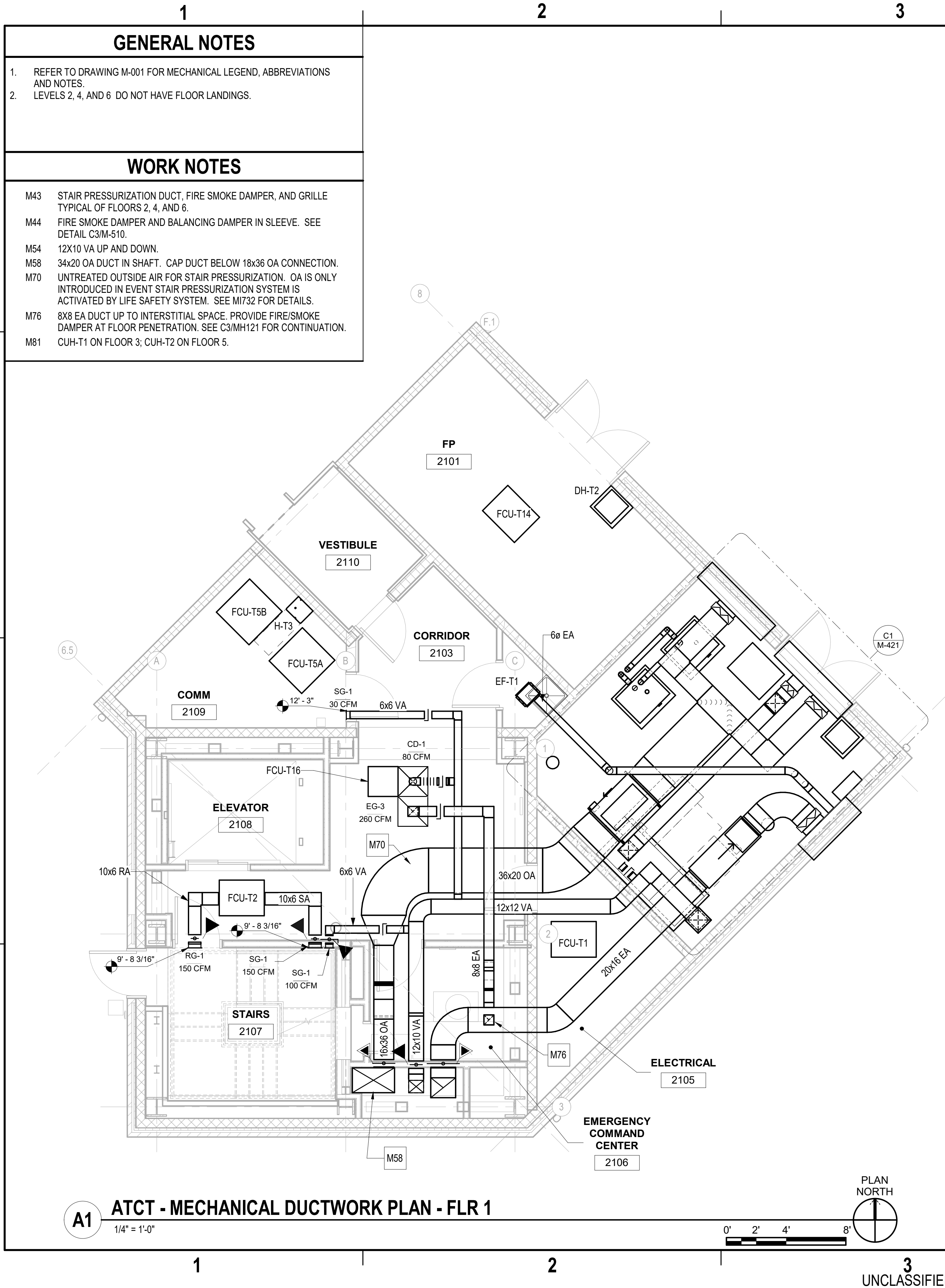
NAVFAC DRAWING NO. 12801705

SHEET 445 OF 686

MH112

DRAWING REVISION: 12 APRIL 2018

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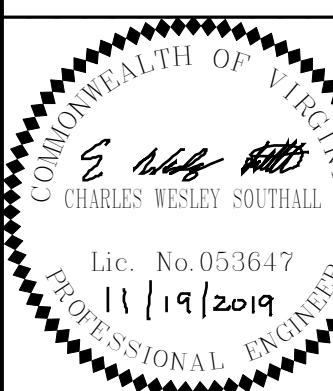
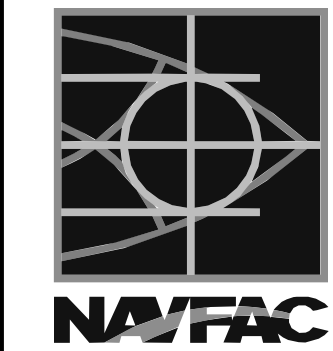
APPROVED	DATE	11/19/19	DATE	11/19/19	APPR
FOR COMMANDER NAVFAC	ACTIVITY	ISSUED FOR CONSTRUCTION	SYM	DESCRIPTION	
 1305 EXECUTIVE BLVD. SUITE 160 CHESAPEAKE, VA 23320					
SATSFACTORY TO DATE					
DES	CWS	BRW	CAW	CHK	RJ
PMIDM	CJC/KDB				
BRANCH MANAGER	JLH				
CHIEF ENGINEER	EJA				
FIRE PROTECTION	HPN				
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC NAVAL STATION - NORFOLK, VA CHERRY POINT, NC					
FY20 P-228 ATC TOWER AND AIR OPERATIONS					
ATCT - MECHANICAL DUCTWORK PLAN - FLRS 1, 2, 3 & 5					
SCALE: AS NOTED					
EPROJCT NO.: 1596300					
CONSTR. CONTR. NO.					
NAVFAC DRAWING NO. 12801706					
SHEET 446 OF 686					
MH121					
DRAWING REVISION: 12 APRIL 2018					

GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- ALL ELEVATIONS SHOWN ARE FROM AFF TO BOTTOM OF OBJECT UNLESS OTHERWISE NOTED.

WORK NOTES

- M4 FIRE SMOKE DAMPER AND BALANCING DAMPER IN SLEEVE. SEE DETAIL C1/M-510.
- M5 20X16 EA FROM BELOW TRANSITIONING TO 12X16 EA UP AFTER TAKEOFF.
- M6 12X10 VA UP AND DOWN.
- M7 16X12 EA UP AND DOWN.
- M8 12X10 VA UP AND DOWN.
- M9 12X16 EA FROM BELOW TRANSITIONING TO 12X10 EA UP AFTER TAKEOFF.
- M10 12X10 VA UP AND DOWN.
- M11 12X10 EA FROM BELOW TRANSITIONING TO 8X8 EA UP AFTER TAKEOFF.
- M12 12X10 VA FROM BELOW TRANSITIONING TO 6X6 VA UP AFTER TAKEOFF.
- M30 8X10 EA UP. PROVIDE FIRE DAMPER AT FLOOR PENETRATION.
- M44 FIRE SMOKE DAMPER AND BALANCING DAMPER IN SLEEVE. SEE DETAIL C3/M-510.
- M45 FIRE DAMPER AND BALANCING DAMPER IN SLEEVE. SEE DETAIL C1/M-510.
- M69 SEE DETAIL A1/M-503.
- M74 CAP 34x20 OA DUCT AT TOP OF SHAFT. CONNECT 12x16 OA DUCT INTO 34x20 CAP.
- M77 PROVIDE BALANCING DAMPER IN VERTICAL.



BURNS & MCDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED:

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES CWS | DRW CAW | CHK RJ

PM/DM CJC/KDB

BRANCH MANAGER

CHIEF ENGINEER

FIRE PROTECTION

HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

ATCT - MECHANICAL DUCTWORK PLAN - FLRS 7, 8, 9, & 10

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12801707

SHEET 447 OF 686

MH122

DRAWING REVISION: 12 APRIL 2018

ELEVATOR

CRAH-T1A

CORRIDOR

COMM

STAIRS

2703

34x20 OA

SG-1

870 CFM

12x16 VA

M44

9' - 8 3/16"

SG-1

30 CFM

6x6 VA

M6

M45

M5

EG-1

260 CFM

IH-T1

9' - 7 3/16"

8x8 EA

M4

M5

M6

M7

M8

M9

M10

M11

M12

M13

M14

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M289

M50	16x16 SA UP TO CAB. SEE SHEET QM-100 FOR CONTINUATION. PROVIDE FIRE DAMPER AT FLOOR PENETRATION. PROVIDE BALANCING DAMPER IN VERTICAL.
M52	18x18 RA UP TO CAB. SEE SHEET QM-100 FOR CONTINUATION. PROVIDE FIRE DAMPER AT FLOOR PENETRATION.
M73	SMACNA TYPE 2 SPLIT TEE FITTING.
M75	DUCTS ARE ABOVE ELEVATOR MACHINE ROOM CAP. SEE SECTION C3/M-321.
M80	FCU SHALL BE MOUNTED ABOVE EA AND RA DUCTS AND DISCHARGE TOWARD AHU-11A/B.
M82	PROVIDE FIRE SMOKE DAMPER IN VERTICAL AT SHAFT PENETRATION. DAMPER SHALL FAIL OPEN ON LOSS OF POWER OR SIGNAL FROM FIRE ALARM CONTROL PANEL.
M83	WRAP DUCT AND DUCT SUPPORTS WITH FIRE WRAP TO SHAFT TO LOUVER PROVIDING 2 HOUR RATING AROUND DUCT.
M84	DUCT MOUNTED CONDENSATE DRAIN PAN. SEE DETAIL A4/M-504.
M85	AIRFLOW MEASURING STATION. INSTALL IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AND WITH SUFFICIENT UNOBSTRUCTED ENTRY AND EXIT LENGTH.



SCALE: AS NOTED	
EPROJECT NO.:	1596300
CONSTR. CONTR. NO.	
NAVFAC DRAWING NO.	
12801708	
SHEET 448	OF 686
MH123	

DRAWFORM REVISION: 12 APRIL 2018

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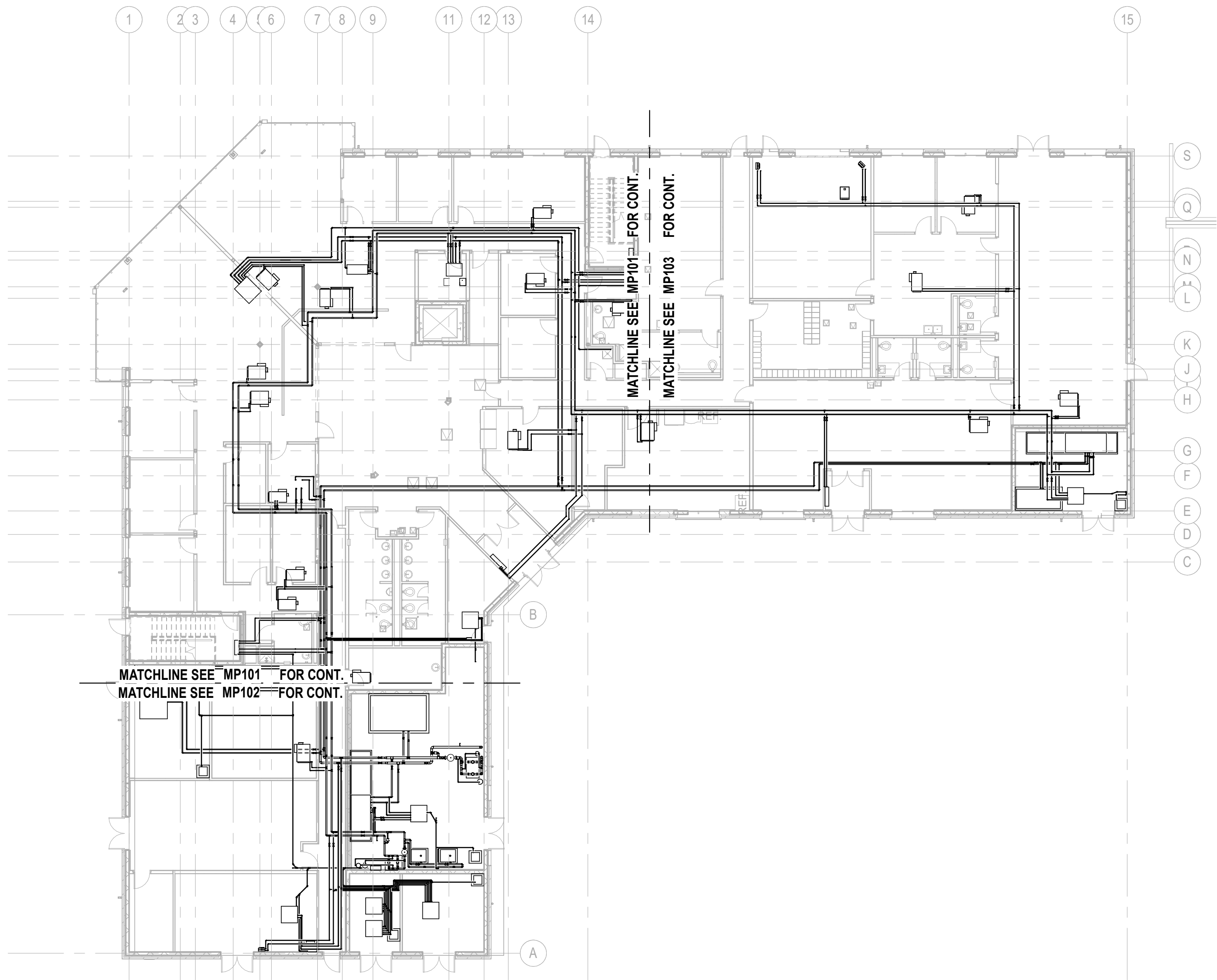
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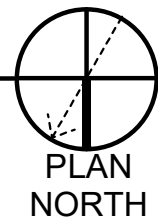
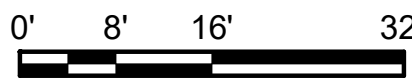
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A1 AIR OPS - FIRST FLOOR MECHANICAL PIPING FLOOR PLAN - OVERALL
1/16" = 1'-0"

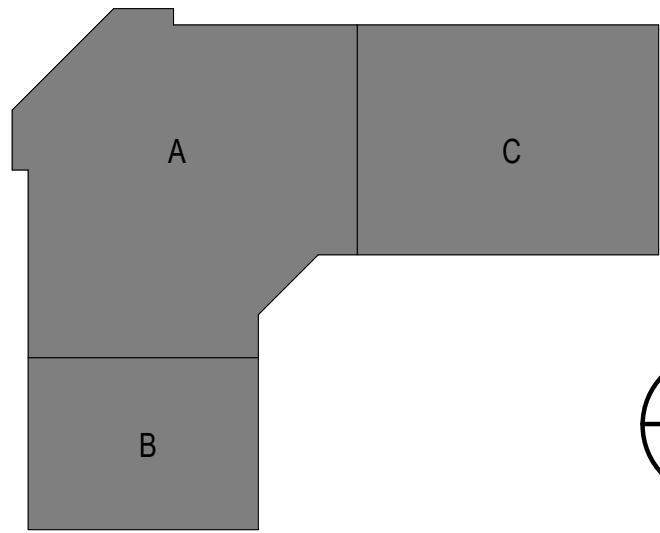
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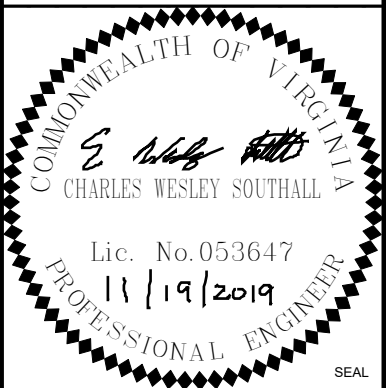
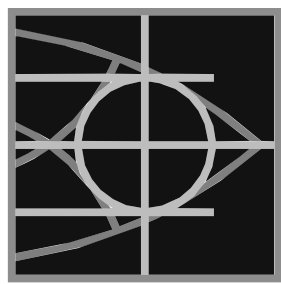
GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

KEY PLAN



SYN	DESCRIPTION	DATE	APPR
0	ISSUED FOR CONSTRUCTION	11/19/19	



BURNS & MCDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED
[Signature]
FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES FYA DRW FYA CHK RJ

PMOM CJC/KDB

BRANCH MANAGER *[Signature]* EJA

CHIEF ENGINEER HPN

FIRE PROTECTION

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

AIR OPS - FIRST FLOOR MECHANICAL PIPING FLOOR PLAN - OVERALL

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12801709

SHEET 449 OF 686

MP100

DRAWING REVISION: 12 APRIL 2018

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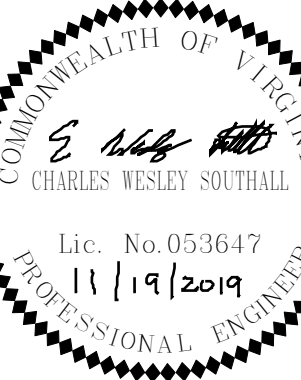
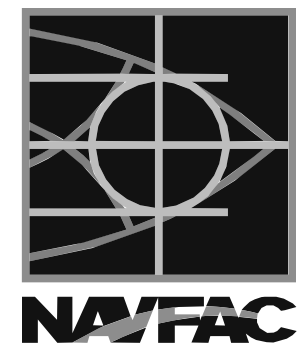
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GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

WORK NOTES

M44 ISOLATION VALVE.



BURNS & MCDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED:
FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DES: FYA | DRW: FYA | CHK: RJ

PMIDW: CJC/KDB

BRANCH MANAGER:

CHIEF ENGINEER: EJA

FIRE PROTECTION: HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

AIR OPS FIRST FLOOR MECHANICAL PIPING FLOOR PLAN - AREA B

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

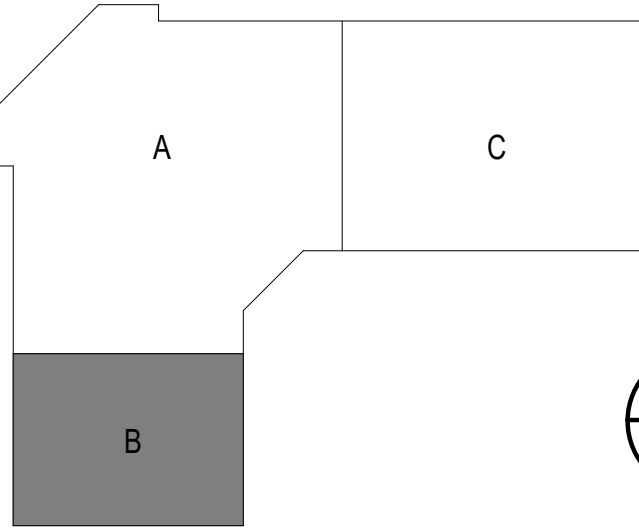
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SHEET 451 OF 686

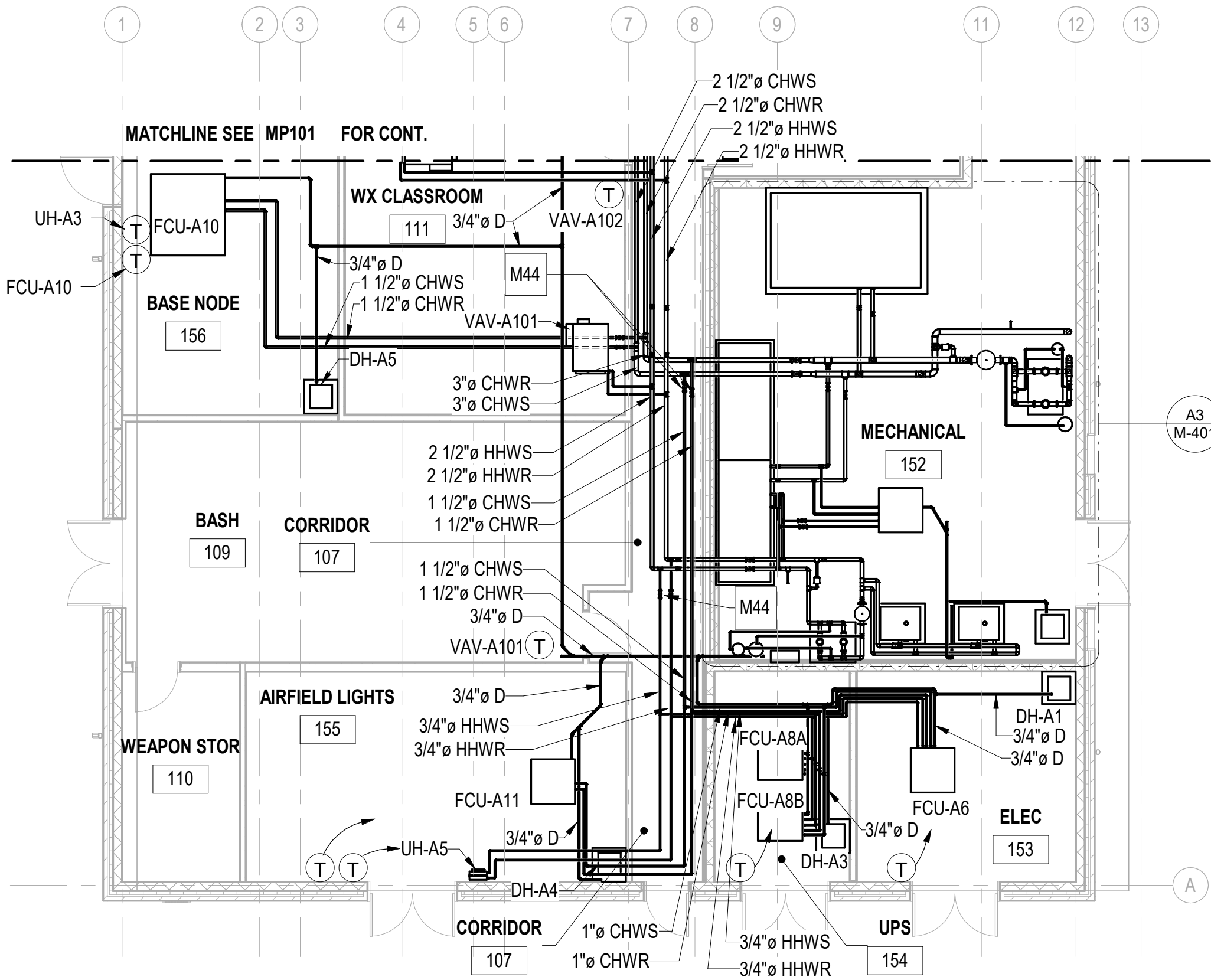
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DRAWING REVISION: 12 APRIL 2018

KEY PLAN



PLAN NORTH

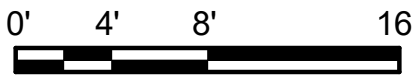


A1

AIR OPS FIRST FLOOR MECHANICAL PIPING FLOOR PLAN - AREA B

1/8" = 1'-0"

1/8" = 1'-0"



PLAN NORTH

UNCLASSIFIED

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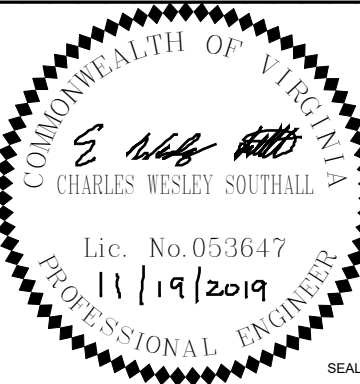
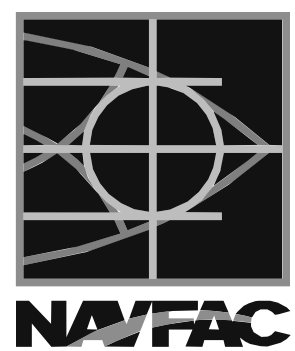
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GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

WORK NOTES

M44 ISOLATION VALVE.



BURNS MEDONNELL
 1305 EXECUTIVE BLVD.
 SUITE 160
 CHESAPEAKE, VA 23320

APPROVED: *[Signature]*
 FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DES FYA DRW FYA CHK RJ

PMOM CJC/KDB

BRANCH MANAGER *[Signature]*

CHIEF ENGINEER EJA

FIRE PROTECTION HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

AIR OPS FIRST FLOOR MECHANICAL PIPING FLOOR PLAN - AREA C

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

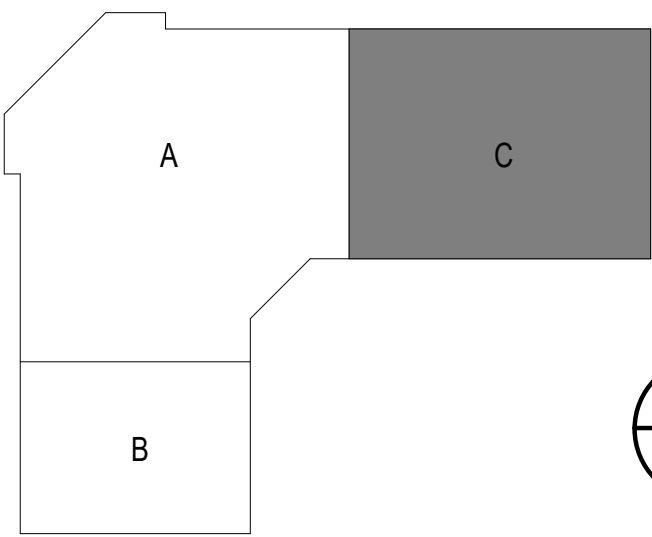
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SHEET 452 OF 686

MP103

DRAWING REVISION: 12 APRIL 2018

KEY PLAN



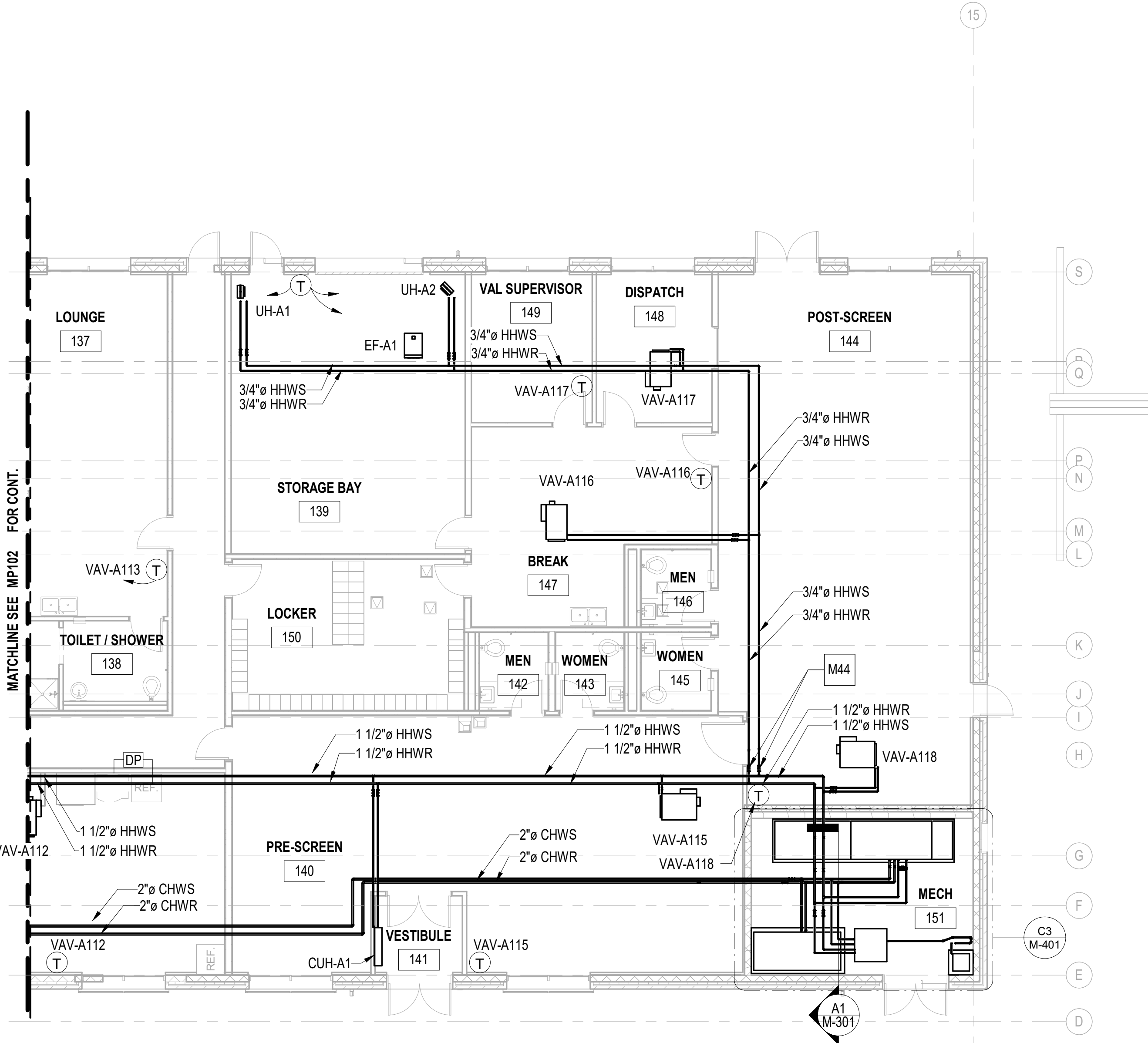
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AIR OPS FIRST FLOOR MECHANICAL PIPING FLOOR PLAN - AREA C

1/8" = 1'-0"

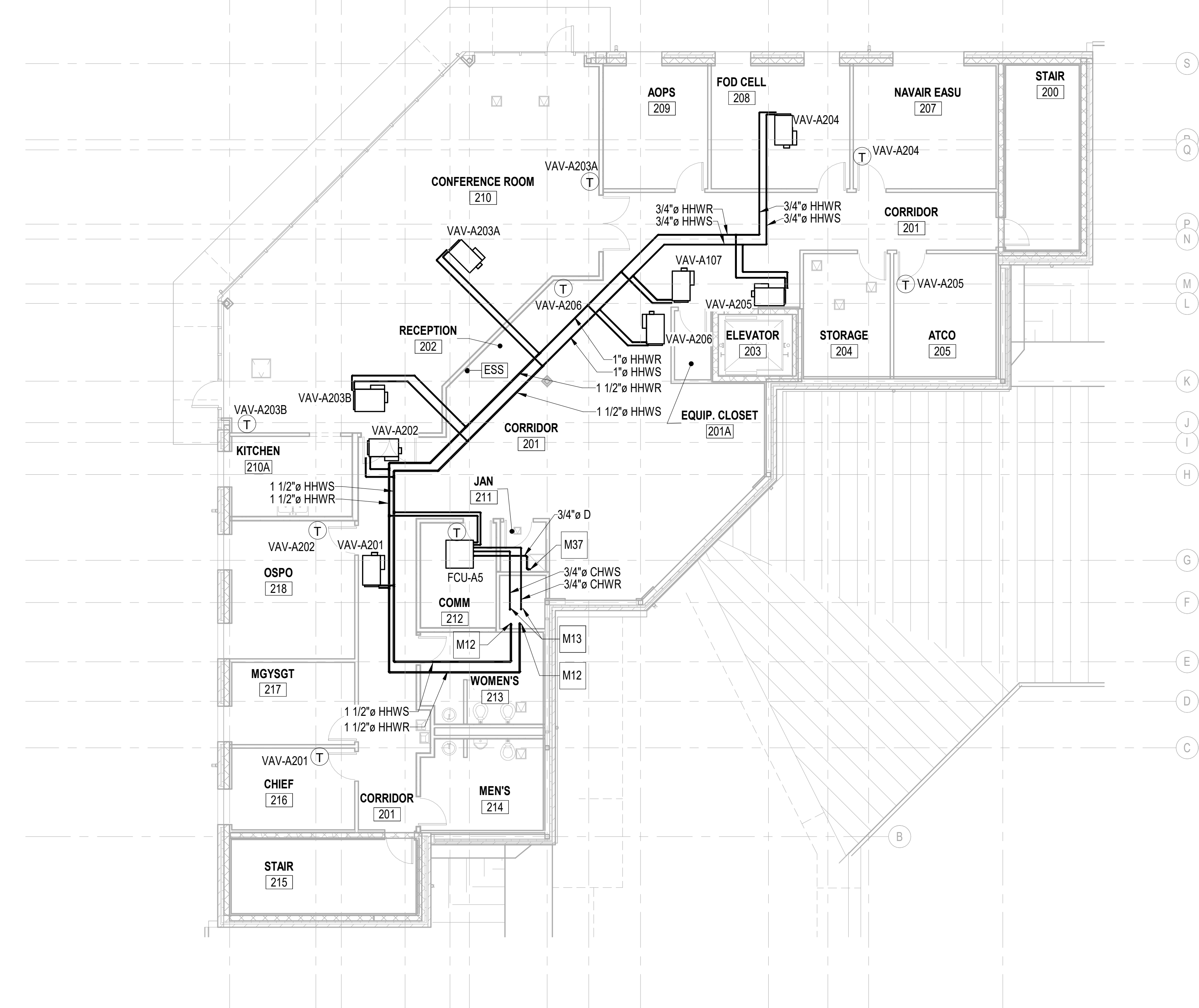
1/8" = 1'-0"

0' 4' 8' 16'



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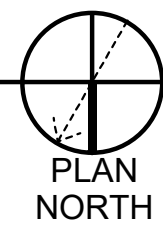
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A1 AIR OPS SECOND FLOOR MECHANICAL PIPING FLOOR PLAN - AREA A
1/8" = 1'-0"

1/8" = 1'-0"

0' 4' 8' 16'

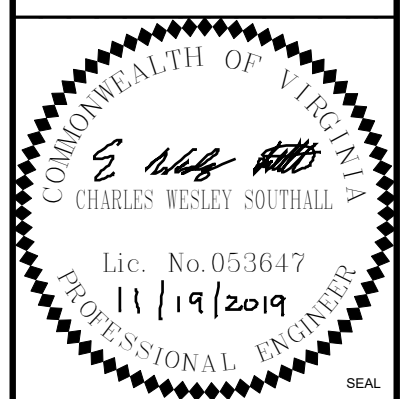
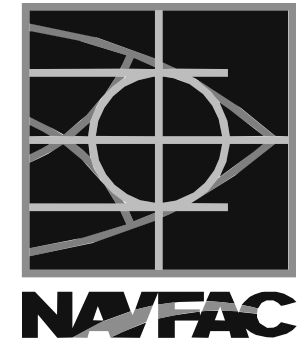


GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

WORK NOTES

- M12 1 1/2" HHWS/R FROM FLOOR BELOW. SEE MP101 FOR CONTINUATION.
M13 3/4" CHWS/R FROM FLOOR BELOW. SEE MP101 FOR CONTINUATION.
M37 CONDENSATE DOWN TO MOP SINK. ROUTE TIGHT TO WALL. TERMINATE DISCHARGE 6 INCHES ABOVE MOP SINK RIM.



BURNS & MCDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED:
FOR COMMANDER NAVFAC
ACTIVITY

SATISFACTORY TO: DATE
DES FYA DRW FYA CHK RJ
PMOM CJC/KDB
BRANCH MANAGER J.L.H.
CHIEF ENGINEER EJA
FIRE PROTECTION HPN

NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
MARINE CORPS IPT
NAVAL STATION - NORFOLK, VA
CHERRY POINT, NC
MCAS CHERRY POINT
FY20 P-228 ATC TOWER AND AIR OPERATIONS
AIR OPS SECOND FLOOR MECHANICAL PIPING FLOOR PLAN - AREA A

SCALE: AS NOTED
PROJECT NO.: 1596300
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12801713
SHEET 453 OF 686

MP104

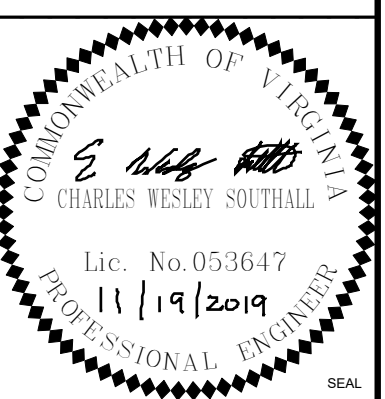
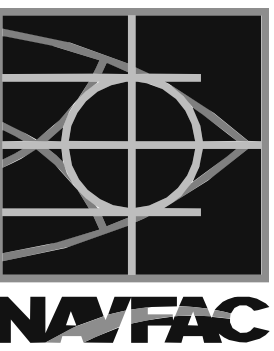
DRAWING REVISION: 12 APRIL 2018

GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- LEVELS 2, 4, AND 6 DO NOT HAVE FLOOR LANDINGS.

WORK NOTES

- | | |
|-----|---|
| M17 | 1 1/2" HHWS/R UP. |
| M18 | 2 1/2" CHWS/R UP. |
| M37 | CHWS/R AND HRHWS/R PIPING TO MTRACON BUILDING. SEE A1/MP111 FOR CONT. |
| M59 | 1" Ø D DOWN TO MOP SINK. |
| M68 | 1 1/2" D FROM ABOVE. |
| M79 | ISOLATION VALVE. |
| M81 | CUH-T1 ON FLOOR 3; CUH-T2 ON FLOOR 5. |

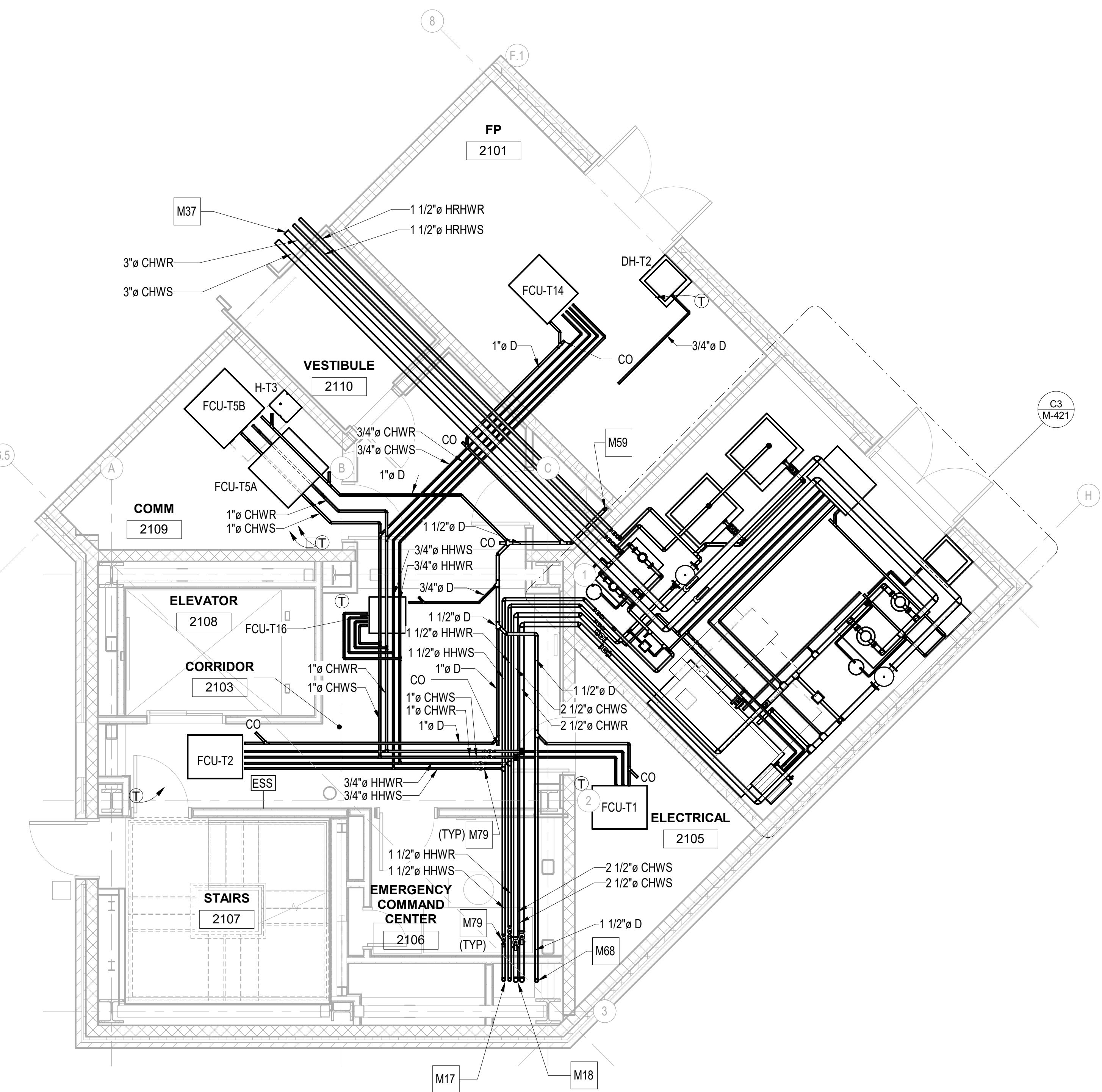


BURNS & MCDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED:
FOR COMMANDER NAVFAC
ACTIVITY
SATISFACTORY TO: DATE
DES: CWS DRW: CAW CHK: RJ
PM/DM: CJC/KDB
BRANCH MANAGER: T.L.H.
CHIEF ENGINEER: EJA
FIRE PROTECTION: HPN

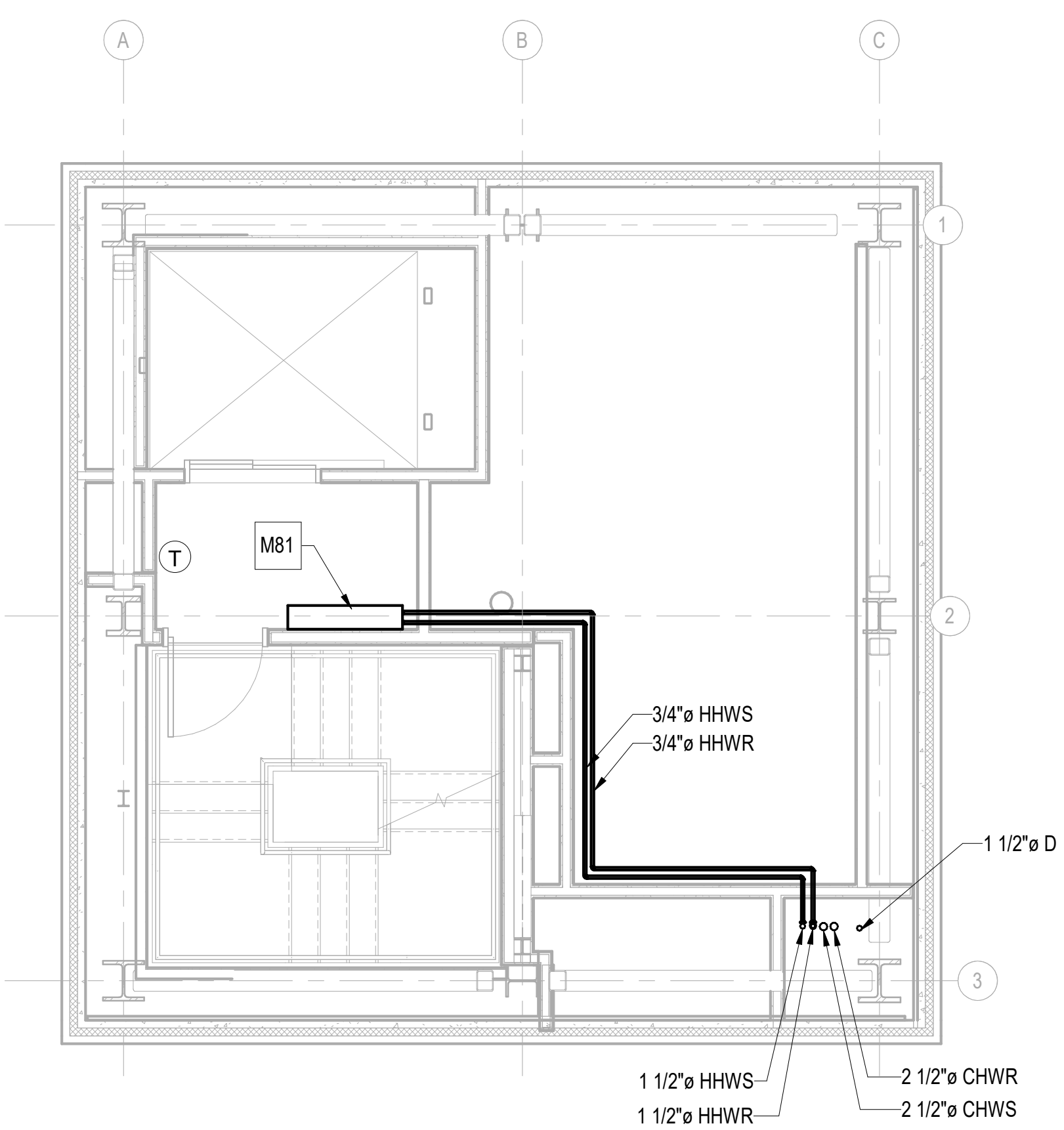
DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
MARINE CORPS IPT
MCAS CHERRY POINT
CHERRY POINT, NC
FY20 P-228 ATC TOWER AND AIR OPERATIONS
ATCT - MECHANICAL PIPING PLAN - 1, 3 & 5

SCALE: AS NOTED
PROJECT NO.: 1596300
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12801715
SHEET 455 OF 686
MP121
DRAWING REVISION: 12 APRIL 2018



A1 ATCT - MECHANICAL PIPING PLAN - FLR 1
1/4" = 1'-0"

0' 2' 4' 8'



A4 ATCT - MECHANICAL PIPING PLAN - FLR 3 & 5
1/4" = 1'-0"

0' 2' 4' 8'



GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- ALL ELEVATIONS SHOWN ARE FROM AFF TO BOTTOM OF OBJECT UNLESS OTHERWISE NOTED.

WORK NOTES

- M19 1 1/2" HHWS/R UP AND DOWN.
M20 2 1/2" CHWS/R UP AND DOWN.
M21 1 1/2" HHWS/R UP AND DOWN.
M22 2 1/2" CHWS/R UP AND DOWN.
M23 1 1/2" HHWS/R UP AND DOWN.
M24 2 1/2" CHWS/R FROM BELOW TRANSITIONING TO 2" CHWS/R ABOVE TAKEOFF.
M25 1 1/2" HHWS/R UP AND DOWN.
M26 2" CHWS/R UP AND DOWN.
M38 CHWS/R, HHWS/R, AND CONDENSATE PIPING UP TO FCU ON LEVEL 10.
M39 CHWS/R, HHWS/R, AND CONDENSATE PIPING DOWN TO LEVEL 9. INSTALL FCU IN ACCORDANCE WITH DETAIL A1/M-503.
M56 CHW DIFFERENTIAL PRESSURE SENSOR. PROVIDE FIRE RATED ACCESS DOOR WITH UNOBSTRUCTED ACCESS TO THE SENSOR. LOCATION OF SENSOR ON PLAN IS SCHEMATIC (SENSOR SHOULD BE INSTALLED WITH UNOBSTRUCTED ACCESS).
M57 HHW DIFFERENTIAL PRESSURE SENSOR. PROVIDE FIRE RATED ACCESS DOOR WITH UNOBSTRUCTED ACCESS TO THE SENSOR. LOCATION OF SENSOR ON PLAN IS SCHEMATIC (SENSOR SHOULD BE INSTALLED WITH UNOBSTRUCTED ACCESS).
M65 1" D DOWN.
M66 1" D FROM ABOVE 1 1/2" D DOWN.
M67 1 1/2" D FROM ABOVE AND DOWN. 45° OFFSET BELOW FLOOR CONNECTION.
M78 PROVIDE SECONDARY DRAIN PAN TO PROTECT EQUIPMENT BELOW.
M79 ISOLATION VALVE.

ELEVATOR
2701COMM
2704

CRAH-T1A

CORRIDOR
2702STAIRS
2703

CRAH-T1B

2" CHWS
2" CHWR

(TYP) M79

H-T1

1 1/4" D

M19

M20

M67

ELEVATOR
2801CORRIDOR
2802STAIRS
2803SIMULATOR
2804

FCU-T7A

H-T4

1 1/4" D

1" D

1" CHWS

1" CHWR

1 1/2" D

FCU-T4

3/4" HHWR

3/4" HHWS

3/4" HHWR

3/4" HHWS

FCU-T7B

1" CHWS

1" CHWR

1" D

(TYP) M79

M21

M22

M66

ELEVATOR
2901CAB ADMIN
2904CORRIDOR
2902STAIRS
2903

FCU-T3

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

ELEVATOR
21001CORRIDOR
21002STAIRS
21003TOILET
21005TOILET
21006BREAKROOM
21004

FCU-T8

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

M39

M39

FCU-T9

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

M39

M39

FCU-T10

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

M39

M39

FCU-T11

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

M39

M39

FCU-T11

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

M39

M39

FCU-T11

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

M39

M39

FCU-T11

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

M39

M39

FCU-T11

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

M39

M39

FCU-T11

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

M39

M39

FCU-T11

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

(TYP) M79

CO

3/4" HHWR

3/4" HHWS

1 1/2" CHWR

1 1/2" CHWS

M57

M23

M56

M24

M65

M38

M39

M39

FCU-T11

1" D

1" CHWR

1" CHWS

3/4" HHWR

3/4" HHWS

1" D

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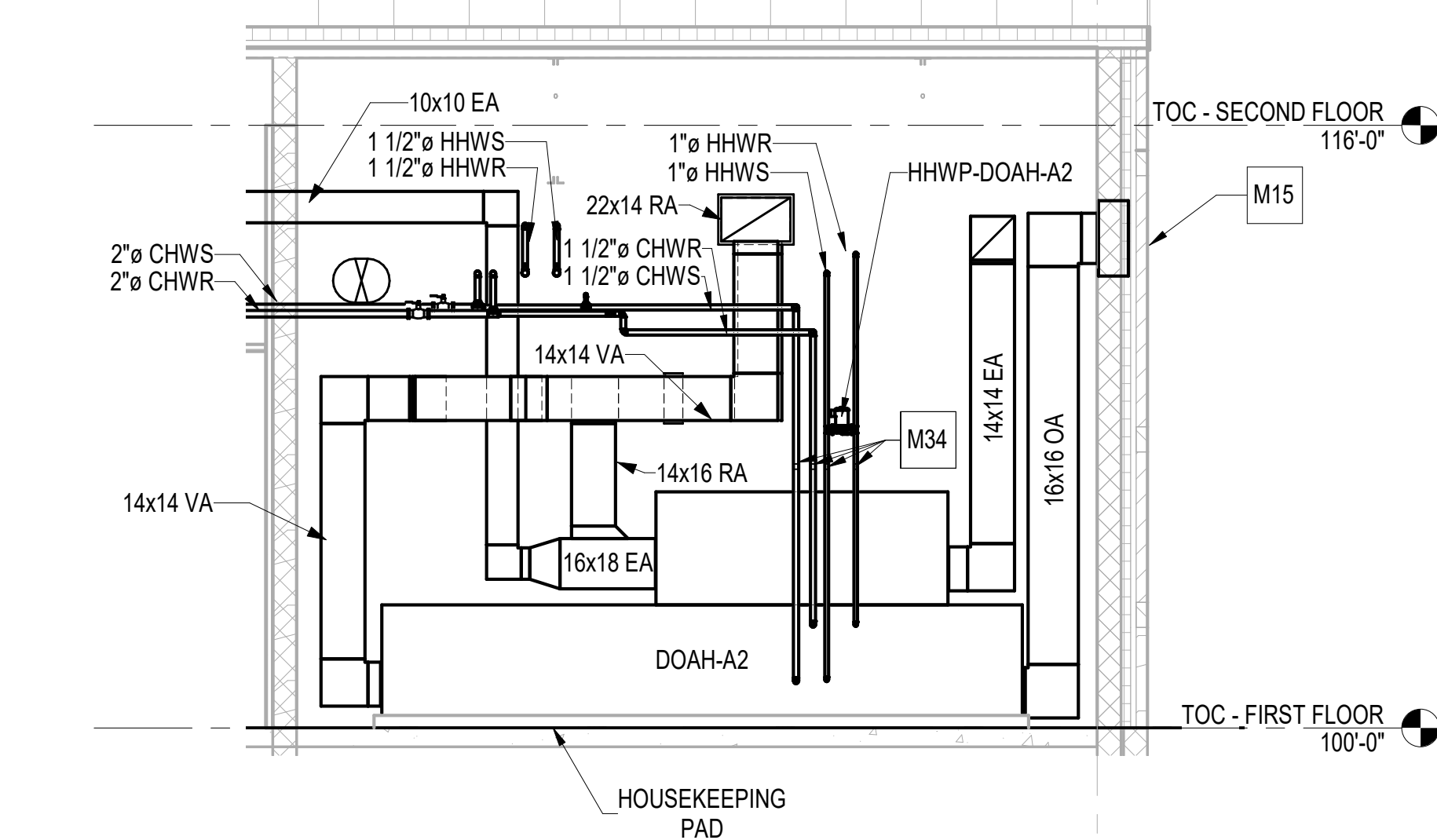
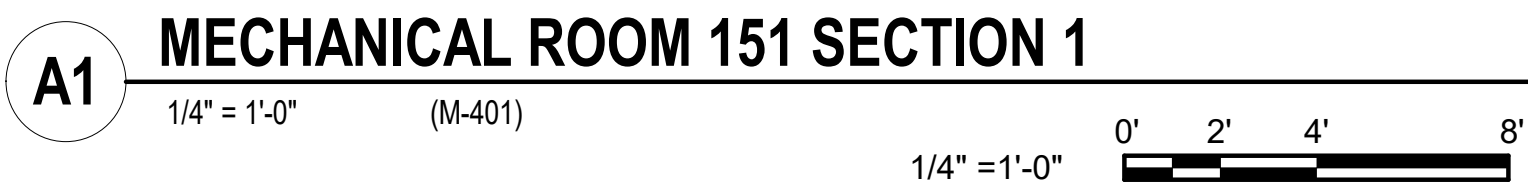
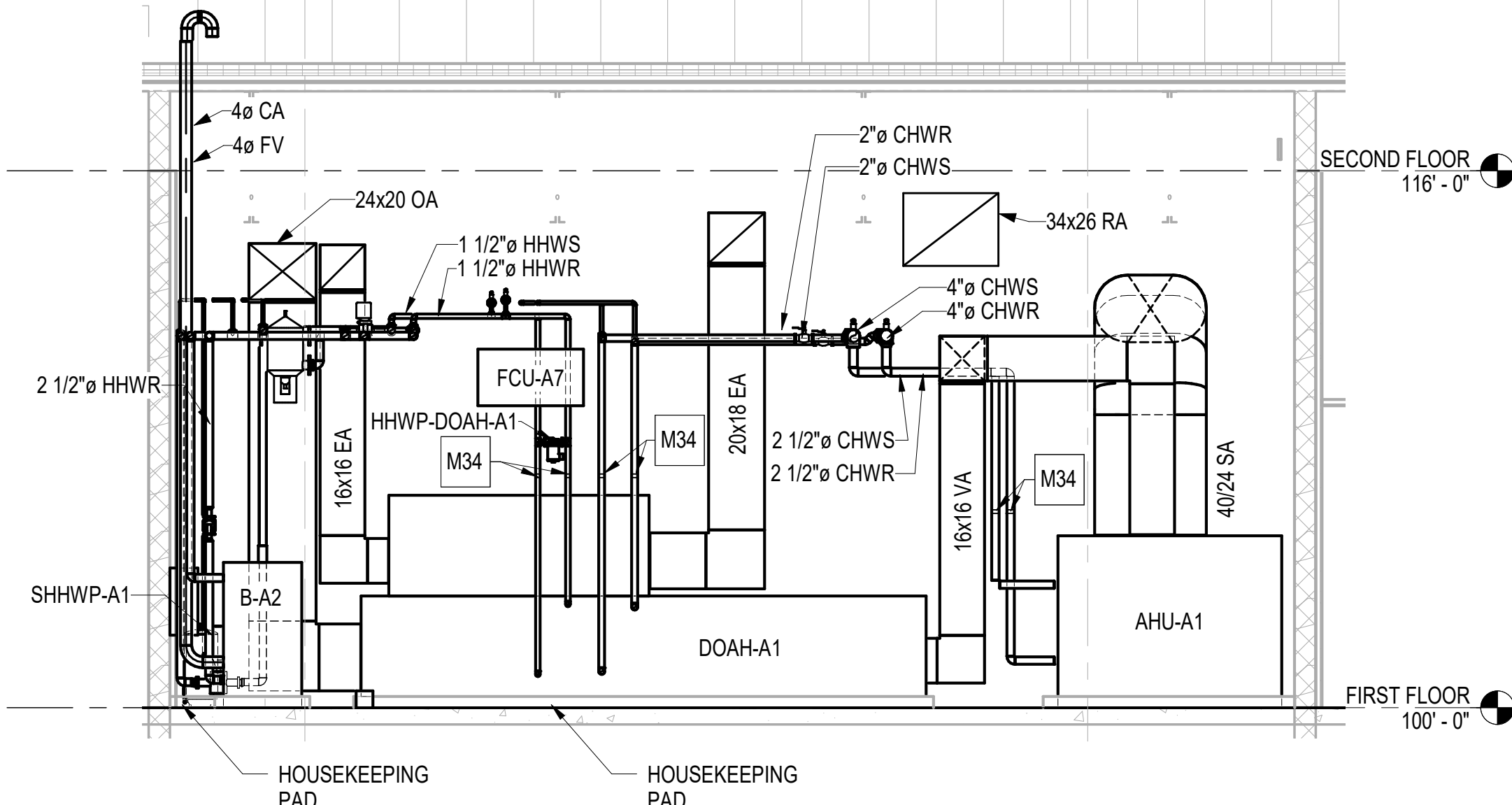
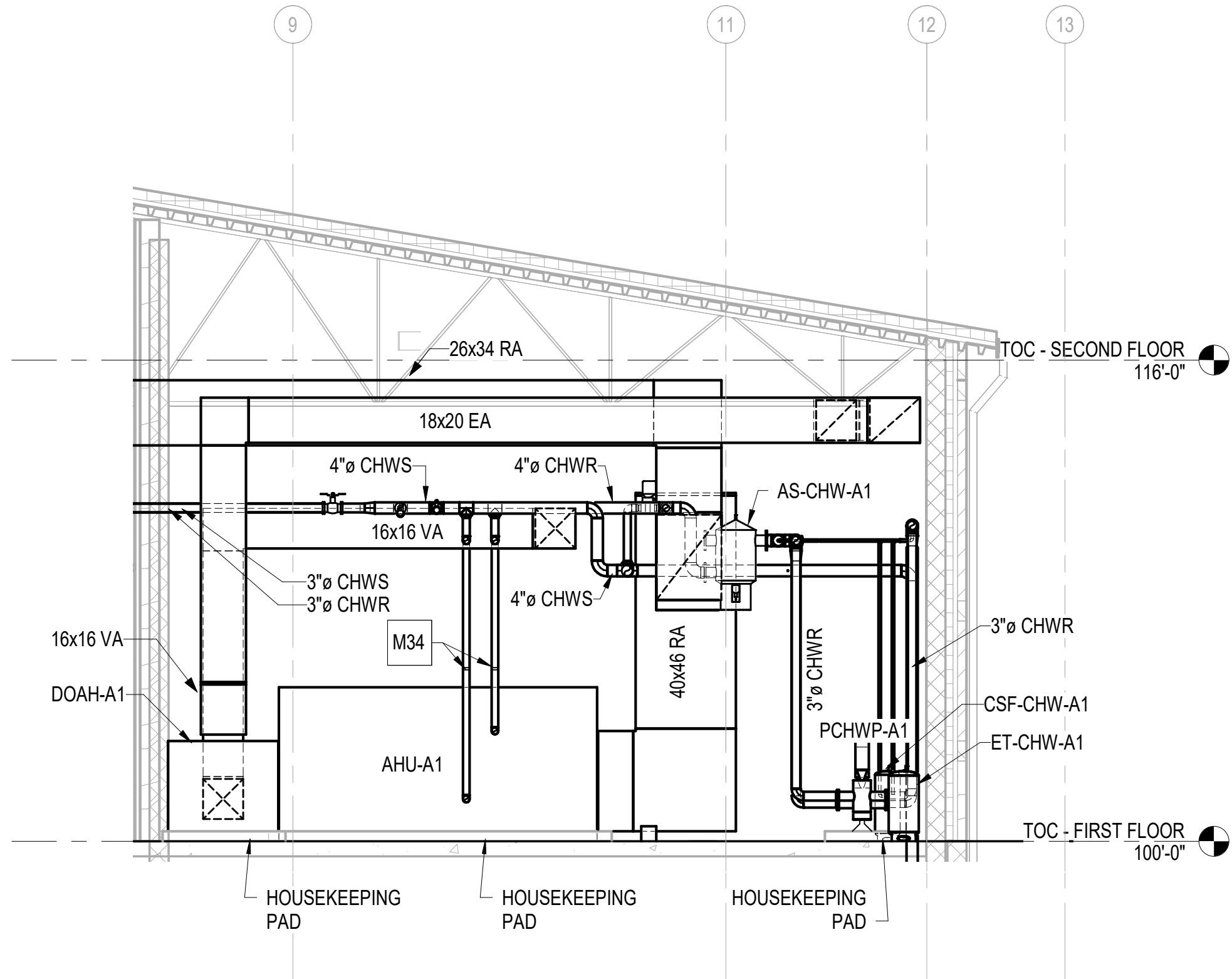
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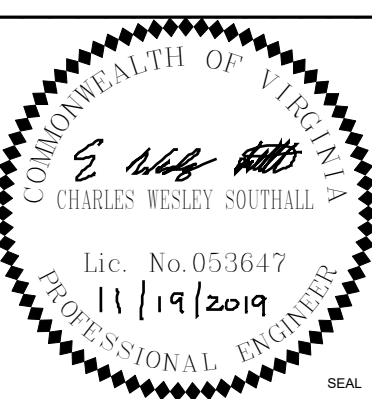
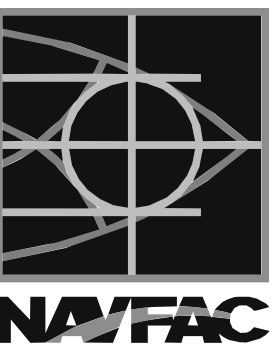


GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

WORK NOTES

M15 36" x 24" INTAKE PLENUM. ATTACH PLENUM TO BACK OF LOUVER.
M34 PROVIDE UNION/FLANGE TO FACILITATE UNIT MAINTENANCE AND COIL REMOVAL.



BURNS & MCDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED: *[Signature]*
FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES FYA DRW FYA CHK RJ

PMIDM CJC/KDB

BRANCH MANAGER *[Signature]*

CHIEF ENGINEER EJA

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC
MARINE CORPS IPT
NAVAL STATION - NORFOLK, VA
CHERRY POINT, NC
FY20 P-228 ATC TOWER AND AIR OPERATIONS

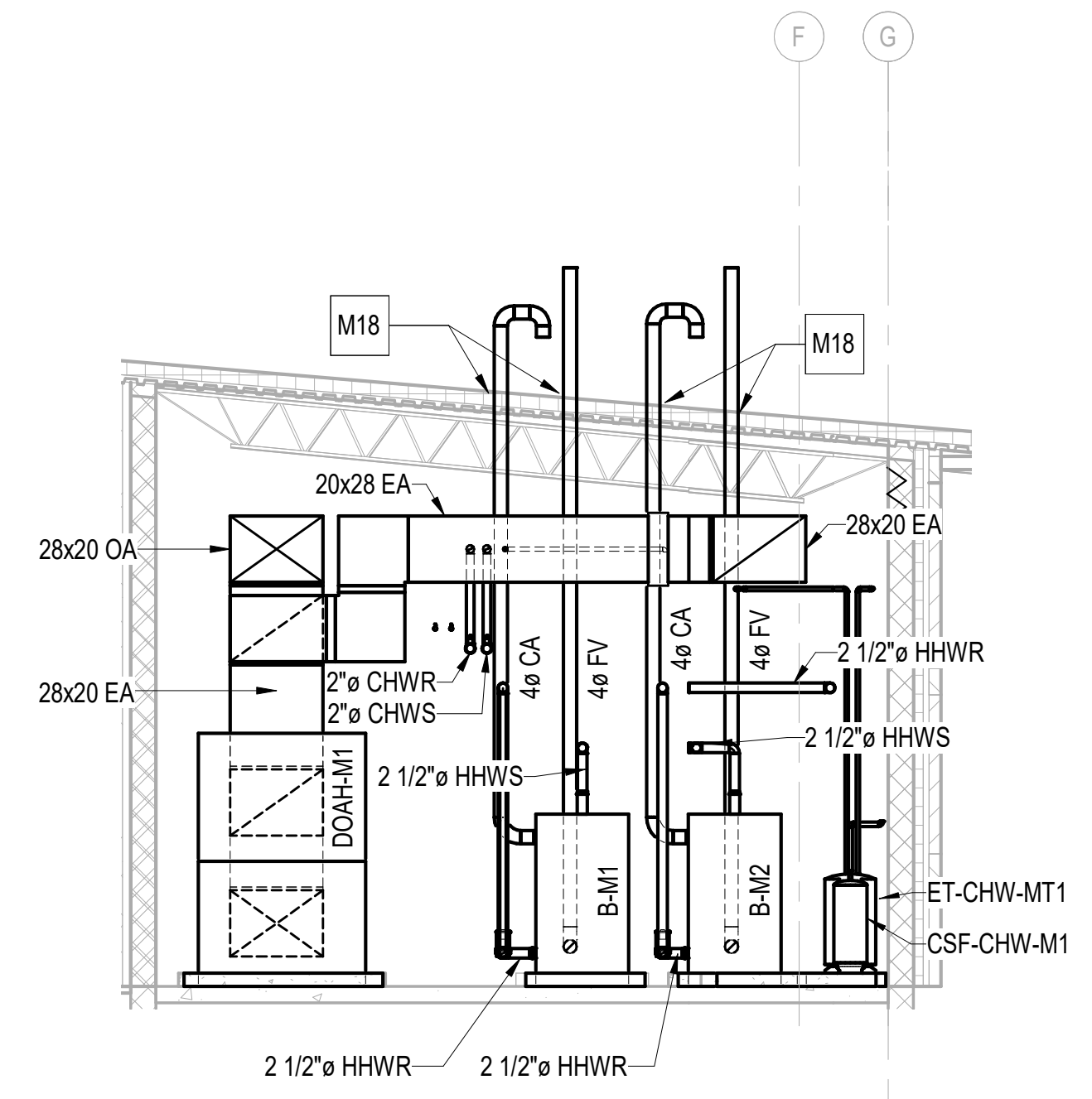
MCAS CHERRY POINT
AIR OPS - MECHANICAL SECTIONS

SCALE: AS NOTED
PROJECT NO.: 1596300
CONSTR. CONTR. NO.

NAVFAC DRAWING NO.
12801718
SHEET 458 OF 686

M-301

DRAWING REVISION: 12 APRIL 2018

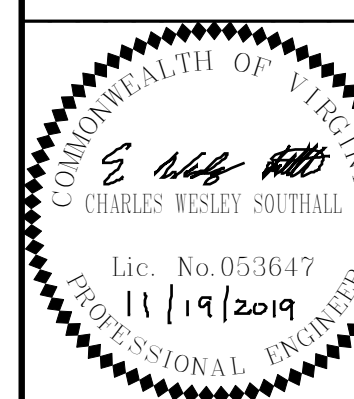


MECHANICAL ROOM 171 SECTION 2



WORK NOTES

- | | |
|-----|---|
| M16 | OPPOSED BLADE MOTORIZED DAMPER WITH INTEGRAL AIRFLOW MEASURING STATION. INSTALL IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AND WITH SUFFICIENT UNOBSTRUCTED ENTRY AND EXIT LENGTH. |
| M18 | BOILER VENT AND COMBUSTION AIR ROOF PENETRATION. SEE A4/M-508 FOR DETAIL. |
| M22 | PROVIDE UNION/FLANGE TO FACILITATE UNIT MAINTENANCE AND COIL REMOVAL. |



**BURNS
MCDONNELL**
1305 EXECUTIVE BLVD
SUITE 160
CHESAPEAKE, VA 23320

APPROVED
JAMES E. HODGINS

FOR COMMANDER NAVFAC
ACTIVITY

SATISFACTORY TO		DATE	
DES	CWS	DRW	TJG
PM/DM		CJC/H	
BRANCH MANAGER		T.L.H.	
CHIEF ENG/ARCH			
FIRE PROTECTION		H.P.M.	

ITIES ENGINEERING COMMAND
~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
CHERRY POINT, NC
OPERATIONS

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
MARINE CORPS IPT
MCAS CHERRY POINT
FY20 P-228 ATC TOWER AND AIR C

SCALE: AS NOTED		
EPROJECT NO.:	1596300	
CONSTR. CONTR. NO.		
NAVFAC DRAWING NO.		
12801719		
SHEET	459	OF 686

M-311

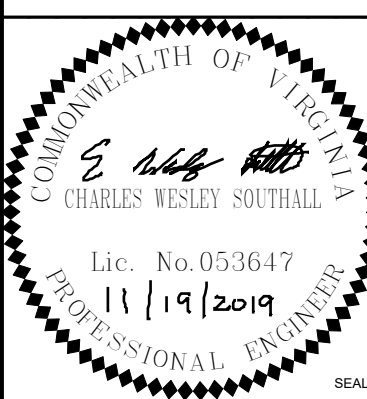
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GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- ALL ELEVATIONS SHOWN ARE FROM AFF TO BOTTOM OF OBJECT UNLESS OTHERWISE NOTED.

WORK NOTES

- M40 FLEXIBLE DUCT CONNECTION.
M61 PROVIDE STRUCTURAL STAND FOR MOUNTING OF UNIT. STAND SHALL BE SEISMICALLY RATED AND SHALL MEET REQUIREMENTS OF SPECIFICATION SECTION 23 05 48.19.
M72 PROVIDE UNION/FLANGE TO FACILITATE UNIT MAINTENANCE AND COIL REMOVAL.
M80 FCU SHALL BE MOUNTED ABOVE EA AND RA DUCTS AND DISCHARGE TOWARD AHU-T1A/B.



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CHESAPEAKE, VA 23320

APPROVED: *[Signature]*
FOR COMMANDER NAVFAC

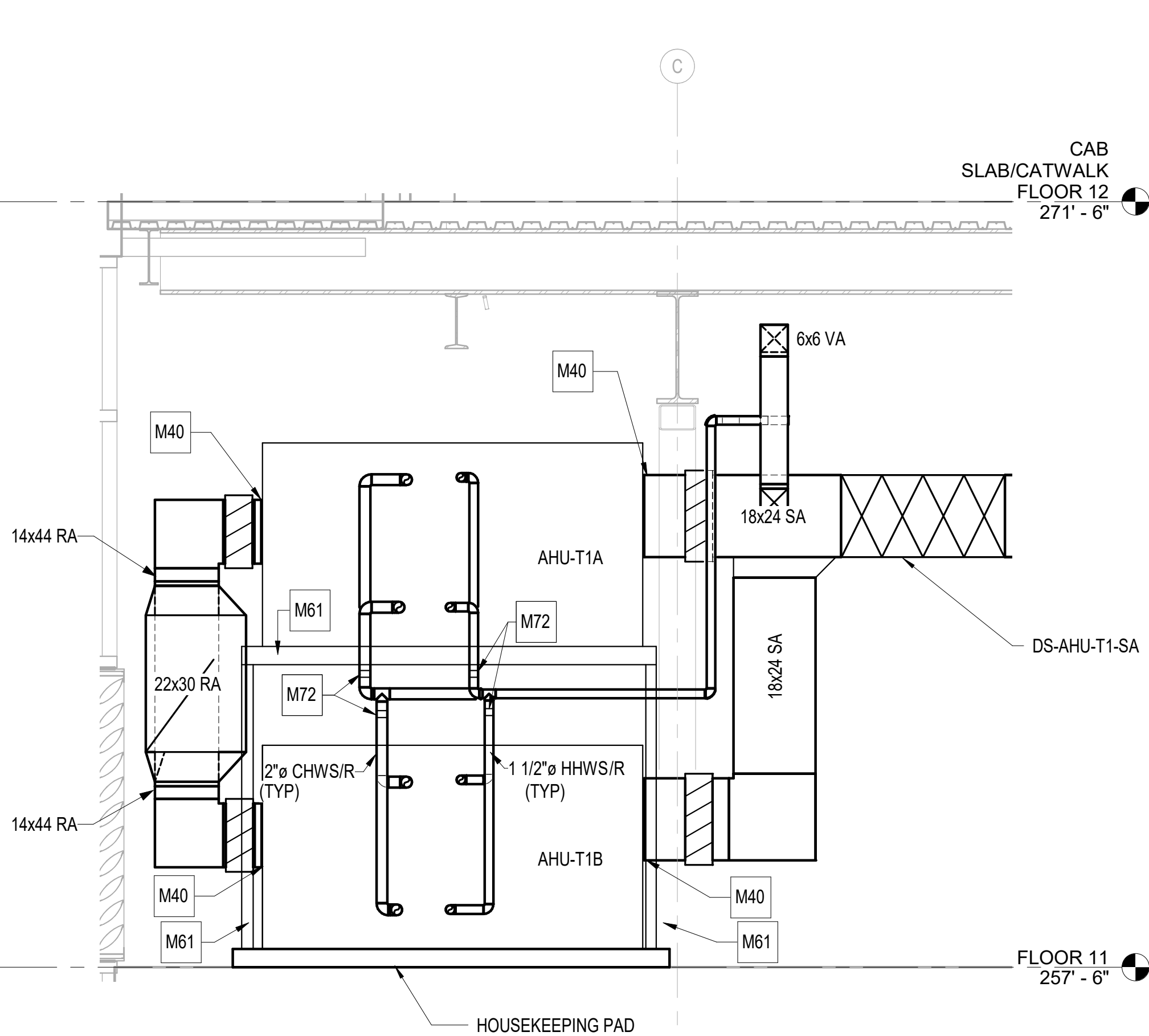
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DES: CWS | DRW: CAW | CHK: RJ
PM/DM: CJC/KDB
BRANCH MANAGER: *[Signature]*
CHIEF ENGINEER: EJA
FIRE PROTECTION: HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL STATION - NORFOLK, VA
CHERRY POINT, NC
FY20 P-228 ATC TOWER AND AIR OPERATIONS
ATOT - MECHANICAL SECTION

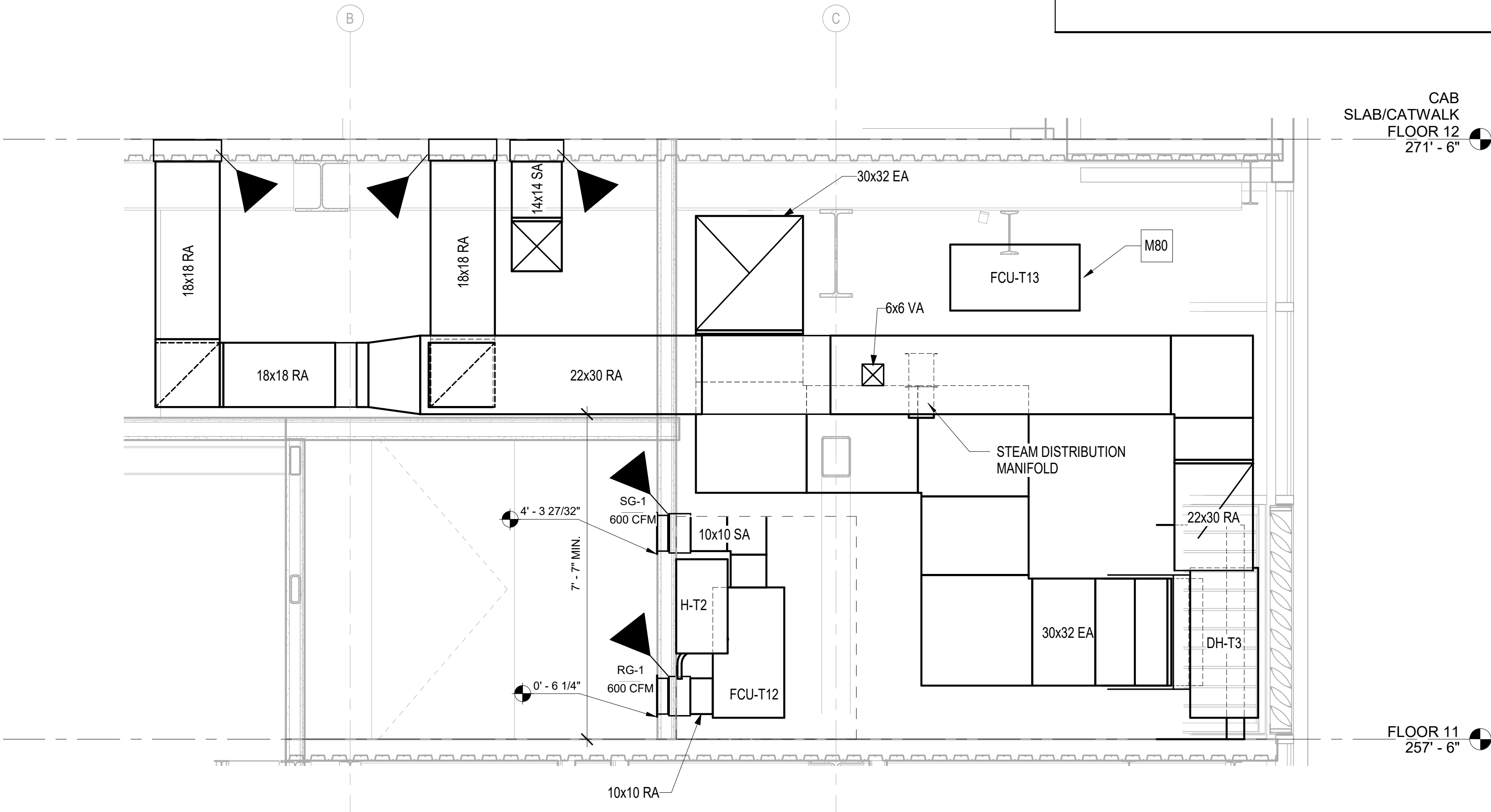
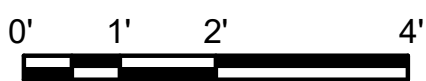
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PROJECT NO.: 1596300
CONSTR. CONTR. NO.:
NAVFAC DRAWING NO.: 12801720
SHEET 460 OF 686

M-321

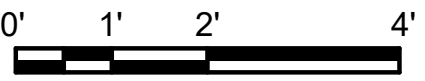
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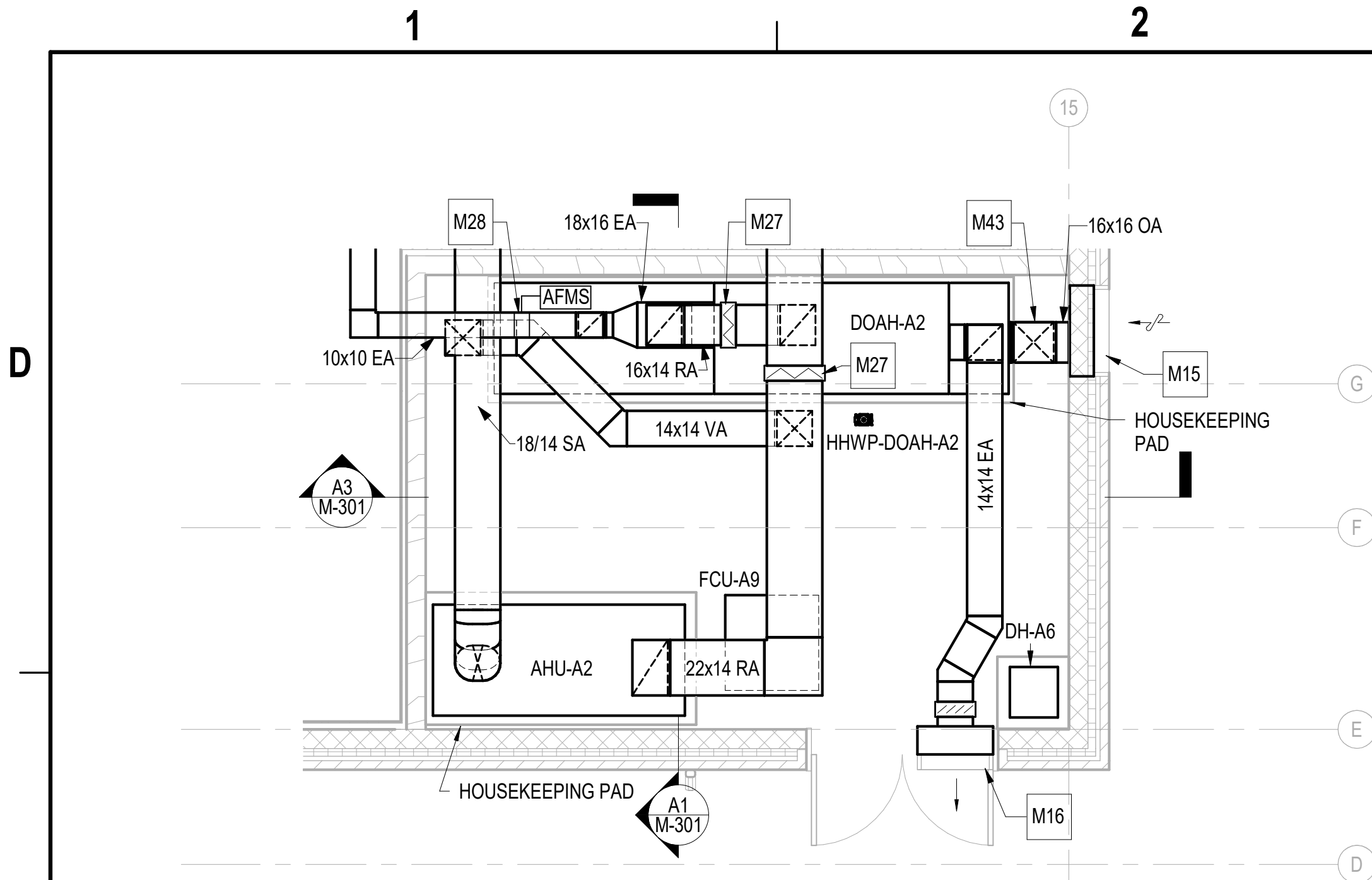


A1 AHU-T1A / AHU-T1B SECTION
1/2" = 1'-0" (MH123)

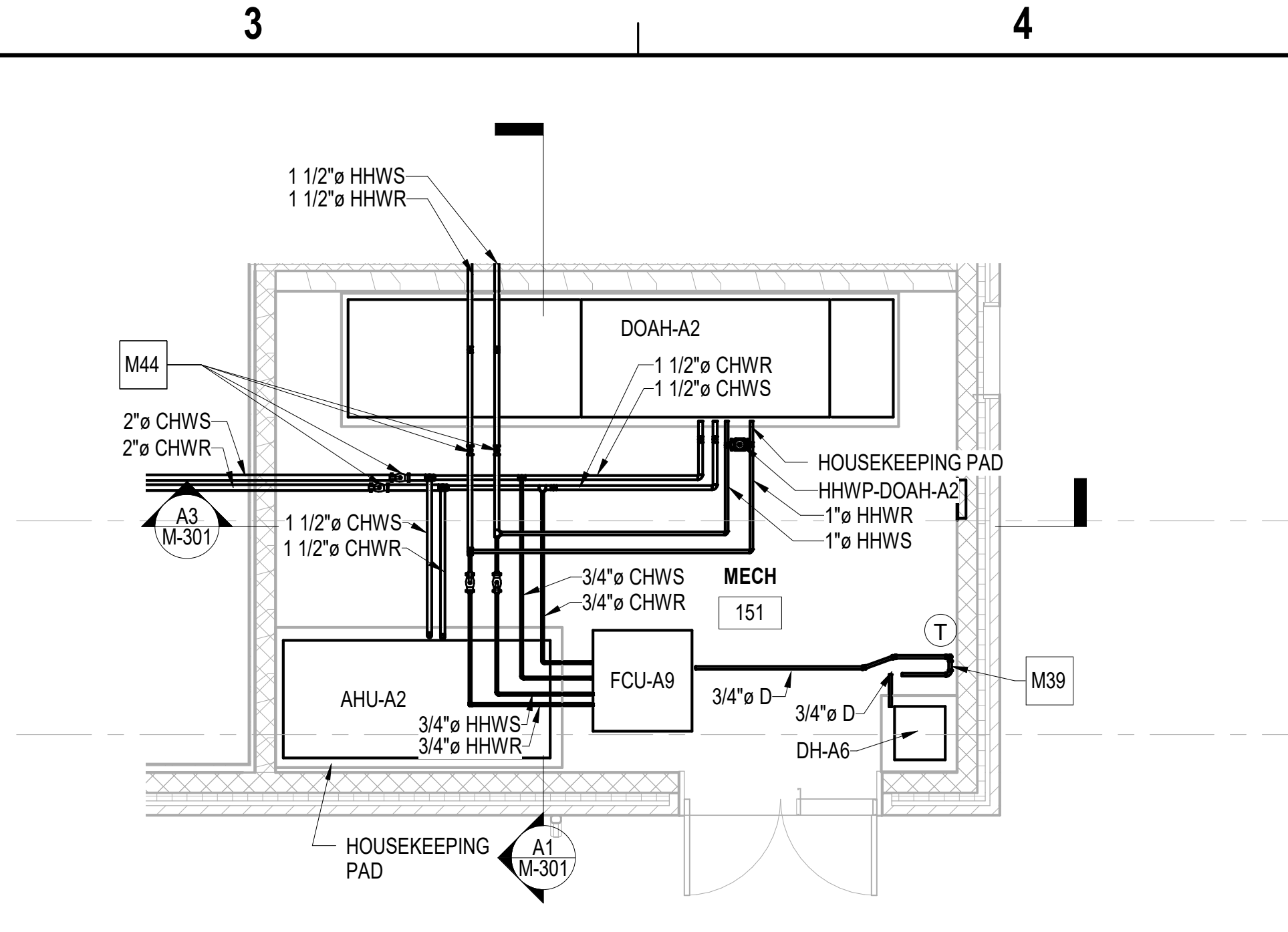


A3 11TH FLOOR SECTION
1/2" = 1'-0" (MH123)

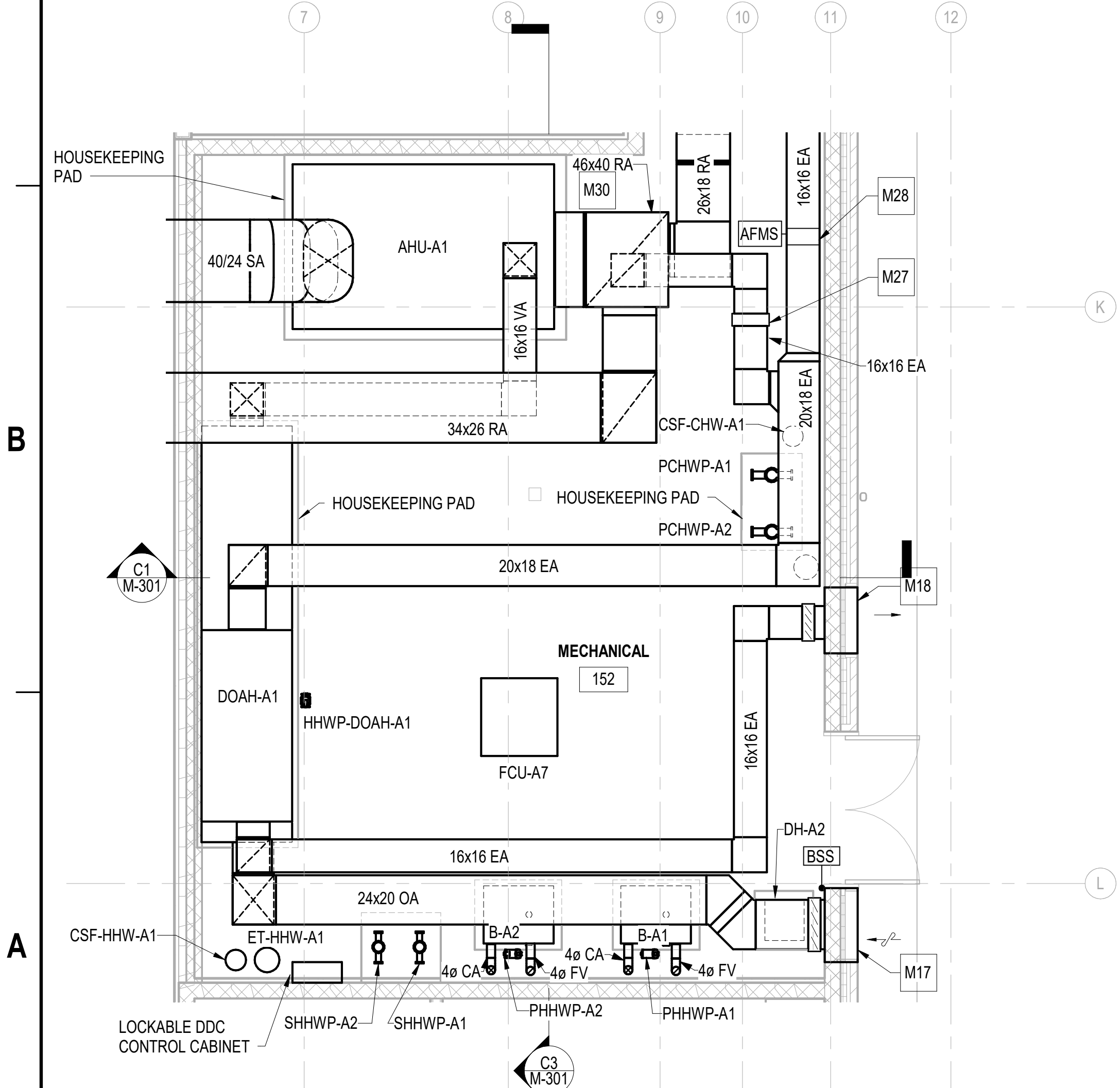




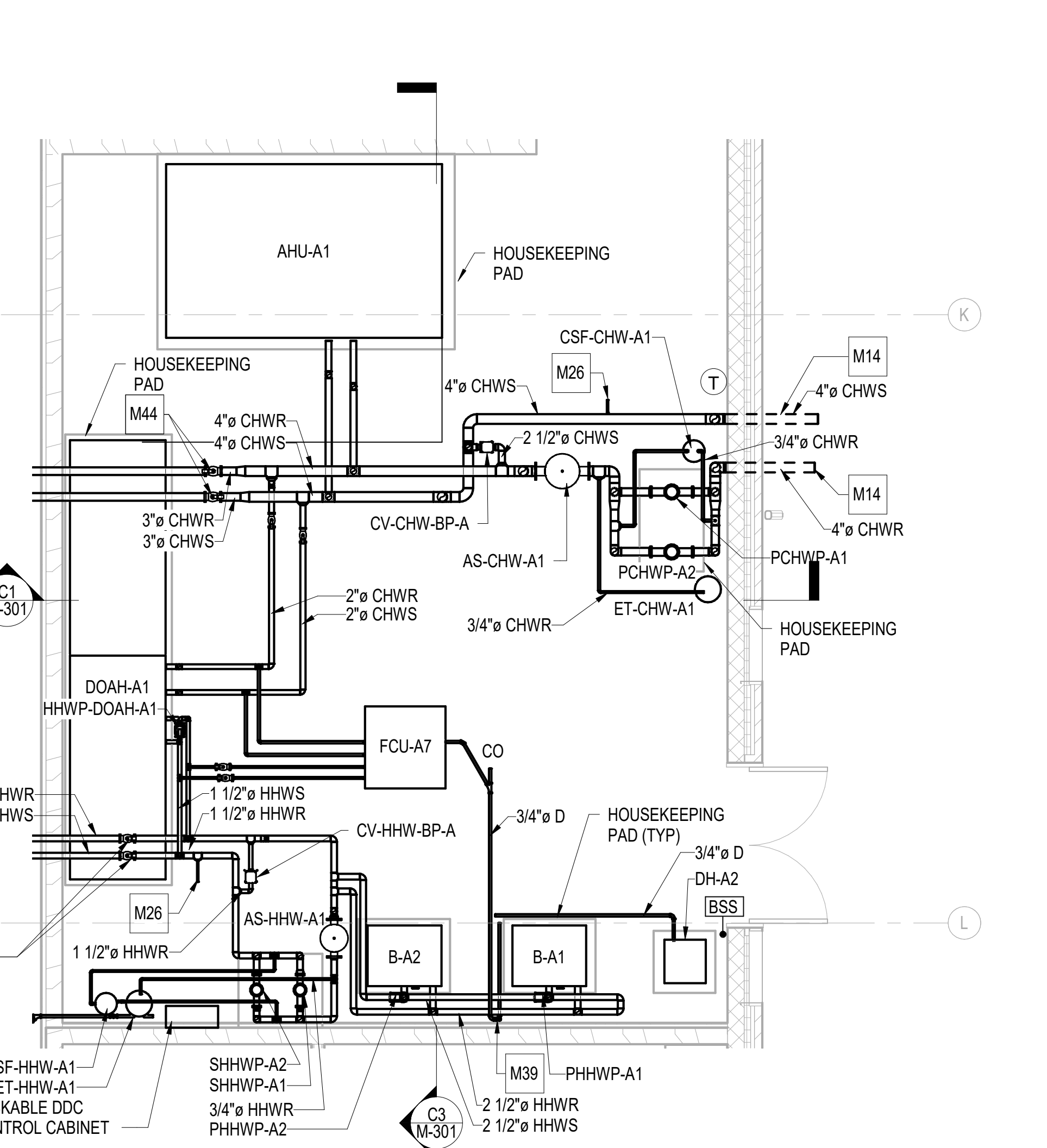
C1 MECHANICAL - ROOM 151 ENLARGED DUCT PLAN



C3 MECHANICAL - ROOM 151 ENLARGED PIPE PLAN



A1 MECHANICAL - ROOM 152 ENLARGED DUCT PLAN



A3 MECHANICAL - ROOM 152 ENLARGED PIPE PLAN

GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

WORK NOTES

- M14 UNDERGROUND PIPING.
- M15 36" x 24" INTAKE PLENUM. ATTACH PLENUM TO BACK OF LOUVER.
- M16 30" x 24" EXHAUST PLENUM. ATTACH PLENUM TO BACK OF LOUVER.
- M17 42" x 36" INTAKE PLENUM. ATTACH PLENUM TO BACK OF LOUVER.
- M18 30" x 36" EXHAUST PLENUM. ATTACH PLENUM TO BACK OF LOUVER.
- M26 DUAL TURBINE INSERTION TYPE FLOWMETER. INSTALL IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AND WITH SUFFICIENT UNOBSTRUCTED ENTRY AND EXIT PIPE LENGTH.
- M27 OPPOSED BLADE MOTORIZED DAMPER WITH INTEGRAL AIRFLOW MEASURING STATION. INSTALL IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AND WITH SUFFICIENT UNOBSTRUCTED ENTRY AND EXIT LENGTH.
- M28 AIRFLOW MEASURING STATION. INSTALL IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS AND WITH SUFFICIENT UNOBSTRUCTED ENTRY AND EXIT LENGTH.
- M30 OPPOSED BLADE MOTORIZED DAMPER IN VERTICAL IN 46x40 RA.
- M39 CONDENSATE DRAIN SHALL DROP DOWN AT WALL AND OVER TO FLOOR DRAIN.
- M43 PROVIDE ISOLATION DAMPER IN VERTICAL.
- M44 ISOLATION VALVE.

DATE		11/19/19	APPR
SYN		DESCRIPTION	ISSUED FOR CONSTRUCTION
0			

1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

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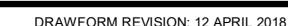
FOR COMMANDER NAVFAC
ACTIVITY

SATISFACTORY TO		DATE	
DES	CWS	BRW	CWS
PMOM			CJC/KDB
BRANCH MANAGER		CHIEF ENGINEER	
FIRE PROTECTION		HPN	

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL STATION - NORFOLK, VA
CHERRY POINT, NC

MCAS CHERRY POINT
FY20 P-228 ATC TOWER AND AIR OPERATIONS
AIR OPS - MECHANICAL ENLARGED PLANS

SCALE: AS NOTED
PROJECT NO.: 1596300
CONSTR. CONTR. NO.: 12801721
NAVJAC DRAWING NO.: 12801721
SHEET 461 OF 686
M-401
DRAWING REVISION: 12 APRIL 2018



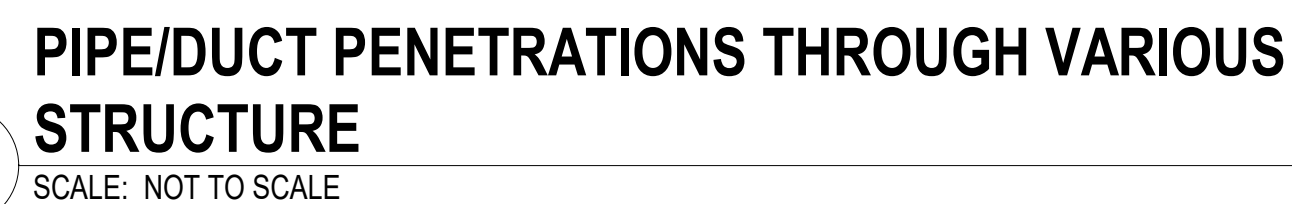
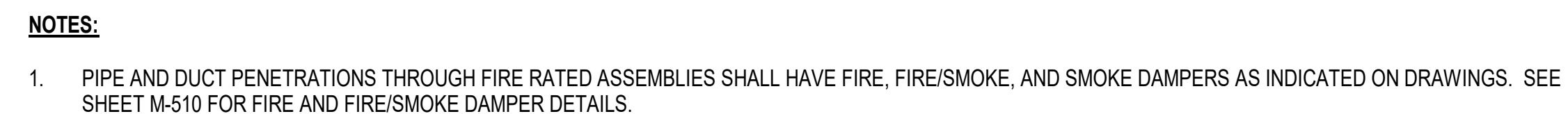
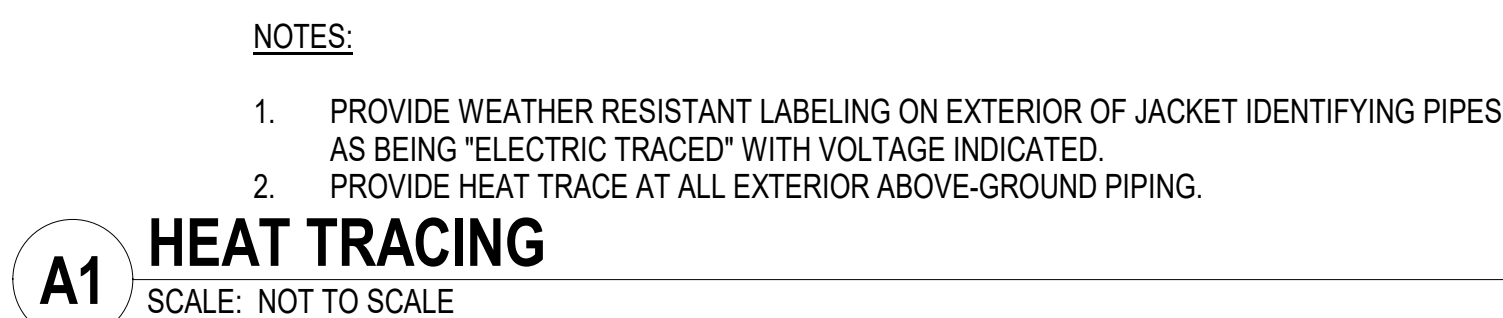


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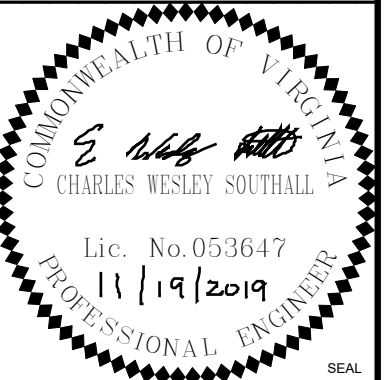
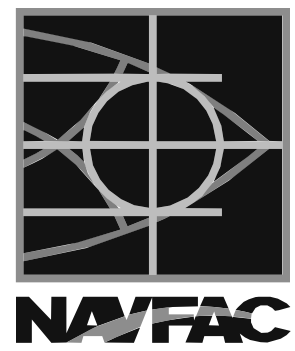
3

A



1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

1. ALL COILS SHALL BE PIPED FOR COUNTER FLOW ARRANGEMENT. (SUPPLY WATER ENTERS NEAREST COIL FACE WHERE AIR LEAVES.)
2. ALL COIL DRAIN AND VENT CONNECTIONS TO BE INSTALLED THROUGH HVAC UNIT CASING TO PERMIT COMPLETE DRAINAGE AND VENTING OF COILS FROM OUTSIDE OF UNIT.
3. LOCATE VALVES AND FLANGES SO THAT PIPING CAN BE DISCONNECTED AND COILS WITHDRAWN IN COIL REMOVAL SPACE.
4. INDIVIDUAL LINES TO EACH COIL SHALL BE SAME SIZE.
5. CONDENSATE DRAIN PIPING SIZE SHALL BE EQUAL TO THE DRAIN PAN CONNECTION SIZE AND HAVE LOOP SEAL TRAPS AS INDICATED.
6. COIL BANKS WITH A SINGLE COIL REQUIRE ONLY A SINGLE CALIBRATED MANUAL BALANCING VALVE.
7. INSTALL MANUAL CALIBRATED BALANCING VALVES IN CONFORMANCE WITH VALVE MANUFACTURER'S RECOMMENDED SPACING UP AND DOWNSTREAM FROM PIPE CHANGES IN DIRECTION AND/OR OTHER VALVES AND COMPONENTS IN THE PIPING.
8. HHWP FOR FREEZE PROTECTION. PUMP IS ONLY FOR PREHEAT COILS (DOAHS). SEE A3/M-510 FOR PUMP DETAIL.
9. CALIBRATED BALANCING VALVES FOR BALANCING WATER FLOW BELOW 1 GPM, SHALL BE 1/2".

[illegible]

**BURNS
MCDONNELL**
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED

James E. Doolittle

James E. Doolittle

SATISFACTORY TO		DATE	
DES	CWS	DRW	CWS
		CHK	RJ
PM/DM		CJC/KDB	
BRANCH MANAGER		724	
CHIEF ENG/ARCH		EJA	
FIRE PROTECTION		HPN	

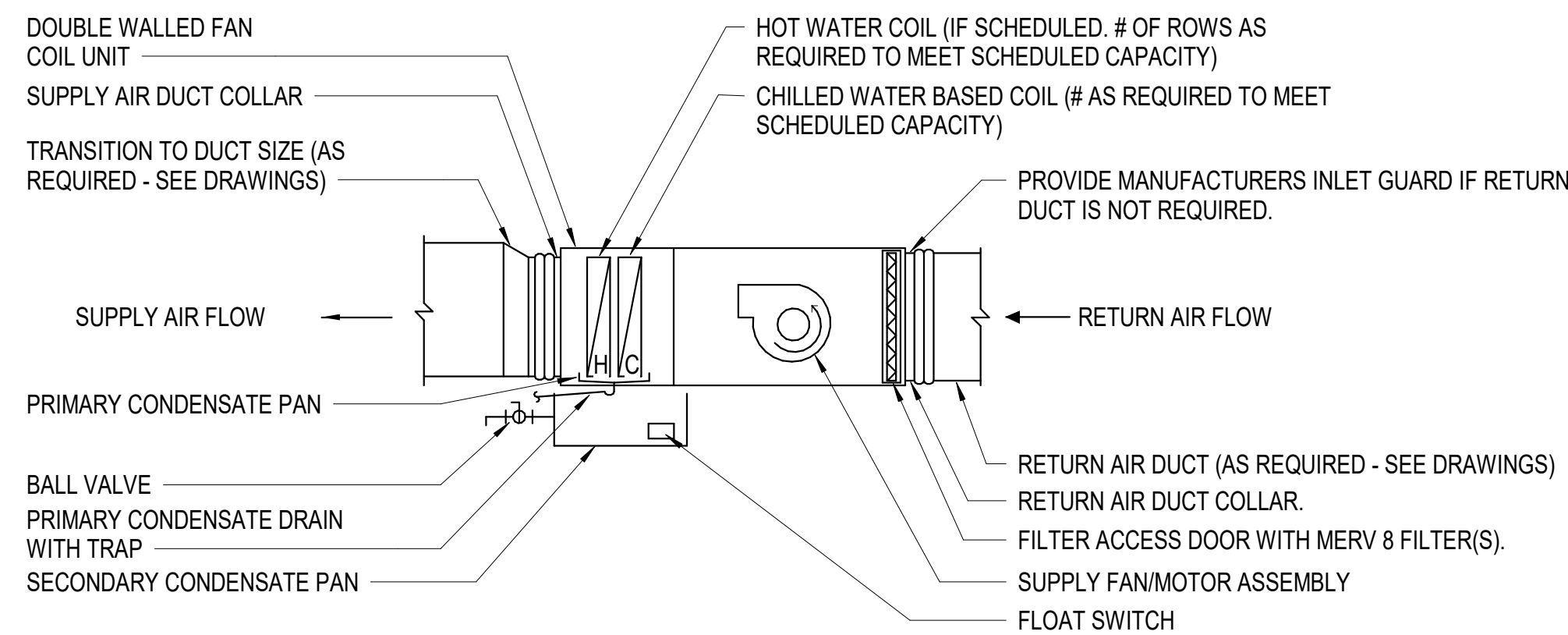
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC NAVAL STATION - NORFOLK, VA MARINE CORPS IPT	CHERRY POINT, NC MCAS CHERRY POINT FY20 P-228 ATC TOWER AND AIR OPERATIONS	MECHANICAL DETAILS
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SCALE: AS NOTED			
EPROJECT NO.:		1596300	
CONSTR. CONTR. NO.			
NAVFAC DRAWING NO.			
		12801725	
SHEET	465	OF	686

M-502

GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

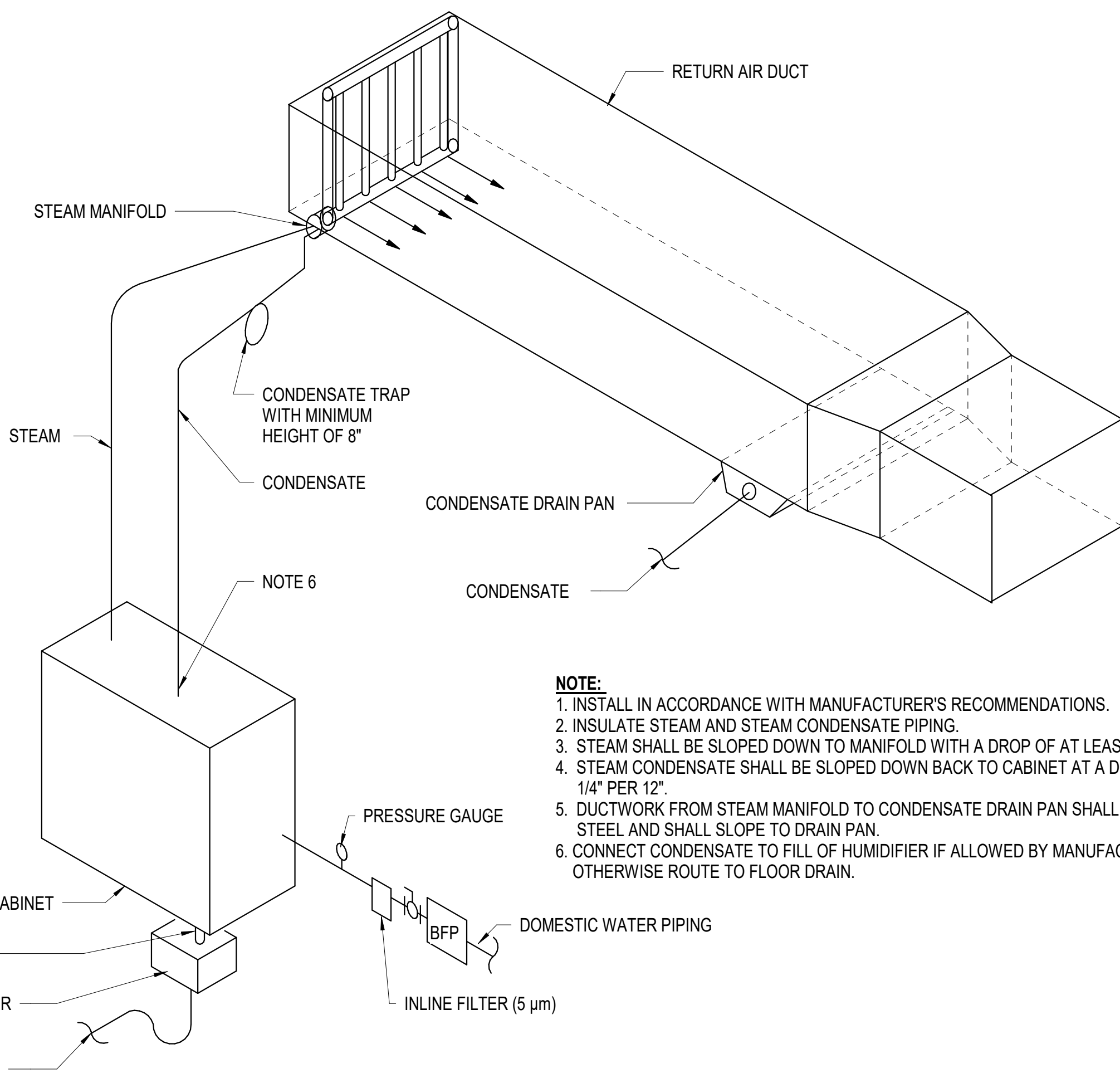


NOTES:

- PROVIDE FLOAT SWITCH TO DISABLE UNIT FAN UPON SENSING WATER IN SECONDARY DRAIN PAN.
- SLOPE PRIMARY CONDENSATE DRAIN AT 1/8" PER FOOT AWAY FROM UNIT. ROUTE TO NEAREST FLOOR DRAIN AS INDICATED ON FLOOR PLANS.
- PROVIDE FULL PORT BALL VALVE (NC) AND ELBOW FOR SECONDARY CONDENSATE PAN. TERMINATE ELBOW IN A CONVENIENT LOCATION FOR DRAINING CONDENSATE PAN INTO BUCKET WITHOUT DAMAGE TO STRUCTURE OR FINISHES.
- SECONDARY DRAIN PAN IS NOT TO INTERFERE WITH ABILITY OF TO CHANGE FILTER AND SERVICE FAN.

B1 HORIZONTAL FCU/CRAH WITH SECONDARY DRAIN PAN

SCALE: NOT TO SCALE

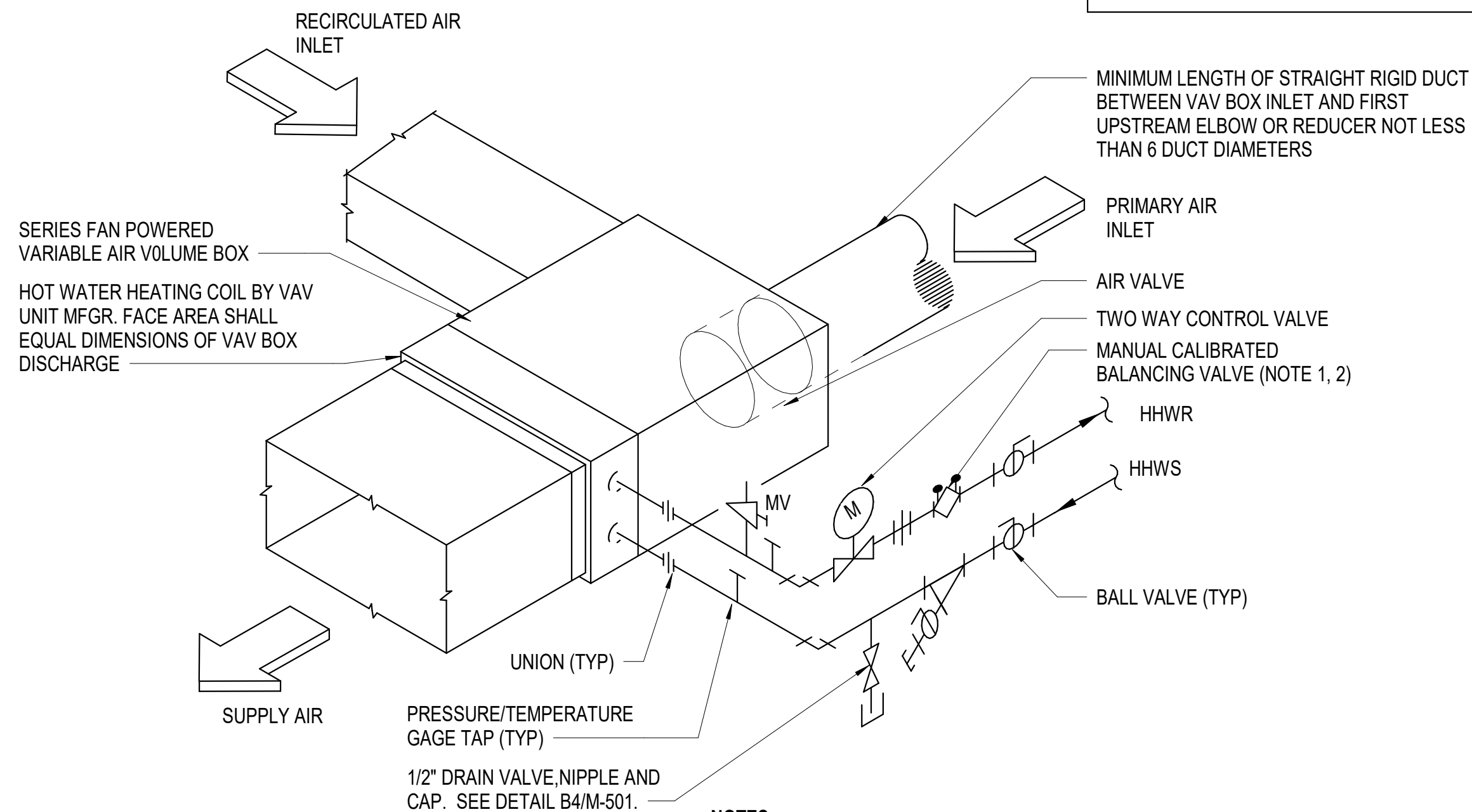


NOTE:

- INSTALL IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
- INSULATE STEAM AND STEAM CONDENSATE PIPING.
- STEAM SHALL BE SLOPED DOWN TO MANIFOLD WITH A DROP OF AT LEAST 1" PER 12".
- STEAM CONDENSATE SHALL BE SLOPED DOWN BACK TO CABINET AT A DROP OF AT LEAST 1/4" PER 12".
- DUCTWORK FROM STEAM MANIFOLD TO CONDENSATE DRAIN PAN SHALL BE STAINLESS STEEL AND SHALL SLOPE TO DRAIN PAN.
- CONNECT CONDENSATE TO FILL OF HUMIDIFIER IF ALLOWED BY MANUFACTURER OTHERWISE ROUTE TO FLOOR DRAIN.

A1 DUCT MOUNTED HUMIDIFIER DETAIL

SCALE: NOT TO SCALE

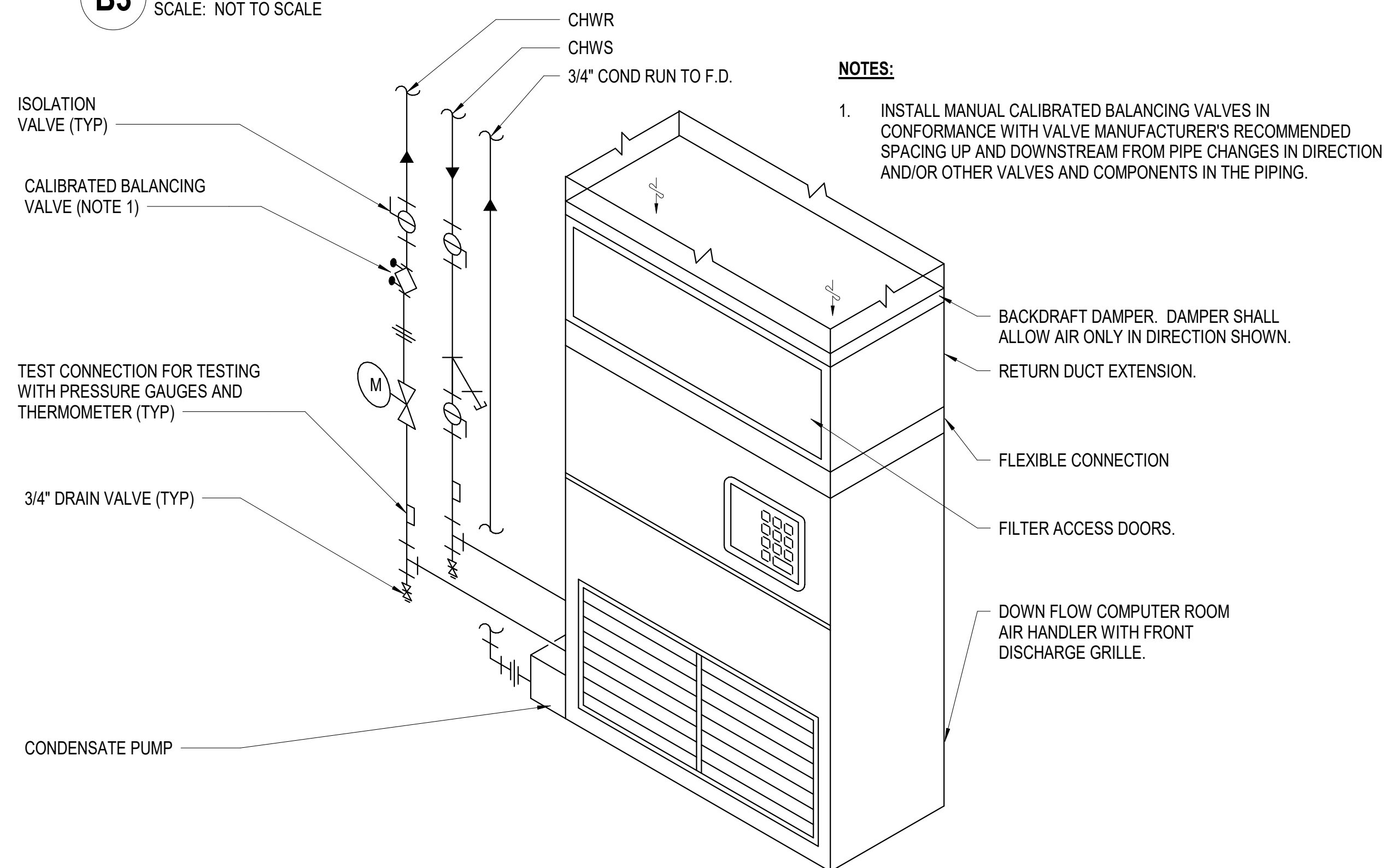


NOTES:

- INSTALL MANUAL CALIBRATED BALANCING VALVES IN CONFORMANCE WITH VALVE MANUFACTURER'S RECOMMENDED SPACING UP AND DOWNSTREAM FROM PIPE CHANGES IN DIRECTION AND/OR OTHER VALVES AND COMPONENTS IN THE PIPING.
- CALIBRATED BALANCING VALVES FOR BALACING WATER FLOW BELOW 1 GPM, SHALL BE 1/2".

B3 SERIES FAN POWERED VAV BOX DETAIL

SCALE: NOT TO SCALE



NOTES:

- INSTALL MANUAL CALIBRATED BALANCING VALVES IN CONFORMANCE WITH VALVE MANUFACTURER'S RECOMMENDED SPACING UP AND DOWNSTREAM FROM PIPE CHANGES IN DIRECTION AND/OR OTHER VALVES AND COMPONENTS IN THE PIPING.

TYPICAL PERIMETER COMPUTER ROOM AIR HANDLER
DETAIL

SCALE: NOT TO SCALE

UNCLASSIFIED

APPR	DATE	11/19/19
SYM	DESCRIPTION	0
BURNS & MCDONNELL 1305 EXECUTIVE BLVD. SUITE 160 CHESAPEAKE, VA 23320		
APPROVED:		
FOR COMMANDER NAVFAC		
ACTIVITY		
SATISFACTORY TO: DATE		
DES	CWS	BRW
CWS	BRW	CWS
CHK	RJ	
PMIDM	CJC/KDB	
BRANCH MANAGER	TJLH	
CHIEF ENGINEER	EJA	
FIRE PROTECTION	HPN	
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC MARINE CORPS IPT NAVAL STATION - NORFOLK, VA MCAS CHERRY POINT CHERRY POINT, NC FY20 P-228 ATC TOWER AND AIR OPERATIONS MECHANICAL DETAILS		
SCALE: AS NOTED		
PROJECT NO.: 1596300		
CONSTR. CONTR. NO.		
NAVFAC DRAWING NO. 12801727		
SHEET 467 OF 686		
M-504		

PLOTTED: 11/19/2019 4:50:07 PM

FILE NAME: C:\Program Files\Autodesk\AutoCAD 2019\Drawings\2019\15963004\OPS-unsaved.dwg

D

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B

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GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
2. DETAILS ARE SHOWN FOR REFERENCE ONLY. SEE SPECIFICATION SECTION 23 05 48.19 SEISMIC BRACING FOR HVAC FOR DESIGN REQUIREMENTS.

APPR

DATE

ISSUED FOR CONSTRUCTION

SYM DESCRIPTION

C

UNCLASSIFIED

B

A



BURNS MEDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED: *[Signature]*

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES CWS BRW CWS CHK RJ

PMOM CJC/KDB

BRANCH MANAGER

CHIEF ENGINEER

FIRE PROTECTION

HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

MECHANICAL DETAILS

SCALE: AS NOTED

PROJECT NO.: 1596300

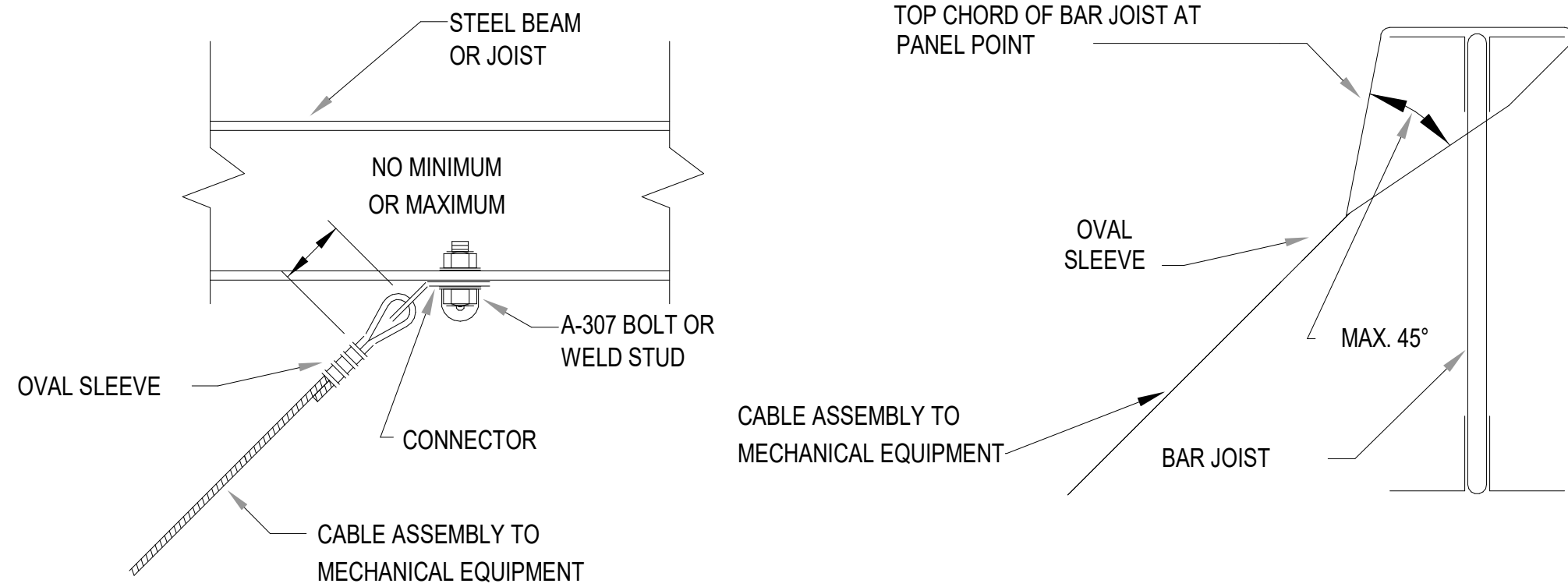
CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12801728

SHEET 468 OF 686

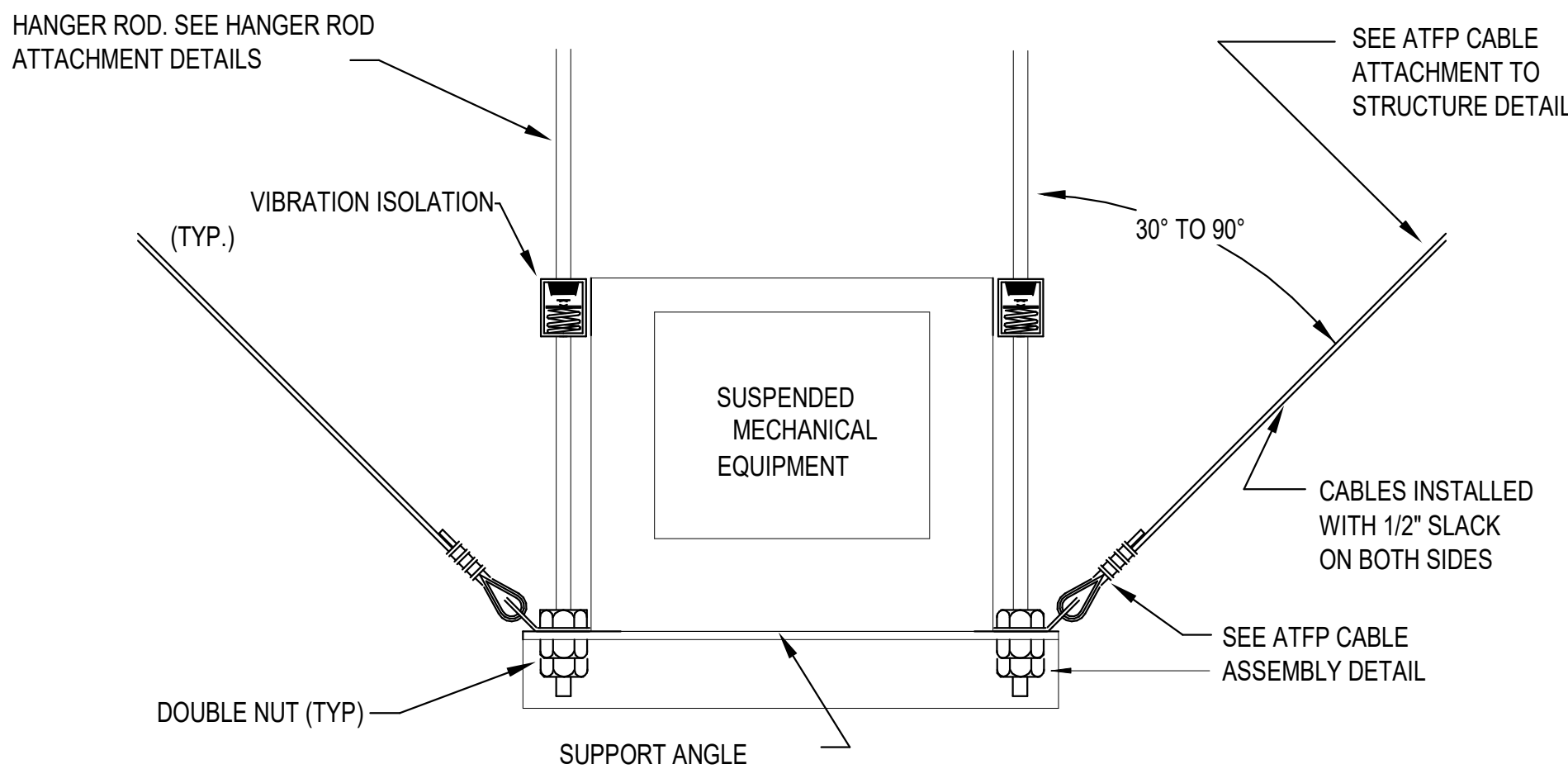
M-505

DRAWING REVISION: 12 APRIL 2018



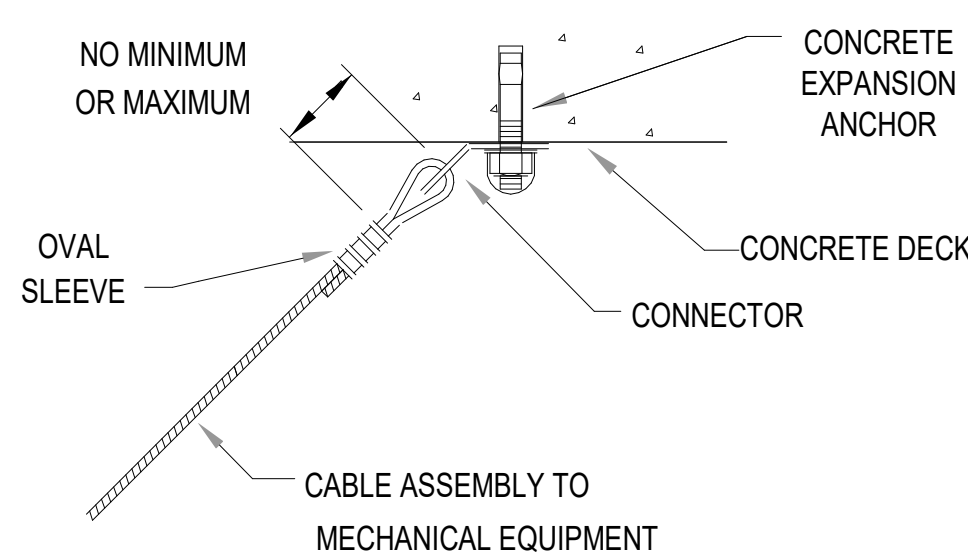
**ATFP CABLE BRACING ATTACHMENT TO STRUCTURE
DETAIL**

C1 SCALE: NOT TO SCALE



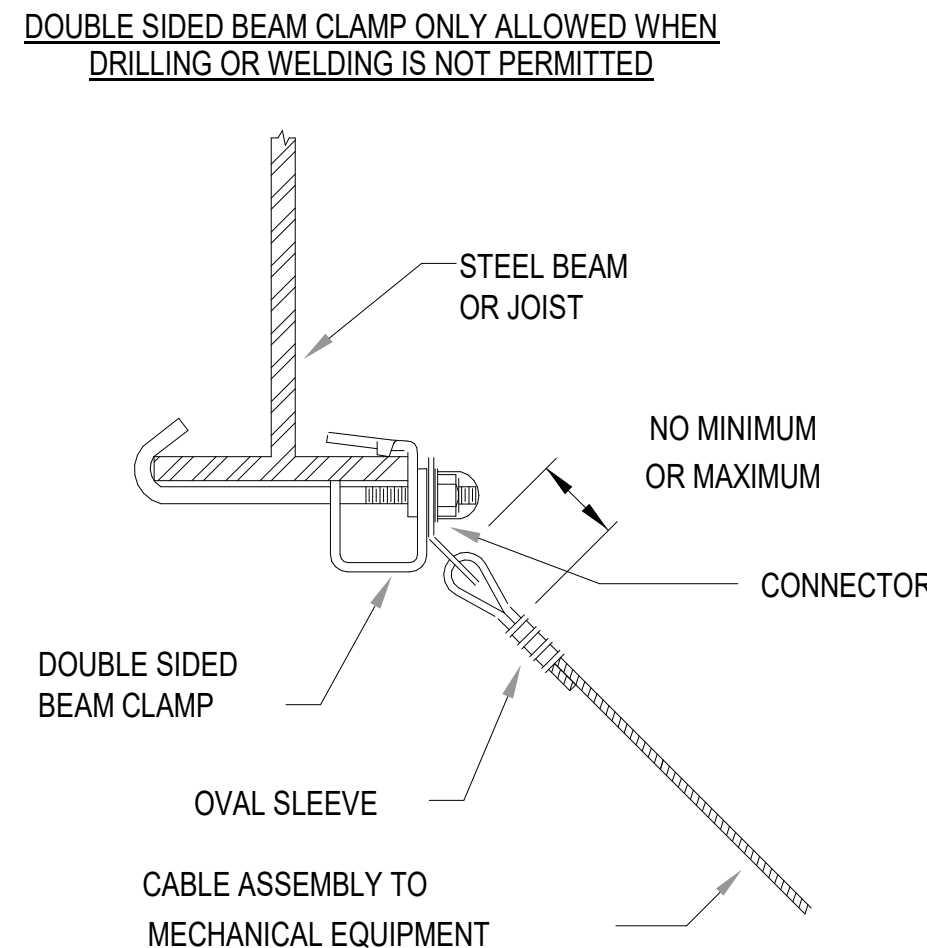
**ATFP BRACING DETAIL FOR SUSPENDED MECHANICAL
EQUIPMENT WITH EXTERNAL VIBRATION ISOLATORS**

A1 SCALE: NOT TO SCALE



**ATFP BRACING DETAIL FOR SUSPENDED INLINE PUMP
WITH VIBRATION ISOLATORS**

A3 SCALE: NOT TO SCALE



UNCLASSIFIED

3

4

5

PLOTTED: 11/19/2019 4:50:09 PM

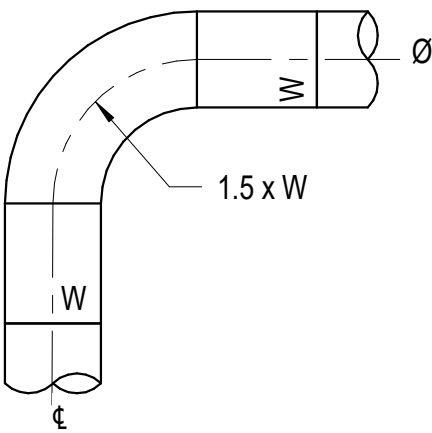
FILE NAME: C:\Program Files\Autodesk\AutoCAD 2019\Drawings\10825_P228-ATC Tower and Air Operations\10825_P228-ATC Tower and Air Operations.dwg

D

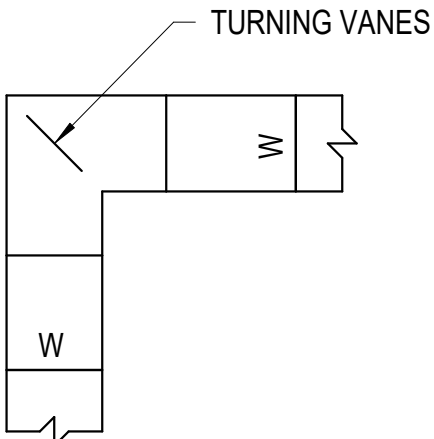
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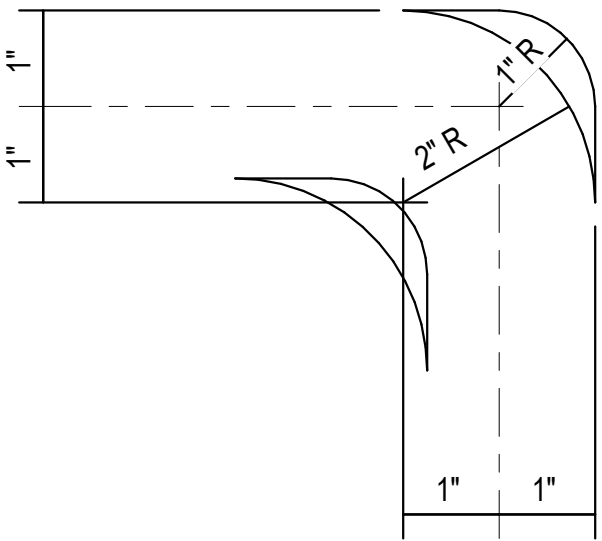
A



ROUND 90 ELBOW

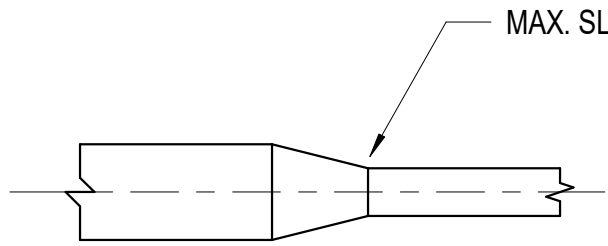


SQUARE 90 ELBOW

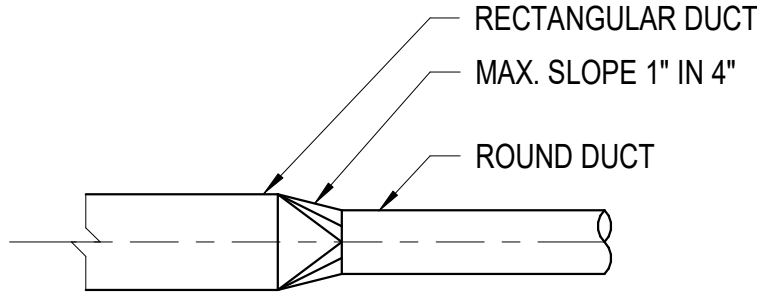


TURNING VANES DETAIL

NOTE:
USE THIS DESIGN WHERE SQUARE 90° ELBOWS ARE SHOWN ON DRAWINGS OR IF SPACE DOES NOT PERMIT ROUND 90° ELBOWS.



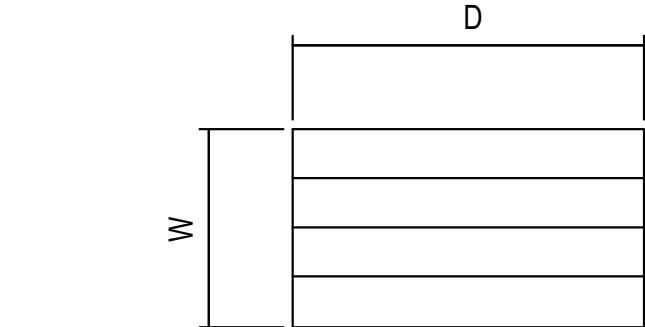
RECTANGULAR TRANSITION



RECTANGULAR TO ROUND TRANSITION

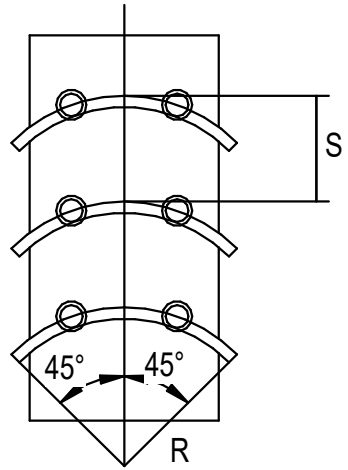
C1 LOW VELOCITY DUCT LAYOUT

SCALE: NOT TO SCALE



S = D/6 (MIN. 2")
W/S = NO. OF SPACES
R = 1 1/4 S
L = 3/4 S

SINGLE VANE SCH.			
	R	S	GA
SMALL	51	38	24
LARGE	114	83	22

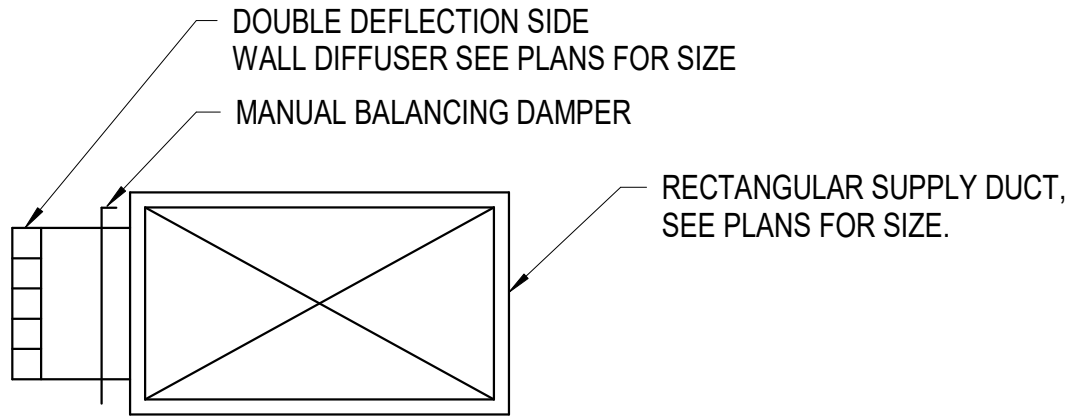


SCREW EVERY SIXTH VANE
BOLT, SCREW OR WELD TO DUCT

BLADES IN TRANSITION ELBOWS SHALL BE DETERMINED ACCORDING TO SMALLEST WIDTH OF ELBOW.

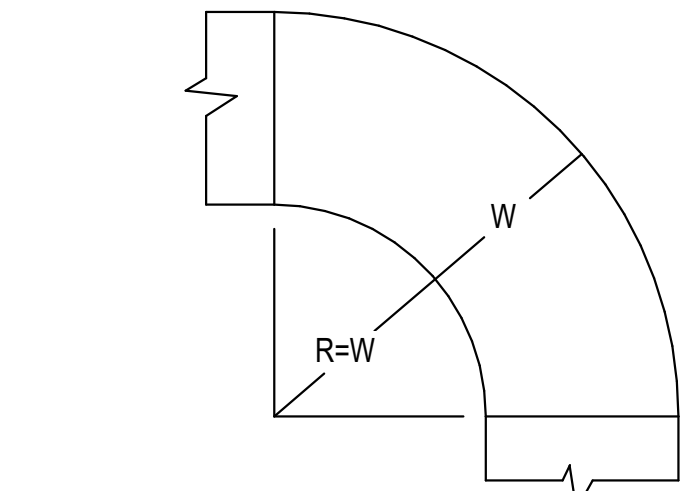
C3 RECTANGULAR AIR DUCT ELBOW TURNING VANES

SCALE: NOT TO SCALE

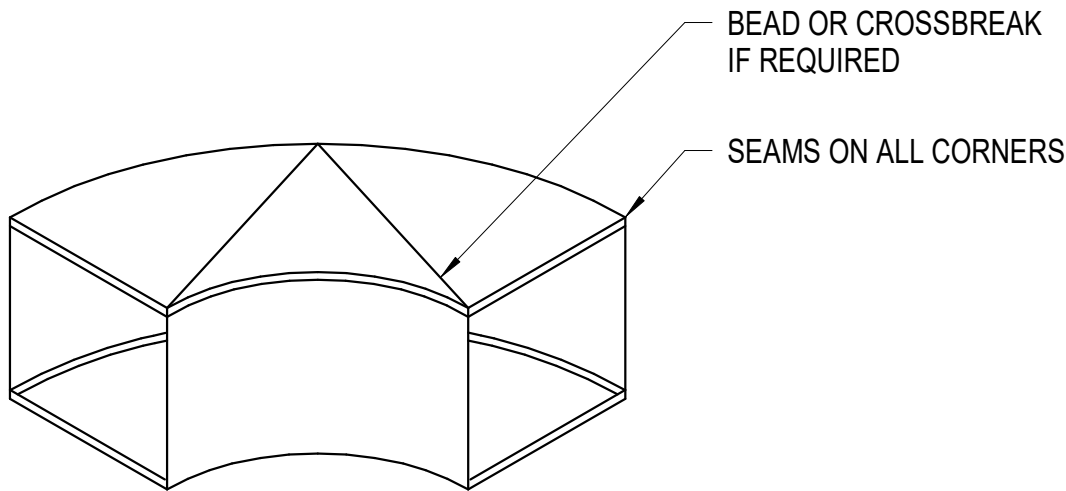


B2 SIDE WALL DIFFUSER

SCALE: NOT TO SCALE

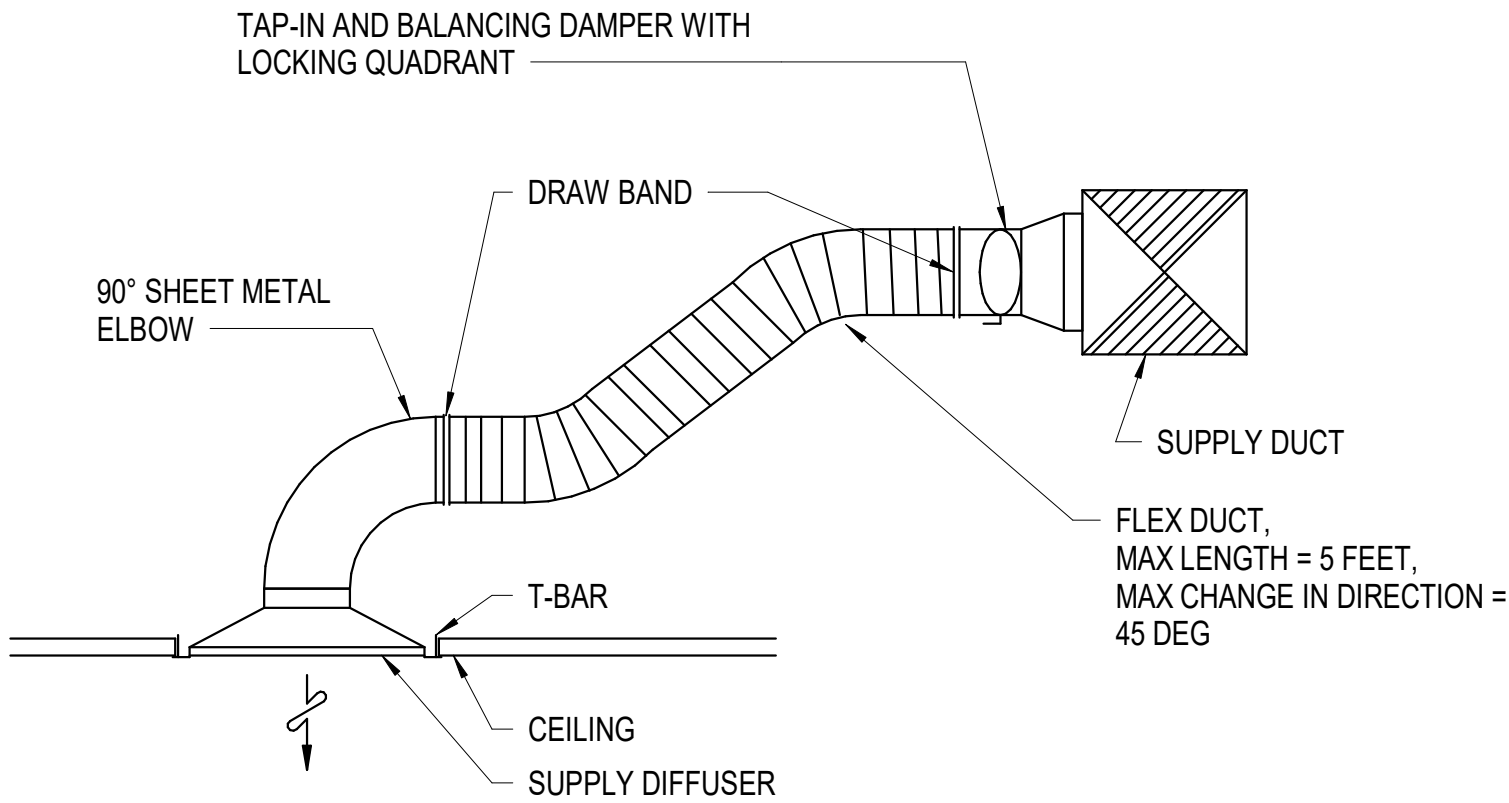


STANDARD RADIUS ELBOW
R=W. CENTERLINE R= 1 1/2 W
ALSO CALLED FULL RADIUS ELBOW.



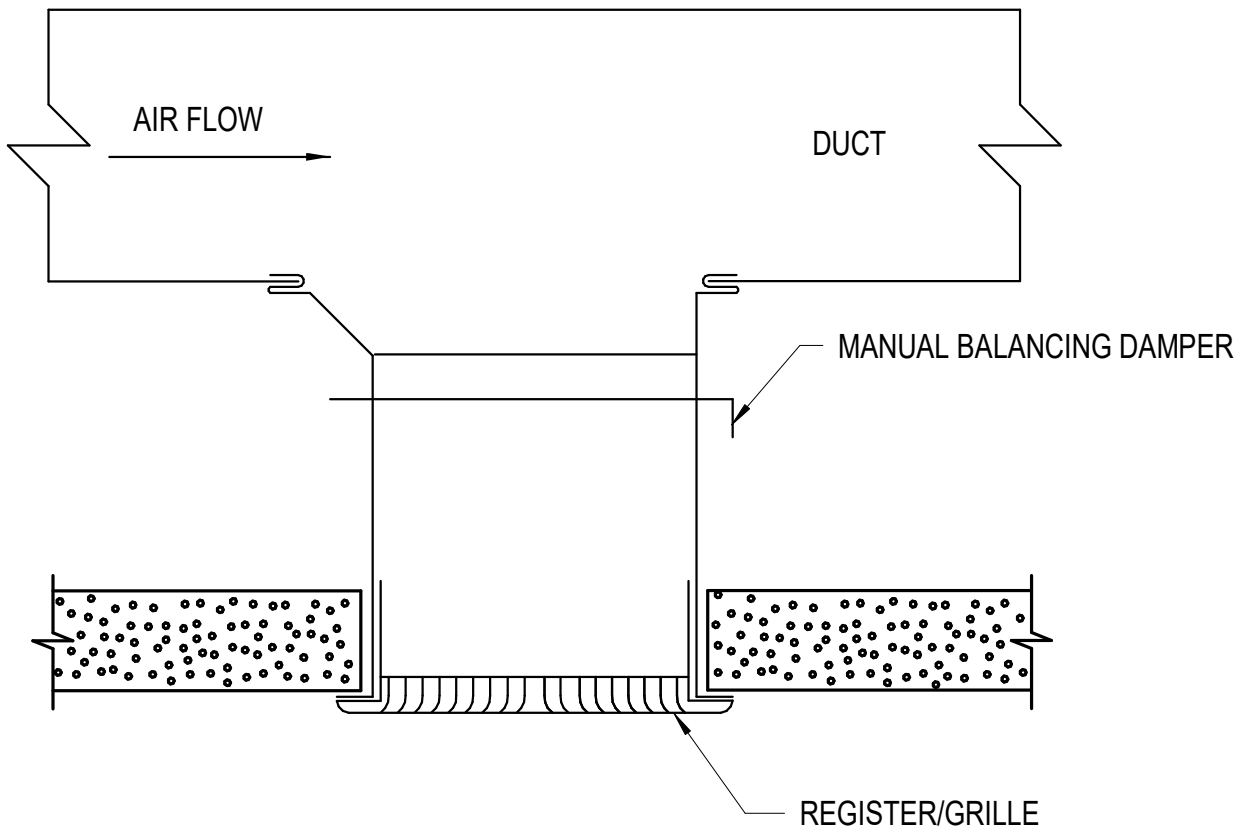
A1 STANDARD CURVE AIR DUCT ELBOW

SCALE: NOT TO SCALE



A2 FLEXIBLE DUCT AT DIFFUSER

SCALE: NOT TO SCALE



NOTE:

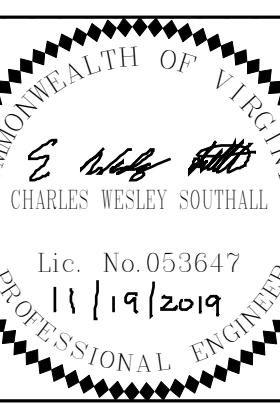
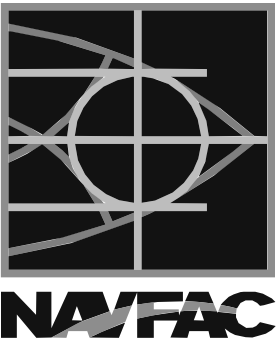
1. GRILLE FLANGES MUST COVER DUCT FLANGES.

A4 VERTICAL REGISTER AND GRILLE CONNECTIONS

SCALE: NOT TO SCALE

GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.



BURNS & MCDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED:

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES CWS BRW CWS CHK RJ

PMOM CJC/KDB

BRANCH MANAGER T.H.

CHIEF ENGINEER EJA

FIRE PROTECTION HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

MECHANICAL DETAILS

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12801729

SHEET 469 OF 686

M-506

DRAWING REVISION: 12 APRIL 2018

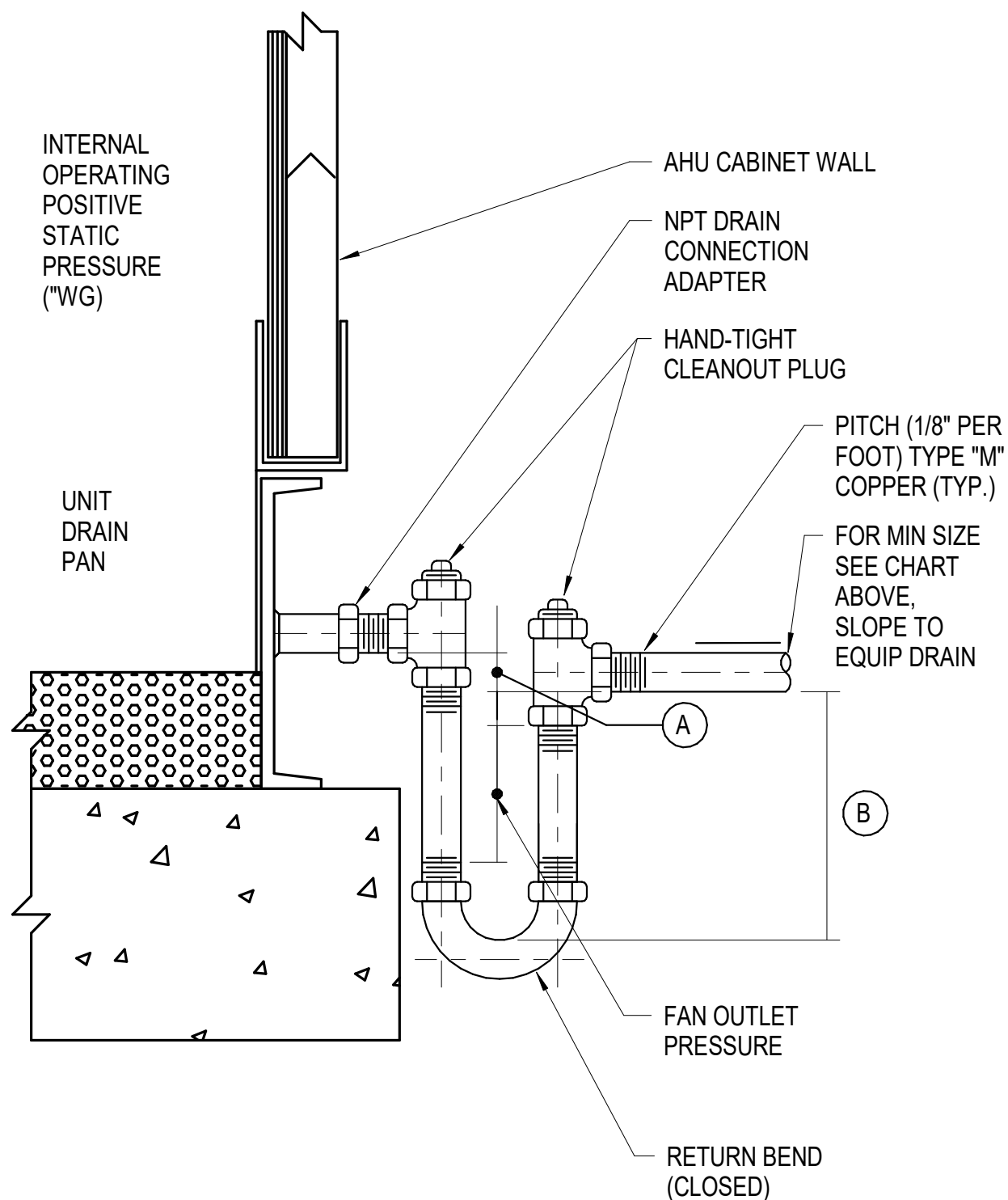
UNCLASSIFIED

A

MINIMUM
CONDENSATE
DRAIN SIZE

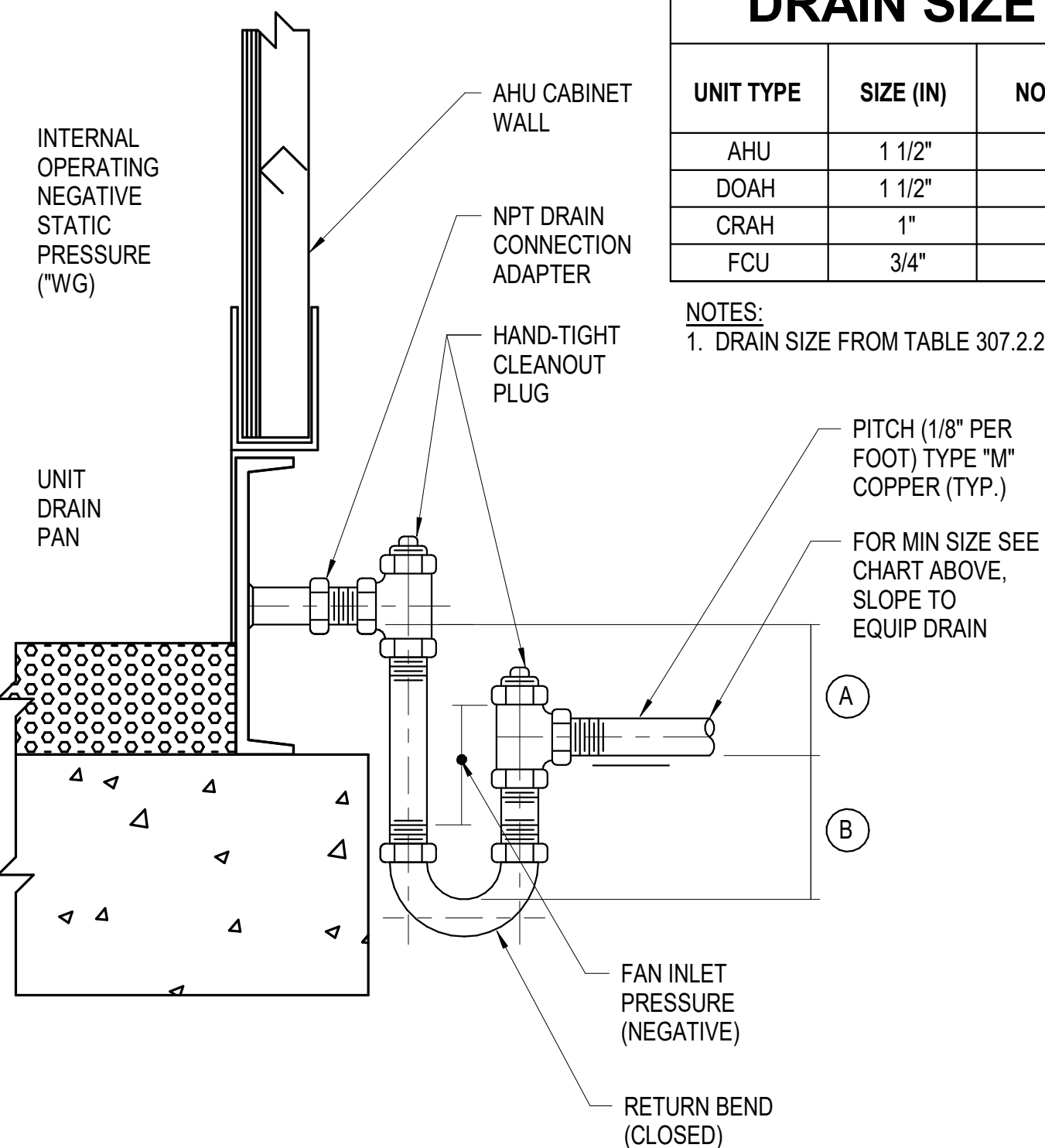
UNIT TYPE	SIZE (IN)	NOTES
AHU	1 1/2"	1
DOAH	1 1/2"	1
CRAH	1"	1
FCU	3/4"	1

NOTES:
1. DRAIN SIZE FROM TABLE 307.2.2 IMC.



NOTES:

1. (A) DIMENSION EQUAL TO 1/2" (MIN.)
2. (B) DIMENSION EQUAL TO MAXIMUM FAN OUTLET PRESSURE + 1/2" (MIN.)

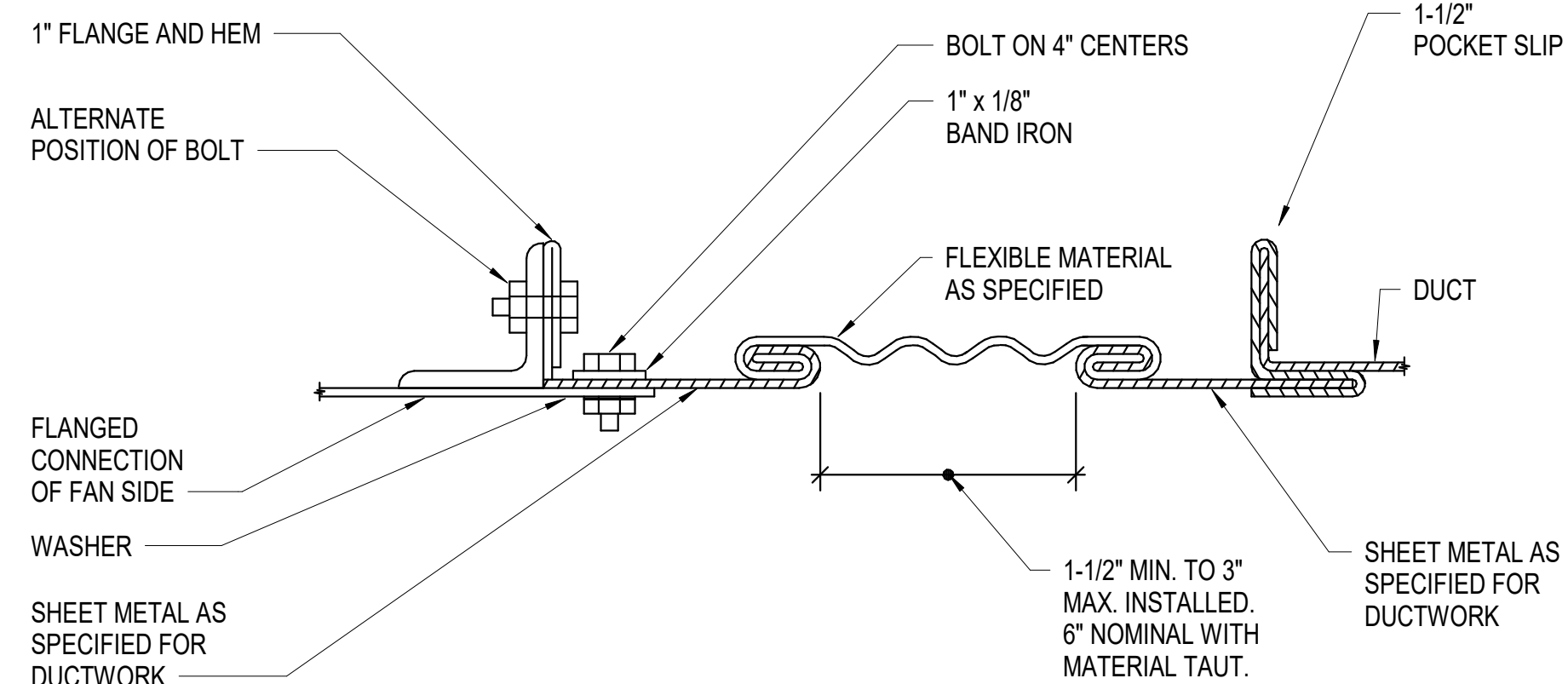


NOTES:

1. (A) DIMENSION EQUAL TO MAXIMUM FAN INLET PRESSURE (INCH WC) + 1" (MIN.)
2. (B) DIMENSION EQUAL TO (A) / 2 (MIN.)

B1 AHU, DOAH, CRAH, FAN CONDENSATE TRAP DRAIN

SCALE: NOT TO SCALE

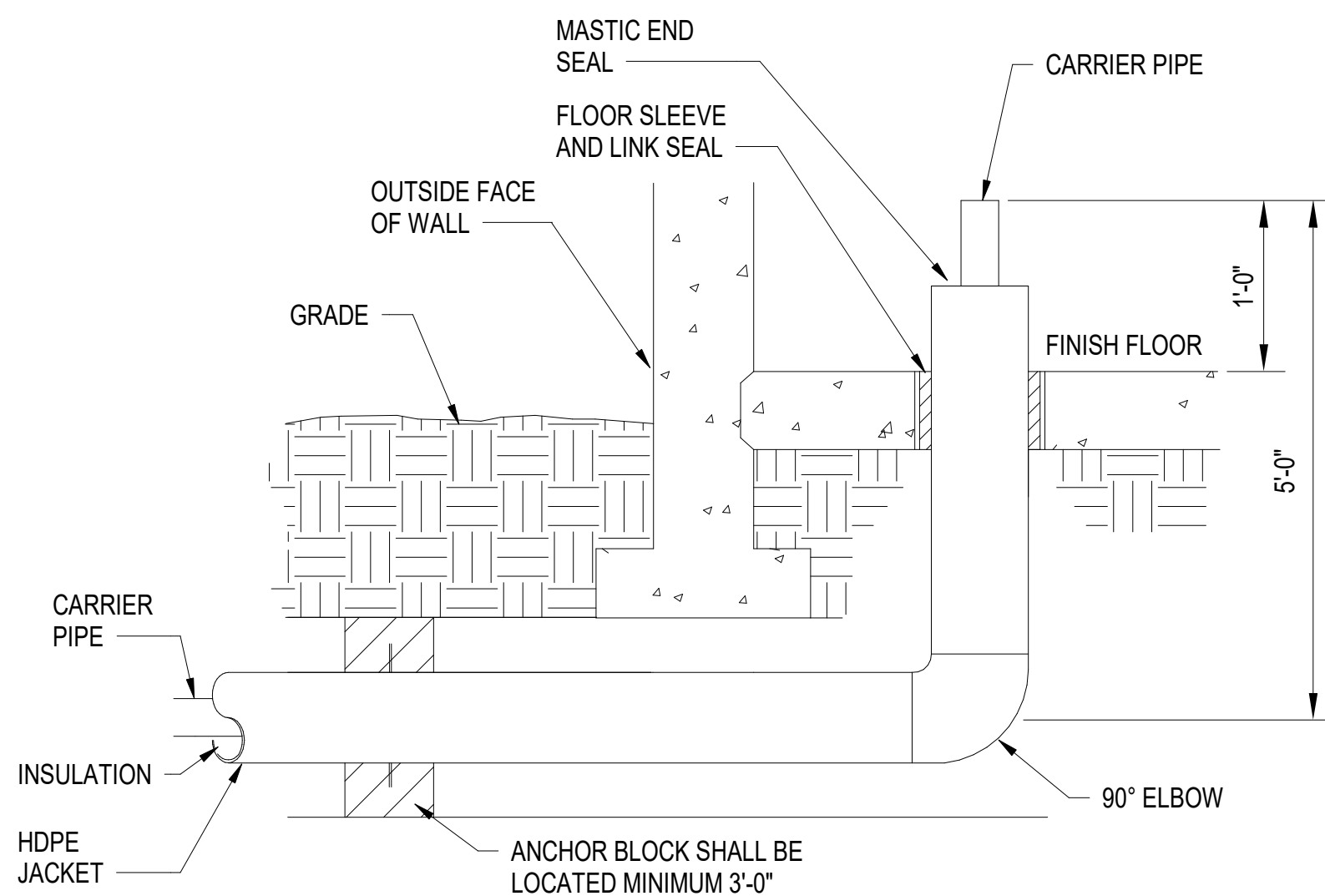


NOTES:

1. PROVIDE FLEXIBLE COPPER GROUNDING STRAPS AT EACH FLEXIBLE CONNECTION.

A1 RECTANGULAR FLEXIBLE CONNECTION

SCALE: NOT TO SCALE

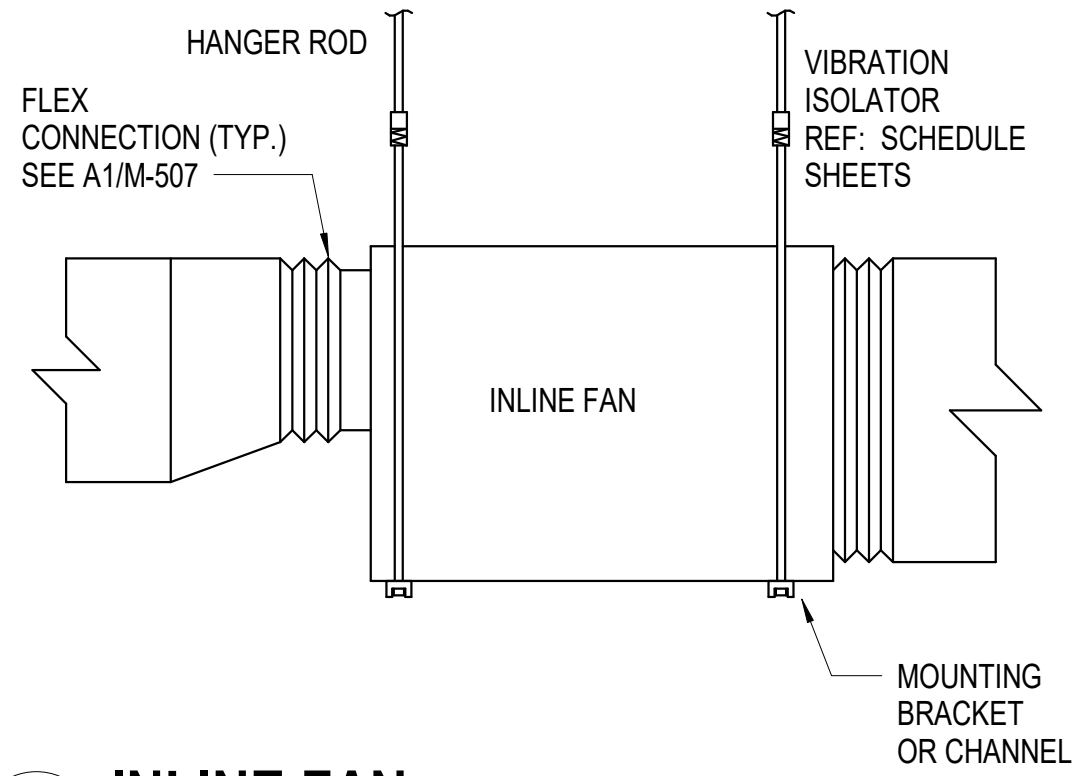


NOTES:

1. INSTALL IN ACCORDANCE WITH THE MANUFACTURER'S RECOMMENDATIONS.

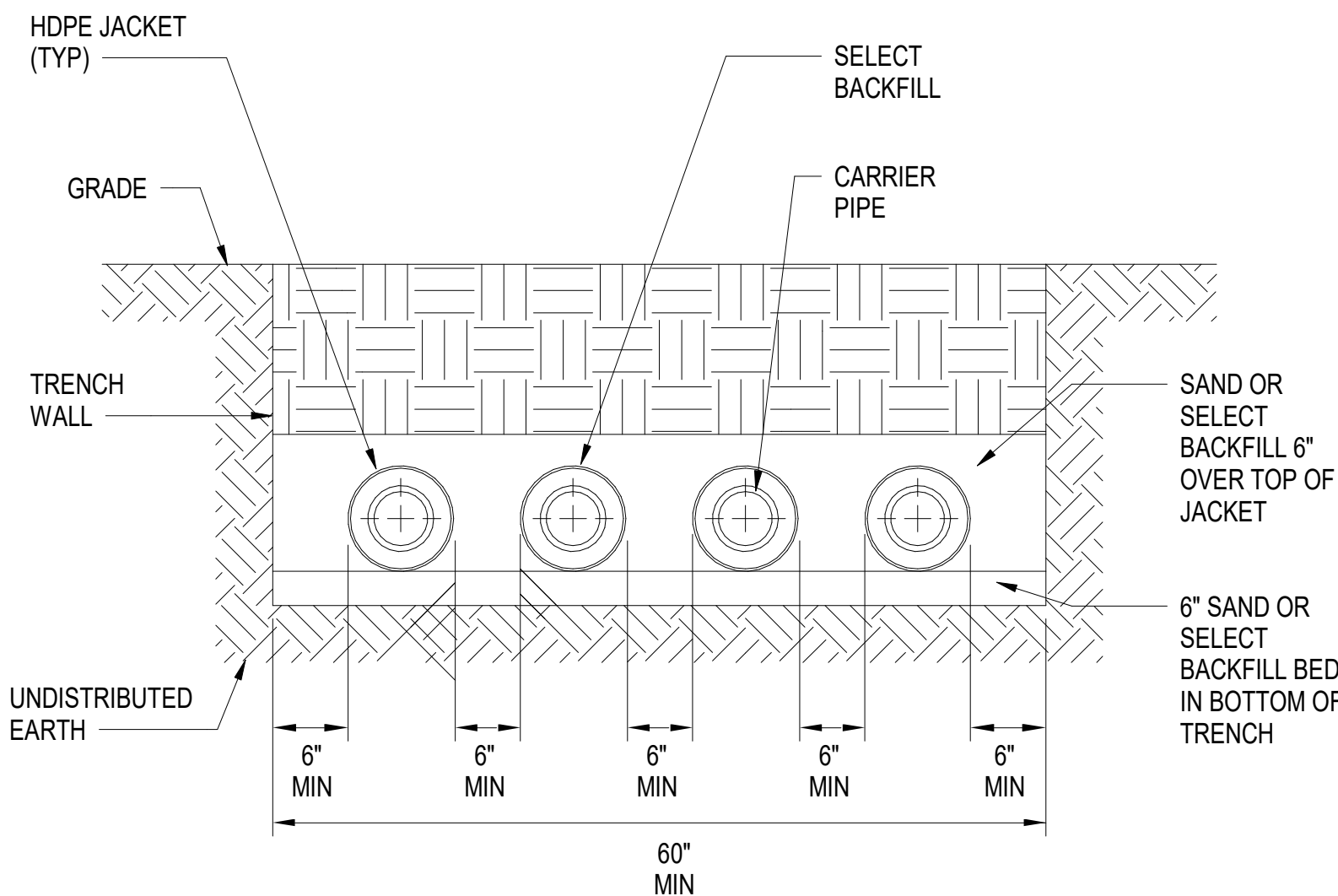
A2 BELOWGRADE PIPING BUILDING ENTRY DETAIL

SCALE: NOT TO SCALE



C3 INLINE FAN

SCALE: NOT TO SCALE

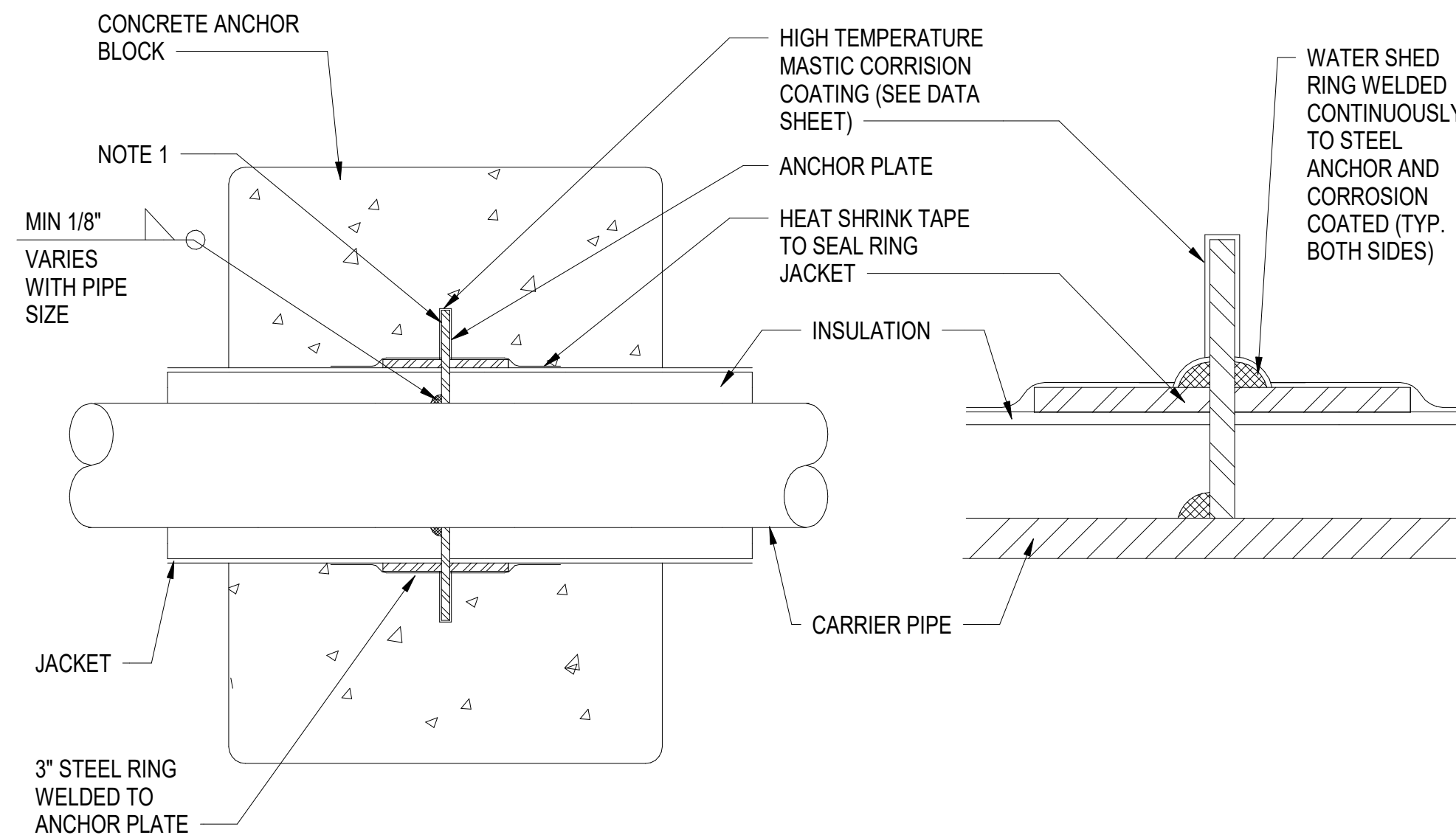


NOTES:

1. INSTALL PIPING IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.

C4 BELOWGRADE PIPING TRENCH DETAIL

SCALE: NOT TO SCALE



NOTES:

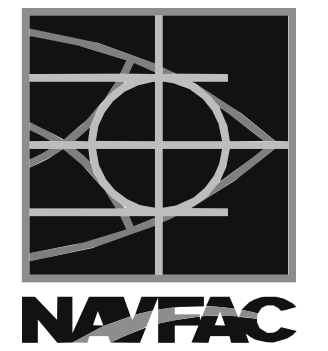
1. STEEL ANCHOR PLATE AND WELDED RINGS FURNISHED BY PIPE MANUFACTURER. ANCHOR PLATE SHALL MEET ASTM A36 AND ON ALL SIZES SHALL BE 1/2" THICK. ANCHOR PLATE SHALL EXTEND 2-1/2" BEYOND THE CASING DIAMETER ON ALL SIDES. ANCHOR PLATE SHALL BE CORROSION COATED WITH A HIGH TEMPERATURE MASTIC MATERIAL AFTER WATERSHED RINGS HAVE BEEN SEALED TO CASING BY HEAT SHRINK TAPE.
2. ANCHOR ASSEMBLY SHALL BE POURED IN A CONCRETE BLOCK BY THE CONTRACTOR IN THE FIELD. GENERALLY, THE ANCHOR BLOCK EXTENDS IN ALL DIRECTIONS FROM THE ANCHOR AND RESTS ON UNDISTURBED EARTH.
3. INSTALL PIPING IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.

A4 BELOWGRADE PIPING ANCHOR DETAIL

SCALE: NOT TO SCALE

GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.



APPROVED:

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DES: CWS | DRW: CWS | CHK: RJ

PM/DM: CJC/KDB

BRANCH MANAGER: T.L.H.

CHIEF ENGINEER: EJA

FIRE PROTECTION: HPN

NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

MECHANICAL DETAILS

SCALE: AS NOTED

EPROJCT NO.: 1596300

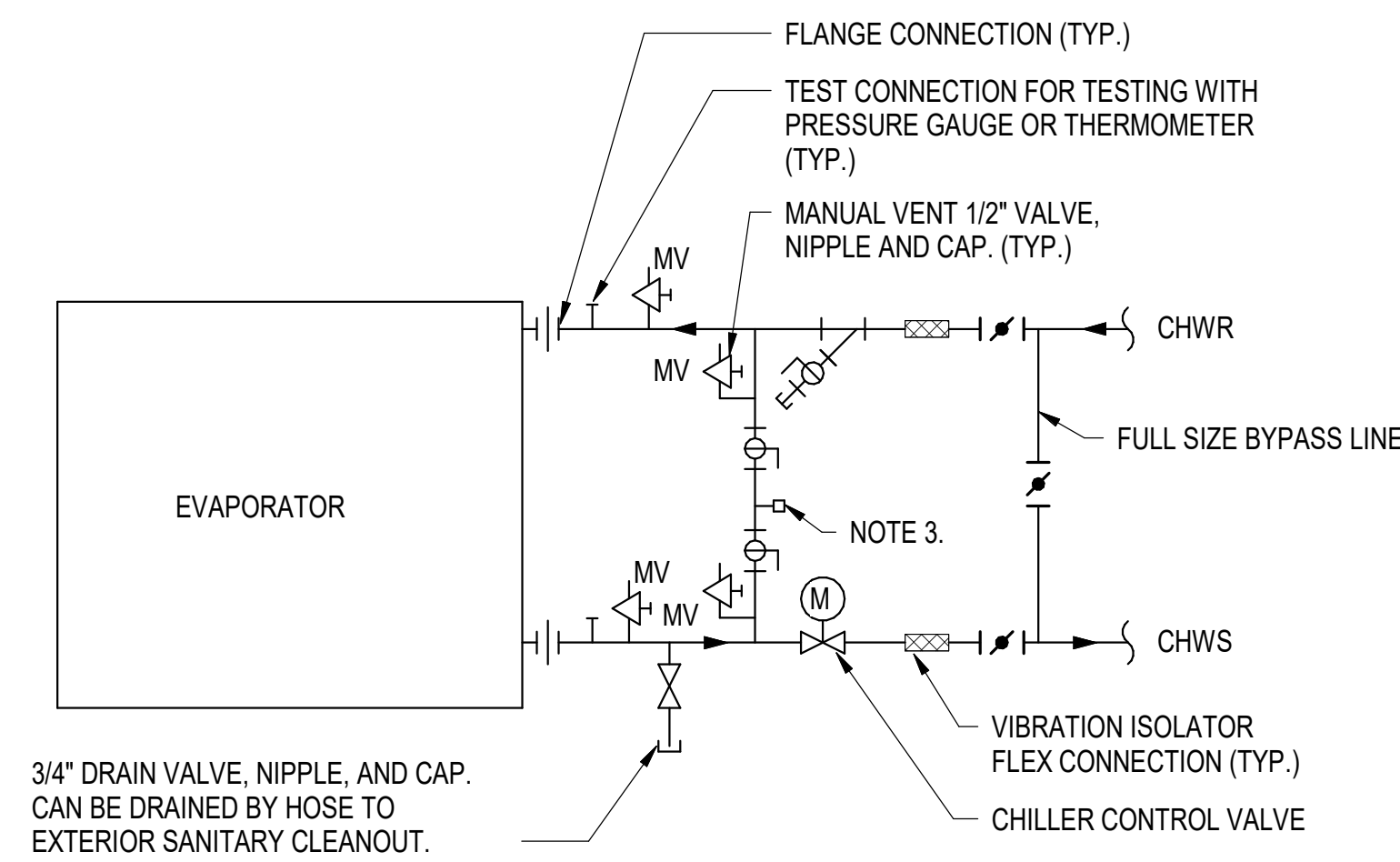
CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12801730

SHEET 470 OF 686

M-507

DRAWING REVISION: 12 APRIL 2018

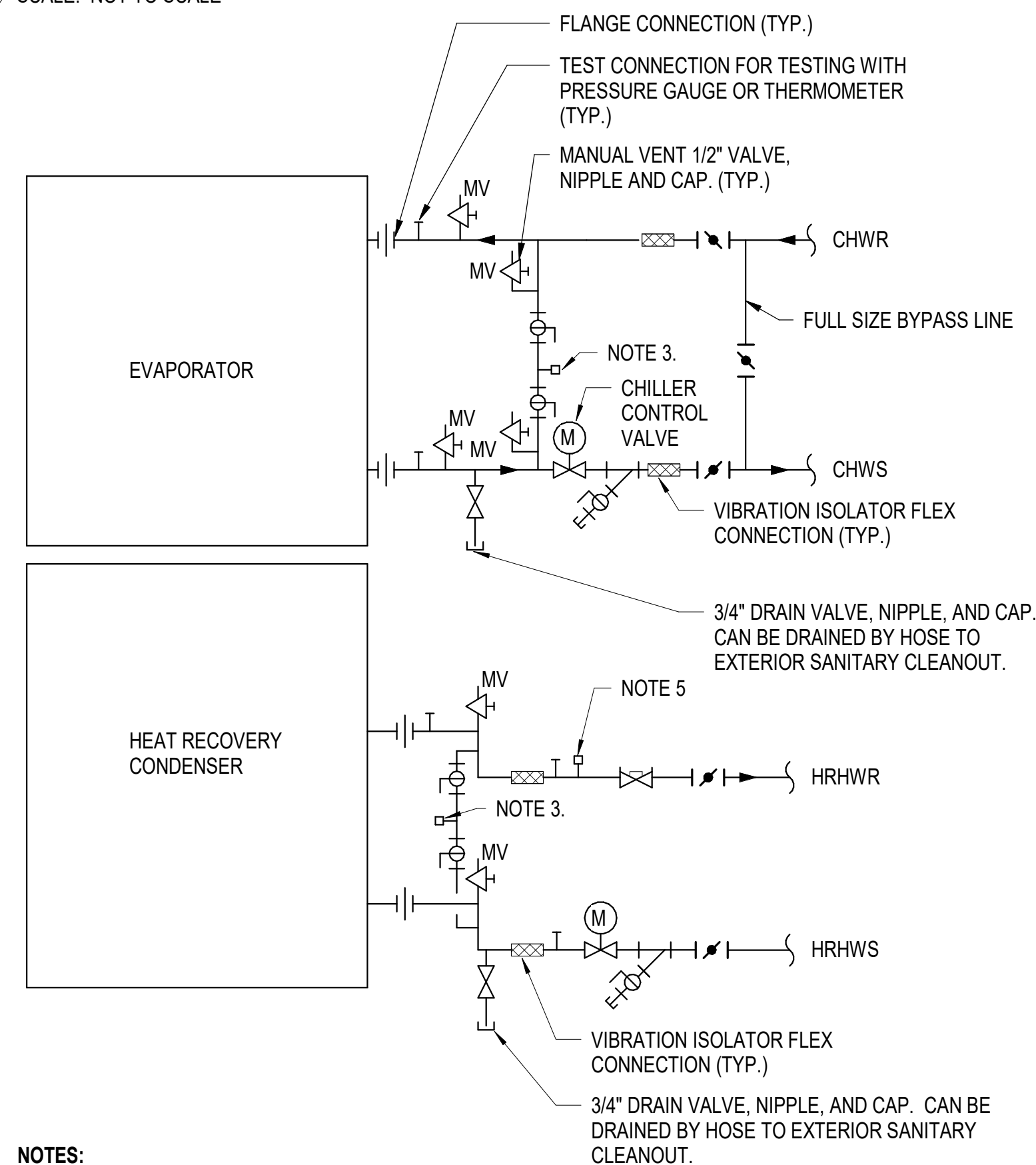


NOTES:

1. INSTALL IN ACCORDANCE WITH CHILLER MANUFACTURER RECOMMENDATIONS.
2. COORDINATE WITH CONTROLS SUBCONTRACTOR TO ACCOMMODATE SENSORS AND SWITCHES.
3. LOCATION OF CHWS/R EVAPORATOR DIFFERENTIAL PRESSURE SENSOR. COORDINATE WITH CHILLER MANUFACTURER AND CONTROL CONTRACTOR AS REQUIRED.
4. CHILLERS CAN BE DRAINED BY HOSE CONNECTION TO EXTERIOR SANITARY CLEANOUT.

C1 AIR COOLED CHILLER CONNECTION DETAIL

SCALE: NOT TO SCALE

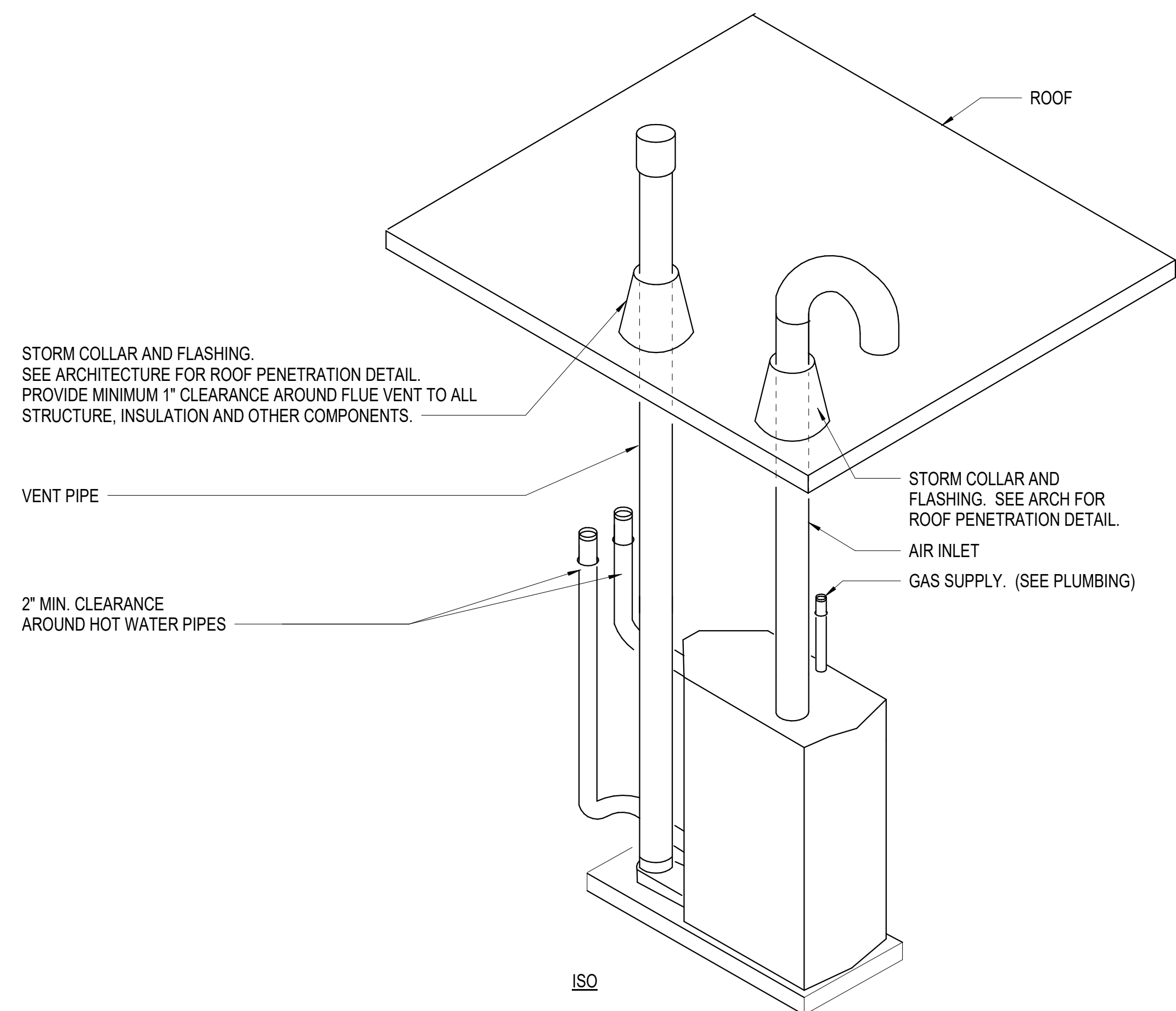


NOTES:

1. INSTALL IN ACCORDANCE WITH CHILLER MANUFACTURER RECOMMENDATIONS.
2. COORDINATE WITH CONTROLS SUBCONTRACTOR TO ACCOMMODATE SENSORS AND SWITCHES.
3. LOCATION OF CHWS/R EVAPORATOR / HRHWS/R CONDENSER DIFFERENTIAL PRESSURE SENSOR. COORDINATE WITH CHILLER MANUFACTURER AND CONTROL CONTRACTOR AS REQUIRED.
4. CHILLERS CAN BE DRAINED BY HOSE CONNECTION TO EXTERIOR SANITARY CLEANOUT.
5. FLOW SWITCH.

HEAT RECOVERY AIR COOLED CHILLER CONNECTION DETAIL

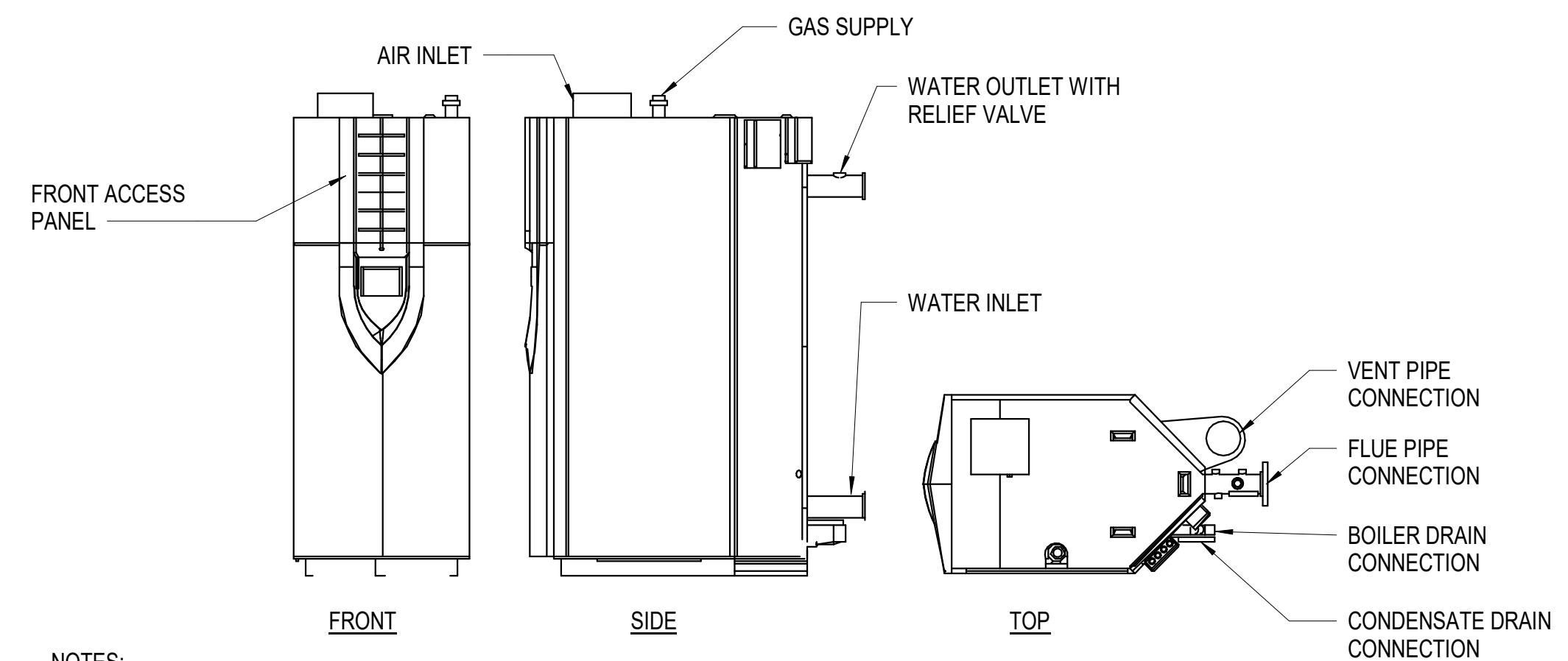
SCALE: NOT TO SCALE



STORM COLLAR AND FLASHING.
SEE ARCHITECTURE FOR ROOF PENETRATION DETAIL.
PROVIDE MINIMUM 1" CLEARANCE AROUND FLUE VENT TO ALL
STRUCTURE, INSULATION AND OTHER COMPONENTS. _____

VENT PIPE

2" MIN. CLEARANCE
AROUND HOT WATER PIPES



NOTES:

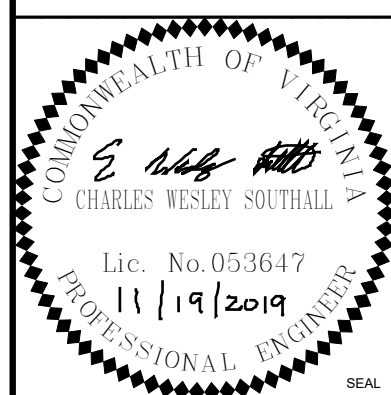
1. INSTALL BOILER IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
2. SUPPLY AIR AND FLUE VENT PENETRATIONS SHALL BE INSTALLED IN ACCORDANCE WITH ARCHITECTURAL DETAILS AND MANUFACTURER'S RECOMMENDATIONS.
3. SUPPLY AIR AND FLUE VENT LOCATIONS OFFSETS AND CLEARANCES SHALL BE IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
4. CONDENSATE SHALL BE ROUTED THROUGH FACTORY SUPPLIED CONDENSATE NEUTRALIZATION RECEIVER BEFORE BEING DISCHARGED TO SANITARY FLOOR DRAIN OR FLOOR SINK.
5. PROVIDE GAS AND WATER PIPING ACCESSORIES AS INDICATED ON SHEET M-802 AND M-804 AND AS RECOMMENDED BY THE BOILER MANUFACTURER. PROVIDE INLET AND OUTLET PRESSURE AND TEMPERATURE GAUGES AND SEPARATE PRESSURE/TEMPERATURE TEST PORTS ON THE INLET AND OUTLET.

CONDENSING BOILER

SCALE: NOT TO SCALE

GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

[illegible]

**BURNS
MCDONNELL**
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED

JAMES H. DUNN

James H. Dunn

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO		DATE	
DES	CWS	DRW	CWS
		CHK	RJ
PM/DM		CJC/KDB	
BRANCH MANAGER		7/24	
CHIEF ENGINEER		EJA	
FIRE PROTECTION		HPN	

DEPARTMENT OF THE NAVY	NAVAL FACILITIES ENGINEERING COMMAND	NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC	NAVAL STATION - NORFOLK VA	
MARINE CORPS IPT	MCAS CHERRY POINT	CHERRY POINT, NC
FY20 P-228 ATC TOWER AND AIR OPERATIONS		
MECHANICAL DETAILS		

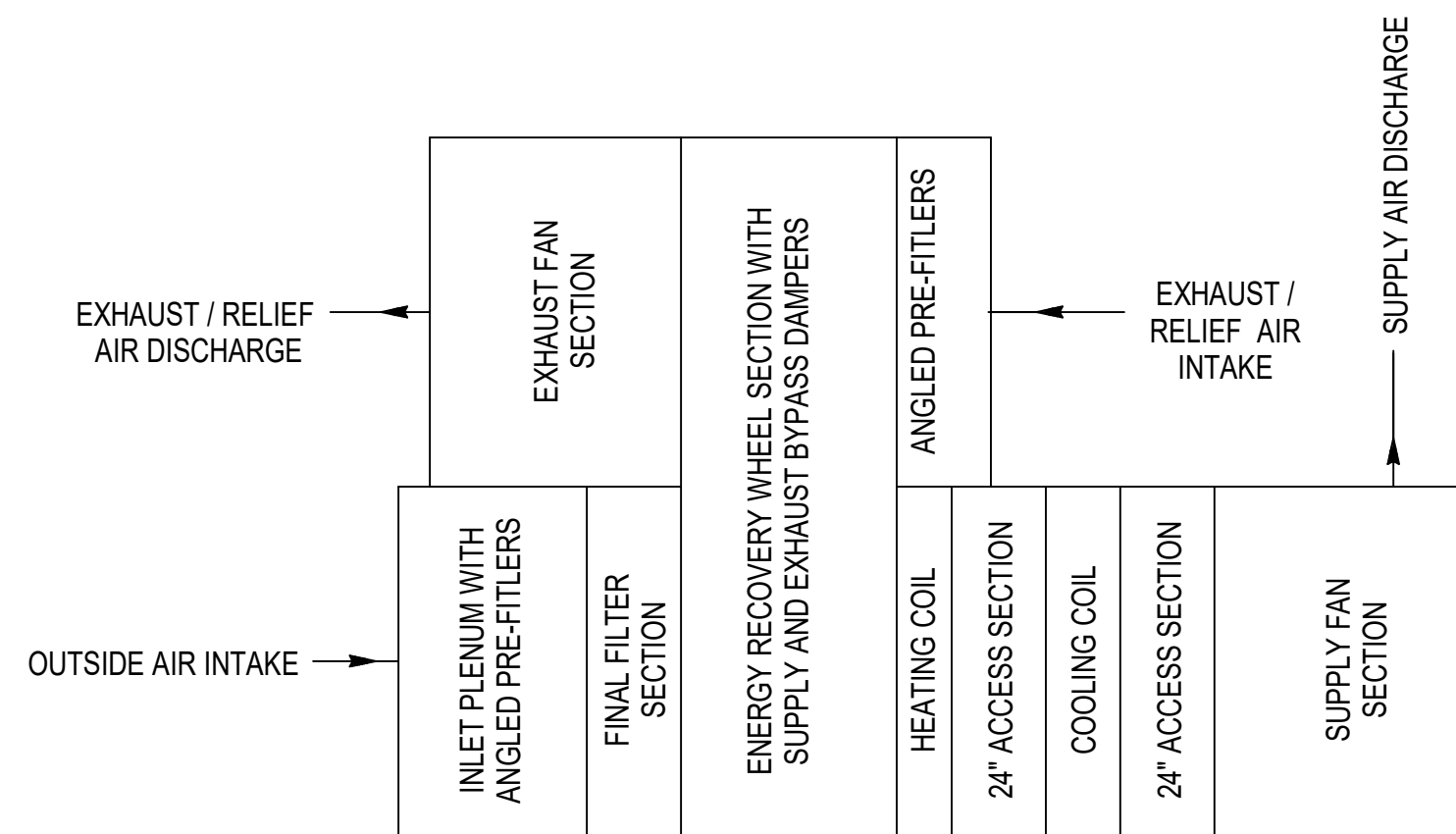
SCALE: AS NOTED		
EPROJECT NO.:		1596300
CONSTR. CONTR. NO.		
NAVFAC DRAWING NO.		
		12801731
SHEET	471	OF 686

M-508

DRAWING REVISION: 12 APRIL 2018

GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

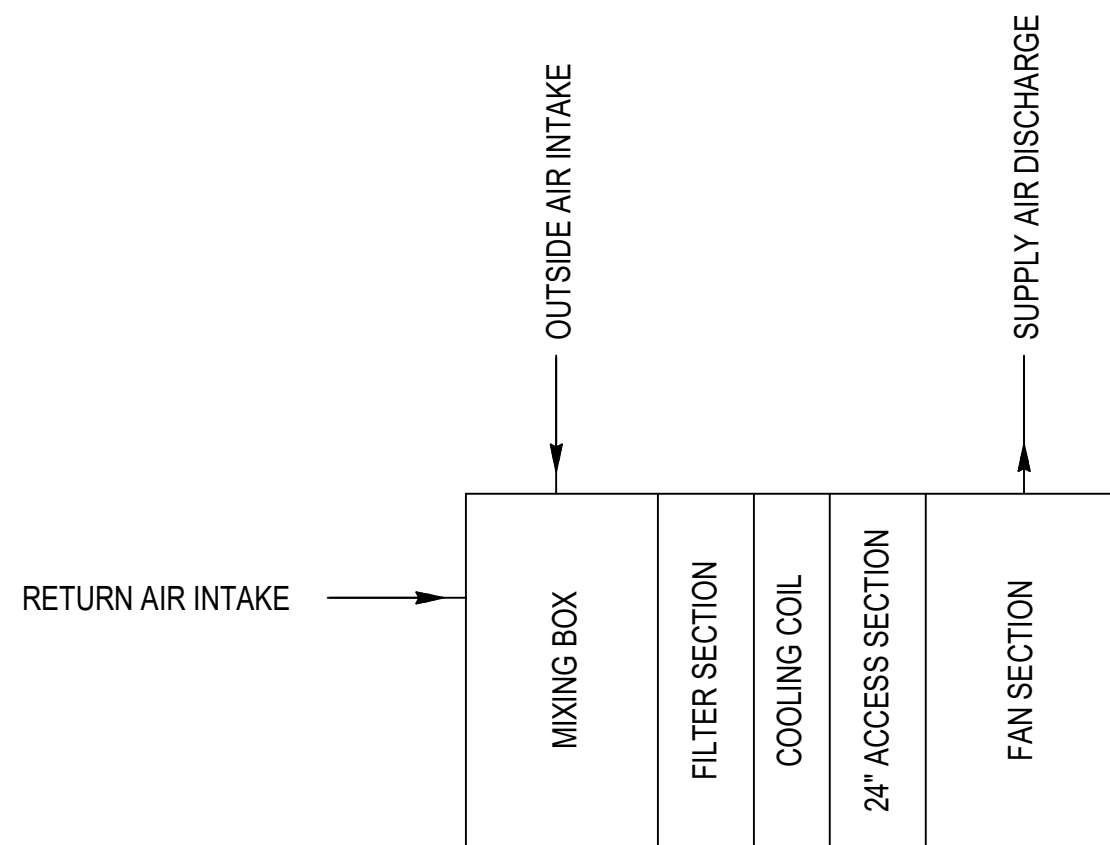


NOTES:

1. SIZE INTAKES AND DISCHARGES TO PROVIDE NO MORE THAN 0.10 IN WG PRESSURE DROP ACROSS THE OPENING AND CONTROL DAMPERS AND TO ALIGN WITH DUCT SIZES INDICATED ON THE DRAWINGS.
2. PROVIDE ISOLATION DAMPERS INTEGRAL TO THE DOAH. SEE CONTROLS DRAWINGS FOR FURTHER INFORMATION.

TYPICAL DEDICATED OUTSIDE AIR UNIT CONFIGURATION DETAIL

SCALE: NOT TO SCALE

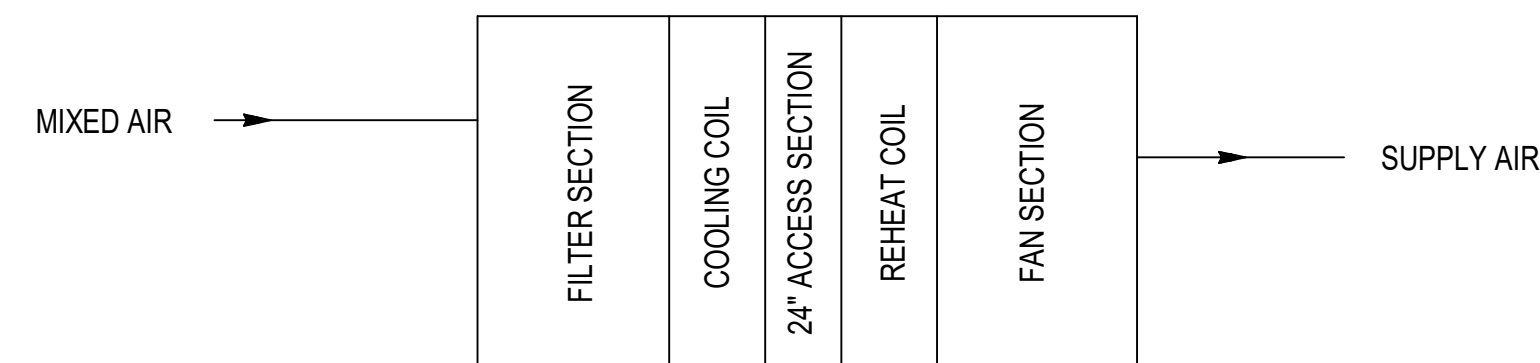


NOTES:

1. SIZE INTAKES AND DISCHARGES TO PROVIDE NO MORE THAN 0.10 IN WG PRESSURE DROP ACROSS THE OPENING AND CONTROL DAMPERS AND TO ALIGN WITH DUCT SIZES INDICATED ON THE DRAWINGS.

AIR HANDLER CONFIGURATION DETAIL (AIR OPS AND MTRACON)

SCALE: NOT TO SCALE

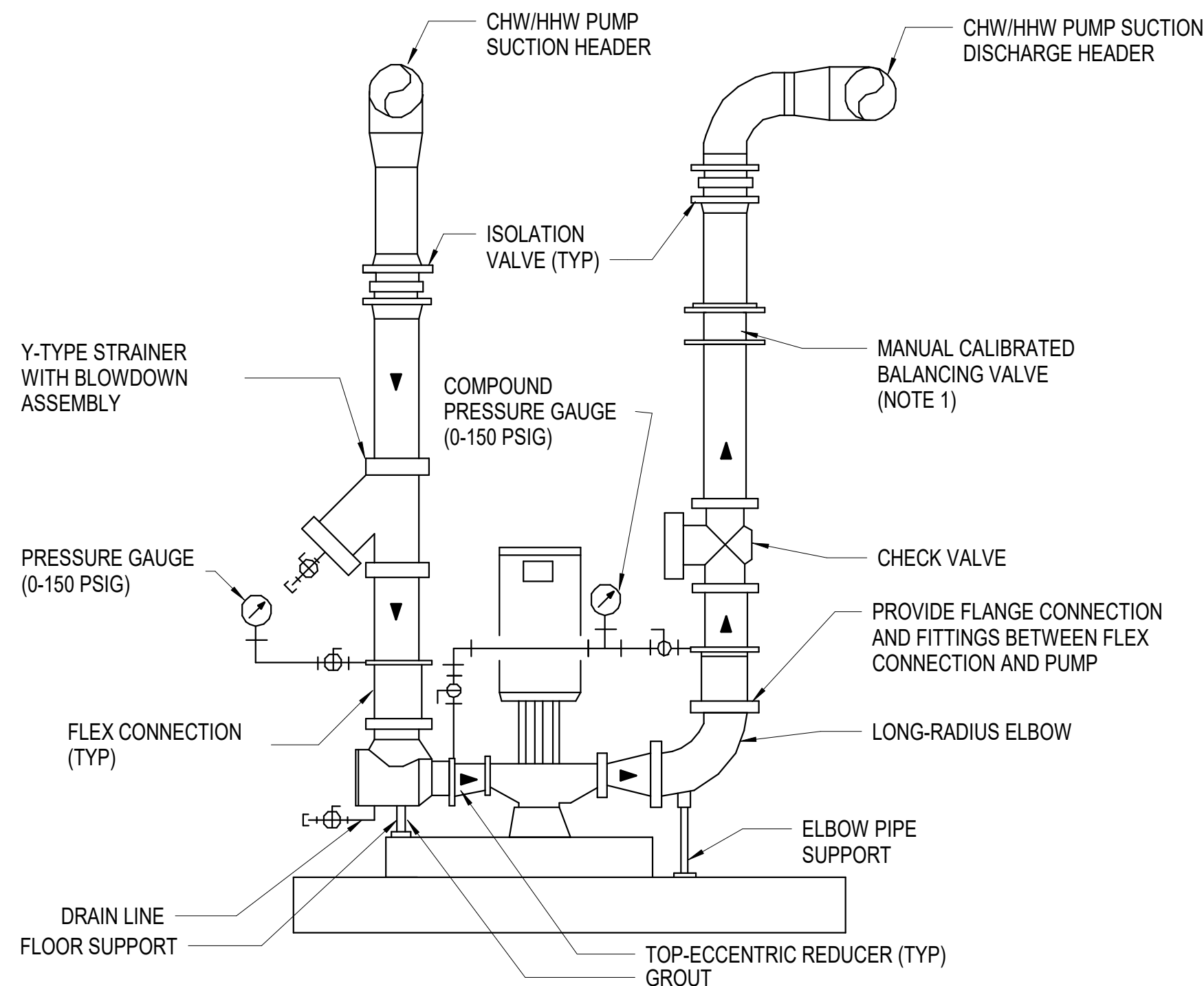


NOTES:

1. SIZE INTAKES AND DISCHARGES TO PROVIDE NO MORE THAN 0.10 IN WG PRESSURE DROP ACROSS THE OPENING AND CONTROL DAMPERS AND TO ALIGN WITH DUCT SIZES INDICATED ON THE DRAWINGS.

AIR HANDLER CONFIGURATION DETAIL (ATCT)

SCALE: NOT TO SCALE

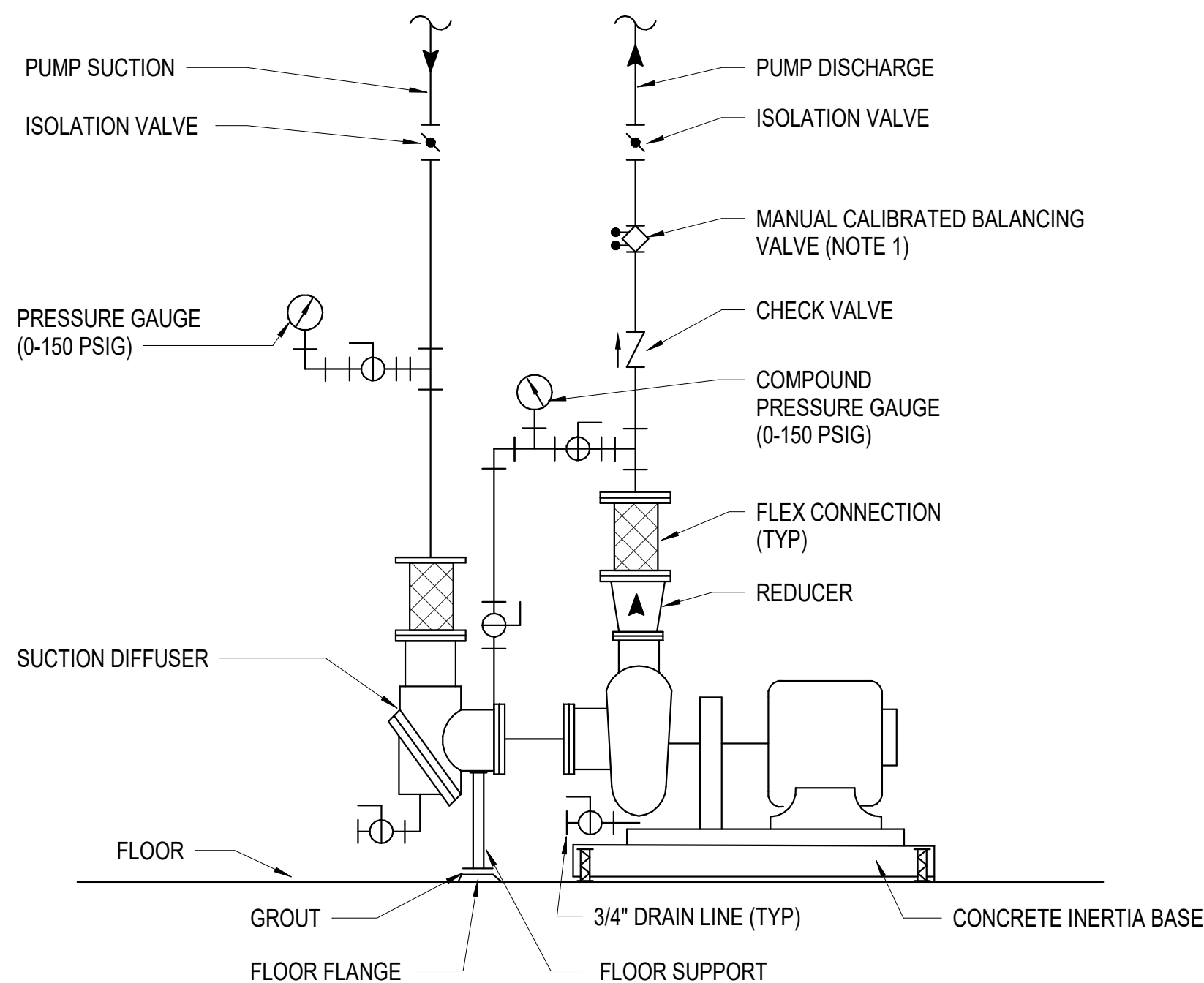


NOTES:

1. INSTALL MANUAL CALIBRATED BALANCING VALVES IN ACCORDANCE WITH VALVE MANUFACTURER'S RECOMMENDED INSTRUCTIONS. VALVE SHALL BE INSTALLED A MINIMUM OF 10 PIPE DIAMETERS FROM PUMP DISCHARGE OR ANY OTHER FITTING OF VALVE.

VERTICAL BASE MOUNTED PUMP DETAIL

SCALE: NOT TO SCALE

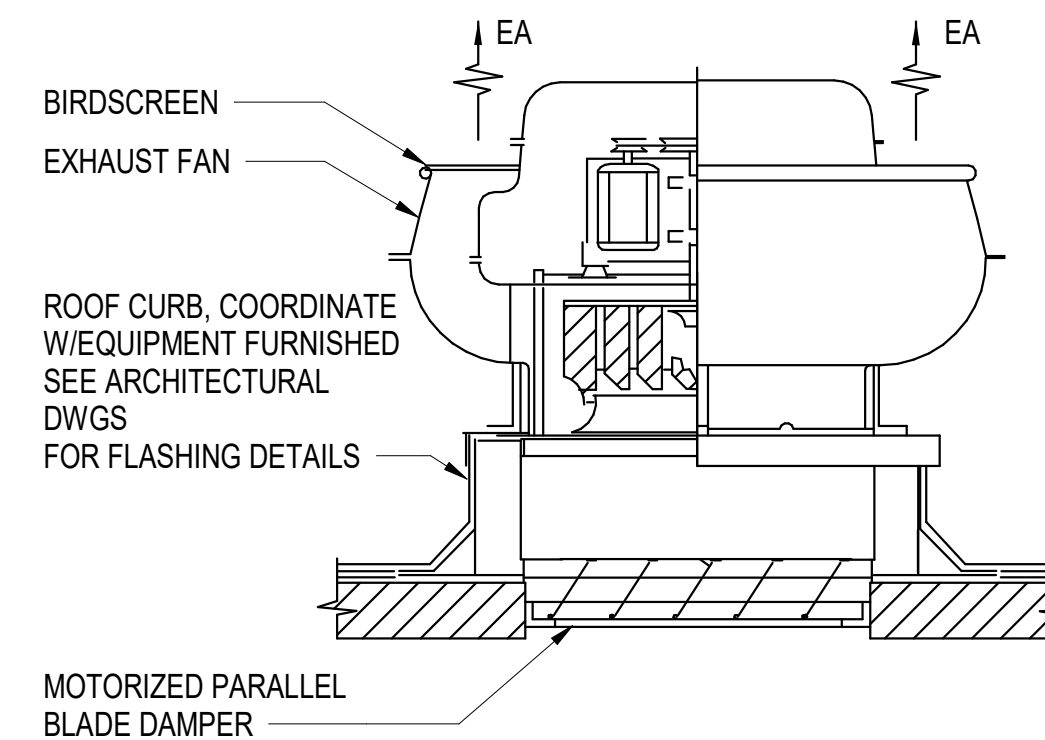


NOTES:

1. INSTALL MANUAL CALIBRATED BALANCING VALVES IN ACCORDANCE WITH VALVE MANUFACTURER'S RECOMMENDED INSTRUCTIONS. VALVE SHALL BE INSTALLED A MINIMUM OF 10 PIPE DIAMETERS FROM PUMP DISCHARGE OR ANY OTHER FITTING OF VALVE.

CLOSE COUPLED END SUCTION PUMP DETAIL

SCALE: NOT TO SCALE



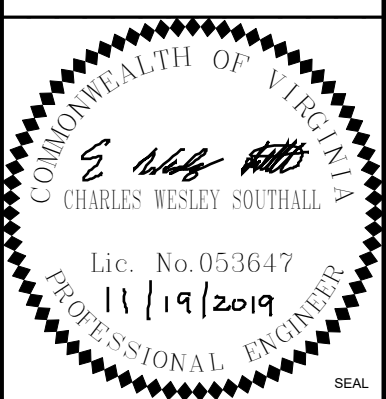
NOTES:

1. INSTALL FAN TO RESIST WIND LOADS INDICATED ON STRUCTURAL DRAWINGS.
2. INSTALL IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.

UPBLAST CENTRIFUGAL FAN DETAIL

SCALE: NOT TO SCALE

DATE	SYN	DESCRIPTION	DATE	APPR
11/19/19	0	ISSUED FOR CONSTRUCTION		



BURNS & MCDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED	DATE
FOR COMMANDER NAVFAC	
ACTIVITY	
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PMOM	CJC/KDB
BRANCH MANAGER	
CHIEF ENGINEER	TLH
FIRE PROTECTION	EJA
	HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
MARINE CORPS IPT
NAVAL STATION - NORFOLK, VA
CHERRY POINT, NC
MCAS CHERRY POINT
FY20 P-228 ATC TOWER AND AIR OPERATIONS
MECHANICAL DETAILS

SCALE: AS NOTED
PROJECT NO.: 1596300
CONSTR. CONTR. NO.
NAVFAC DRAWING NO. 12801732
SHEET 472 OF 686
M-509

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AIR HANDLING UNIT SCHEDULE																						
MARK	LOCATION	FAN DATA				ELECTRICAL DATA				COOLING COIL DATA										NOTES		
		S.A. CFM		OA CFM	E.S.P. (IN WG)	FAN (QTY) H.P.	VOLTS/PH/Hz	FLA	MCA	MOCp	TOTAL (MBH)	SENSIBLE (MBH)	EWT °F	EAT °FDB	EAT °FWB	LAT °FDB	LAT °FWB	MAX. COIL FV (FPM)	# OF ROWS		GPM	MAX WATER PRESSURE DROP (FT WC)
		(MAX)	(MIN)																			
AHU-A1	152 MECH	10,260	4,250	2,200	4.6	(2) @ 10.0	460/3/60	28.65	32.15	45 A	307.3	269.7	45	75.0	61.7	51.1	51.1	500	8	61.5	10	1,2,3,5
AHU-A2	151 MECH	2,640	1,530	1,530	3.0	(1) @ 5.0	460/3/60	8.85	10.90	20 A	54.4	44.1	45	67.2	59.3	52.0	52.0	500	8	10.9	10	1,2,4,5

- NOTES:
- PROVIDE MERV 8 AND MERV 13 FLAT FILTERS UPSTREAM OF COILS.
 - PROVIDE WITH VFDs. ONE VFD PER FAN. PROVIDE WITH SINGLE POINT ELECTRICAL CONNECTION.
 - MAXIMUM UNIT DIMENSIONS ARE 126" LONG 84" WIDE 60" HIGH. UNIT SHALL BE MODULAR AND SHALL BE ABLE TO FIT THROUGH 84"x72" OPENING.
 - MAXIMUM UNIT DIMENSIONS ARE 100" LONG 48" WIDE 36" HIGH. UNIT SHALL BE MODULAR AND SHALL BE ABLE TO FIT THROUGH 84"x72" OPENING.
 - ESP INCLUDES 0.5 IN WG FOR MERV 8 & 0.7 IN WG FOR MERV 13 LOADING.

DEDICATED OUTDOOR AIR UNIT SCHEDULE 1 OF 1																										
MARK	SUPPLY FAN			EXHAUST FAN			COOLING COIL DATA										HEATING COIL DATA									
	CFM	EXTERNAL SP (IN WG)	MOTOR (QTY) HP	CFM	EXTERNAL SP (IN WG)	MIN. MOTOR HP	TOTAL (MBH)	SENSIBLE (MBH)	EAT °FDB	EAT °FWB	LAT °FDB	LAT °FWB	EWT °F	LWT °F	GPM	WTR P.D. FT. H2O	MAX. COIL FV (FPM)	MBH	EAT °FDB	LAT °FDB	EWT °F	LWT °F	GPM	WTR P.D. FT. H2O	MAX. COIL FV (FPM)	
DOAH-A1	2,200	2.0	(1) 5	1,280	1.2	3	127.5	62.9	81.0	72.4	55.0	55.0	45	55	25.5	10	500	66.5	27.0	55.0	130	110	6.7	5.0	500	
DOAH-A2	1,530	1.9	(1) 5	1,380	1.2	3	88.7	43.8	81.0	72.4	55.0	55.0	45	55	17.7	10	500	46.3	27.0	55.0	130	110	4.6	5.0	500	

DEDICATED OUTDOOR AIR UNIT SCHEDULE 2 OF 2																				
MARK	ENERGY WHEEL DATA															UNIT ELECTRICAL DATA				NOTES
	SUMMER CONDITIONS								WINTER CONDITIONS							V / HZ / PH	VFD PROVIDED BY	FLA	MOCP	
	OUTSIDE AIR CFM	OUTSIDE AIR EAT (DB/WB)	OUTSIDE AIR LAT (DB/WB)	OUTSIDE AIR PRESSURE DROP (IN WG)	ENTERING EXHAUST CFM	EXHAUST EAT (DB / WB)	EXHAUST LAT (DB / WB)	EXHAUST AIR PRESSURE DROP (IN WG)	OUTSIDE AIR EAT (DB / WB)	OUTSIDE AIR LAT (DB/WB)	OUTSIDE AIR PRESSURE DROP (IN WG)	ENTERING EXHAUST CFM	EXHAUST EAT (DB / WB)	EXHAUST LAT (DB / WB)	EXHAUST AIR PRESSURE DROP (IN WG)					
DOAH-A1	2,200	84.0 / 78.7	80.5 / 71.8	0.5	1,280	78.0 / 65.0	82.9 / 75.9	0.5	27.0 / 25.0	50.7 / 41.4	0.5	1,280	68.0 / 51.5	34.0 / 30.4	0.5	460/60/3	DIVISION 23	15 A	20 A	1,2,3,5
DOAH-A2	1,530	84.0 / 78.7	79.7/ 70.1	0.5	1,380	78.0 / 65.0	82.8 / 75.6	0.5	27.0 / 25.0	56.1 / 45.8	0.5	1,380	68.0 / 53	35.2 / 31.8	0.5	460/60/3	DIVISION 23	15 A	20 A	1,2,4,5

- NOTES:
- PROVIDE MERV 8 AND MERV 13 FLAT FILTERS UPSTREAM OF WHEEL FOR SUPPLY, AND MERV 8 UPSTREAM OF WHEEL FOR EXHAUST.
 - PROVIDE SUPPLY AND EXHAUST FANS WITH VFDs. PROVIDE WITH SINGLE POINT ELECTRICAL CONNECTION FOR ENTIRE UNIT.
 - MAXIMUM UNIT DIMENSIONS ARE 202.1" LONG 44" WIDE 70.8" HIGH. UNIT SHALL BE MODULAR AND SHALL BE ABLE TO FIT THROUGH 84"x72" OPENING.
 - MAXIMUM UNIT DIMENSIONS ARE 204.0" LONG 44" WIDE 70.8" HIGH. UNIT SHALL BE MODULAR AND SHALL BE ABLE TO FIT THROUGH 84"x72" OPENING.
 - ESP INCLUDES 0.5 IN WG FOR MERV 8 & 0.7 IN WG FOR MERV 13 LOADING.
 - DOAH COOLING COIL SHALL BE SIZED AT 50% SENSIBLE AND TOTAL EFFECTIVENESS. ENERGY WHEEL SHALL PERFORM AS SCHEDULED.
 - ENERGY WHEEL SHALL BE 460V, 3-PHASE.

FAN COIL UNIT SCHEDULE																												
MARK	TYPE	FAN DATA			COOLING COIL DATA										HEATING COIL DATA										V/HZ/PH	MAX ECM FLA	STARTER/DISCONNECT PROVIDED BY	NOTES
		CFM (CLG/HTG)	ESP (IN. W.G.)	NOMINAL H.P.	SENSIBLE MBH	TOTAL MBH	ENT. AIR		LVG. AIR		ENT. WTR TEMP (°F)	EXT. WTR TEMP (°F)	GPM	MAX. COIL FV (FPM)	W. P.D. FT. HD.	PIPING RUNOUT SIZE (IN)	MBH	ENT. AIR (°F) DB	ENT. WTR TEMP (°F)	WTR TEMP ΔT (°F)	GPM	MAX. COIL FV (FPM)	W. P.D. FT. HD.	PIPING RUNOUT SIZE (IN)				
							(°F) DB	(°F) WB	(°F) DB	(°F) WB																		
FCU-A1	VERTICAL CABINET	150 / 150	-	107 W	2.7	4.2	80.0	70.0	64.4	62.1	45	55	0.8	500	10	3/4	5.0	50.0	130	20	0.5	500	10	3/4	120/60/1	1.2	DIV 23	1,2,3,5,6
FCU-A2	VERTICAL CABINET	250 / 250	-	107 W	3.8	5.8	80.0	70.0	66.3	63.4	45	55	1.2	500	10	3/4	5.9	50.0	130	20	0.6	500	10	3/4	120/60/1	1.2	DIV 23	1,2,3,5,6
FCU-A3	SUSPENDED HORIZONTAL CABINET	620 / -	-	124 W	19.5	24.5	90.0	70.5	61.6	58.7	45	55	4.9	500	10	1	-	-	-	-	-	-	-	-	120/60/1	6.4	DIV 23	1,2,3,4
FCU-A4	SUSPENDED HORIZONTAL CABINET	1220 / -	-	1/2	22.8	31.7	75.0	65.0	58.1	56.6	45	55	6.3	500	10	1	-	-	-	-	-	-	-	-	120/60/1	6.4	DIV 23	1,2,4,8
FCU-A5	SUSPENDED HORIZONTAL CABINET	1280 / -	-	1/2	23.7	32.8	75.0	65.0	58.2	56.7	45	55	6.5	500	10	1	-	-	-	-	-	-	-	-	120/60/1	4.8	DIV 23	1,2,4,8
FCU-A6	SUSPENDED HORIZONTAL CABINET	310 / 190	-	107 W	9.4	11.8	90.0	70.5	62.9	59.4	45	55	2.4	500	10	3/4	5.6	50.0	130	20	0.6	500	10	3/4	120/60/1	1.5	DIV 23	1,2,3,4
FCU-A7	SUSPENDED HORIZONTAL CABINET	480 / 360	-	124 W	14.6	18.4	90.0	70.5	62.3	59.0	45	55	3.7	500	10	1	11.2	50.0	130	20	1.1	500	10	3/4	120/60/1	1.9	DIV 23	1,2,3,4
FCU-A8A	SUSPENDED HORIZONTAL CABINET	800 / 500	-	1/2	18.8	23.7	80.0	66.5	58.7	57.1	45	55	4.7	500	10	1	15.1	50.0	130	20	1.5	500	10	3/4	120/60/1	6.4	DIV 23	1,2,4,8
FCU-A8B	SUSPENDED HORIZONTAL CABINET	800 / 500	-	1/2	18.8	23.7	80.0	66.5	58.7	57.1	45	55	4.7	500	10	1	15.1	50.0	130	20	1.5	500	10	3/4	120/60/1	6.4	DIV 23	1,2,4,8
FCU-A9	SUSPENDED HORIZONTAL CABINET	160 / 160	-	107 W	5.0	6.4	90.0	70.5	61.0	58.2	45	55	1.3	500	10	3/4	5.0	50.0	130	20	0.5	500	10	3/4	120/60/1	1.3	DIV 23	1,2,3,4
FCU-A10	SUSPENDED HORIZONTAL DUCTED	5500 / -	0.3	5 HP	127.8	137.9	80.0	66.5	58.9	58.9	45	58.9	19.9	500	10	1 1/2	-	-	-	-	-	-	-	-	460/60/3	8.4	DIV 23	1,2,3,7
FCU-A11	SUSPENDED HORIZONTAL DUCTED	830 / -	0.3	149 W	23.6	28.7	90.0	70.5	64.4	60.4	45	55	5.7	500	10	1	-	-	-	-	-	-	-	-	120/60/1	3.5	DIV 23	1,2,3,4
FCU-A12	SUSPENDED HORIZONTAL DUCTED	660 / 500	0.3	1/2	12.5	16.9	75.0	65.0	57.8	56.7	45	55	3.8	500	10	1	10.4	70	130	20	1.0	500	10	3/4	120/60/1	6.4	DIV 23	1,2,4,8
FCU-A13	SUSPENDED HORIZONTAL DUCTED	670 / 420	0.3	1/2	13.1	17.8	75.0	65.0	57.3	56.4	45	58.6	2.7	500	10	1	10.6	70	130	20	1.1	500	10	3/4	120/60/1	7.5	DIV 23	1,2,4,8

- NOTES:
- PROVIDE UNIT WITH UNIT MOUNTED DISCONNECT AND 120V/60HZ/1PH CONTROL VOLTAGE TRANSFORMER.
 - NUMBER OF MOTORS IN PARENTHESIS.
 - PROVIDE UNITS WITH FILTER RACK ON THE UNIT INTAKE WITH A 2" MERV 8 FILTER.
 - PROVIDE WITH EC TYPE MOTOR. EC MOTORS SHALL BE PROVIDED WITH EQUIPMENT TO PREVENT ADDING HARMONICS TO ELECTRICAL SYSTEM (INDUCTOR, ETC.) AND SHALL BE PROGRAMMED TO PREVENT EC MOTOR OVERSPEED.
 - PROVIDE WITH CONDENSATE PUMP. PUMP CAPACITY SHALL BE 10 GPH AT 20 FT WC. CONDENSATE PUMP SHALL BE POWERED BY FCU ELECTRICAL CONNECTION.
 - MOTOR CAN BE EITHER EC TYPE OR TWO SPEED. IF EC TYPE, PROVIDE MOTOR WITH EQUIPMENT TO PREVENT ADDING HARMONICS TO ELECTRICAL SYSTEM (INDUCTOR, ETC.) AND SHALL BE PROGRAMMED TO PREVENT EC MOTOR OVERSPEED.
 - PROVIDE WITH EITHER MOTOR WITH VFD OR EC TYPE MOTOR. EC MOTORS SHALL BE PROVIDED WITH EQUIPMENT TO PREVENT ADDING HARMONICS TO ELECTRICAL SYSTEM (INDUCTOR, ETC.) AND SHALL BE PROGRAMMED TO PREVENT EC MOTOR OVERSPEED.
 - PROVIDE UNITS WITH FILTER RACK ON THE UNIT INTAKE WITH BOTH A 2" MERV 8 PRE-FILTER, AND A MERV 11 FINAL FILTER.

PUMP SCHEDULE												
MARK	SERVICE	TYPE	FLOW (GPM)	TDH (FT WC)	MOTOR RPM	ELECTRICAL					VFD / DISCONNECT PROVIDED BY	REMARKS
						MIN. MOTOR POWER (HP)	SPEED CONTROL	VOLTS	PHASE	HZ		
HHWP-DOAH-A1	DOAH CIRCULATOR	INLINE CIRCULATOR	6.7	22	3250	1/8	-	120	1	60	DIVISION 23	
HHWP-DOAH-A2	DOAH CIRCULATOR	INLINE CIRCULATOR	4.6	20	3250	1/8	-	120	1	60	DIVISION 23	
PCHWP-A1	PRIMARY CHILLED WATER PUMP	VERTICAL INLINE	171	109	1760	10	VFD	460	3	60	DIVISION 23	2
PCHWP-A2	PRIMARY CHILLED WATER PUMP	VERTICAL INLINE	171	109	1760	10	VFD	460	3	60	DIVISION 23	2
PHHWP-A1	PRIMARY HEATING HOT WATER PUMP	INLINE CIRCULATOR	33	21	1750	1/4	VFD	120	1	60	DIVISION 23	1,2
PHHWP-A2	PRIMARY HEATING HOT WATER PUMP	INLINE CIRCULATOR	33	21	1750	1/4	VFD	120	1	60	DIVISION 23	1,2
SHHWP-A1	SECONDARY HEATING HOT WATER PUMP	VERTICAL INLINE	51	68	1760	2	VFD	460	3	60	DIVISION 23	2
SHHWP-A2	SECONDARY HEATING HOT WATER PUMP	VERTICAL INLINE	51	68	1760	2	VFD	460	3	60	DIVISION 23	2

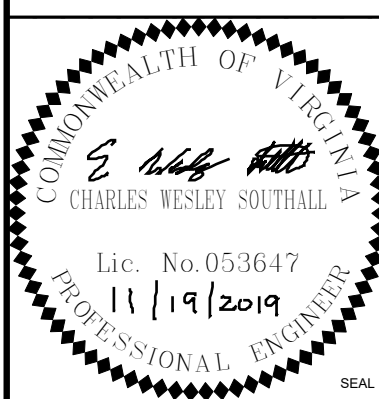
- NOTES:
- PROVIDE WITH BOILER. COORDINATE PRESSURE DROP WITH BOILER. SELECTED PRESSURE DROP SHALL BE BOILER WPD + 10.2 FT WC.
 - PROVIDE WITH VFD & INTEGRAL DISCONNECT.

BOILER SCHEDULE									
MARK	LOCATION	TYPE	LVG. WATER TEMP (°F)	ENT. WATER TEMP (°F)	WATER FLOW (GPM)	NATURAL GAS INPUT (MBH)	NATURAL GAS OUTPUT (MBH)	V/PH/Hz	NOTES
B-A1	152 MECH	NATURAL GAS CONDENSING	130	110	33	350	330	120/1/60	1,2
B-A2	152 MECH	NATURAL GAS CONDENSING	130	110	33	350	330	120/1/60	1,2

- NOTES:
- BOILER AHRI CERTIFIED EFFICIENCY SHALL BE MINIMUM OF 95%.
 - PROVIDE BOILER WITH PHHWP, SEE HVAC PUMP SCHEDULE FOR PUMP INFORMATION.

GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.



APPROVED:

FOR COMMANDER NAVFAC
ACTIVITY

SATISFACTORY TO: DATE
DES FYA DRW FYA CHK RJ

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VARIABLE AIR VOLUME TERMINAL BOX

MARK	UNIT TYPE	SERVED BY	# PEOPLE	DESIGN OA (CFM)	FAN AIRFLOW (CFM)	MAX PRIMARY AIRFLOW (CFM)	MIN PRIMARY AIRFLOW (CFM)	HOT WATER HEATING COIL DATA							VAV BOX INLET DIAMETER (IN)	MIN INLET PRESS (IN WC)	SOUND LEVEL (MAX NC)	ELECTRICAL			NOTES
								CAPACITY (MBh)	EAT (°F)	LAT (°F)	EWT (°F)	FLOWRATE (GPM)	MAX WATER PRESSURE DROP (FT WC)	HHW INLET SIZE (IN)				MOTOR HP	VOLTS/PH/HZ	FLA	
VAV-M101	SERIES FAN POWERED	AHU-M1	5	71	510	420	140	10.0	63	85	130	1.0	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M102	SERIES FAN POWERED	AHU-M1	3	38	220	220	70	5.0	64	85	130	0.6	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M103	SERIES FAN POWERED	AHU-M1	6	76	660	620	250	14.9	63	85	130	1.5	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M104	SERIES FAN POWERED	AHU-M1	21	219	820	720	410	22.6	56	85	130	2.3	10	3/4	10	0.5	30	1/2	120/1/60	7.1	1,2,3,4,5
VAV-M105	SERIES FAN POWERED	AHU-M1	7	77	480	400	120	9.2	64	85	130	1.0	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M106	SERIES FAN POWERED	AHU-M1	12	76	370	370	120	8.5	64	85	130	0.9	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M107	SERIES FAN POWERED	AHU-M1	6	71	640	640	200	14.6	64	85	130	1.5	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M108	SERIES FAN POWERED	AHU-M1	4	32	250	250	80	5.7	64	85	130	0.6	10	3/4	5	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M109	SERIES FAN POWERED	AHU-M1	20	238	2250	2250	680	50.9	64	85	130	5.1	10	1.0	16	0.5	30	1	460/3/60	12.0	1,2,3,4,5
VAV-M110	SERIES FAN POWERED	AHU-M1	7	57	260	240	80	5.8	63	85	130	0.6	10	3/4	5	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M111	SERIES FAN POWERED	AHU-M1	17	178	590	580	340	16.7	58	85	130	1.7	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M112	SERIES FAN POWERED	AHU-M1	25	232	830	770	470	22.0	59	85	130	2.2	10	3/4	10	0.5	30	1/2	120/1/60	7.1	1,2,3,4,5
VAV-M113	SERIES FAN POWERED	AHU-M1	30	202	610	590	300	15.6	61	85	130	1.6	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M114	SERIES FAN POWERED	AHU-M1	30	276	600	590	420	17.6	57	85	130	1.8	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M115	SERIES FAN POWERED	AHU-M1	1	11	230	210	70	6.8	55	85	130	0.7	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M116	SERIES FAN POWERED	AHU-M1	1	12	170	130	40	2.9	64	85	130	0.5	10	3/4	5	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M117	SERIES FAN POWERED	AHU-M1	10	86	360	290	140	8.1	59	85	130	0.9	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M118	SERIES FAN POWERED	AHU-M1	6	77	510	440	210	12.3	59	85	130	1.3	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M119	SERIES FAN POWERED	AHU-M1	2	20	130	130	40	2.9	64	85	130	0.5	10	3/4	6	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M120	SERIES FAN POWERED	AHU-M1	3	23	210	210	70	4.8	64	85	130	0.5	10	3/4	6	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M121	SERIES FAN POWERED	AHU-M1	10	65	270	270	90	6.5	63	85	130	0.7	10	3/4	6	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M122	SERIES FAN POWERED	AHU-M1	4	41	260	260	80	5.9	64	85	130	0.6	10	3/4	6	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M123	SERIES FAN POWERED	AHU-M1	9	120	880	650	200	14.7	64	85	130	1.5	10	3/4	8	0.5	30	1/2	120/1/60	7.1	1,2,3,4,5
VAV-M124	SERIES FAN POWERED	AHU-M1	0	48	300	300	100	9.7	55	85	130	1.0	10	3/4	6	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5
VAV-M125	SERIES FAN POWERED	AHU-M1	4	98	540	540	180	17.5	55	85	130	1.8	10	3/4	8	0.5	30	1/3	120/1/60	4.6	1,2,3,4,5

NOTES:

- PROVIDE WITH 120V/60HZ/1PH CONTROL VOLTAGE TRANSFORMER.
- PROVIDE WITH UNIT MOUNTED DISCONNECT.
- MOTOR SHALL BE EC TYPE. EC MOTORS SHALL BE PROVIDED WITH EQUIPMENT TO PREVENT ADDING HARMONICS TO ELECTRICAL SYSTEM (INDUCTOR, ETC) AND SHALL BE PROGRAMMED TO PREVENT EC MOTOR OVERSPEED.
- NUMBER OF OCCUPANTS SHOWN FOR INFORMATION ONLY.
- PROVIDE WITH ATTENUATOR IF REQUIRED TO ACHIEVE DESIRED NC LEVELS.

DEDICATED OUTDOOR AIR UNIT SCHEDULE 1 OF 2

MARK	SUPPLY FAN			EXHAUST FAN			COOLING COIL DATA										HEATING COIL DATA								
	CFM	EXTERNAL SP (IN WG)	MOTOR (QTY) HP	CFM	EXTERNAL SP (IN WG)	MOTOR (QTY) HP	TOTAL (MBH)	SENSIBLE (MBH)	EAT °FDB	EAT °FWB	LAT °FDB	LAT °FWB	EWT °F	LWT °F	GPM	WTR P.D. FT. H2O	MAX. COIL FV (FPM)	MBH	EAT °FDB	LAT °FDB	EWT °F	LWT °F	GPM	WTR P.D. FT. H2O	MAX. COIL FV (FPM)
DOAH-M1	3,730	1.9	(1) 7.5	3,400	1.1	(1) 3	216.2	106.7	81.0	72.4	55.0	54.9	45.0	55.0	43.2	10	500	112.8	27.0	55.0	130.0	110.0	11.3	5.0	500

DEDICATED OUTDOOR AIR UNIT SCHEDULE 2 OF 2

MARK	ENERGY WHEEL															UNIT ELECTRICAL DATA					NOTES
	SUMMER CONDITIONS								WINTER CONDITIONS							V / HZ / PH	FLA	MCA	MOCp	VFD(s) PROVIDED BY	
	OUTSIDE AIR CFM	OUTSIDE AIR EAT (DB/WB)	OUTSIDE AIR LAT (DB/WB)	OUTSIDE AIR PRESSURE DROP (IN WG)	ENTERING EXHAUST CFM	EXHAUST EAT (DB / WB)	EXHAUST LAT (DB / WB)	EXHAUST AIR PRESSURE DROP (IN WG)	OUTSIDE AIR EAT (DB / WB)	OUTSIDE AIR LAT (DB/WB)	OUTSIDE AIR PRESSURE DROP (IN WG)	ENTERING EXHAUST CFM	EXHAUST EAT (DB / WB)	EXHAUST LAT (DB / WB)	EXHAUST AIR PRESSURE DROP (IN WG)						
DOAH-M1	3,730	84.0 / 78.7	79.8 / 71.4	0.85	3,400	77.1 / 65.0	81.9 / 73.9	0.75	27.0 / 25.0	51.8 / 42.1	0.85	3,400	68.0 / 51.5	41.3 / 35.7	0.75	460/60/3	17.11	19.86	30 A	VFD BY DIVISION 23	1 - 5

NOTES:

- PROVIDE MERV 8 AND MERV 13 FLAT FILTERS UPSTREAM OF WHEEL FOR SUPPLY, AND MERV 8 UPSTREAM OF WHEEL FOR EXHAUST.
- PROVIDE SUPPLY AND EXHAUST FANS WITH VFDs. PROVIDE WITH SINGLE POINT ELECTRICAL CONNECTION FOR ENTIRE UNIT.
- MAXIMUM UNIT DIMENSIONS ARE 216" LONG 51" WIDE 76" HIGH. UNIT SHALL BE MODULAR AND SHALL BE ABLE TO FIT THROUGH 84"x72" OPENING.
- ESP INCLUDES 0.5 IN WG FOR MERV 8 & 0.7 IN WG FOR MERV 13 LOADING.
- DOAH COOLING COIL SHALL BE SIZED AT 50% SENSIBLE AND TOTAL EFFECTIVENESS. ENERGY WHEEL SHALL PERFORM AS SCHEDULED.

DUCT CONSTRUCTION SCHEDULE

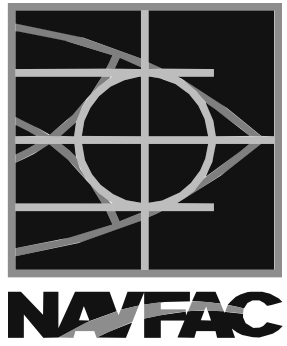
SYSTEM	LOCATION	DESCRIPTION	PRESSURE CLASS (IN WG)	LEAKAGE CLASS		SEAL CLASS	DALt/ QTY	DUCT TEST PRESSURE (IN WG)
				ROUND	RECTANGULAR			
SUPPLY AIR - UPSTREAM OF VAV BOXES	CONCEALED / MECH AREA	SINGLE WALL	+ 4.0	3	6	A	YES / 25%	4.0
SUPPLY AIR - DOWNSTREAM OF VAV BOXES	CONCEALED / MECH AREA	SINGLE WALL	+ 2.0	3	6	A	YES / 25%	2.0
RETURN AIR	CONCEALED / MECH AREA	SINGLE WALL	- 2.0	3	6	A	YES / 25%	2.0
EXHAUST AIR	CONCEALED / MECH AREA	SINGLE WALL	- 2.0	3	6	A	YES / 25%	2.0
TRANSFER AIR	CONCEALED	SINGLE WALL	+/- 1.0	3	6	A	-	-
TREATED OUTSIDE AIR (VA)	CONCEALED / MECH AREA	SINGLE WALL	+ 2.0	3	6	A	YES / 100%	2.0
UNTREATED OUTSIDE AIR (OA)	MECH AREA	SINGLE WALL	- 2.0	3	6	A	YES / 100%	2.0
COMBUSTION AIR INTAKE	MECH AREA	SINGLE WALL	- 6.0	3	-	A	-	-
BOILER COMBUSTION AIR VENT	MECH AREA	DOUBLE WALL	+ 10.0	3	-	A	-	-

PIPING CONSTRUCTION SCHEDULE

SYSTEM	MATERIAL	INSULATION	INSULATION JACKET	NOTES
CHILLED WATER	TYPE L COPPER, SCHEDULE 40 STEEL	CELLULAR GLASS, POLYISOCYANURATE	SEE SPECIFICATION SECTION 23 07 00	
CONDENSATE DRAIN	TYPE L COPPER	FLEXIBLE ELASTOMERIC	SEE SPECIFICATION SECTION 23 07 00	
HEATING HOT WATER	TYPE L COPPER, SCHEDULE 40 STEEL	MINERAL FIBER	SEE SPECIFICATION SECTION 23 07 00	
HEAT RECOVERY HOT WATER	TYPE L COPPER, SCHEDULE 40 STEEL	MINERAL FIBER	SEE SPECIFICATION SECTION 23 07 00	
BURIED CHILLED WATER	BURIED FACTORY-PREFABRICATED PREINSULATED WATER PIPING		SEE SPECIFICATION SECTION 33 61 14	
BURIED HEAT RECOVERY HOT WATER	BURIED FACTORY-PREFABRICATED PREINSULATED WATER PIPING		SEE SPECIFICATION SECTION 33 61 14	

GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.



BURNS MEDONNELL
 1305 EXECUTIVE BLVD. SUITE 160
 CHESAPEAKE, VA 23320

APPROVED:

Charles Southall

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES CWS | DRW CWS | CHK RJ

PMIDM CJC/KDB

BRANCH MANAGER

CHIEF ENGINEER

FIRE PROTECTION

HPN

NAVFACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

MTRACON - MECHANICAL SCHEDULES

DEPARTMENT OF THE NAVY

NAVFAC DRAWING NO.

12801738

SHEET 478 OF 686

M-612

DRAWING REVISION: 12 APRIL 2018

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EXPANSION TANK SCHEDULE						
MARK	SYSTEM TYPE	TOTAL VOLUME (GAL)	ACCEPTANCE VOLUME (GAL)	MAX PRESSURE (PSI)	MAX TEMP (°F)	REMARKS
ET-CHW-MT1	CHILLED WATER	34	19.0	125	90	
ET-HW-MT1	HEATING HOT WATER	34	19.0	125	130	

- NOTES:
- PROVIDE FAN WITH AMCA 99-0401 TYPE B SPARK RESISTANT CONSTRUCTION.
 - UPBLAST FAN SHALL BE DESIGNED TO WITHSTAND WIND LOADS INDICATED ON STRUCTURAL DRAWINGS.
 - FAN SHALL BE ANCHORED TO RESIST UPLIFT LOADS INDICATED ON STRUCTURAL DRAWINGS.
 - FAN SHALL BE PROVIDED WITH AUTOMATIC BELT TENSIONER.

FAN SCHEDULE											
MARK	SERVICE	TYPE	CFM	RPM	ESP (IN WG)	MAX SONES	HP	ELECTRICAL DATA			NOTES
								VOLTS	PH	FLA	
EF-M1	UPS EXHAUST	BELT DRIVEN CENTRIFUGAL UPBLAST	150	1,560	0.4	4.0	1/15	120	1	1.3	1,2,3,4

FAN SOUND POWER LEVEL SCHEDULE

MARK	INLET/RETURN SOUND POWER (dB)								OUTLET SOUND POWER (dB)							
	62.5	125	250	500	1000	2000	4000	8000	62.5	125	250	500	1000	2000	4000	8000
AHU-M1	73	72	76	79	64	66	57	49	90	83	88	92	85	85	79	70
DOAH-M1	77	76	88	80	71	70	64	57	76	76	82	78	74	72	67	59
EF-M1	67	69	63	48	42	40	36	34	-	-	-	-	-	-	-	-

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HHW CONTROL VALVE SCHEDULE

UNIT MARK	VALVE MARK	FLOWRATE (GPM)	VALVE SIZE (IN)	CONFIGURATION	Cv	NOTES
CUH-M1	CV-CUH-M1-HHW	0.5	1/2	2-WAY	0.2	1
CV-CH-H-MT-1	CV-CH-H-MT-1	40.0	1	2-WAY	57.0	2,3
CV-CH-H-MT-2	CV-CH-H-MT-2	40.0	1	2-WAY	57.0	2,3
CV-HHW-BP-M	CV-HHW-BP-M	12.7	3/4	2-WAY	5.7	1
DOAH-M1	CV-DOAH-M1-HHW	11.3	1/2	2-WAY	5.0	1
FCU-M3	CV-FCU-M3-HHW	0.6	1/2	2-WAY	0.3	1
FCU-M5	CV-FCU-M5-HHW	1.1	1/2	2-WAY	0.5	1
FCU-M6	CV-FCU-M6-HHW	1.0	1/2	2-WAY	0.4	1
FCU-M7	CV-FCU-M7-HHW	2.8	1/2	2-WAY	1.3	1
UH-M1	CV-UH-M1-HHW	0.5	1/2	2-WAY	0.2	1
VAV-M101	CV-VAV-M101-HHW	1.0	1/2	2-WAY	0.4	1
VAV-M102	CV-VAV-M102-HHW	0.6	1/2	2-WAY	0.3	1
VAV-M103	CV-VAV-M103-HHW	1.5	1/2	2-WAY	0.7	1
VAV-M104	CV-VAV-M104-HHW	2.3	1/2	2-WAY	1.0	1
VAV-M105	CV-VAV-M105-HHW	1.0	1/2	2-WAY	0.4	1
VAV-M106	CV-VAV-M106-HHW	0.9	1/2	2-WAY	0.4	1
VAV-M107	CV-VAV-M107-HHW	1.5	1/2	2-WAY	0.7	1
VAV-M108	CV-VAV-M108-HHW	0.6	1/2	2-WAY	0.3	1
VAV-M109	CV-VAV-M109-HHW	5.1	1/2	2-WAY	2.3	1
VAV-M110	CV-VAV-M110-HHW	0.6	1/2	2-WAY	0.3	1
VAV-M111	CV-VAV-M111-HHW	1.7	1/2	2-WAY	0.8	1
VAV-M112	CV-VAV-M112-HHW	2.2	1/2	2-WAY	1.0	1
VAV-M113	CV-VAV-M113-HHW	1.6	1/2	2-WAY	0.7	1
VAV-M114	CV-VAV-M114-HHW	1.8	1/2	2-WAY	0.8	1
VAV-M115	CV-VAV-M115-HHW	0.7	1/2	2-WAY	0.3	1
VAV-M116	CV-VAV-M116-HHW	0.5	1/2	2-WAY	0.2	1
VAV-M117	CV-VAV-M117-HHW	0.9	1/2	2-WAY	0.4	1
VAV-M118	CV-VAV-M118-HHW	1.3	1/2	2-WAY	0.6	1
VAV-M119	CV-VAV-M119-HHW	0.5	1/2	2-WAY	0.2	1
VAV-M120	CV-VAV-M120-HHW	0.5	1/2	2-WAY	0.2	1
VAV-M121	CV-VAV-M121-HHW	0.7	1/2	2-WAY	0.3	1
VAV-M122	CV-VAV-M122-HHW	0.6	1/2	2-WAY	0.3	1
VAV-M123	CV-VAV-M123-HHW	1.5	1/2	2-WAY	0.7	1
VAV-M124	CV-VAV-M124-HHW	1.0	1/2	2-WAY	0.4	1
VAV-M125	CV-VAV-M125-HHW	1.8	1/2	2-WAY	0.8	1

- NOTES:
- VALVE SHALL BE MODULATING TYPE.
 - VALVE SHALL BE TWO POSITION TYPE.
 - COORDINATE CONTROL VALVE SELECTION WITH CHILLER PROVIDED. GPM SHALL BE GREATER OF EITHER GPM SHOWN OR MINIMUM CHILLER FLOWRATE.

CHW CONTROL VALVE SCHEDULE

UNIT MARK	VALVE MARK	FLOWRATE (GPM)	VALVE SIZE (IN)	CONFIGURATION	Cv	NOTES
AHU-M1	CV-AHU-M1-CHW	70	1 1/4	2-WAY	31.3	1
CRAH-M1A	CV-CRAH-M1A-CHW	16.5	1/2	2-WAY	7.4	1
CRAH-M1B	CV-CRAH-M1B-CHW	16.5	1/2	2-WAY	7.4	1
CRAH-M2A	CV-CRAH-M2A-CHW	15.1	1/2	2-WAY	6.7	1
CRAH-M2B	CV-CRAH-M2B-CHW	15.1	1/2	2-WAY	6.7	1
CV-CH-C-MT1	CV-CH-C-MT1	215	4	2-WAY	305.0	2,3
CV-CH-C-MT2	CV-CH-C-MT2	215	4	2-WAY	305.0	2,3
CV-CHW-BP-M	CV-CHW-BP-M	56.7	1 1/2	2-WAY	25.3	1
DOAH-M1	CV-DOAH-M1-CHW	43.2	1	2-WAY	19.3	1
FCU-M1A	CV-FCU-M1A-CHW	3.1	1/2	2-WAY	1.4	1
FCU-M1B	CV-FCU-M1B-CHW	3.1	1/2	2-WAY	1.4	1
FCU-M2A	CV-FCU-M2A-CHW	6	1/2	2-WAY	2.7	1
FCU-M2B	CV-FCU-M2B-CHW	6	1/2	2-WAY	2.7	1
FCU-M3	CV-FCU-M3-CHW	2.3	1/2	2-WAY	1.0	1
FCU-M4A	CV-FCU-M4A-CHW	13.3	1/2	2-WAY	5.9	1
FCU-M4B	CV-FCU-M4B-CHW	13.3	1/2	2-WAY	5.9	1
FCU-M5	CV-FCU-M5-CHW	3.4	1/2	2-WAY	1.5	1
FCU-M6	CV-FCU-M6-CHW	2.7	1/2	2-WAY	1.2	1
FCU-M7	CV-FCU-M7-CHW	19.3	1/2	2-WAY	8.6	1

- NOTES:
- VALVE SHALL BE MODULATING TYPE.
 - VALVE SHALL BE TWO POSITION TYPE.
 - COORDINATE CONTROL VALVE SELECTION WITH CHILLER PROVIDED. GPM SHALL BE GREATER OF EITHER GPM SHOWN OR MINIMUM CHILLER FLOWRATE.

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NOTES:

- IF PROVIDED, EC MOTORS SHALL BE PROVIDED WITH EQUIPMENT TO PREVENT ADDING HARMONICS TO ELECTRICAL SYSTEM (INDUCTOR, ETC) AND SHALL BE PROGRAMMED TO PREVENT EC MOTOR OVERSPEED.

CABINET UNIT HEATER SCHEDULE													
MARK	TYPE	MOUNTING STYLE	FAN DATA			HEATING COIL DATA					V/PHz	VFD/STARTER/ DISCONNECT BY	NOTES
			CFM	ESP (IN WG)	MOTOR (HP)	TOTAL CAPACITY (MBH)	EWT (°F) / LWT(°F)	WATER FLOW (GPM)	MAX PRESSURE DROP (FT WC)	HHW ROUNTOUT SIZE (IN)			
CUH-M1	CABINET	VERTICAL FLOOR MOUNTED	140	0.1	25 W	5.0	130/110	0.5	10	0.75	120/1/60	DIVISION 23	1

CABINET UNIT HEATER SCHEDULE

DEHUMIDIFIER SCHEDULE

MARK	TYPE MARK	°F DB	% RH	CAPACITY PINTS/HR	SENSIBLE MBH	FLA	V/PH/Hz	NOTES
DH-M1	DH-1	90	38	5.3	0.9	13.1	120/1/60	2
DH-M2	DH-1	80	50	6.5	0.9	13.1	120/1/60	1, 2
DH-M3	DH-1	90	38	5.3	0.9	13.1	120/1/60	1, 2

- NOTES:
- PROVIDE WITH INTERNAL CONDENSATE PUMP. PUMP SHALL BE CAPABLE OF LIFTING DEHUMIDIFIER CAPACITY BY 17 FT WC.
 - UNITS SHALL BE PLUG IN. PROVIDE WITH CORD AND PLUG.

HUMIDIFIER SCHEDULE

MARK	LB/HR	TYPE	ELECTRIC INPUT (kW)	VOLT/PHASE/ HZ	DISCONNECT PROVIDED BY	NOTES
H-M1	3.0	ELECTRODE STEAM	1.9	120/1/60	DIVISION 23	1,2
H-M2	2.0	ELECTRODE STEAM	1.9	120/1/60	DIVISION 23	1,2
H-M3	8.0	ELECTRODE STEAM	3.7	208/1/60	DIVISION 23	1,2
H-M4	2.0	ELECTRODE STEAM	1.9	120/1/60	DIVISION 23	1,2
H-M5	1.0	ELECTRODE STEAM	1.9	120/1/60	DIVISION 23	1,2
H-M6	4.0	ELECTRODE STEAM	1.9	120/1/60	DIVISION 23	1,2

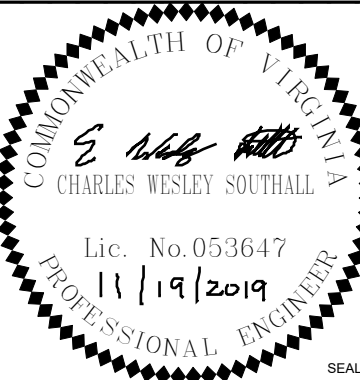
- NOTES:
- PROVIDE WITH UNIT MOUNTED FAN PACK, DRAIN COOLER AND HIGH TEMPERATURE CONDENSATE PUMP SUITABLE FOR 140°F WATER. SINGLE POINT POWER CONNECTION.
 - MOUNT UNITS ON WALL WITH BOTTOM OF THE UNIT AT 4'-0" ABOVE FINISH FLOOR.

GRILLE, REGISTER, AND DIFFUSER SCHEDULE									
MARK	TYPE	MOUNTING	FACE SIZE (IN)	NECK SIZE (IN)	AIRFLOW		MAX NC	THROW PATTERN	NOTES
					MIN	MAX			
CD-1	3-CONE SUPPLY	ACT	24x24	6ø	0 CFM	155 CFM	20	4-WAY	1
CD-2	3-CONE SUPPLY	ACT	24x24	8ø	156 CFM	275 CFM	20	4-WAY	1
CD-3	3-CONE SUPPLY	ACT	24x24	10ø	276 CFM	380 CFM	20	4-WAY	1
CD-4	3-CONE SUPPLY	ACT	24x24	12ø	381 CFM	710 CFM	20	4-WAY	1
CD-5	3-CONE SUPPLY	ACT	12x12	6ø	0 CFM	155 CFM	20	4-WAY	1
EG-1	EGGCRATE EXHAUST GRILLE	SURFACE	8x8	6x6	0 CFM	150 CFM	30	-	1
EG-2	EGGCRATE EXHAUST GRILLE	SURFACE	12x12	8x8	151 CFM	300 CFM	30	-	1
EG-3	EGGCRATE EXHAUST GRILLE	ACT	24x24	10x10	300 CFM	500 CFM	30	-	1
RG-1	RETURN GRILLE	ACT	24x24	6x6	0 CFM	200 CFM	30	-	1
RG-2	RETURN GRILLE	ACT	24x24	8x8	201 CFM	350 CFM	30	-	1
RG-3	RETURN GRILLE	ACT	24x24	10x10	351 CFM	550 CFM	30	-	1
RG-4	RETURN GRILLE	ACT	24x24	12x12	551 CFM	700 CFM	30	-	1
RG-5	RETURN GRILLE	ACT	24x24	18x18	701 CFM	1500 CFM	30	-	1
RG-6	FILTERED RETURN GRILLE	ACT	24x24	6x6	0 CFM	200 CFM	30	-	1,2
RG-7	FILTERED RETURN GRILLE	ACT	24x24	8x8	201 CFM	350 CFM	30	-	1,2
RG-8	FILTERED RETURN GRILLE	ACT	24x24	10x10	351 CFM	550 CFM	30	-	1,2
RG-9	FILTERED RETURN GRILLE	ACT	24x24	12x12	551 CFM	700 CFM	30	-	1,2
TG-1	EGGCRATE TRANSFER GRILLE	ACT / SURFACE	24x24	6x6	0 CFM	100 CFM	30	-	1
TG-2	EGGCRATE TRANSFER GRILLE	ACT / SURFACE	24x24	10x10	105 CFM	350 CFM	30	-	1
TG-3	EGGCRATE TRANSFER GRILLE	ACT	24x24	16x16	350 CFM	1000 CFM	30	-	1

- NOTES:
- DEVICE RUNOUTS SHALL MATCH NECK SIZE INDICATED ON SCHEDULE, UNLESS OTHERWISE INDICATED ON DRAWINGS.
 - PROVIDE WITH MERV 8 24x24 FILTER.

GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.



BURNS & MCDONNELL
 1305 EXECUTIVE BLVD.
 SUITE 160
 CHESAPEAKE, VA 23320

APPROVED: *[Signature]*

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DES: CWS | DRW: CWS | CHK: RJ

PMCM: CJC/KDB

BRANCH MANAGER: *[Signature]*

CHIEF ENGINEER: EJA

FIRE PROTECTION: HPN

DEPARTMENT OF THE NAVY
 NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
 MARINE CORPS IPT
 MCAS CHERRY POINT
 CHERRY POINT, NC
 FY20 P-228 ATC TOWER AND AIR OPERATIONS
 MTRACON - MECHANICAL SCHEDULES

SCALE: AS NOTED
 EPROJECT NO.: 1596300
 CONSTR. CONTR. NO.

NAVFAC DRAWING NO.
 12801739
 SHEET 479 OF 686

M-613

DRAWING REVISION: 12 APRIL 2018

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AIR HANDLING UNIT SCHEDULE																																
MARK	LOCATION	FAN DATA							COOLING COIL DATA										HEATING COIL DATA							FILTER DATA			NOTES			
		S.A. CFM		OA CFM	E.S.P. (IN WG)	FAN (QTY) H.P.	VOLTS/PH/Hz	FLA	TOTAL (MBH)	SENSIBLE (MBH)	EWT °F	LWT °F	EAT °FDB	EAT °FWB	LAT °FDB	LAT °FWB	MAX. COIL FV (FPM)	# OF ROWS	GPM	MAX WATER PRESSURE DROP (FT WC)	CAPACITY (MBH)	EWT °F	EAT °FDB	LAT °FDB	MAX. COIL FV (FPM)	# OF ROWS	GPM	MAX WATER PRESSURE DROP (FT WC)		TYPE	PREFILTER MERV	FINAL FILTER MERV
		(MAX)	(MIN)																													
AHU-T1A	11TH FLOOR MECHANICAL	3850	3850	150	2.5	(1) 5.0	460/60/3	8.5	160	100	45	55	75.5	66.1	52.3	52.3	500	4	32	5	65.3	130	69.3	85	500	1	6.5	5	PLEATED	8	13	1,2,3,4,5,6
AHU-T1B	11TH FLOOR MECHANICAL	3850	3850	150	2.5	(1) 5.0	460/60/3	8.5	160	100	45	55	75.5	66.1	52.3	52.3	500	4	32	5	65.3	130	69.3	85	500	1	6.5	5	PLEATED	8	13	1,2,3,4,5,6

NOTES:

- PROVIDE MERV 8 AND MERV 13 FLAT FILTER UPSTREAM OF COILS.
- PROVIDE WITH VFD.
- MAXIMUM UNIT DIMENSIONS ARE 96" LONG 64" WIDE 48" HIGH. UNIT SHALL BE MODULAR AND SHALL BE ABLE TO FIT THROUGH 45" X 60" OPENING.
- ESP INCLUDES 1.2 IN WG OF FILTER LOADING.
- UNITS ARE TO BE STACKED. SEE M-321 FOR SECTION VIEW OF UNITS.
- UNIT SHALL MEET SEISMIC REQUIREMENTS OUTLINED IN DIVISION 23 05 48.19 OF SPECIFICATIONS. UNIT SHALL MEET ASCE 7 CHAPTER 12.2.2.1 REQUIREMENTS FOR EQUIPMENT THAT MUST REMAIN OPERATIONAL.

FAN COIL UNIT SCHEDULE

MARK	TYPE	LOCATION	FAN DATA			COOLING COIL DATA											HEATING COIL DATA										V/HZ/PH	FLA	STARTER/ DISCONNECT PROVIDED BY	NOTES
			CFM	ESP (IN. W.G.)	(QTY) H.P.	SENSIBLE MBH	TOTAL MBH	ENT. AIR		LVG. AIR		ENT. WTR TEMP (°F)	EXT. WTR TEMP (°F)	GPM	MAX. COIL FV (FPM)	W. P.D. FT. HD.	PIPING RUNOUT SIZE (IN)	MBH	ENT. AIR (°F) DB	ENT. WTR TEMP (°F)	WTR TEMP ΔT (°F)	GPM	MAX. COIL FV (FPM)	W. P.D. FT. HD.	PIPING RUNOUT SIZE					
								(°F) DB	(°F) WB	(°F) DB	(°F) WB																			
FCU-T1	SUSPENDED HORIZONTAL CABINET	2105 ELECTRICAL	230	-	(1) 107W	5.5	6.0	90	70.5	68.3	62.7	45	55	1.2	500	10.0	3/4	-	-	-	-	-	-	-	115/60/1	1.3	DIVISION 23	1,2,3,4		
FCU-T2	SUSPENDED HORIZONTAL DUCTED	2103 CORRIDOR	150	0.3	(1) 107W	3.4	4.5	80	70	58.7	56.9	45	55	0.9	500	10.0	3/4	5.0	50	130	110	0.5	500	10.0	3/4	115/60/1	1.3	DIVISION 23	1,2,3,4	
FCU-T3	SUSPENDED HORIZONTAL DUCTED	2902 CORRIDOR	150	0.3	(1) 107W	3.4	4.5	80	70	58.7	56.9	45	55	0.9	500	10.0	3/4	5.0	50	130	110	0.5	500	10.0	3/4	115/60/1	1.3	DIVISION 23	1,2,3,4	
FCU-T4	SUSPENDED HORIZONTAL DUCTED	2802 CORRIDOR	150	0.3	(1) 107W	3.4	4.5	80	70	58.7	56.9	45	55	0.9	500	10.0	3/4	5.0	50	130	110	0.5	500	10.0	3/4	115/60/1	1.3	DIVISION 23	1,2,3,4	
FCU-T5A	SUSPENDED HORIZONTAL CABINET	2109 COMM	960	-	1/2	21.6	33.8	75	65.3	54.6	53.6	45	55	6.7	500	10.0	1	-	-	-	-	-	-	-	115/60/1	6.4	DIVISION 23	1,3,4,6		
FCU-T5B	SUSPENDED HORIZONTAL CABINET	2109 COMM	960	-	1/2	21.6	33.8	75	65.3	54.6	53.6	45	55	6.7	500	10.0	1	-	-	-	-	-	-	-	115/60/1	6.4	DIVISION 23	1,3,4,6		
FCU-T6	SUSPENDED HORIZONTAL DUCTED	2904 CAB ADMIN	900/550	0.3	(1) 124W	21.5	27.7	78	65	56.4	54.9	45	55	5.5	500	10.0	1	14.2	68	130	110	1.4	500	10.0	3/4	115/60/1	4.8	DIVISION 23	1,2,3,4	
FCU-T7A	SUSPENDED HORIZONTAL DUCTED	2804 SIMULATOR	2100/1800	0.3	(1) 1.0	38.7	54.1	75	65.3	58.3	57.0	45	55	10.7	500	10.0	1	31.6	71	130	110	3.2	500	10.0	3/4	460/60/3	1.8	DIVISION 23	1,3,4,6	
FCU-T7B	SUSPENDED HORIZONTAL DUCTED	2804 SIMULATOR	2100/1800	0.3	(1) 1.0	38.7	54.1	75	65.3	58.3	57.0	45	55	10.7	500	10.0	1	31.6	71	130	110	3.2	500	10.0	3/4	460/60/3	1.8	DIVISION 23	1,3,4,6	
FCU-T8	VERTICAL CABINET	21004 BREAKROOM	220	0.1	(1) 25W	4.9	6.0	78	65	57.4	55.9	45	55	1.2	500	10.0	3/4	5.5	68	130	110	0.6	500	10.0	3/4	115/60/1	0.83	DIVISION 23	1,2,3,5	
FCU-T9	VERTICAL CABINET	21004 BREAKROOM	220	0.1	(1) 25W	4.9	6.0	78	65	57.4	55.9	45	55	1.2	500	10.0	3/4	5.5	68	130	110	0.6	500	10.0	3/4	115/60/1	0.83	DIVISION 23	1,2,3,5	
FCU-T10	VERTICAL CABINET	21004 BREAKROOM	220	0.1	(1) 25W	4.9	6.0	78	65	57.4	55.9	45	55	1.2	500	10.0	3/4	5.5	68	130	110	0.6	500	10.0	3/4	115/60/1	0.83	DIVISION 23	1,2,3,5	
FCU-T11	VERTICAL CABINET	21004 BREAKROOM	220	0.1	(1) 25W	4.9	6.0	78	65	57.4	55.9	45	55	1.2	500	10.0	3/4	5.5	68	130	110	0.6	500	10.0	3/4	115/60/1	0.83	DIVISION 23	1,2,3,5	
FCU-T12	VERTICAL BLOWER COIL	11TH FLOOR MECH	600	0.3	(1) 1/4	16.0	34.0	90	78.5	66.2	64.1	45	55	6.7	500	10.0	1	16.9	50	130	110	1.7	500	10.0	3/4	115/60/1	5.2	DIVISION 23	1,2,3,5	
FCU-T13	SUSPENDED HORIZONTAL CABINET	11TH FLOOR MECH	230	-	(1) 107W	8.1	10.9	90	70.5	58.0	56.0	45	55	2.2	500	10.0	3/4	6.4	50	130	110	0.6	500	10.0	3/4	115/60/1	1.5	DIVISION 23	1,2,3,4	
FCU-T14	SUSPENDED HORIZONTAL CABINET	2101 FP	140	-	(1) 15W	3.8	5.1	90	70.5	62.3	59.1	45	55	1.0	500	10.0	3/4	5.3	50	130	110	0.5	500	10.0	3/4	115/60/1	2.75	DIVISION 23	1,2,3,4	
FCU-T15	SUSPENDED HORIZONTAL CABINET	2102 MECHANICAL	200	-	(1) 107W	6.0	7.6	90	70.5	62.3	59.1	45	55	1.5	500	10.0	3/4	5.3	50	130	110	0.5	500	10.0	3/4	115/60/1	1.3	DIVISION 23	1,2,3,4	
FCU-T16	CEILING CASSETTE	2103 CORRIDOR	260	-	(1) 93W	5.8	6.0	78	65	57.7	57.6	45	55	1.2	500	10.0	3/4	13.1	68	130	110	1.3	500	10.0	3/4	115/60/1	0.7	DIVISION 23	1,2,3,4	

NOTES:

- PROVIDE UNIT WITH UNIT MOUNTED DISCONNECT AND 120V/60HZ/1 PH CONTROL VOLTAGE TRANSFORMER.
- PROVIDE UNITS WITH FILTER RACK ON THE UNIT INTAKE AND 2" MERV 8 FILTER.
- UNIT SHALL MEET SEISMIC REQUIREMENTS OUTLINED IN DIVISION 23 05 48.19 OF SPECIFICATIONS. UNIT SHALL MEET ASCE 7 CHAPTER 12.2.2.1 REQUIREMENTS FOR EQUIPMENT THAT MUST REMAIN OPERATIONAL.
- PROVIDE WITH EC TYPE MOTOR. EC MOTOR SHALL BE PROVIDED WITH EQUIPMENT TO PREVENT ADDING HARMONICS TO ELECTRICAL SYSTEM (INDUCTOR, ETC) AND SHALL BE PROGRAMMED TO PREVENT EC MOTOR OVERSPEED.
- PROVIDE WITH EITHER EC MOTOR OR TWO SPEED MOTOR. IF EC MOTOR, PROVIDE WITH EQUIPMENT TO PREVENT ADDING HARMONICS TO ELECTRICAL SYSTEM (INDUCTOR, ETC) AND SHALL BE PROGRAMMED TO PREVENT EC MOTOR OVERSPEED.
- PROVIDE UNITS WITH FILTER RACK ON THE UNIT INTAKE AND 2" MERV 8 PRE FILTER AND 2" MERV 11 FILTER.

COMPUTER ROOM AIR HANDLER TYPE SCHEDULE

MARK	TYPE	FAN				COOLING COIL													VFD/STARTER/ DISCONNECT PROVIDED BY	NOTES
		TOTAL AIRFLOW (CFM)	EXT. STATIC PRESSURE (IN. W.C.)	FAN MOTOR (QTY) HP	V/Hz/PH	MAX. COIL FACE VELOCITY (FPM)	SENSIBLE CAPACITY (MBTUH)	TOTAL CAPACITY (MBTUH)	EAT DB (°F)	EAT WB (°F)	LAT DB (°F)	LAT WB (°F)	EWT (°F)	LWT (°F)	CHILLED WATER FLOW (GPM)	MAX. WATER PRESSURE DROP (FT. W.C.)	MIN. ROWS / MAX. FPI	PIPING RUNOUT SIZE		
CRAH-T1A	HORIZONTAL	5000	0.5	1 (5.0)	460/60/3	500	113.2	143.0	75	62.5	54.0	52.5	45	56.5	27	25	4	2	DIVISION 23	1,2,3,4
CRAH-T1B	HORIZONTAL	5000	0.5	1 (5.0)	460/60/3	500	113.2	143.0	75	62.5	54.0	52.5	45	56.5	27	25	4	2	DIVISION 23	1,2,3,4

NOTES:

- UNIT SHALL MEET SEISMIC REQUIREMENTS OUTLINED IN DIVISION 23 05 48.19 OF SPECIFICATIONS. UNIT SHALL MEET ASCE 7 CHAPTER 12.2.2.1 REQUIREMENTS FOR EQUIPMENT THAT MUST REMAIN OPERATIONAL.
- MOTOR SHALL BE EC TYPE OF INVERTER RATED WITH VFD. EC MOTORS SHALL BE PROVIDED WITH EQUIPMENT TO PREVENT ADDING HARMONICS TO ELECTRICAL SYSTEM (INDUCTOR, ETC) AND SHALL BE PROGRAMMED TO PREVENT EC MOTOR OVERSPEED.
- COOLING CAPACITIES ARE NET (CAPACITY IS DE-RATED FOR FAN HEAT).
- PROVIDE UNITS WITH FILTER RACK ON THE UNIT INTAKE AND 2" MERV 8 PRE FILTER AND 2" MERV 11 FILTER.

FAN SCHEDULE

MARK	SERVICE	TYPE	CFM	RPM	ESP (IN WG)	MAX SONES	HP	V/PH/Hz	FLA	VFD/STARTER /DISCONNECT PROVIDED BY	REMARKS
EF-T1	JANITOR EXHAUST	CEILING MOUNT CABINET	70	900	0.3	0.5	15 W	115/60/1	0.34	DIVISION 23	2
EF-T2	10TH FLOOR EXHAUST	INLINE CABINET	210	1050	0.6	4.0	92 W	115/60/1	0.71	DIVISION 23	2
SPF-T1	STAIRWELL PRESSURIZATION	BELT DRIVE INLINE	9000	1250	2.5	20.0	7.5	460/60/3	11	DIVISION 23	1,3,4
SPF-T2	PRESSURIZATION RELIEF	BELT DRIVE CENTRIFUGAL INLINE	2275	1910	2.0	19.5	2.0	460/60/3	3.4	DIVISION 23	1,3,4

NOTES:

- PROVIDE WITH VFD.
- MOTORS SHALL BE EC TYPE. EC MOTORS SHALL BE PROVIDED WITH EQUIPMENT TO PREVENT ADDING HARMONICS TO ELECTRICAL SYSTEM (INDUCTOR, ETC) AND SHALL BE PROGRAMMED TO PREVENT EC MOTOR OVERSPEED.
- FAN SHALL BE RATED FOR SMOKE CONTROL UNDER UL (POWER VENTILATOR FOR SMOKE CONTROL).
- FAN SHALL MEET SEISMIC REQUIREMENTS OUTLINED IN DIVISION 23 05 48.19 OF SPECIFICATIONS. UNIT SHALL MEET ASCE 7 CHAPTER 12.2.2.1 REQUIREMENTS FOR EQUIPMENT THAT MUST REMAIN OPERATIONAL.

BOILER SCHEDULE

MARK	ROOM	TYPE	LVG. WATER TEMP (°F)	ENT. WATER TEMP (°F)	WATER FLOW (GPM)	NATURAL GAS INPUT (MBH)	MINIMUM GROSS OUTPUT (MBH)	V/PH/Hz	FLA	STARTER/DISCONNECT PROVIDED BY	NOTES
B-T1	2102 MECH	FLOOR MOUNTED CONDENSING BOILER	130	110	16.3	172	163	120/1/60	12	DIVISION 23	1,2
B-T2	2102 MECH	FLOOR MOUNTED CONDENSING BOILER	130	110	16.3	172	163	120/1/60	12	DIVISION 23	1,2

NOTES:

- PROVIDE BOILER WITH PHHWP, SEE HVAC PUMP SCHEDULE FOR PUMP INFORMATION.
- 95% MINIMUM EFFICIENCY

GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- ALL ELEVATIONS SHOWN ARE FROM AFF TO BOTTOM OF OBJECT UNLESS OTHERWISE NOTED.

HUMIDIFIER SCHEDULE

MARK	LB/HR	TYPE	ELECTRIC INPUT (kW)	VOLT/PH	VFD/STARTER/DISCONNECT PROVIDED BY	NOTES
H-T1	1.0	WALL MOUNT ELECTRODE STEAM HUMIDIFIER	1.9	120/1/60	DIVISION 23	1,4,6
H-T2	8.0	DUCT MOUNTED ELECTRODE STEAM HUMIDIFIER	3.7	208/1/60	DIVISION 23	2,3
H-T3	1.0	WALL MOUNT ELECTRODE STEAM HUMIDIFIER	1.9	120/1/60	DIVISION 23	1,4
H-T4	2.0	WALL MOUNT ELECTRODE STEAM HUMIDIFIER	1.9	120/1/60	DIVISION 23	1,4,6
H-T5	1.0	WALL MOUNT ELECTRODE STEAM HUMIDIFIER	1.9	120/1/60	DIVISION 23	1,5

NOTES:

- PROVIDE UNIT MOUNTED FAN PACK AND DRAIN COOLER.
- PROVIDE WITH STEAM DISTRIBUTION DUCT MOUNTED MANIFOLD. MANIFOLD SHALL BE DESIGNED TO BE PLACED IN 24"x18" DUCT.
- PROVIDE WITH MANUFACTURER FLOOR STAND.
- MOUNT HUMIDIFIER WITH BOTTOM AT 4'-0" ABOVE FINISHED FLOOR.
- MOUNT HUMIDIFIER WITH BOTTOM AT 3'-0" ABOVE FINISHED FLOOR.



1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
2. REFER TO DRAWING MI700 FOR MECHANICAL CONTROLS LEGEND, ABBREVIATIONS AND NOTES.

AIR OPS DOAH DDC POINTS LIST			
MARK	DESCRIPTION	ALARM	TREND
ANALOG INPUT			
AI01	HIGH LIMIT STATIC PRESSURE	Yes	No
AI02	HIGH LIMIT STATIC PRESSURE	Yes	No
AI03	ERW DIFFERENTIAL PRESSURE - OA SIDE	No	Yes
AI04	ERW BYPASS DAMPER POSITION (FEEDBACK) - OA SIDE	No	Yes
AI05	ERW LEAVING AIR HUMIDITY - OA SIDE	No	Yes
AI06	ERW LAT - OA SIDE	No	Yes
AI07	HWR VALVE POSITION (FEEDBACK)	No	Yes
AI08	HEATING COIL LAT	No	Yes
AI09	CHW VALVE POSITION (FEEDBACK)	No	Yes
AI10	COOLING COIL LAT	No	Yes
AI11	TOA FAN DIFFERENTIAL PRESSURE	No	Yes
AI12	TOA TEMPERATURE	No	Yes
AI13	TOA HUMIDITY	No	Yes
AI14	TOA AIRFLOW	No	Yes
AI15	EA FAN DIFFERENTIAL PRESSURE	No	Yes
AI16	EA AIRFLOW	No	Yes
AI17	ERW LEAVING AIR HUMIDITY - EA SIDE	No	Yes
AI18	ERW LAT - EA SIDE	No	Yes
AI19	ERW DIFFERENTIAL PRESSURE - EA SIDE	No	Yes
AI20	ERW BYPASS DAMPER POSITION (FEEDBACK) - EA SIDE	No	Yes
AI21	ERW ENTERING AIR HUMIDITY - EA SIDE	No	Yes
AI22	ERW EAT - EA SIDE	No	Yes
AI23	DOAH SF SPEED (FEEDBACK)	No	No
AI24	DOAH EF SPEED (FEEDBACK)	No	No
AI25	BATHROOM EXHAUST AIRFLOW	No	Yes
AI26	RELIEF DAMPER POSITION (FEEDBACK)	No	No
AI27	RELIEF AIRFLOW	No	Yes
ANALOG OUTPUT			
AO01	ERW BYPASS DAMPER CONTROL - OA SIDE	No	Yes
AO02	HHW VALVE CONTROL	No	Yes
AO03	CHW VALVE CONTROL	No	Yes
AO04	ERW BYPASS DAMPER CONTROL - EA SIDE	No	Yes
AO05	DOAH SF SPEED (FEEDBACK)	No	Yes
AO06	DOAH EF SPEED (FEEDBACK)	No	Yes
AO07	RELIEF DAMPER CONTROL	No	Yes
BINARY INPUT			
BI01	DOAH OA DAMPER POSITION OPEN (FEEDBACK)	Yes	No
BI02	DOAH OA DAMPER POSITION CLOSED (FEEDBACK)	Yes	No
BI03	MERV 8 UOA FILTER STATUS	Yes	No
BI04	MERV 13 UOA FILTER STATUS	Yes	No
BI06	FREEZESTAT	Yes	No
BI07	HIGH LIMIT STATIC PRESSURE	Yes	No
BI08	TOA DUCT SMOKE DETECTOR	Yes	No
BI09	DOAH EA DISCHARGE DAMPER POSITION OPEN (FEEDBACK)	Yes	Yes
BI10	DOAH EA DISCHARGE DAMPER POSITION CLOSED (FEEDBACK)	Yes	Yes
BI11	MERV 8 EA FILTER STATUS	Yes	No
BI12	TREATED OA DAMPER POSITION OPEN (FEEDBACK)	Yes	Yes
BI13	TREATED OA DAMPER POSITION CLOSED (FEEDBACK)	Yes	Yes
BI14	EA INLET DAMPER POSITION OPEN (FEEDBACK)	Yes	Yes
BI15	EA INLET DAMPER POSITION CLOSED (FEEDBACK)	Yes	Yes
BI16	DOAH SF STATUS	No	Yes
BI17	DOAH SF ALARM	Yes	No
BI18	ERW STATUS (FEEDBACK)	Yes	Yes
BI19	DOAH EF ALARM	Yes	No
BI20	DOAH EF STATUS	No	Yes
BI21	ESS (ATFP EMERGENCY SHUTDOWN) *GLOBAL POINT	Yes	No
BI22	HHWP-DOAH-X CIRCULATOR PUMP STATUS	No	Yes
BI23	DOAH DDC HARDWARE RESET	No	No
BI24	24/7 EA ISOLATION DAMPER POSITION OPEN (FEEDBACK)	Yes	Yes
BI25	24/7 EA DAMPER POSITION CLOSED (FEEDBACK)	Yes	Yes
BINARY OUTPUT			
BO01	DOAH OA DAMPER CONTROL	No	Yes
BO02	DOAH SF ENABLE	No	No
BO03	DOAH EA DAMPER CONTROL	No	Yes
BO04	TREATED OA DAMPER CONTROL	No	Yes
BO05	EA INLET DAMPER CONTROL	No	Yes
BO06	ERW CONTROL	No	Yes
BO07	DOAH EF ENABLE	No	No
BO08	HHWP-DOAH-X CIRCULATOR PUMP ON/OFF	No	Yes
BO09	24/7 EA DAMPER CONTROL	No	Yes



APPROVED					
<i>[Signature]</i>					
FOR COMMANDER NAVFAC					
ACTIVITY					
SATISFACTORY TO DATE					
DES FYA		DRW FYA		CHK	RJ
PM/DIM			CJC/KDB		
BRANCH MANAGER			<i>T/LH</i>		
CHIEF ENGR/AIRC			EJA		
FIRE PROTECTION			HPN		

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC NAVAL STATION - NORFOLK, VA	CHERRY POINT, NC
MARINE CORPS IPT MICAS CHERRY POINT	CHERRY POINT, NC
DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC NAVAL STATION - NORFOLK, VA	CHERRY POINT, NC
AIR OPS / ITRACON - DOAH CONTROL SCHEMATIC AND POINTS LIST	CHERRY POINT, NC

SCALE: AS NOTED		
EPROJECT NO.:	1596300	
CONSTR. CONTR. NO.		
NAVFAC DRAWING NO.		
12801744		
SHEET	484	OF 686
MI702		
DRAWING REVISION: 12 APRIL 2018		

PLOTTED: 11/19/2019 4:58 PM

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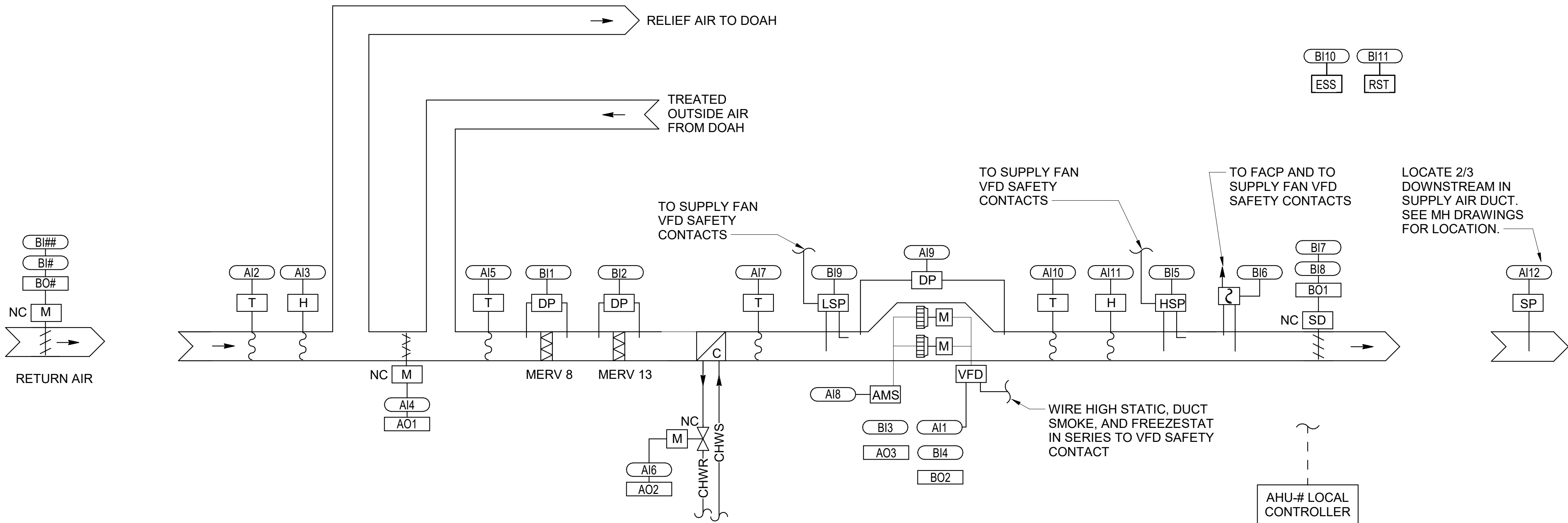
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AIR OPS / MTRACON - AHU SCHEMATIC

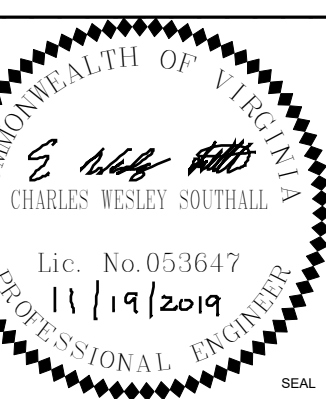


GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- REFER TO DRAWING M1700 FOR MECHANICAL CONTROLS LEGEND, ABBREVIATIONS AND NOTES.

AIR OPS AHU DDC POINTS LIST

MARK	DESCRIPTION	ALARM	TREND
ANALOG INPUT			
AI1	SF SPEED (FEEDBACK) *TYPICAL OF 2	Yes	Yes
AI2	MA TEMPERATURE	No	Yes
AI3	MA HUMIDITY	No	Yes
AI4	RA DAMPER POSITION (FEEDBACK)	No	Yes
AI5	MA TEMPERATURE	No	Yes
AI6	CHW VALVE POSITION (FEEDBACK)	No	No
AI7	COOLING COIL LAT	No	Yes
AI8	SA FAN AIRFLOW	No	Yes
AI9	SA DIFFERENTIAL PRESSURE	No	Yes
AI10	SA TEMPERATURE	No	Yes
AI11	SA HUMIDITY	No	Yes
AI12	SUPPLY AIR PRESSURE	No	Yes
ANALOG OUTPUT			
AO1	RA DAMPER CONTROL	No	No
AO2	CHW VALVE CONTROL	No	No
AO3	SF SPEED *TYPICAL OF 2	No	No
BINARY INPUT			
BI1	MERV 8 FILTER STATUS	Yes	No
BI2	MERV 13 FILTER STATUS	Yes	No
BI3	SF STATUS *TYPICAL OF 2	No	Yes
BI4	SF ALARM * TYPICAL OF 2	Yes	No
BI5	HIGH LIMIT STATIC PRESSURE	Yes	No
BI6	SA DUCT SMOKE DETECTOR	Yes	No
BI7	SA SMOKE DAMPER POSITION OPEN (FEEDBACK)	Yes	No
BI8	SA SMOKE DAMPER POSITION CLOSED (FEEDBACK)	Yes	No
BI9	LOW LIMIT STATIC PRESSURE	Yes	No
BI10	ESS (ATFP EMERGENCY SHUTDOWN) *GLOBAL POINT	Yes	No
BI11	AHU DDC RESET	No	No
BI#	RA 24/7 ISOLATION DAMPER POSITION OPEN (FEEDBACK) *TYPICAL	Yes	No
BI##	RA 24/7 ISOLATION DAMPER POSITION CLOSED (FEEDBACK) *TYPICAL	Yes	No
BINARY OUTPUT			
BO1	SA SMOKE DAMPER CONTROL	No	No
BO2	SF ENABLE *TYPICAL OF 2	Yes	No
BO#	RA 24/7 ISOLATION DAMPER CONTROL *TYPICAL	No	No



BURNS & MCDONNELL
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED:

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DES FYA DRW FYA CHK RJ

PMIDM CJC/KDB

BRANCH MANAGER

CHIEF ENGINEER

FIRE PROTECTION

HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

AIR OPS / MTRACON - AHU CONTROL SCHEMATIC AND POINTS LIST

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

NAVFAC DRAWING NO.

12801746

SHEET 486 OF 686

M1704

DRAWING REVISION: 12 APRIL 2018

UNCLASSIFIED

B

A

GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- REFER TO DRAWING MI700 FOR MECHANICAL CONTROLS LEGEND, ABBREVIATIONS AND NOTES.

MTRACON/ATCT - CHW SYSTEM POINT LIST

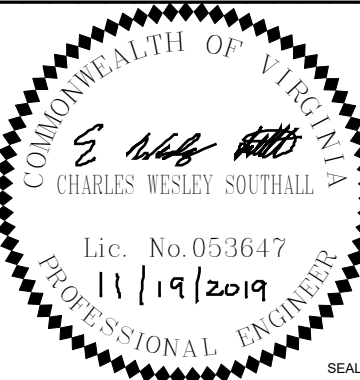
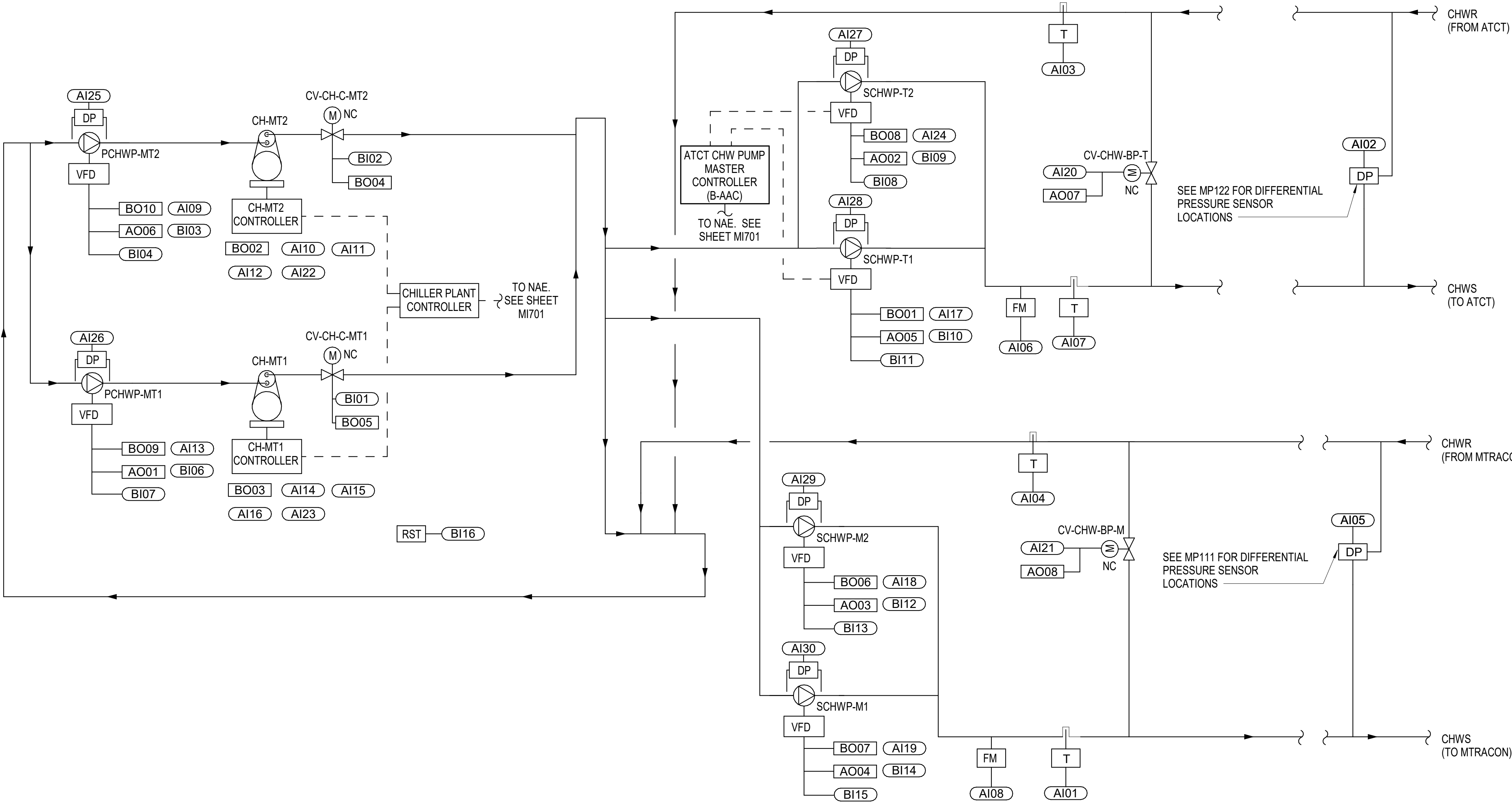
MARK	DESCRIPTION	ALARM	TREND
ANALOG INPUT			
AI01	MTRACON CHW SUPPLY TEMPERATURE	Yes	Yes
AI02	TOWER CHW DIFFERENTIAL PRESSURE	No	Yes
AI03	TOWER CHW RETURN TEMPERATURE	Yes	Yes
AI04	MTRACON CHW RETURN TEMPERATURE	Yes	Yes
AI05	MTRACON CHW DIFFERENTIAL PRESSURE	No	Yes
AI06	TOWER SECONDARY CHW FLOWRATE	No	Yes
AI07	TOWER CHW SUPPLY TEMPERATURE	Yes	Yes
AI08	MTRACON SECONDARY CHW FLOWRATE	No	Yes
AI09	PCHWP-MT2 SPEED (FEEDBACK)	No	Yes
AI10	CH-MT2 CHW DIFFERENTIAL PRESSURE	No	Yes
AI11	CH-MT2 FLOW	Yes	Yes
AI12	CH-MT2 CHW SUPPLY TEMPERATURE	Yes	Yes
AI13	PCHWP-A1 SPEED (FEEDBACK)	No	Yes
AI14	CH-MT1 CHW DIFFERENTIAL PRESSURE	No	Yes
AI15	CH-MT1 FLOW	Yes	Yes
AI16	CH-MT1 CHW SUPPLY TEMPERATURE	Yes	Yes
AI17	SCHWP-T1 SPEED (FEEDBACK)	No	Yes
AI18	SCHWP-M2 SPEED (FEEDBACK)	No	Yes
AI19	SCHWP-M1 SPEED (FEEDBACK)	No	Yes
AI20	ATCT CHW BYPASS POSITION (FEEDBACK)	No	Yes
AI21	MTRACON CHW BYPASS POSITION (FEEDBACK)	No	Yes
AI22	CH-MT2 CHW RETURN TEMPERATURE	Yes	Yes
AI23	CH-MT1 CHW RETURN TEMPERATURE	Yes	Yes
AI24	SCHWP-T2 SPEED (FEEDBACK)	No	Yes
AI25	PCHWP-MT2 DIFFERENTIAL PRESSURE	No	Yes
AI26	PCHWP-MT1 DIFFERENTIAL PRESSURE	No	Yes
AI27	SCHWP-T2 DIFFERENTIAL PRESSURE	No	Yes
AI28	SCHWP-T1 DIFFERENTIAL PRESSURE	No	Yes
AI29	SCHWP-M2 DIFFERENTIAL PRESSURE	No	Yes
AI30	SCHWP-M1 DIFFERENTIAL PRESSURE	No	Yes

ANALOG OUTPUT			
AO01	PCHWP-MT1 SPEED CONTROL	No	Yes
AO02	SCHWP-T2 SPEED CONTROL	No	Yes
AO03	SCHWP-M2 SPEED CONTROL	No	Yes
AO04	SCHWP-M1 SPEED CONTROL	No	Yes
AO05	SCHWP-T1 SPEED CONTROL	No	Yes
AO06	PCHWP-MT2 SPEED CONTROL	No	Yes
AO07	ATCT CHW BYPASS VALVE CONTROL	No	No
AO08	MTRACON CHW BYPASS VALVE CONTROL	No	No

BINARY INPUT			
BI01	CV-CHW-MT1 POSITION (FEEDBACK)	Yes	No
BI02	CV-CHW-MT2 POSITION (FEEDBACK)	Yes	No
BI03	PCHWP-A1 VFD STATUS	No	Yes
BI04	PCHWP-A1 VFD ALARM	Yes	No
BI06	PCHWP-A1 VFD STATUS	No	Yes
BI07	PCHWP-A1 VFD ALARM	Yes	No
BI08	SCHWP-T2 VFD ALARM	Yes	No
BI09	SCHWP-T2 VFD STATUS	No	Yes
BI10	SCHWP-T1 VFD STATUS	No	Yes
BI11	SCHWP-T1 VFD ALARM	Yes	No
BI12	SCHWP-M2 VFD STATUS	No	Yes
BI13	SCHWP-M2 VFD ALARM	Yes	No
BI14	SCHWP-M1 VFD STATUS	No	Yes
BI15	SCHWP-M1 VFD ALARM	Yes	No
BI16	AIR OPS/MTRACON CHILLER PLANT DDC RESET	No	No

BINARY OUTPUT			
BO01	SCHWP-T1 START/STOP	No	No
BO02	CH-MT2 CHILLER CONTROL	No	No
BO03	CH-MT1 CHILLER CONTROL	No	No
BO04	CV-CHW-MT2 VALVE CONTROL	No	No
BO05	CV-CHW-MT1 VALVE CONTROL	No	No
BO06	SCHWP-M2 START/STOP	No	No
BO07	SCHWP-M1 START/STOP	No	No
BO08	SCHWP-T2 START/STOP	No	No
BO09	PCHWP-MT1 START/STOP	No	No
BO10	PCHWP-MT2 START/STOP	No	No

MTRACON/ATCT - CHW SYSTEM SCHEMATIC



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APPROVED: *[Signature]*
FOR COMMANDER NAVFAC

SATISFACTORY TO: _____ DATE: _____
DES: CWS | DRW: T/JG | CHK: RJ
PMIDM: CJC/KDB
BRANCH MANAGER: *[Signature]*
CHIEF ENGINEER: EJA
FIRE PROTECTION: HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL STATION - NORFOLK, VA
CHERRY POINT, NC
MCAS CHERRY POINT
FY20 P-228 ATC TOWER AND AIR OPERATIONS
MTRACON/ATCT - CHW SYSTEM CONTROL SCHEMATIC AND POINTS LIST

SCALE: AS NOTED
PROJECT NO.: 1596300
CONSTR. CONTR. NO.:
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SHEET 492 OF 686
MI720
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MTRACTON/ATCT - CHW SYSTEM SEQUENCE OF OPERATION

3.4.3 MTRACON AND ATCT CHILLED WATER SYSTEM

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL MUST BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

THE CHILLED WATER (CHW) SYSTEM CONSISTS OF (2) AIR COOLED CHILLERS, (2) VFD-DRIVEN CHILLED WATER CONSTANT VOLUME PRIMARY PUMPS, AND (4) VFD-DRIVEN CHILLED WATER VARIABLE VOLUME SECONDARY PUMPS. THE CHW SYSTEM IS CONFIGURED IN A CONSTANT-SPEED PRIMARY/VARIABLE-SPEED SECONDARY FLOW CONFIGURATION. THE SECONDARY CHW PUMPS ARE EACH SIZED FOR THE BUILDING LOAD (N+1), BUT CAN ALSO OPERATE IN PARALLEL TO PROVIDE HIGHER EFFICIENCY. A MASTER CHILLER PLANT CONTROLLER SHALL BE PROVIDED TO ENABLE AND SEQUENCE THE CHILLERS. EACH CHILLER SHALL BE FURNISHED WITH FACTORY CONTROLS AND A BACNET COMMUNICATION BOARD TO INTEGRATE WITH THE MASTER CHILLER PLANT DDC CONTROLLER. EACH CHILLER SHALL BE PROVIDED WITH THE FOLLOWING FACTORY CONTROLS: PRIMARY PUMP CONTROLS INCLUDING START/STOP AND FAILURE DETECTION, FLOW SWITCH, FLOWMETER, ENTERING AND LEAVING CHW TEMPERATURE SENSORS. THE MASTER CHILLER PLANT DDC CONTROLLER SHALL BE ABLE TO MONITOR THE CHILLER'S SYSTEM PERFORMANCE AND ADJUST SETPOINTS.

3.4.3.1 SYSTEM ENABLE AND LOOP ENABLE
THE CHILLER PLANT CONTROLLER SHALL BE ENABLED TO RUN AT ALL TIMES.
UPON DEMAND FOR CHILLED WATER AS INDICATED ON OPENING OF CHW CONTROL VALVE (AHU, DOAH, FCU - IN EITHER MTRACON OR ATCT) , THE CHILLER PLANT CONTROLLER SHALL ENABLE THE CHILLER CONTROL AND CHILLED WATER TEMPERATURE CONTROL LOOPS. EACH BUILDING'S SECONDARY PUMP CONTROL LOOPS SHALL BE ENABLED WHEN THE CHILLER CONTROL LOOP IS ENABLED AND ANY OF THE ASSOCIATED BUILDING'S CHW CONTROL VALVES ARE OPENED (AHU,DOAH,FCU).

3.4.3.2 INTEGRAL H-O-A SWITCH:
THE PRIMARY AND SECONDARY CHILLED WATER PUMP VFDS SHALL HAVE INTEGRAL H-O-A SWITCHES.

3.4.3.2.1 **HAND**
WITH THE H-O-A SWITCH IN HAND POSITION, THE PUMP STARTS AND RUNS CONTINUOUSLY. PUMP SPEED SHALL BE UNDER MANUAL-OPERATOR CONTROL.

3.4.3.2.2 OFF
WITH THE H-O-A SWITCH IN OFF POSITION, THE PUMP STOPS.

3.4.3.2.3 **AUTO**
WITH THE H-O-A SWITCH IN AUTO POSITION, THE PUMP SHALL RUN SUBJECT TO THE CHILLED WATER PUMP START/STOP (PCHW-PMP-SS OR SCHW-PMP-SS) COMMAND AND PUMP SPEED SHALL BE UNDER CONTROL OF THE DDC SYSTEM.

3.4.3.3 PROOFS AND SAFETIES
DDC HARDWARE SHALL MONITOR ALL PROOFS AND SAFETIES.

3.4.3.3.1 PROOFS

- CHILLERS
- CONTROL VALVES
- PRIMARY CHILLED WATER PUMPS
- SECONDARY CHILLED WATER PUMPS

3.4.3.3.2 SAFETIES

IF ANY PIECE OF EQUIPMENT FAILS TO PROVE STATUS AFTER IT IS COMMANDED, WITH A 20 SEC. (ADJ.) TIME DELAY, AN ALARM SHALL BE TRIGGERED. THIS INCLUDES THE FOLLOWING:

- CHILLERS
- CONTROL VALVES
- PRIMARY CHILLED WATER PUMPS
- SECONDARY CHILLED WATER PUMPS

IF THE CHW SUPPLY TEMPERATURE DRIFTS 3°F (ADJ.) ABOVE OR BELOW SETPOINT FOR 10 MINUTES (ADJ.), AN ALARM SHALL BE SENT TO THE DDC. THE DDC SYSTEM SHALL SUM CHILLER LOADING AND REPORT TO THE OPERATOR WORKSTATIONS IN UNITS OF TONS.

3.4.3.3.3 DDC HARDWARE RESET
DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE OR BOTH (WHERE THE CONTRACTOR PROVIDES BOTH RESET FUNCTIONS AND THE OPERATOR CAN USE EITHER ONE TO PERFORM THE RESET), AS SHOWN ON THE POINTS SCHEDULE DRAWING.

3.4.3.4 CHILLER AND CHILLED WATER PUMP CONTROL

3.4.3.4.1 CHILLER CONTROL
UPON A CALL FOR CHILLED WATER, THE DDC CHILLER PLANT CONTROL PANEL SHALL ENABLE THE LEAD CHILLER'S PRIMARY PUMP VIA THE LEAD CHILLER'S CONTROL PANEL. THE LEAD CHILLER SHOULD BE SELECTED AS THE CHILLER WITH THE LEAST RUN HOURS. AFTER PRIMARY PUMP IS PROVEN OPERATIONAL AND RAMPS TO THE SPEED ASSOCIATED WITH FULL WATER FLOW, THE CHILLER CONTROL PANEL SHALL ENERGIZE THE LEAD CHILLER. THE LEAD CHILLER'S INTERNAL CONTROLS SHALL OPERATE TO MAINTAIN THE CHILLED WATER SUPPLY TEMPERATURE SETPOINT. WHEN THE LEAD CHILLER REACHES 75% CAPACITY OR CAPACITY DETERMINED FOR OPTIMAL EFFICIENCY AND ADDITIONAL CAPACITY IS NEEDED OR THE SECONDARY FLOW RATE IS 98% OF THE PRIMARY FLOW RATE, THE MASTER CHILLER PLANT CONTROLLER SHALL ENABLE THE NEXT CHILLER AND PRIMARY PUMP IN THE OPERATIONAL ORDER. THE SECOND CHILLER SHOULD HAVE THE NEXT LOWEST RUN HOURS. THE MASTER CHILLER PLANT CONTROLLER SHALL BALANCE LOAD BETWEEN THE TWO OPERATIONAL CHILLERS AND MAINTAIN THE PRIMARY LOOP FLOW RATE ABOVE THE SECONDARY LOOP FLOW RATE. WHEN THE BUILDING DEMAND DROPS AND ALLOWS THE LOAD TO MET BY A SINGLE CHILLER WHILE KEEPING THE PRIMARY FLOW RATE ABOVE THE SECONDARY FLOW RATE, DE-ENERGIZE THE CHILLER WITH THE MOST RUN HOURS. AFTER THE CHILLER CONTROLLER INDICATES THE CHILLER WITH THE MOST RUN HOURS IS DE-ENERGIZED, THE ASSOCIATED PRIMARY CHW PUMP SHALL BE DE-ENERGIZED FOLLOWING A 30 SEC DELAY. THE FINAL CHILLER SHALL BE DE-ENERGIZED BY THE MASTER CHILLER PLANT CONTROLLER WHEN THERE IS NO LONGER CHILLED WATER DEMAND. AFTER THE CHILLER CONTROLLER INDICATES THE FINAL CHILLER IS DE-ENERGIZED, THE REMAINING PRIMARY CHW PUMP SHALL BE DE-ENERGIZED AFTER A 30 SEC DELAY (ADJ).

3.4.3.4.2 **PRIMARY CHW PUMP CONTROL**
THE PRIMARY CHW PUMPS SHALL BE ENABLED AS DESCRIBED UNDER THE CHILLER CONTROL PARAGRAPH. PRIMARY CHW PUMPS SHALL BE CONTROLLED BY ITS CHILLER CONTROL PANEL. THE OPERATIONAL SPEED FOR THE PRIMARY CHW PUMPS TO PROVIDE THE DESIGN CHILLER FLOW RATE (CONSTANT VOLUME) SHALL BE DETERMINED DURING TESTING AND BALANCING (TAB).

3.4.3.4.3 SECONDARY CHW PUMP CONTROL

THE SECONDARY CHW PUMP CONTROL LOOP SHALL BE ENABLED ONCE THE CHILLER PLANT MASTER CONTROL INDICATES A CHILLER AND ASSOCIATED PRIMARY PUMP IS OPERATIONAL AND THE BUILDING THE SECONDARY PUMPS SERVE HAS CHILLED WATER DEMAND (CHW VALVE OPEN). THE SECONDARY CHW PUMP VFD SPEED SHALL BE CONTROLLED TO MAINTAIN BUILDING LOOP DIFFERENTIAL SETPOINTS. THE DIFFERENTIAL PRESSURE SETPOINT SHALL BE RESET TO KEEP THE CRITICAL CONTROL VALVE BETWEEN 90-95% OPEN. THE CRITICAL CONTROL VALVE SHALL BE THE CONTROL VALVE THAT IS THE MOST OPEN AT ANY GIVEN TIME. THE DIFFERENTIAL PRESSURE SETPOINTS SHALL BE RESET DOWN IF CRITICAL CONTROL VALVE IS LESS THAN 90% OPEN, AND THE SETPOINT SHALL BE RESET UP IF CRITICAL CONTROL VALVE IS GREATER THAN 95% OPEN. TO ENSURE STABLE SYSTEM FLOW, THE RATE AT WHICH CHW PUMP SPEED MAY CHANGE SHALL BE LIMITED TO 10% PER MINUTE (ADJ). THE MINIMUM NUMBER OF SECONDARY CHW PUMPS SHALL OPERATE TO REMAIN WITHIN THE OPTIMAL FLOW (TYP. 50-75%), HEAD, AND PUMP-EFFICIENCY RANGE, AS DETERMINED BY THE DDC PROGRAM FROM EVALUATION OF THE SPECIFIC MODELS' SECONDARY CHW PUMP CURVES. BUILDING LOOP DIFFERENTIAL-PRESSURE ERROR SHALL BE TAKEN INTO ACCOUNT, WHEN ADDING SECONDARY CHW PUMPS.

3.4.3.5 MINIMUM SECONDARY FLOW CONTROL VALVE
THE DDC SYSTEM SHALL MODULATE THE CHW BYPASS VALVE TO ENSURE MINIMUM FLOW OF 30% OF THE TOTAL CHW FLOW THROUGH THE CHILLED WATER LOOP. THIS IS ACCOMPLISHED BY MEASURING THE SYSTEM FLOW AND USING THIS AS INPUT TO A FEEDBACK CONTROL ALGORITHM WHICH MODULATES THE POSITION OF THE CHW BYPASS VALVE.

3.4.3.6 FAILURE
THE MASTER CHILLER PLANT DDC CONTROLLER SHALL ALARM AND STAGE CHILLERS, ASSOCIATED PRIMARY CHW PUMPS, AND SECONDARY CHILLED WATER PUMPS BASED UPON FAILURE. IN EVENT OF CHILLER FAILURE ENABLE NON-CRITICAL LOAD SHED SEQUENCE DESCRIBED IN SECTION "CHILLED WATER LOAD SHED". UPON LOSS OF POWER, CHILLERS AND PUMPS SHOULD AUTOMATICALLY RESTART AS SOON AS POWER IS RESTORED.

3.4.3.6.1 STAGING
IF OPERATING CHW PUMP OR CHILLER FAILS TO PROVE STATUS FOR 30 SECONDS (ADJ), THE OPERATING CHW PUMP OR CHILLER SHALL BE COMMANDED TO OFF AND THE DDC CONTROLLER SHALL START STANDBY CHW PUMP OR CHILLER (IF AVAILABLE).

3.4.3.6.2 CHILLED WATER LOAD SHED
IF OPERATING CHILLER FAILS AND REMAINING CHILLER LOADING IS < 95%, ENABLE CHILLER FAILURE LOAD SHED. SEE DOAH, AND FCU SEQUENCES OF OPERATION FOR UNITS THAT WILL HAVE COOLING MODES DISABLED. LOAD SHED SEQUENCE SHALL BE MANUALLY DISABLED THRU MASTER CHILLER CONTROLLER ONCE FAILED CHILLER IS REPAIRED.

3.4.3.6.2 ALARMS

ALARMS SHALL BE TRANSMITTED FOR THE FOLLOWING CONDITIONS:

- LOW TEMPERATURE ALARM - CHW SUPPLY TEMPERATURE IS BELOW SETPOINT FOR 15 MINS (ADJ)
- HIGH TEMPERATURE ALARM - CHW SUPPLY TEMPERATURE IS ABOVE SETPOINT FOR 15 MINS (ADJ)
- PCHWP PUMP STATUS - PUMP COMMANDED ON, BUT ASSOCIATED CHILLER FLOW SWITCH REPORTS NO FLOW VIA BACNET MSTP
- SCHWP PUMP STATUS - PUMP COMMANDED ON, BUT ASSOCIATED DIFFERENTIAL PRESSURE TRANSMITTER REPORTS NO FLOW.
- ACC STATUS - CHILLER COMMANDED ON, BUT FAILS TO PROVE STATUS VIA BACNET MSTP.

3.4.3.7 REPORTING

CUSTOMIZED REPORTS SHALL BE GENERATED TO PRINT ALARMS, TRENDS, AND MESSAGES, THE PARAMETERS TO BE MONITORED INCLUDE

- TOTAL CHW LOOP FLOW AND LOAD
- MTRACON BUILDING CHW LOOP FLOW AND LOAD
- ATCT BUILDING CHW LOOP FLOW AND LOAD (DERIVED FROM SUBTRACTING HQ LOAD AND FLOW FROM TOTAL LOAD AND FLOW)
- CHWS TEMPERATURE VERSUS CHWS SETPOINT
- INDIVIDUAL PRIMARY CHW PUMP RUN-HOURS
- INDIVIDUAL SECONDARY CHW PUMP RUN-HOURS
- INDIVIDUAL CHILLER RUN-HOURS

3.4.3.8 SETPOINTS
CHW SUPPLY TEMPERATURE = 45°F (ADJ)
MATRACON CHILLED WATER PRESSURE SETPOINT = 25 FT (INITIAL SETPOINT), CONFIRM DURING TAB
ATCT CHILLED WATER PRESSURE SETPOINT = 50 FT (INITIAL SETPOINT), CONFIRM DURING TAB

GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
2. REFER TO DRAWING MI700 FOR MECHANICAL CONTROLS LEGEND, ABBREVIATIONS AND NOTES.

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APPROVED



FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES CWS DRW TJG CHK RJ

PM/DM CJC/KDB

BRANCH MANAGER J. L. H.

CHIEF ENGINEER EJA

FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
NAVAL STATION - NORFOLK, VA
MARINE CORPS IPT
MCAS CHERRY POINT
CHERRY POINT, NC
FY20 P-228 ATC TOWER AND AIR OPERATIONS
MTRACON/ATCT - CHW SYSTEM SEQUENCE OF OPERATION

SCALE: AS NOTED			
EPROJECT NO.:		1596300	
CONSTR. CONTR. NO.			
NAVFAC DRAWING NO.			
		12801753	
SHEET	493	OF	686

MI721



1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
2. REFER TO DRAWING MI700 FOR MECHANICAL CONTROLS LEGEND, ABBREVIATIONS AND NOTES.

MTRACON/ATCT - HHW SYSTEM POINT LIST			
MARK	DESCRIPTION	ALARM	TREND
AO10	PHHWP-M2 SPEED CONTROL	No	Yes
AO11	HHW BYPASS VALVE POSITION - MTRACON	No	No
AO12	HHW BYPASS VALVE POSITION - ATCT	No	No
BINARY INPUT			
BI01	CV-HR-MT1 POSITION (FEEDBACK)	Yes	No
BI02	CV-HR-MT2 POSITION (FEEDBACK)	Yes	No
BI03	HRHWP-MT2 VFD STATUS	No	Yes
BI04	HRHWP-MT2 VFD ALARM	Yes	No
BI05	HRHWP-MT1 VFD ALARM	Yes	No
BI06	HRHWP-MT1 VFD STATUS	No	Yes
BI07	SHHWP-M1 VFD STATUS	No	Yes
BI08	SHHWP-M1 VFD ALARM	Yes	No
BI09	SHHWP-M2 VFD STATUS	No	Yes
BI10	SHHWP-M2 VFD ALARM	Yes	No
BI11	B-M1 STATUS	No	No
BI12	PHHWP-M1 VFD STATUS	No	Yes
BI13	PHHWP-M1 VFD ALARM	Yes	No
BI14	B-M2 STATUS	No	No
BI15	PHHWP-M2 VFD STATUS	No	Yes
BI16	PHHWP-M2 VFD ALARM	Yes	No
BI17	B-T1 STATUS	No	No
BI18	PHHWP-T1 VFD STATUS	No	Yes
BI19	PHHWP-T1 VFD ALARM	Yes	No
BI20	B-T2 STATUS	No	No
BI21	PHHWP-T2 VFD STATUS	No	Yes
BI22	PHHWP-T2 VFD ALARM	Yes	No
BI23	SHHWP-T1 VFD STATUS	No	Yes
BI24	SHHWP-T1 VFD ALARM	Yes	No
BI25	SHHWP-T2 VFD STATUS	No	Yes
BI26	SHHWP-T2 VFD ALARM	Yes	No
BI27	BOILER B-T1/T2 REMOTE SHUTDOWN SWITCH	Yes	No
BI28	BOILER B-M1/M2 REMOTE SHUTDOWN SWITCH	Yes	No
BI29	MTRACON/ATCT BOILER PLANT DDC RESET	No	No
BINARY OUTPUT			
BO01	HRHWP-MT1 START/STOP	No	No
BO02	CH-MT1 CHILLER HEAT RECOVERY	No	No
BO03	CV-HR-MT1 VALVE CONTROL	No	No
BO04	CV-HR-MT2 VALVE CONTROL	No	No
BO05	CH-MT2 CHILLER HEAT RECOVERY	No	No
BO06	HRHWP-MT2 START/STOP	No	No
BO07	SHHWP-M1 START/STOP	No	No
BO08	SHHWP-M2 START/STOP	No	No
BO09	B-M1 ON/OFF	No	No
BO10	PHHWP-M1 START/STOP	No	No
BO11	B-M2 ON/OFF	No	No
BO12	PHHWP-M2 START/STOP	No	No
BO13	B-T1 ON/OFF	No	No
BO14	PHHWP-T1 START/STOP	No	No
BO15	B-T2 ON/OFF	No	No
BO16	PHHWP-T2 START/STOP	No	No
BO17	SHHWP-T1 START/STOP	No	No
BO18	SHHWP-T2 START/STOP	No	No

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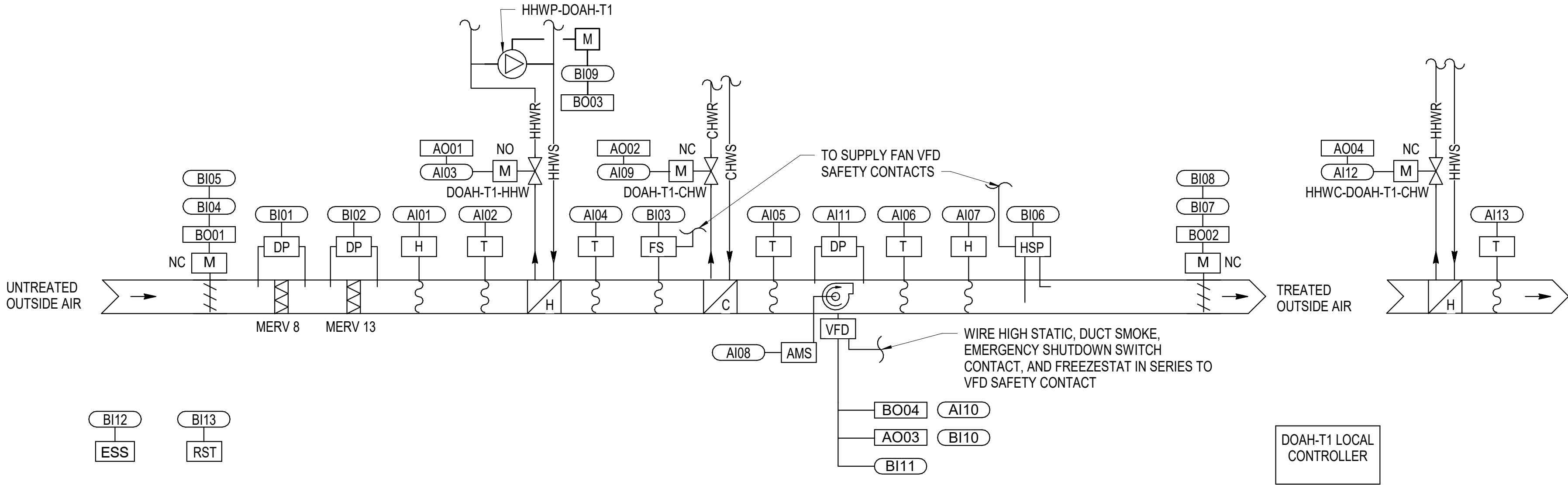
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ATCT - DOAH SCHEMATIC

3.2.3 SEQUENCE OF OPERATION – DOAH-T1

DOAH SHALL BE ENABLED THROUGH THE FACILITY DDC SYSTEM. LOCAL DOAH CONTROLLER SHALL CONTROL AND MONITOR INDIVIDUAL DOAH OPERATION. DOAH SHALL HAVE A VARIABLE SPEED SUPPLY FAN, PRE-HEAT COIL, CHILLED WATER COOLING COIL, CONTROLLER, AND REQUIRED DAMPERS AND SENSORS. SYSTEM PROVIDES ROOM NEUTRAL OUTSIDE AIR FOR BUILDING PRESSURIZATION AND CODE REQUIRED OUTSIDE AIR TO OCCUPIED SPACES.

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL SHALL BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

3.2.3.1 HAND-OFF-AUTO SWITCHES
SUPPLY FAN VARIABLE FREQUENCY DRIVES (VFDs) SHALL ACCEPT A FIRE ALARM PANEL (FAP) SIGNAL THAT TAKES PRECEDENCE OVER ALL OTHER VFD INPUTS AND SWITCHES AND SHALL CAUSE THE VFDs TO BE DE-ENERGIZED. THE VFD SHALL ACCEPT AN OCCUPANT ACCESSIBLE EMERGENCY SHUTOFF SWITCH AS SHOWN. THE SUPPLY FAN VARIABLE FREQUENCY DRIVES (VFDs) SHALL HAVE AN INTEGRAL H-O-A SWITCH:

3.2.3.1.1 HAND
WITH THE H-O-A SWITCH IN HAND POSITION, THE SUPPLY FAN SHALL START AND RUN CONTINUOUSLY, SUBJECT TO SAFETIES. FAN SPEED SHALL BE UNDER MANUAL-OPERATOR CONTROL.

3.2.3.1.2 OFF
WITH THE H-O-A SWITCH IN OFF POSITION, THE SUPPLY FAN SHALL STOP.

3.2.3.1.3 AUTO
WITH THE H-O-A SWITCH IN AUTO POSITION, THE SUPPLY FAN SHALL RUN SUBJECT TO THE SUPPLY FAN START/STOP SIGNAL (SF-SS) AND SAFETIES. FAN SPEED SHALL BE UNDER CONTROL OF THE DDC HARDWARE.

3.2.3.2 OCCUPANCY MODES
THE SYSTEM SHALL OBTAIN ITS OCCUPANCY MODE INPUT FROM THE SYSTEM SCHEDULER AS SPECIFIED AND SHOWN. THE SYSTEM SHALL BE OCCUPIED 24/7/365.

3.2.3.3 PROOFS AND SAFETIES
THE SUPPLY FAN AND ALL DDC HARDWARE CONTROL LOOPS SHALL BE SUBJECT TO PROOFS AND SAFETIES. SAFETIES SHALL BE DIRECT-HARDWIRE INTERLOCKED TO THE VFD AS SHOWN. DDC HARDWARE SHALL MONITOR ALL PROOFS AND SAFETIES AND FAILURE OF ANY PROOF OR ACTIVATION OF ANY SAFETY SHALL RESULT IN ALL CONTROL LOOPS BEING DISABLED AND THE FAN BEING COMMANDED OFF UNTIL RESET.

3.2.3.3.1 PROOFS
A. DOAH SUPPLY FAN STATUS (DOAHSF-S)

3.2.3.3.2 SAFETIES
A. PREHEAT COIL DISCHARGE AIR TEMPERATURE LOW LIMIT (FREEZESTAT) (PH-DA-T-LL).
B. SUPPLY AIR DUCT PRESSURE HIGH LIMIT (SA-P-HL)

3.2.3.3.3 DDC HARDWARE RESET
DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE, AS SHOWN ON THE POINTS SCHEDULE DRAWING.

3.2.3.4 SYSTEM ENABLE AND LOOP ENABLE

3.2.3.4.1 OCCUPIED MODE
THE OUTSIDE AIR DAMPERS SHALL BE COMMANDED OPEN, ONCE THE DAMPERS ARE PROVEN OPEN. THE DOAH SUPPLY FAN SHALL BE ENABLED (SYS-ENA) AND COMMANDED TO RUN (DOAHSF-SS). ALL CONTROL LOOPS SHALL BE ENABLED. SUPPLY AIRFLOW SHOULD BE MAINTAINED AT SCHEDULED AIFLOW.

3.2.3.5 FAN CAPACITY CONTROL

3.2.3.5.1 SUPPLY DUCT STATIC PRESSURE CONTROL
WHEN THIS LOOP IS ENABLED THE DDC HARDWARE SHALL MODULATE THE SUPPLY FAN VARIABLE FREQUENCY DRIVE UNIT TO MAINTAIN THE SUPPLY AIRFLOW AT SCHEDULED AIRFLOWS (DOAHSA-F). WHEN THIS LOOP IS DISABLED, THE DDC HARDWARE CAPACITY MODULATION OUTPUT TO THE VFD SHALL BE ZERO PERCENT AND DOAH DAMPERS SHALL BE CLOSED.

3.2.3.6 PREHEAT MODE CONTROL
THE UNIT SHALL BE IN PREHEAT MODE WHEN THE OUTDOOR AIR TEMPERATURE IS BELOW THE DOAH PREHEAT COIL DISCHARGE SETPOINT (DOAHHC-T-SP). PREHEAT MODE LOOP SHALL BE DISABLED WHEN OUTSIDE AIR TEMPERATURE RISES ABOVE PREHEAT DOAH DISCHARGE SETPOINT, THE HHW VALVE SHALL MODULATE TO MAINTAIN PREHEAT DOAH DISCHARGE SETPOINT AS MEASURED BY HEATING COIL LAT TEMPERATURE SENSOR.

IF PREHEAT COIL ENTERING AIR TEMPERATURE DROPS BELOW 35°F AS INDICATED BY ERW LAT - OA SIDE SENSORS, INLINE CIRCULATOR PUMP (HHWP-DOAH-T1) SHALL BE COMMANDED ON. WHEN THE PREHEAT COIL ENTERING AIR TEMPERATURE SENSOR INDICATES A RISE IN DRY BULB TEMPERATURE ABOVE 40°F, THE PUMP SHALL BE COMMANDED OFF.

WHEN THE DOAH IS DISABLED THE HHW VALVE SHALL MODULATE TO MAINTAIN A UNIT CASING TEMPERATURE OF 40°F AS MEASURED BY THE DOAH HEATING HOT WATER DISCHARGE COIL TEMPERATURE SENSOR.

3.2.3.7 COOLING MODE CONTROL
THE UNIT SHALL BE IN COOLING MODE WHEN THE OUTDOOR AIR TEMPERATURE IS ABOVE THE DOAH COOLING COIL DISCHARGE SETPOINT (DOAHCC-T-SP). COOLING MODE LOOP SHALL BE DISABLED WHEN OUTSIDE AIR TEMPERATURE DROPS BELOW DOAH COOLING COIL DISCHARGE SETPOINT (DOAHCC-T-SP). THE CHW VALVE SHALL MODULATE TO MAINTAIN THE DOAH COOLING COIL DISCHARGE SETPOINT (DOAHCC-T-SP) AS MEASURED BY COOLING COIL LAT SENSOR.

3.2.3.8 REHEAT MODE CONTROL
THE UNIT SHALL BE IN REHEAT MODE WHEN OPERATIONAL. THE REHEAT HHW VALVE SHALL MODULATE TO MAINTAIN DOAH DISCHARGE SETPOINT (DOAHSA-T-SP) AS MEASURED BY DOAH DISCHARGE TEMPERATURE SENSOR.

3.2.3.9 FREEZE PROTECTION CONTROL
WHEN THE UNIT IS DISABLED AND GLOBAL OA TEMPERATURE SENSOR INDICATES A DRY BULB TEMPERATURE OF LESS THAN 40°F, HHW VALVE SHALL MODULATE TO MAINTAIN A CASING TEMPERATURE OF 40°F AS MEASURED BY THE HEATING COIL LAT TEMPERATURE SENSOR. ALSO ENERGIZE INLINE CIRCULATOR PUMP (HHWP-DOAH-T1). WHEN THE GLOBAL OA TEMPERATURE SENSOR INDICATES A RISE IN DRY BULB TEMPERATURE ABOVE 40°F, THE PUMP SHALL BE COMMANDED OFF.

3.2.3.10 HVAC EMERGENCY SHUTDOWN SWITCH (ESS) SEQUENCE
UPON RECEIVING A SHUTDOWN SIGNAL FROM ANY EMERGENCY SHUTDOWN SWITCH, DISABLE DOAH AND CLOSE INTAKE DAMPERS.

3.2.3.11 ALARMS
LOW TEMPERATURE AS MEASURED BY (DOAHHW-T) < 40°F
FREEZESTAT < 40°F
SUPPLY AIR DUCT PRESSURE HIGH LIMIT (SA-P-HL): 1.5" WG INITIAL (CONFIRM DURING TAB)
SUPPLY FAN FAILURE: COMMANDED ON, STATUS IS OFF
CIRCULATOR PUMP FAILURE: COMMANDED ON, STATUS IS OFF
DAMPER COMMAND AND POSITION FEEDBACK MISMATCH
CONTROLLER FAILURE
LOSS OF COMMUNICATIONS
OUTSIDE AIR FLOW OUT OF RANGE: +/-10% OR MODE FROM SCHEDULED VALUE
HIGH DOAH HUMIDITY: DEW POINT TEMP ≥ 55°F AS MEASURED BY DOAH HUMIDITY SENSOR.

MAINTENANCE ALARMS:
HIGH DISCHARGE COOLING COIL TEMPERATURE: DOAHCC-T ≥ 58°F
LOW DISCHARGE COOLING COIL TEMPERATURE: DOAHCC-T ≤ 42°F
SUPPLY AIR DIRTY MERV 8 FILTER SWITCH: 0.75" WG
SUPPLY AIR DIRTY MERV 13 FILTER SWITCH: 1.00" WG

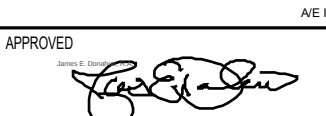
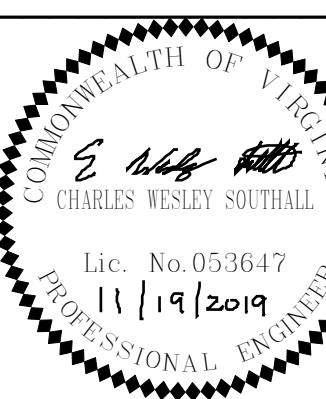
3.2.3.12 SETPOINTS
DOAH PREHEAT COIL SETPOINT (DOAHHC-T-SP): 53°F
DOAH COOLING COIL SETPOINT (DOAHCC-T-SP): 55°F
DOAH DISCHARGE SETPOINT (DOAHSA-T-SP): 68°F

3.2.3.12.1 DOAH-A1 SETPOINTS
AIRFLOWS: 860 CFM SUPPLY

GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- REFER TO DRAWING M1700 FOR MECHANICAL CONTROLS LEGEND, ABBREVIATIONS AND NOTES.

POINTS LIST DOAH-T1			
MARK	DESCRIPTION	ALARM	TREND
ANALOG INPUT			
AI01	UOA RELATIVE HUMIDITY	No	Yes
AI02	UOA TEMPERATURE	No	Yes
AI03	HHW VALVE POSITION (FEEDBACK)	No	Yes
AI04	HEATING COIL LAT	No	Yes
AI05	COOLING COIL LAT	Yes	Yes
AI06	TOA TEMPERATURE	No	Yes
AI07	TOA HUMIDITY	Yes	Yes
AI08	TOA AIRFLOW	Yes	Yes
AI09	CHW VALVE POSITION (FEEDBACK)	No	Yes
AI10	FAN SPEED (FEEDBACK)	No	Yes
AI11	TOA FAN DIFFERENTIAL PRESSURE	No	Yes
AI12	REHEAT HHW VALVE POSITION (FEEDBACK)	No	Yes
AI13	DOAH DISCHARGE TEMPERATURE	No	Yes
ANALOG OUTPUT			
AO01	HHW VALVE CONTROL	No	Yes
AO02	CHW VALVE CONTROL	No	Yes
AO03	FAN SPEED CONTROL	No	Yes
AO04	REHEAT HHW VALVE CONTROL	No	Yes
BINARY INPUT			
BI01	MERV 8 UOA FILTER STATUS	Yes	No
BI02	MERV 13 UOA FILTER STATUS	Yes	No
BI03	FREEZESTAT	Yes	No
BI04	UOA DAMPER POSITION CLOSED (FEEDBACK)	Yes	No
BI05	UOA DAMPER POSITION OPEN (FEEDBACK)	Yes	No
BI06	HIGH LIMIT STATIC PRESSURE	Yes	No
BI07	TREATED OA DAMPER POSITION CLOSED (FEEDBACK)	Yes	No
BI08	TREATED OA DAMPER POSITION OPEN (FEEDBACK)	Yes	No
BI09	HHWP-DOAH-T1 CIRCULATOR PUMP STATUS	Yes	Yes
BI10	FAN VFD STATUS	No	Yes
BI11	FAN VFD ALARM	Yes	No
BI12	ESS (ATFP EMERGENCY SHUTDOWN) *GLOBAL POINT	Yes	No
BI13	ATCT DOAH DDC RESET	No	No
BINARY OUTPUT			
BO01	UOA DAMPER CONTROL	No	No
BO02	TREATED OA DAMPER CONTROL	No	No
BO03	HHWP-DOAH-T1 CIRCULATOR PUMP ON/OFF	No	No
BO04	FAN START/STOP	No	No



FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DES CWS | DRW T.J.G | CHK R.J.

PMIDM CJC/KDB

BRANCH MANAGER

CHIEF ENGINEER

FIRE PROTECTION

HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL STATION - NORFOLK, VA

CHERRY POINT, NC

MCAS CHERRY POINT

FY20 P-228 ATC TOWER AND AIR OPERATIONS

ATCT - DOAH CONTROL SCHEMATIC AND POINT LIST

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

NAVFAC DRAWING NO. 12801756

SHEET 496 OF 686

M1730

DRAWING REVISION: 12 APRIL 2018

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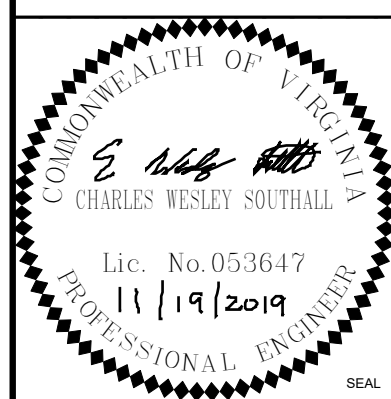
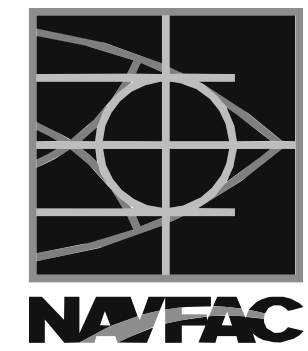
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1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
2. REFER TO DRAWING MI700 FOR MECHANICAL CONTROLS LEGEND, ABBREVIATIONS AND NOTES.

ATCT STAIR PRESSURIZATION DDC POINTS LIST			
MARK	DESCRIPTION	ALARM	TREND
ANALOG INPUT			
AI01	SPF-T1 SPEED (FEEDBACK)	No	Yes
AI02	SPF-T2 SPEED (FEEDBACK)	No	Yes
ANALOG OUTPUT			
AO01	SPF-T1 SPEED	No	Yes
AO02	SPF-T2 SPEED	No	Yes
BINARY INPUT			
BI01	SPF-T1 PRESSURE SWITCH	No	Yes
BI02	SPF-T1 ALARM	Yes	No
BI03	SPF-T1 STATUS	No	Yes
BI04	SPF-T2 PRESSURE SWITCH	No	Yes
BI05	SPF-T2 ALARM	Yes	No
BI06	SPF-T2 STATUS	No	Yes
BINARY OUTPUT			
BO01	SPF-T1 ENABLE	No	No
BO02	SPF-T2 ENABLE	No	No

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**BURNS
MCDONNELL**
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO		DATE	
DES	CAW	DRW	CWS
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PM/DM		CJC/KDB	
BRANCH MANAGER		-TLH	
CHIEF ENG/ARCH		EJA	
FIRE PROTECTION		HPN	

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SCALE: AS NOTED

PROJECT NO: 1596300

CONSTR. CONTR. NO.

NAVFAC DRAWING NO.
12801758

SHEET 498 OF 686

MI732

MI752

DRAWFORM REVISION: 12 APRIL 2018

FAN COIL UNIT WITH COOLING COILS

FAN COIL UNIT WITH HEATING AND COOLING COILS

UNIT HEATER

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COMPUTER ROOM AIR HANDLING UNITS

3.3.7.7 SETPOINTS
RANGE EQPT COOLING SETPOINT (RE-T-SP): 72°F
IFR EQPT COOLING SETPOINT (IFREQ-T-SP): 75°F
ATCT COMM (TCOMM-T-SP): 75°F

THE DUTY UNIT SHALL BE ROTATED EVERY 40 HOURS.

3.3.7.5 DUTY/STANDBY
IN EVENT OF DUTY UNIT FAILURE, DUTY UNIT SHALL BE DE-ENERGIZED AND ALARM SHALL BE SENT TO DDC. DUTY UNIT FAILURE SHALL BE DUE TO FAN FAILURE OR HIGH/LOW TEMPERATURE FOR 15 MINUTES OR LONGER. THE STANDBY UNIT SHALL BE ENERGIZED AND SHALL BECOME THE DUTY UNIT.

3.3.7.4 COOLING COIL CONTROL
WHEN THIS LOOP IS ENABLED THE CRAH CONTROLLER SHALL ENABLE CRAH COOLING COIL CONTROL MODE. IN COOLING COIL CONTROL MODE, CHW VALVES TO MAINTAIN ZONE TEMPERATURE (ZN-T) AT SETPOINT (ZN-T-SP). THE ZONE TEMPERATURE SHALL BE MAINTAINED BY MODULATING CHW VALVES FOR THE FIRST STAGE OF COOLING. THE CONTROLLER SHALL MODULATE FAN SPEED AS THE SECOND STAGE OF COOLING. THE CONTROLLER SHALL OPERATE FAN AT LOW SPEED IF POSSIBLE TO SAVE ENERGY. WHEN THIS LOOP IS DISABLED, THE CHW VALVE SHALL BE CLOSED.

3.3.7.3.3 DDC HARDWARE
DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE, AS SHOWN ON THE POINTS SCHEDULE DRAWING.

3.3.7.3.2 SAFETIES
A. SMOKE DETECTION. REFER TO FIRE ALARM SHEETS. IN THE EVENT OF SMOKE DETECTION, THE CRAHS SHALL BE SHUTDOWN. (CRITICAL)
B. HIGH CONDENSATE LEVEL (CRITICAL)

3.3.7.3.1 PROOFS
SUPPLY FAN STATUS (PROOF) (SF-S) (CRITICAL)

3.3.7.3 PROOFS AND SAFETIES
THE SUPPLY FAN AND ALL DDC HARDWARE CONTROL LOOPS SHALL BE SUBJECT TO PROOFS AND SAFETIES. SAFETIES SHALL BE DIRECT-HARDWIRE INTERLOCKED TO THE FAN STARTER CIRCUIT AS SHOWN. DDC HARDWARE SHALL MONITOR ALL PROOFS AND SAFETIES AND FAILURE OF ANY PROOF OR ACTIVATION OF ANY CRITICAL SAFETY SHALL RESULT IN ALL CONTROL LOOPS BEING DISABLED AND THE CRAH FAN BEING COMMANDED OFF UNTIL RESET. FAILURE TO PROVE FAN STATUS OR ACTIVATION OF SAFETY SHALL CONSTITUTE UNIT FAILURE.

3.3.7.2.1 OPERATING MODE
WHILE THE ZONE TEMPERATURE (ZN-T) IS BELOW THE ZONE TEMPERATURE SETPOINT (ZN-T-SP), COOLING COIL CONTROL LOOP SHALL BE DISABLED AND THE DUTY UNIT SUPPLY FAN SHALL RUN AT MINIMUM SPEED TO CIRCULATE AIR IN THE SPACE 24/7/365. WHEN ZN-T RISES ABOVE ZN-T-SP (WITH A 2 DEGREES F DEADBAND) THE LOCAL CRAH CONTROLLER SHALL ENABLE COOLING MODE.

3.3.7.1.3 AUTO
WITH THE H-O-A SWITCH IN AUTO POSITION, THE SUPPLY FAN SHALL RUN SUBJECT TO THE SUPPLY FAN START/STOP (SF-SS) COMMAND AND SAFETIES.

3.3.7.1.2 OFF
WITH THE H-O-A SWITCH IN OFF POSITION, THE SUPPLY FAN SHALL STOP.

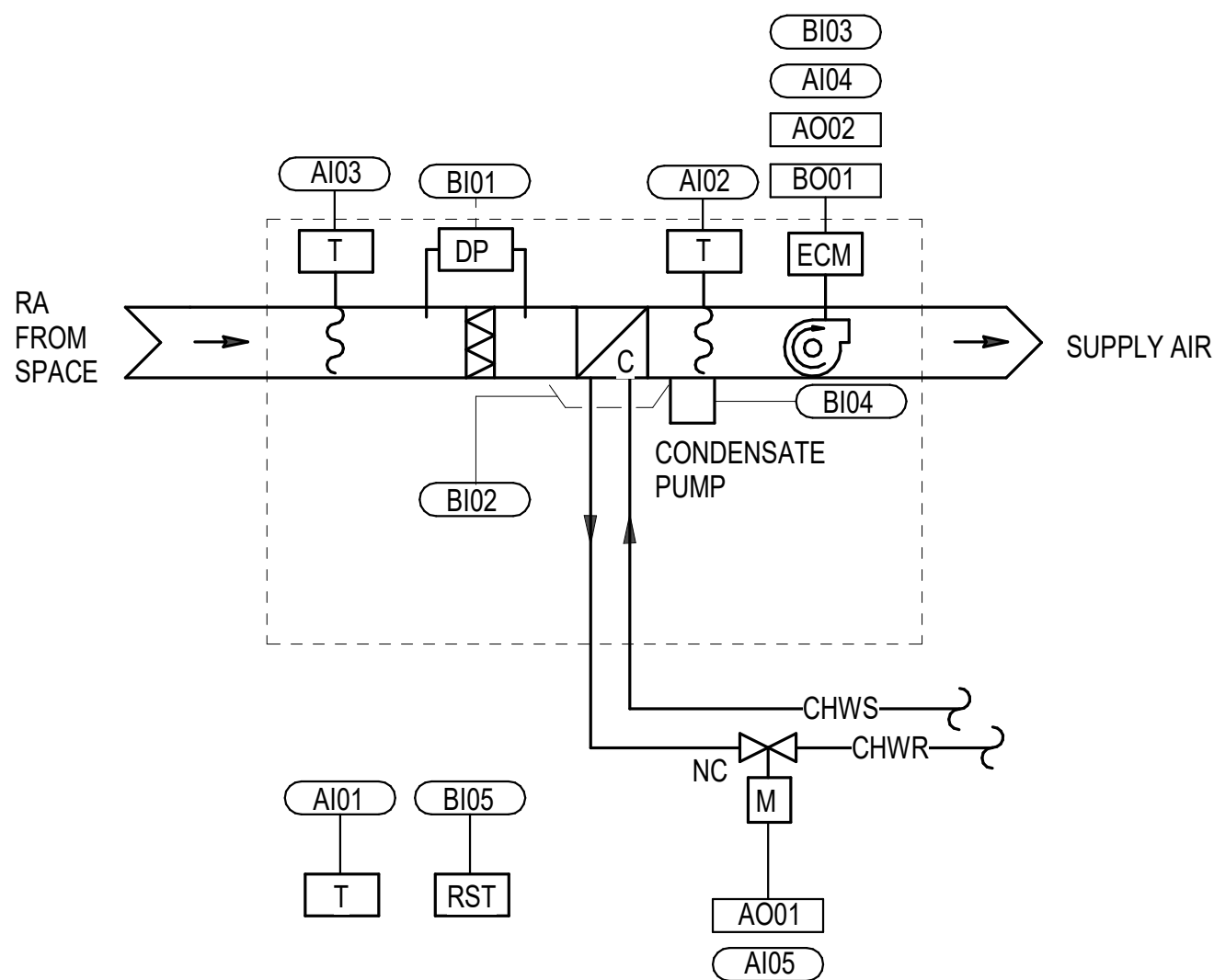
3.3.7.1.1 HAND
WITH THE H-O-A SWITCH IN HAND POSITION, THE SUPPLY FAN SHALL START AND RUN CONTINUOUSLY, SUBJECT TO SAFETIES.

3.3.7.1 HAND-OFF-AUTO SWITCH
SUPPLY FAN MOTOR STARTER SHALL ACCEPT A FIRE ALARM PANEL (FAP) SIGNAL THAT TAKES PRECEDENCE OVER ALL OTHER STARTER INPUTS AND SWITCHES AND SHALL STOP THE FAN. THE FAN MOTOR STARTER SHALL ACCEPT AN OCCUPANT ACCESSIBLE EMERGENCY SHUTOFF SWITCH AS INDICATED. THE SUPPLY FAN MOTOR STARTER SHALL HAVE AN H-O-A SWITCH:

*WHERE MULTIPLE TEMPERATURE SENSORS ARE USED (CRAH-M1A/B - ATCM SERVER ROOM 121), ZONE TEMPERATURE SHALL BE DETERMINED BY AVERAGING EACH TEMPERATURE SENSOR. IF ANY TEMPERATURE SENSOR VARIES BY 5°F OR MORE, AN ALARM SHALL BE GENERATED BY THE MASTER CRAH CONTROLLER.

3.3.7 COMPUTER ROOM AIR HANDLING (CRAH) UNITS

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. THE CRAH UNITS ARE TO BE CONTROLLED THROUGH A LOCAL CONTROLLER. CONTROLLER SHALL UTILIZE "TEAM" MODE TO PROVIDE PRIMARY/STANDBY OPERATION. CONTROLLER SHALL CONNECT TO DDC SYSTEM. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL SHALL BE PROPORTIONAL-INTEGRAL (PI) CONTROL.



CRAH DDC POINTS LIST			
MARK	DESCRIPTION	ALARM	TREND
ANALOG INPUT			
AI01	SPACE TEMPERATURE	Yes	Yes
AI02	SUPPLY AIR TEMPERATURE	No	Yes
AI03	RA TEMPERATURE	No	Yes
AI04	FAN SPEED (FEEDBACK)	No	No
AI05	CONTROL VALVE (FEEDBACK)	No	No
ANALOG OUTPUT			
AO01	CHW VALVE CONTROL	No	No
AO02	FAN SPEED CONTROL	No	No
BINARY INPUT			
BI01	DIRTY FILTER SWITCH	Yes	No
BI02	DRAIN PAN OVERFLOW SWITCH	Yes	No
BI03	FAN STATUS	Yes	No
BI04	CONDENSATE PUMP STATUS	Yes	No
BI05	CRAH DDC RESET	No	No
BINARY OUTPUT			
BO01	FAN START/STOP	No	No

SERIES-FAN POWERED VAV BOX SEQUENCE OF OPERATION

3.3.1 SERIES-FAN POWERED VAV BOX

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL SHALL BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

3.3.1.1 SYSTEM ENABLE AND LOOP ENABLE
SERIES-FAN POWERED BOXES SHALL BE OPERABLE DURING WITH OCCUPIED PERIODS DETERMINED BY DDC SYSTEM SCHEDULER OR DURING UNOCCUPIED MODE WHEN COOLING OR HEATING IS REQUIRED.

3.3.1.1.1 OCCUPIED MODE
WHEN THE DDC SYSTEM SCHEDULER INDICATES UNIT IS IN OCCUPIED MODE, THE FAN SHALL BE ENERGIZED AND SHALL MAINTAIN A CONSTANT DISCHARGE AIRFLOW. BOX SHALL MODULATE PRIMARY AIR DAMPER BETWEEN MINIMUM AND MAXIMUM TO MAINTAIN SPACE BELOW OCCUPIED COOLING SETPOINT. IF SPACE TEMPERATURE FALLS BELOW OCCUPIED HEATING SETPOINT, MODULATE BOX DAMPER TO MINIMUM POSITION AND MODULATE HHW VALVE TO MAINTAIN SPACE AT OCCUPIED HEATING SETPOINT.

3.3.1.1.2 UNOCCUPIED MODE
WHEN THE DDC SYSTEM SCHEDULER INDICATES UNIT IS IN UNOCCUPIED MODE, THE FAN SHALL BE DE-ENERGIZED IF THE SPACE TEMPERATURE IS WITHIN UNOCCUPIED HEATING AND UNOCCUPIED COOLING SETPOINTS.

WHEN THE SPACE TEMPERATURE FALLS BELOW THE UNOCCUPIED HEATING SETPOINT, THE FAN SHALL BE ENERGIZED AND SHALL MAINTAIN A CONSTANT DISCHARGE AIRFLOW AND THE HHW VALVE SHALL BE MODULATED TO MAINTAIN SPACE AT UNOCCUPIED HEATING SETPOINT.

WHEN THE SPACE TEMPERATURE RISES ABOVE THE UNOCCUPIED HEATING SETPOINT, THE ASSICOATED AHU SHALL BE ENERGIZED AND THE BOX FAN SHALL BE ENERGIZED TO MAINTAIN A CONSTANT DISCHARGE AIRFLOW. BOX SHALL THEN MODEULATE PRIMARY AIR DAMPER BETWEEN MINIMUM AND MAXIMUM TO MAINTAIN SPACE AT UNOCCUPIED COOLING SETPOINT.

3.3.1.2 DDC HARDWARE
DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE, AS SHOWN ON THE POINTS SCHEDULE DRAWING.

3.3.1.3 ALARMS
HIGH/LOW SPACE TEMPERATURE (SETPOINT +/- 3°F)
FAN FAILURE (INDICATED BY STATUS MISMATCH)

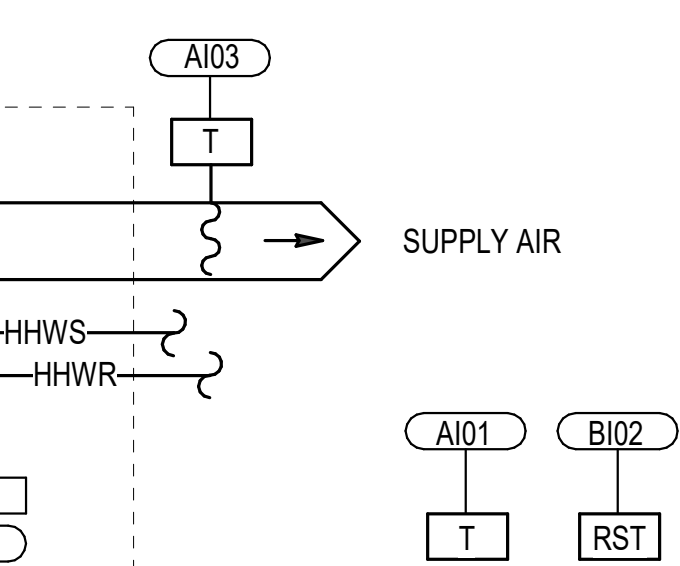
3.3.1.4 SETPOINTS
ATC CRITICAL SPACES (AIR OPS - WX FLOOR, F/C. MTRACON - IFR ROOM)
OCCUPIED COOLING SETPOINT: 75°F (SPACE IS CONTINUOUSLY OCCUPIED)
OCCUPIED HEATING SETPOINT: 71°F (SPACE IS CONTINUOUSLY OCCUPIED)

RANGE CONTROL (RANGE CONTROL)
OCCUPIED COOLING SETPOINT: 72°F (SPACE IS CONTINUOUSLY OCCUPIED)
OCCUPIED HEATING SETPOINT: 70°F (SPACE IS CONTINUOUSLY OCCUPIED)

TYPICAL 24/7 SPACES
OCCUPIED COOLING SETPOINT: 78°F (SPACE IS CONTINUOUSLY OCCUPIED)
OCCUPIED HEATING SETPOINT: 68°F (SPACE IS CONTINUOUSLY OCCUPIED)

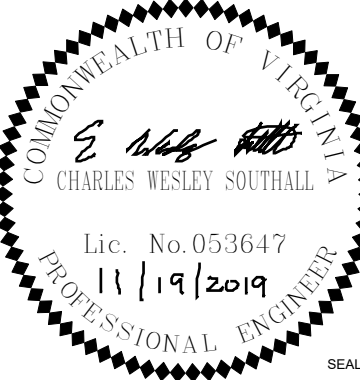
TYPICAL NON 24/7 SPACES
OCCUPIED COOLING SETPOINT: 78°F (SPACE IS CONTINUOUSLY OCCUPIED)
OCCUPIED HEATING SETPOINT: 68°F (SPACE IS CONTINUOUSLY OCCUPIED)
UNOCCUPIED COOLING SETPOINT: 80°F (SPACE IS CONTINUOUSLY OCCUPIED)
UNOCCUPIED HEATING SETPOINT: 60°F (SPACE IS CONTINUOUSLY OCCUPIED)

SFPVAV DDC POINTS LIST			
MARK	DESCRIPTION	ALARM	TREND
ANALOG INPUT			
AI01	SPACE TEMPERATURE	Yes	Yes
AI02	VELOCITY PRESSURE SENSOR	No	Yes
AI03	SUPPLY AIR TEMPERATURE	No	Yes
AI04	HHW VALVE CONTROL (FEEDBACK)	No	Yes
ANALOG OUTPUT			
AO01	HW VALVE CONTROL	No	No
AO02	DAMPER CONTROL	No	No
BINARY INPUT			
BI01	FAN STATUS	Yes	No
BI02	VAV DDC RESET	No	No
BINARY OUTPUT			
BO01	FAN START/STOP	No	No



GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- REFER TO DRAWING M1700 FOR MECHANICAL CONTROLS LEGEND, ABBREVIATIONS AND NOTES.



BURNS & MCDONNELL
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SUITE 160
CHESAPEAKE, VA 23320

APPROVED: *[Signature]*
FOR COMMANDER NAVFAC

SATISFACTORY TO: DATE
DES CWS DRW FYA CHK RJ
PMIDM CJC/KDB
BRANCH MANAGER
CHIEF ENGINEER
FIRE PROTECTION HPN

DEPARTMENT OF THE NAVY
NAVAL FACILITIES ENGINEERING COMMAND
NAVAL FACILITIES ENGINEERING COMMAND ~ MID-ATLANTIC
MARINE CORPS IPT
NAVAL STATION - NORFOLK, VA
MCAS CHERRY POINT
CHERRY POINT, NC
FY20 P-228 ATC TOWER AND AIR OPERATIONS
MECHANICAL CONTROLS

SCALE: AS NOTED
PROJECT NO.: 1596300
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NAVFAC DRAWING NO. 12801760
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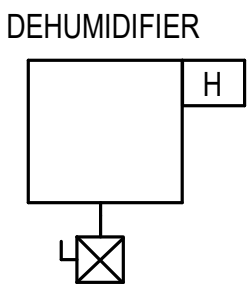
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3.2.13 DEHUMIDIFIER CONTROLS

DEHUMIDIFIER OPERATIONS SHALL BE BY MANUFACTURER INTERNAL CONTROLS.

3.2.13.1 SYSTEM ENABLE AND LOOP ENABLE

3.2.13.1.1 DEHUMIDIFICATION MODE
WHILE THE ZONE HUMIDITY (ZN-RH) IS BELOW THE SETPOINT (ZN-RH-HL), HUMIDIFIER SHALL REMAIN OFF. WHEN ZN-RH RISES ABOVE ZN-RH-HL (WITH A 3% RH DEADBAND) DEHUMIDIFIER SHALL BE ENABLED.

3.2.13.2 SETPOINTS
SPACE HIGH HUMIDITY SETPOINT (ZN-RH-HL): 38% RH (DH-A1,A2,A4,M1,M3,T1,T2,T3)
SPACE HIGH HUMIDITY SETPOINT (ZN-RH-HL): 50% RH (DH-A5,M2)
SPACE HIGH HUMIDITY SETPOINT (ZN-RH-HL): 60% RH (DH-A6)

A

DEHUMIDIFIER CONTROLS

D

3.3.8 HUMIDIFIER CONTROLS

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL MUST BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

3.3.8.1 SYSTEM ENABLE AND LOOP ENABLE

3.3.8.1.1 HUMIDIFICATION MODE
WHILE THE ZONE HUMIDITY (ZN-RH) IS ABOVE THE SETPOINT (ZN-RH-LL) DISABLE ALL CONTROL LOOPS AND THE HUMIDIFIER DOES NOT RUN. WHEN ZN-RH DROPS BELOW ZN-RH-LL (WITH A 3% RH DEADBAND) ENABLE THE HUMIDIFIER (SYS-ENA) AND COMMAND TO RUN (H-SS) AND ENABLE THE HUMIDIFIER CONTROL LOOP.

3.3.8.2 PROOFS AND SAFETIES
THE HUMIDIFIER AND ALL DDC HARDWARE CONTROL LOOPS ARE SUBJECT TO PROOFS AND SAFETIES. SAFETIES MUST BE DIRECT-HARDWARE INTERLOCKED TO THE HUMIDIFIER CIRCUIT AS INDICATED. DDC HARDWARE MUST MONITOR ALL PROOFS AND SAFETIES AND FAILURE OF ANY PROOF OR ACTIVATION OF ANY SAFETY RESULT IN ALL CONTROL LOOPS BEING DISABLED AND HUMIDIFIER BEING COMMANDED OFF UNTIL RESET.

3.3.8.2.1 PROOFS
HUMIDIFIER STATUS (PROOF) (H-S)

3.3.8.2.2 DDC HARDWARE
DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE, AS SHOWN ON THE POINTS SCHEDULE DRAWING.

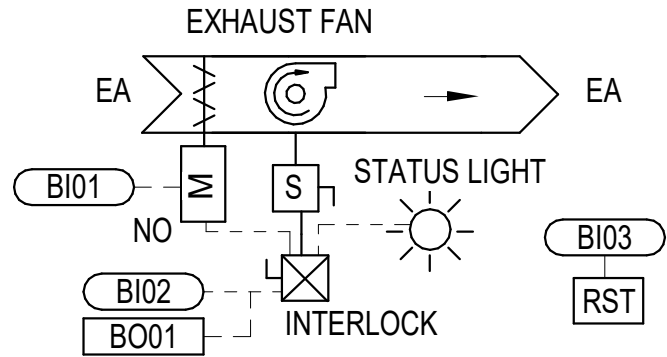
3.3.8.3 HUMIDIFIER CONTROL
WHEN THIS LOOP IS ENABLED THE DDC HARDWARE SHALL ENABLE THE HUMIDIFIER TO MAINTAIN ZONE HUMIDITY (ZN-RH) AT SETPOINT (ZN-RH-LL) AS SHOWN. WHEN THIS LOOP IS DISABLED, THE HUMIDIFIER SHALL BE DISABLED.

3.3.8.4 ALARM
HUMIDIFIER FAILURE, HUMIDIFIER COMMANDED ON, STATUS SHOWS HUMIDIFIER IS OFF.

3.3.8.5 SETPOINTS
SPACE LOW HUMIDITY SETPOINT (ZN-RH-LL): 35% RH - TYP
SPACE LOW HUMIDITY SETPOINT (ZN-RH-LL): 30% RH - H-M5&H-M6 ONLY

HUMIDIFIER CONTROLS

2



3.3.9 UPS BATTERY EXHAUST FAN CONTROLS (EF-A2, EF-A3, EF-M1)

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL MUST BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

INDICATOR LIGHT SHALL BE PLACED NEAR DOOR AND SHALL BE ILLUMINATED GREEN WHEN FAN IS OPERATIONAL. LIGHT SHALL TURN RED UPON FAN FAILURE.

3.3.9.1 SYSTEM ENABLE

3.3.9.1.1 VENTILATION MODE
THE UPS EXHAUST FAN IS TO OPERATE 24/7/365.

3.3.9.2 PROOFS AND SAFETIES

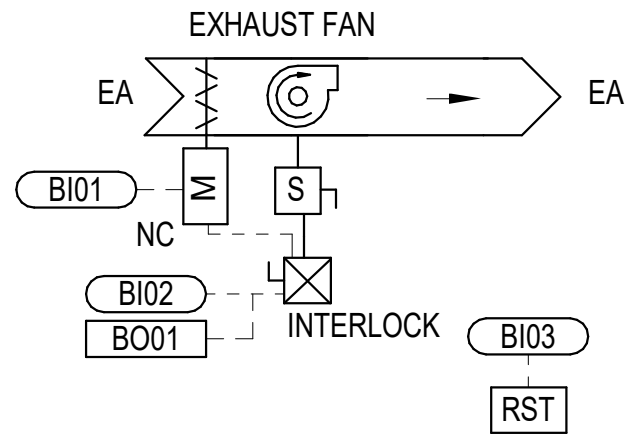
THE FAN AND ALL DDC HARDWARE CONTROL LOOPS ARE SUBJECT TO PROOFS AND SAFETIES. SAFETIES MUST BE DIRECT-HARDWARE INTERLOCKED TO THE FAN STARTER CIRCUIT AS INDICATED. DDC HARDWARE MUST MONITOR ALL PROOFS AND SAFETIES AND FAILURE OF ANY PROOF OR ACTIVATION OF ANY SAFETY RESULT IN ALL CONTROL LOOPS BEING DISABLED AND THE FAN BEING COMMANDED OFF UNTIL RESET.

3.3.9.2.1 PROOFS
FAN STATUS (PROOF) (EF-S)

3.3.9.2.2 DDC HARDWARE
DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE, AS SHOWN ON THE POINTS SCHEDULE DRAWING.

3.3.9.3 ALARM
ALARM ON FAN FAILURE (CRITICAL - POTENTIAL FOR EXPLOSIVE CONCENTRATION OF H2 GAS).

EXHAUST FAN CONTROLS - UPS EXHAUST



3.3.11 BATHROOM EXHAUST FAN CONTROLS

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL MUST BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

3.3.11.1 SYSTEM ENABLE

3.3.11.1.1 VENTILATION MODE
THE BATHROOM EXHAUST FAN IS TO OPERATE 24/7/365.

3.3.11.2 PROOFS AND SAFETIES

THE FAN AND ALL DDC HARDWARE CONTROL LOOPS ARE SUBJECT TO PROOFS AND SAFETIES. SAFETIES MUST BE DIRECT-HARDWARE INTERLOCKED TO THE FAN STARTER CIRCUIT AS INDICATED. DDC HARDWARE MUST MONITOR ALL PROOFS AND SAFETIES AND FAILURE OF ANY PROOF OR ACTIVATION OF ANY SAFETY RESULT IN ALL CONTROL LOOPS BEING DISABLED AND THE FAN BEING COMMANDED OFF UNTIL RESET.

3.3.11.2.1 PROOFS
FAN STATUS (PROOF) (EF-S)

3.3.11.2.2 DDC HARDWARE
DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE, AS SHOWN ON THE POINTS SCHEDULE DRAWING.

3.3.11.3 ALARM
ALARM ON FAN FAILURE.

EXHAUST FAN CONTROLS - BATHROOM EXHAUST

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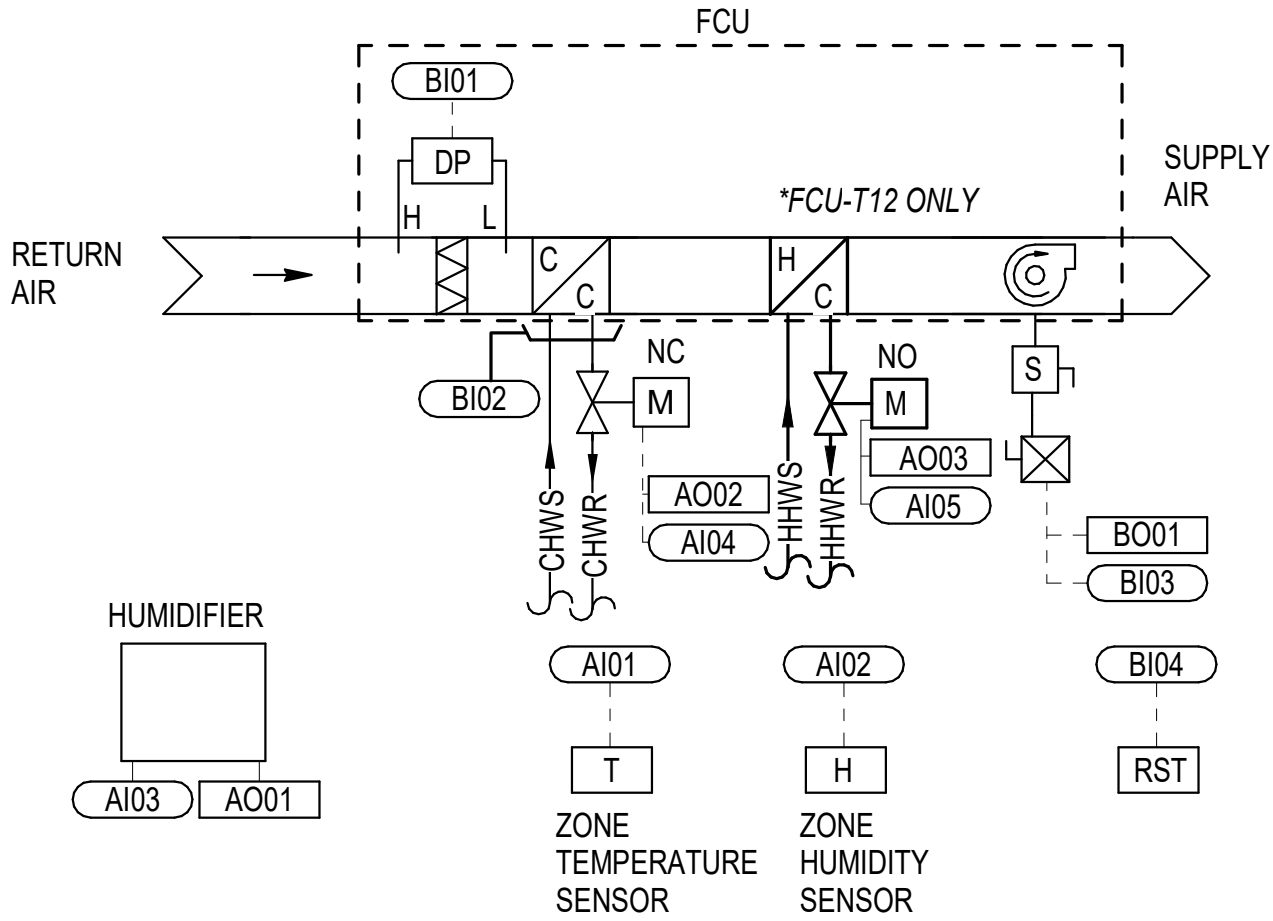
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3.3.10 ELEVATOR MACHINE ROOM CONTROLS (FCU-A3, FCU-T12)

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL MUST BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

3.3.10.1 SYSTEM ENABLE AND LOOP ENABLE

3.3.10.1.1 COOLING-AND-DEHUMIDIFICATION MODE
WHILE THE ZONE TEMPERATURE (ZN-T) AND ZONE HUMIDITY (ZN-RH) IS BELOW THE ZONE HIGH LIMIT TEMPERATURE SETPOINT (ZN-T-HL) AND ZONE HUMIDITY SETPOINT (ZN-RH-HL) RESPECTIVELY, COOLING-AND-DEHUMIDIFICATION CONTROL LOOPS SHALL BE DISABLED. WHEN ZN-T RISES ABOVE ZN-T-HL (WITH A 2 DEGREES F DEADBAND) OR ZN-RH RISES ABOVE ZN-RH-HL (3% RH DEADBAND), ENABLE THE SUPPLY FAN (SYS-ENA) AND COMMAND TO RUN (SF-SS) AND ENABLE THE COOLING-AND-DEHUMIDIFICATION COIL CONTROL LOOP.

3.3.10.1.2 HUMIDIFICATION MODE
WHILE THE ZONE HUMIDITY (ZN-RH) IS ABOVE THE SETPOINT (ZN-RH-LL) DISABLE HUMIDIFIER CONTROL LOOP. WHEN ZN-RH DROPS BELOW ZN-RH-LL (WITH A 3% RH DEADBAND) ENABLE THE HUMIDIFIER CONTROL LOOP.

3.3.10.1.3 HEATING MODE (FCU-T12 ONLY)
WHEN THE ZONE TEMPERATURE (ZN-T) IS ABOVE THE ZONE TEMPERATURE SETPOINT (ZN-T-HL), HEATING CONTROL LOOPS SHALL BE DISABLED. WHEN ZN-T FALLS BELOW ZN-T-HL (WITH A 2 DEGREES F DEADBAND), ENABLE THE HEATING CONTROL LOOP.

3.3.10.2 PROOFS AND SAFETIES
THE SUPPLY FAN, HUMIDIFIER AND ALL DDC HARDWARE CONTROL LOOPS ARE SUBJECT TO PROOFS AND SAFETIES. SAFETIES MUST BE DIRECT-HARDWARE INTERLOCKED TO THE FAN STARTER CIRCUIT AND THE HUMIDIFIER CIRCUIT AS INDICATED. DDC HARDWARE MUST MONITOR ALL PROOFS AND SAFETIES AND FAILURE OF ANY PROOF OR ACTIVATION OF ANY SAFETY RESULT IN ALL CONTROL LOOPS BEING DISABLED AND THE FAN COIL UNIT FAN AND HUMIDIFIER BEING COMMANDED OFF UNTIL RESET. FAILURE TO PROVE FAN STATUS OR ACTIVATION OF HIGH CONDENSATE SAFETY SHALL CONSTITUTE FAN COIL FAILURE. FAILURE TO PROVE HUMIDIFIER STATUS SHALL CONSTITUTE HUMIDIFIER FAILURE.

3.3.10.2.1 PROOFS
SUPPLY FAN STATUS (PROOF) (FCU-SF-S)
HUMIDIFIER STATUS (PROOF) (H-S)

3.3.10.2.2 SAFETIES
A. HIGH CONDENSATE LEVEL (CRITICAL)

3.3.10.2.3 DDC HARDWARE
DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE, AS SHOWN ON THE POINTS SCHEDULE DRAWING.

3.3.10.3 COOLING-AND-DEHUMIDIFICATION COIL CONTROL
WHEN THIS LOOP IS ENABLED THE FAN COIL CONTROLLER SHALL ENERGIZE FAN COIL UNIT FAN, ONCE FAN STATUS IS CONFIRMED MODULATE CHW CONTROL VALVE TO MAINTAIN ZONE TEMPERATURE (ZN-T) AT SETPOINT (ZN-T-HL) OR RELATIVE HUMIDITY (ZN-RH) AT SETPOINT (ZN-RH-HL). WHEN THIS LOOP IS DISABLED, THE COIL VALVE SHALL BE CLOSED.

3.3.10.4 HEATING CONTROL
WHEN THIS LOOP IS ENABLED THE FAN COIL CONTROLLER SHALL ENERGIZE FAN COIL UNIT FAN, ONCE FAN STATUS IS CONFIRMED MODULATE HHW CONTROL VALVE TO MAINTAIN ZONE TEMPERATURE (ZN-T) AT SETPOINT (ZN-T-HL). WHEN THIS LOOP IS DISABLED, THE COIL VALVE SHALL BE CLOSED.

3.3.10.5 HUMIDIFIER CONTROL
WHEN THIS LOOP IS ENABLED THE DDC HARDWARE SHALL ENABLE THE HUMIDIFIER TO MAINTAIN ZONE HUMIDITY (ZN-RH) AT SETPOINT (ZN-RH-LL) AS SHOWN. WHEN THIS LOOP IS DISABLED, THE HUMIDIFIER SHALL BE DISABLED.

3.3.10.6 CHILLED WATER LOAD SHED
FCU-A3, FCU-A12
UPON RECIEVING SIGNAL FROM DDC DISABLE COOLING MODE UNTIL FAILED CHILLER IS RESTORED TO SERVICE.

3.3.10.7 ALARMS
HIGH SPACE TEMPERATURE: 95°F
LOW SPACE TEMPERATURE: 65°F
HIGH SPACE HUMIDITY: 65% RH
LOW SPACE HUMIDITY: 30% RH
HUMIDIFIER FAILURE (FEEDBACK INDICATES FAILED UNIT)
FAN COIL FAILURE (FAN STATUS MISMATCH)
HIGH LEVEL CONDENSATE
DIRTY FILTER (0.50" WG) - ADJ.

3.3.10.8 SETPOINTS
COOLING SETPOINT (ZN-T-HL): 90°F
HEATING SETPOINT (ZN-T-HL): 70°F (FCU-T12 ONLY)
SPACE HIGH HUMIDITY SETPOINT (ZN-RH-HL): 60% RH
SPACE LOW HUMIDITY SETPOINT (ZN-RH-LL): 35% RH

ELEVATOR MACHINE ROOM TEMPERATURE CONTROLS

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GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- REFER TO DRAWING MI700 FOR MECHANICAL CONTROLS LEGEND, ABBREVIATIONS AND NOTES.

ELEVATOR MACHINE ROOM DDC POINTS LIST

MARK	DESCRIPTION	ALARM	TREND
ANALOG INPUT			
AI01	TEMPERATURE SENSOR	Yes	Yes
AI02	HUMIDITY SENSOR	Yes	Yes
AI03	HUMIDIFIER OUTPUT	Yes	Yes
AI04	CHW VALVE POSITION (FEEDBACK)	No	Yes
AI05	HHW VALVE CONTROL (FEEDBACK)	No	Yes
ANALOG OUTPUT			
AO01	HUMIDIFIER CONTROL	No	No
AO02	CHW VALVE CONTROL	No	No
AO03	HHW VALVE CONTROL	No	No
BINARY INPUT			
BI01	FILTER STATUS	Yes	No
BI02	SECONDARY DRAIN PAN OVERFLOW SWITCH	Yes	No
BI03	SUPPLY FAN STATUS	Yes	No
BI04	ELEVATOR MACHINE EQUIPMENT RESET	No	No
BINARY OUTPUT			
BO01	SUPPLY FAN START/STOP	No	No

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NAVFAC DRAWING NO. 12801761

CONSTR. CONTR. NO.

PROJECT NO.: 1596300

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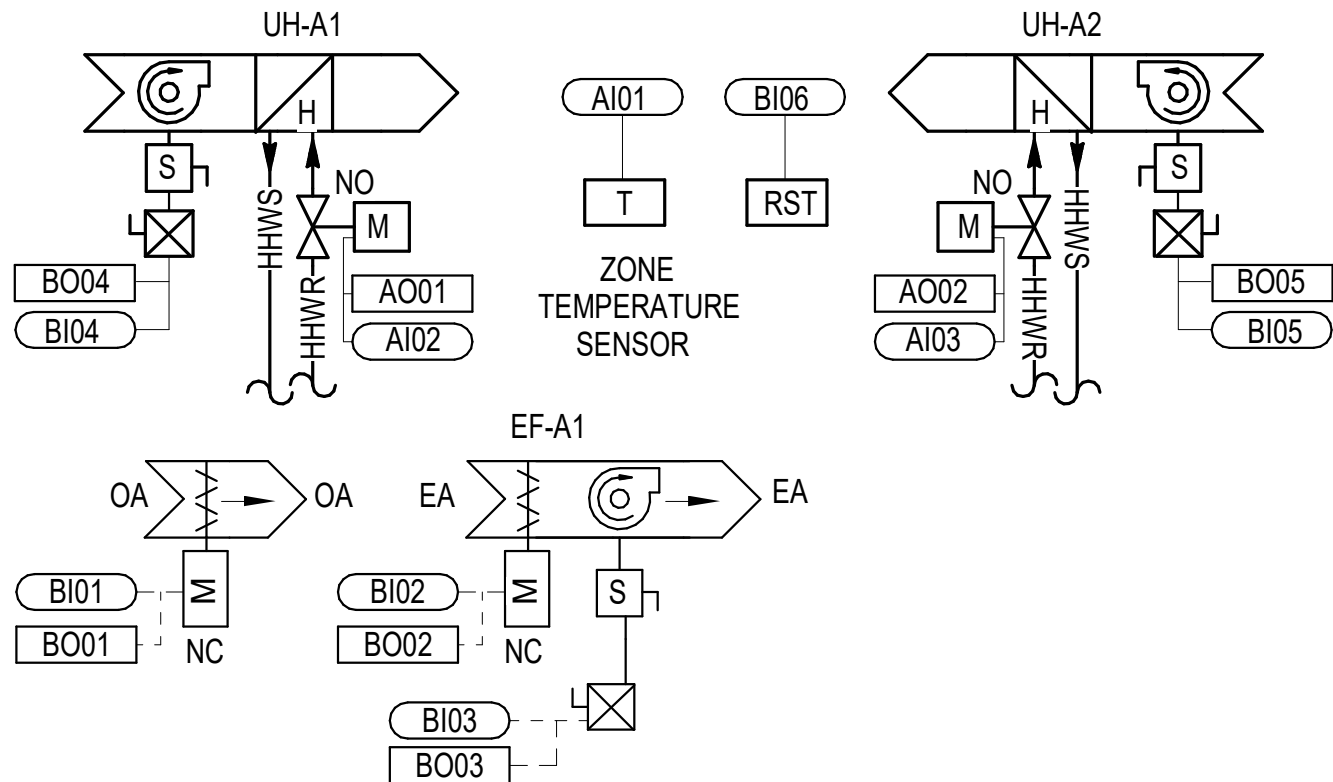
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3.3.12 AIR OPS - STORAGE BAY HEATING & VENTILATING CONTROLS

INSTALL DDC HARDWARE TO PERFORM THIS SEQUENCE OF OPERATION AND TO PROVIDE INPUTS AND OUTPUTS AS SPECIFIED AND SHOWN ON THE POINTS SCHEDULE. UNLESS OTHERWISE SPECIFIED, ALL MODULATING CONTROL SHALL BE PROPORTIONAL-INTEGRAL (PI) CONTROL.

3.3.12.1 SYSTEM ENABLE AND LOOP ENABLE

3.3.12.1.1 VENTILATION COOLING MODE
WHILE THE ZONE TEMPERATURE (ZN-T) IS BELOW THE ZONE TEMPERATURE HIGH LIMIT SETPOINT (ZN-T-SP-HL), VENTILATION COOLING LOOP SHALL BE DISABLED. WHEN ZN-T RISES ABOVE ZN-T-SP-HL (WITH A 2 DEGREES F DEADBAND), ENABLE THE VENTILATION COOLING CONTROL LOOP.

3.3.12.1.2 HEATING MODE
WHILE THE ZONE TEMPERATURE (ZN-T) IS ABOVE THE ZONE TEMPERATURE LOW LIMIT SETPOINT (ZN-T-SP-LL), THE HEATING LOOP SHALL BE DISABLED. WHEN ZN-T DROPS BELOW ZN-T-SP-LL (WITH A 3 DEGREES F DEADBAND), ENABLE THE HEATING CONTROL LOOP.

3.3.12.2 PROOFS AND SAFETIES
THE SUPPLY FAN, HUMIDIFIER AND ALL DDC HARDWARE CONTROL LOOPS ARE SUBJECT TO PROOFS AND SAFETIES. SAFETIES MUST BE DIRECT-HARDWIRE INTERLOCKED TO THE FAN STARTER CIRCUIT AND THE HUMIDIFIER CIRCUIT AS INDICATED. DDC HARDWARE MUST MONITOR ALL PROOFS AND SAFETIES AND FAILURE OF ANY PROOF OR ACTIVATION OF ANY SAFETY RESULT IN ALL CONTROL LOOPS BEING DISABLED AND THE FAN COIL UNIT FAN AND HUMIDIFIER BEING COMMANDED OFF UNTIL RESET.

3.3.12.2.1 PROOFS
EXHAUST FAN STATUS (PROOF) (EF-S)
OA DAMPER STATUS (PROOF)
EA DAMPER STATUS (PROOF)
UNIT HEATER FAN (PROOF)

3.3.12.2.2 DDC HARDWARE
DDC HARDWARE RESET OF ALL PROOFS AND SAFETIES SHALL BE VIA A LOCAL BINARY PUSH-BUTTON (RST-BUT) INPUT TO THE DDC HARDWARE, VIA A REMOTE COMMAND TO THE DDC HARDWARE, AS SHOWN ON THE POINTS SCHEDULE DRAWING.

3.3.12.3 VENTILATION COOLING CONTROL
WHEN THIS LOOP IS ENABLED THE DDC HARDWARE SHALL OPEN BOTH THE OA AND EA DAMPER. ONCE THE DAMPER POSITION IS PROVEN OPEN, ENERGIZE EXHAUST FAN. WHEN THIS LOOP IS DISABLED, THE EXHAUST FAN SHALL BE DE-ENERGIZED AND THE AFTER A 15 SECOND DELAY BOTH THE OA AND EA DAMPER SHALL CLOSE.

3.3.12.4 HEATING CONTROL
WHEN THIS LOOP IS ENABLED THE DDC HARDWARE SHALL ENERGIZE THE UNIT HEATER FANS AND MODULATE THE HHW VALVE TO MAINTAIN ZONE TEMPERATURE (ZN-T) AT SPACE LOW LIMIT SETPOINT (ZN-T-SP-LL) AS SHOWN. WHEN THIS LOOP IS DISABLED, THE UNIT HEATERS SHALL BE DE-ENERGIZED AND THE HHW VAVLES SHALL CLOSE.

3.3.12.5 ALARMS
LOW TEMPERATURE: 40°F
EXHAUST FAN FAILURE (FAN STATUS MISMATCH)
UNIT HEATER FAN FAILURE (FAN STATUS MISMATCH)
OA DAMPER FAILURE
EA DAMPER FAILURE

3.3.12.6 SETPOINTS
HIGH LIMIT SETPOINT: 80°F
LOW LIMIT SETPOINT: 55°F

AIR OPS - HEATING AND VENTILATING

AIR OPS H&V DDC POINTS LIST			
MARK	DESCRIPTION	ALARM	TREND
ANALOG INPUT			
AI01	SPACE TEMPERATURE	Yes	Yes
AI02	UH-A1 HHW VALVE CONTROL	No	Yes
AI03	UH-A2 HHW VALVE CONTROL	No	Yes
ANALOG OUTPUT			
AO01	UH-A1 HHW VALVE CONTROL	No	No
AO02	UH-A2 HHW VALVE CONTROL	No	No
BINARY INPUT			
BI01	OA DAMPER END SWITCH	Yes	No
BI02	EA DAMPER END SWITCH	Yes	No
BI03	EF-A1 STATUS (FEEDBACK)	Yes	No
BI04	UH-A1 FAN STATUS	Yes	No
BI05	UH-A2 FAN STATUS	Yes	No
BI06	AIR OPS H&V DDC RESET	Yes	Yes
BINARY OUTPUT			
BO01	OA DAMPER OPEN/CLOSE	No	No
BO02	EA DAMPER OPEN/CLOSE	No	No
BO03	EF-A1 START/STOP	No	No
BO04	UH-A1 FAN START/STOP	No	No
BO05	UH-A2 FAN START/STOP	No	No

GENERAL NOTES

- REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.
- REFER TO DRAWING MI700 FOR MECHANICAL CONTROLS LEGEND, ABBREVIATIONS AND NOTES.



**BURNS
MCDONNELL**
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED:

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO: DATE

DES CWS | DRW CWS | CHK RJ

PMCM CJC/KDB

BRANCH MANAGER

CHIEF ENGINEER

FIRE PROTECTION

HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL STATION - NORFOLK, VA

CHERRY POINT, NC

CHERRY POINT, NC

MECHANICAL CONTROLS

FY20 P-228 ATC TOWER AND AIR OPERATIONS

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

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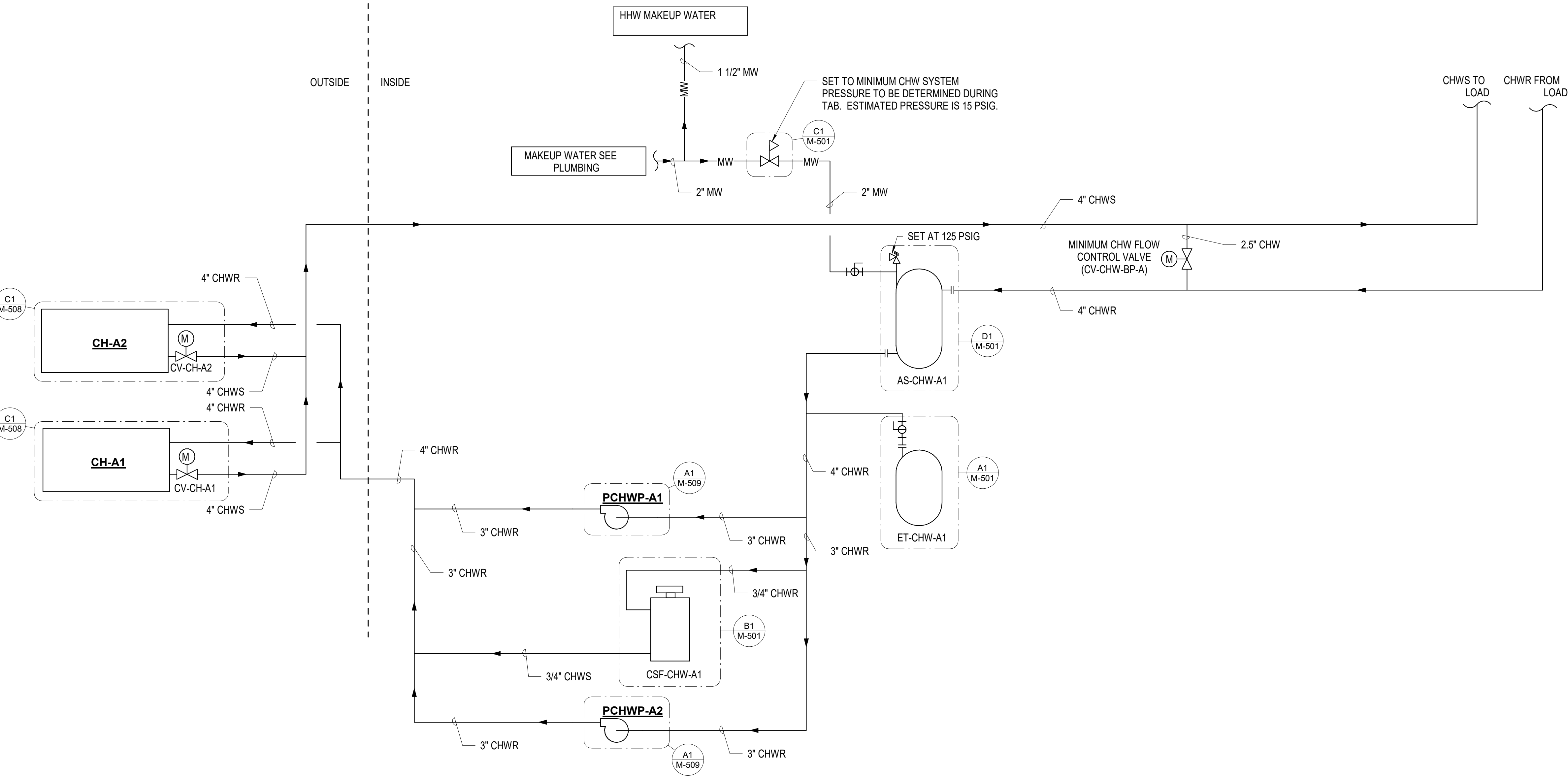
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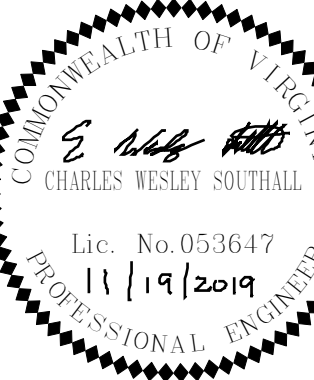
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A1 AIR OPS - CHILLED WATER PIPING SCHEMATIC
SCALE: NOT TO SCALE

GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.



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ACTIVITY

SATISFACTORY TO: DATE

DES CWS BRW FYA CHK RJ

PMIDM CJC/KDB

BRANCH MANAGER *[Signature]*

CHIEF ENGINEER EJA

FIRE PROTECTION HPN

NAVAL FACILITIES ENGINEERING COMMAND

NAVAL FACILITIES ENGINEERING COMMAND - MID-ATLANTIC

NAVAL STATION - NORFOLK, VA

MCAS CHERRY POINT

CHERRY POINT, NC

FY20 P-228 ATC TOWER AND AIR OPERATIONS

AIR OPS - CHILLED WATER PIPING SCHEMATIC

SCALE: AS NOTED

PROJECT NO.: 1596300

CONSTR. CONTR. NO.

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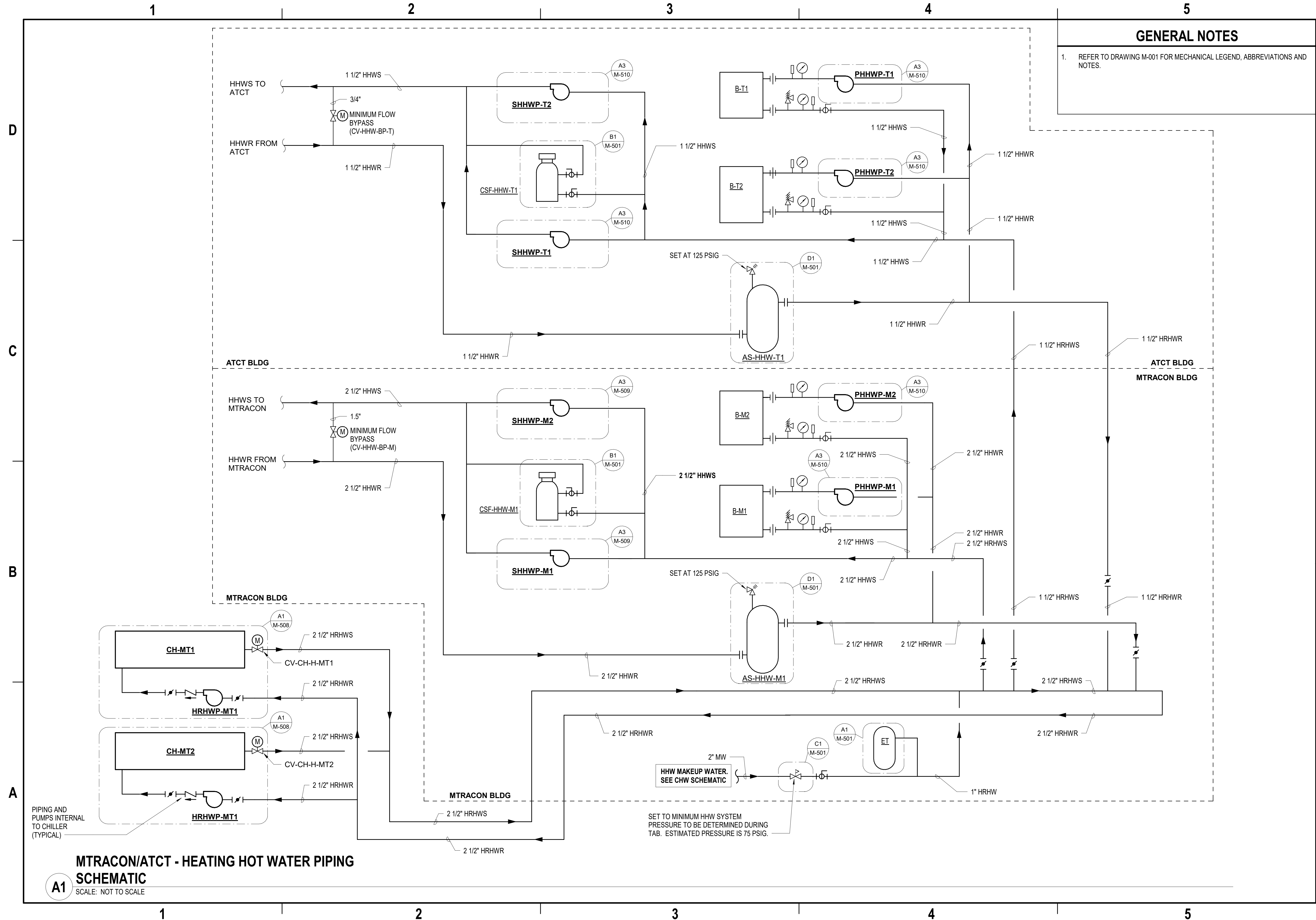
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1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.

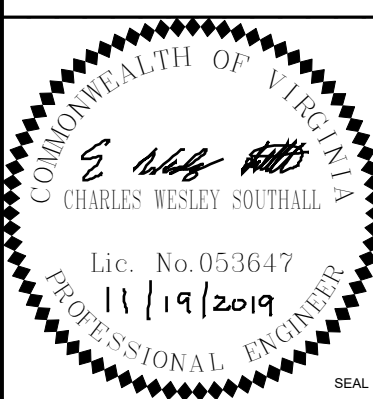
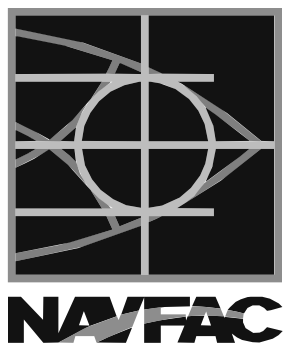
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GENERAL NOTES

1. REFER TO DRAWING M-001 FOR MECHANICAL LEGEND, ABBREVIATIONS AND NOTES.



**BURNS
McDONNELL**
1305 EXECUTIVE BLVD.
SUITE 160
CHESAPEAKE, VA 23320

APPROVED

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ACTIVITY

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CHK		RJ	
PM/DM		CJC/KDB	
BRANCH MANAGER		7/24	
CHIEF ENGI/ARCH		EJA	
FIRE PROTECTION		HPN	

ENGINEERING COMMAND
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